

WIND ENERGY: ADVANTAGES AND DISADVANTAGES

What is wind energy?

Wind, that is moving air, possesses some kinetic energy due to its high speed.

Wind is a result of the solar energy, as heating of land results in movement of air.

The most common application of wind could probably be kite flying! Or even paragliding, sail boats etc. Well these activities are definitely not possible without wind. Ever wondered what else it could be useful for?? Wind energy can be harnessed and used for generating electricity or for other smaller purposes by a windmill. In olden times, windmills were used to draw water out of wells or to grind flour etc. It is the rotatory motion of the shaft in a windmill that is used to rotate the turbine and convert it to the form of energy we need it in.

The main advantage of wind energy is that harnessing it doesn't disrupt natural processes or harm the environment, unlike a lot of other energy sources. To generate electricity on a large scale, a number of windmills are set up over a large area, called a wind energy farm. Such areas need a wind speed of 15kmph.

Lets discuss advantages and disadvantages of Wind Energy.

Advantages of Wind Energy

- 1) Wind Energy is an inexhaustible source of energy and is virtually a limitless resource.
- 2) Energy is generated without polluting environment.
- 3) This source of energy has tremendous potential to generate energy on large scale.
- 4) Like solar energy and hydropower, wind power taps a natural physical resource.
- 5) Windmill generators **don't** emit any emissions that can lead to acid rain or greenhouse effect.
- 6) Wind Energy can be used directly as mechanical energy.
- 7) In remote areas, wind turbines can be used as great resource to generate energy.
- 8) In combination with Solar Energy they can be used to provide reliable as well as steady supply of electricity.
- 9) Land around wind turbines can be used for other uses, e.g. Farming.

Disadvantages of Wind Energy

- 1) Wind energy requires expensive storage during peak production time.
- 2) It is unreliable energy source as winds are uncertain and unpredictable.
- 3) There is visual and aesthetic impact on region.

- 4) Requires large open areas for setting up wind farms.
- 5) Noise pollution problem is usually associated with wind mills.
- 6) Wind energy can be harnessed only in those areas where wind is strong enough and weather is windy for most parts of the year.
- 7) Usually places, where wind power set-up is situated, are away from the places where demand of electricity is there. Transmission from such places increases cost of electricity.
- 8) The average efficiency of wind turbine is very less as compared to fossil fuel power plants. We might require many wind turbines to produce similar impact.
- 9) It can be a threat to wildlife. Birds do get killed or injured when they fly into turbines.
- 10) Maintenance cost of wind turbines is high as they have mechanical parts which undergo wear and tear over the time.

Even though there are advantages of wind energy, the limitations make it extremely difficult for it to be harnessed and prove to be a setback.

Source: <http://www.ianswer4u.com/2012/02/wind-energy-advantages-and.html#axzz3Qs5RV0Sb>

CHAPTER – 1 ENERGY SOURCES

1.1. Introduction to energy

The energy of a body is its capacity to do work. It is measured the total amount of work that the body can do.

Energy is the primary and most universal measure of all kinds work by human beings and nature. Every thing what happens the world is the expression of flow of energy in one of its forms.

1.2. Different forms of energy

The different forms of energy are:

1. Mechanical energy (kinetic and potential)
2. Thermal (or) Heat energy
3. Chemical energy
4. Electrical energy
5. Nuclear energy
6. Electromagnetic energy
7. Gravitational energy

The S.I unit of energy is Joule or KJ or Watt.h.

Primary Energy Sources

Primary energy sources can be defined as sources which provide a net supply of energy Coal, Oil, Uranium etc., are examples of this type. The energy required to obtain these fuels is much use than what they can produce by combustion or nuclear reaction. The supply of primary fuels is limited. It becomes very essential to use these fuels sparingly.

Examples

Coal, natural gas, oil and nuclear energy.

Secondary Energy sources

Secondary fuels produce no net energy. Though it may by necessary for the economy, these may not yield net energy.

Secondary sources are like sun, wind, water (tides), etc. Solar energy can be used through plants, solar cells, solar heaters and solar collectors.

1.3. Energy sources and their availability

Introduction

Today every country draws its energy needs from a variety of sources. We can broadly categorize these sources as commercial and non-commercial. The commercial sources include the fossil fuels (coal, oil and natural gas), hydro-electric power and nuclear power, while the non-commercial sources include wood, animal waste and agricultural wastes. In an industrialized country like U.S.A. most of the energy requirements are met from commercial sources, while in an industrially less developed country like India, the use of commercial and non-commercial sources are about equal.

1.4. Conventional and non-conventional sources energy

1.4.1. Conventional energy sources

Conventional energy sources are:

- (a) Fossil fuel energy
- (b) Hydraulic energy
- (c) Nuclear energy

(a) Fossil fuel energy

Coal, petroleum, and natural gas are called Fossil fuel as these are formed by the decomposition of the remains of dead plants and animals buried under the earth for a long time. These are non-renewable sources of energy, which, if exhausted, can not be replenished in a short time. Their reserves are limited and are considered very precious. These should be used with care and caution to let them last long. These are also contributing to the global environmental pollution.

(i) Coal

Since the advent of industrialization coal has been most common source of energy. In the last three decades, the world switched over from

coal to oil as a Major source of energy because it is simpler and cleaner to obtain useful energy from oil.

Coal is a complex mixture of compounds of carbon, hydrogen and oxygen. Small amounts of nitrogen and Sulphur compounds are also present in coal. It is mainly available in Bihar, West Bengal, Orissa and Madhya Pradesh. The big coal mines in our country are at Jharia and Bokaro in Bihar and at Raniganj in West Bengal. It is considered as the backbone of the energy sector for its use in industry, transportation and electric power generation.

Depending upon the carbon contents of coal, it is classified as follows:

Type of coal	Carbon content (%)
1. Peat	60
2. Lignite (soft coal)	70
3. Bituminous (house hold coal)	80
4. Anthracite (hard coal)	90

It is clear that peat is the most inferior quality of coal. Whereas anthracite is the most superior quality of coal. Bituminous is most commonly used in households and industry.

On strong heating, coal breaks up to produce coal gas, ammonia, coaltar and coke. Coke is 98% carbon, obtained after losing all its volatile constituents during destructive distillation of coal. It can be used as smoke free fuel.

(ii). Petroleum

It is a dark coloured, viscous and foul smelling crude oil. The petroleum means rock oil. It is normally found under the crust of earth trapped in rocks. The crude oil is a complex mixture of several solid liquid gaseous hydrocarbons mixed with water, salt and earth particles. It is a natural product obtained from oil wells.

Some of the crude oil producing locations in our country are:

- (i) Ankleshwar and Kalol in Gujarat
- (ii) Rudrasagar and Lakwa in Assam; and
- (iii) Bombay high (off-shore area)

The oil wells of Bombay high are producing about 22 million tons of crude petroleum oil per year, which is little less than half of the total requirement of the country. The efforts are also being made to search oil well in off-shore deltas to Godavari, Kaveri, and Rajasthan.

The crude petroleum is refined by the process of fractional distillation to obtain more useful petroleum products. The crude petroleum is heated to a temperature of about 400°C in a furnace and vapors thus formed are passed into a tall fractioning column from near its bottom. As the mixture of hot vapours rises in the column, it starts getting cooled gradually.

The products obtained from crude petroleum as follows:

- (i) Petroleum gas (below 40°C) used as LPG.
- (ii) Petrol (40°C to 170°C) for light vehicles.
- (iii) Kerosene (170 to 250°C) for household and industrial use.
- (iv) Diesel oil (250 to 350°C) for heavy vehicles.
- (v) Residual oil ; (a) Lubrication oils (b) paraffin wax and (c) asphalt
- (vi) Fuel oil (350 to 400°C) for boilers and furnaces.

(iii) Natural gas

It consists about 95% Methane and rest ethane and propane. It occurs deep under the crust of the earth either alone or along with oil above the petroleum deposits. It is a product of petroleum mining.

The gas is available in Tripura, Jaisalmer, off-shore areas of Bombay High and in the Krishna – Godavari delta. It is used as a domestic and industrial fuel. The natural gas is now also available as CNG (Compressed Natural Gas) a substitution of petrol in automobiles.

(b) Hydraulic energy (or) Water power

Water power is developed by allowing water to fall under the force of gravity. It is used almost exclusively for electric power generation, in fact, the generation of water power on a large scale became possible around the beginning of the twentieth century only with the development of electrical power plants or Hydro electric plants were usually of small capacities usually less than 100 KW.

Potential energy of water is converted into Mechanical energy by using prime moves known as hydraulic turbines. Water power is quite cheap where water is available in abundance. Although capital cost of hydro electric power plants is higher as compared to other types of power plants but their operating Costs are quite low, as no fuel is required in this case. The development rate of hydropower is still low, due to the following problems:

1. In developing a project, it will take about 6-10 years time for planning, investigation and construction.
2. High capital investment is needed, and some parts of the investment have to be designed from foreign sources.
3. There are growing problems on relocation of villages, involved, compensation for damage, selecting the suitable resettlement area and environmental impact.

Because of long transmission line to the villages with low load factor, the electric power will be available to the people in rural areas may not be economical. This leads to the development of Mini or Micro hydroelectric projects to supply the electric power to remote areas. The Importance of Micro hydroelectric projects have been observed in some parts of the country with availability of river flow through out the year.

In order to reduce the cost of development several Measures have been considered as follows:

- (a) Development of low cost turbines and generators.
- (b) Participation of villages in the development and operation of the project.
- (c) Using the appropriate technology and tolerable substandard requirement and project civil work component at the beginning stage.

(c) Nuclear energy

According to modern theories of atomic structure, matter consists of minute particles known as atoms. Heavier unstable atoms such as U^{235} , Th^{239} , liberate large amount of heat energy. The energy released by the complete fission of one Kg of Uranium (U^{235}), is equal to the heat energy obtained by burning 4500 tonnes of coal (or) 220 tons of oil. The heat produced by nuclear fission of the atoms of fissionable material is utilized in special heat exchangers for the production of steam which is then used to drive turbogenerators as in the conventional power plants.

However there are some limitations in the use of nuclear energy namely high capital cost of nuclear power plants, limited availability of raw materials, difficulties associated with disposal of radio active waste and shortage of well trained personnel to handle the nuclear power plants.

The Uranium reserves in the world at present are small. These reserves are recoverable but are expensive. The presently working power plants are:

1. Tarapur atomic power station in Maharashtra
2. Ranapratap sagar atomic power station near Tota, Rajasthan
3. Kalpakkam atomic power station near Madras, Tamilnadu.
4. Narora atomic power station in U.P.

About 3% of the energy produced in India is obtained from nuclear power plants.

1.4.2. Non-Conventional Energy Sources

The sources of energy which are being produced continuously in nature and are in exhaustible are called renewable sources of energy (or) non-conventional energy.

Some of these sources are:

- (a) Wind energy
- (b) Tidal energy
- (c) Solar energy

(a) Wind energy

Winds are caused because of two factors.

1. The absorption of solar energy on the earth's surface and in the atmosphere.
2. The rotation of the earth about its axis and its motion around the Sun.

A wind mill converts the kinetic energy of moving air into Mechanical energy that can be either used directly to run the Machine or to run the generator to produce electricity.

(b) Tidal energy

Tides are generated primarily by the gravitational attraction between the earth and the Moon. They arise twice a day in Mid-Ocean. The tidal range is only a Meter.

Basically in a tidal power station water at high tide is first trapped in a artificial basin and then allowed to escape at low tide. The escaping water is used to drive water turbines, which in turn drive electrical generators.

(c) Solar energy

Brief history of solar energy (or) Importance of solar energy:

Energy from the sun is called solar energy. The Sun's energy comes from nuclear fusion reaction that take place deep in the Sun. Hydrogen nucleus fuse into helium nucleus. The energy from these reactions flow out from the sun and escape into space.

QUESTIONS

Short Questions

1. Define energy?
2. What are the different forms of energy?
3. What are the compounds present in coal?
4. What products can obtain from crude oil?
5. What are the conventional sources of energy?
6. What are non-conventional sources of energy?

Essay Type Questions

1. What are the Conventional sources of energy and explain briefly?
2. What are the non-conventional sources of energy and explain briefly?

CHAPTER – 2 SOLAR ENERGY

2.1. Introduction

Energy from the sun is called solar energy. The Sun's energy comes from nuclear fusion reaction that takes place deep in the sun. Hydrogen nucleus fuse into helium nucleus. The energy from these reactions flow out from the sun and escape into space.

Solar energy is some times called radiant energy. These are different kinds of radiant energy emitted by sun. The most important are light infrared rays. Ultra violet rays, and X- Rays.

The sun is a large sphere of very hot gases. It's diameter is 1.39×10^6 KM. While that of the earth is 1.27×10^4 KM. The mean distance between the two is 1.5×10^8 KM. The beam radiation received from the sun on the earth is reflected in to space, another 15% is absorbed by the earth atmosphere and the rest is absorbed by the earth's surface. This absorbed radiation consists of light and infrared radiation with out which the earth would be barren.

All life on the earth depends on solar energy. Green plants make food by means of photosynthesis. Light is essential from in this process to take place. This light usually comes from sun. Animal get their food from plants or by eating other animals that feed on plants. Plants and animals also need some heat to stay alive. Thus plants are store houses of solar energy.

The solar energy that falls on India in one minute is enough to supply the energy needs of our country for one day. Man has made very little use of this enormous amount of solar energy that reaches the earth.

2.2 Solar Constant

The sun is a large sphere of very hot gases, the heat being generated by various kinds of fusion reactions. Its diameter is 1.39×10^6 KM. While that

of the earth is 1.27×10^4 KM. The mean distance between the two is 1.50×10^8 KM. Although the sun is large, it subtends an angle of only 32 minutes at the earth's surface. This is because it is also a very large distance. Thus the beam radiation received from the sun on the earth is almost parallel. The brightness of the sun varies from its center to its edge. However for engineering calculations, it is customary to assume that the brightness all over the solar disc is uniform. As viewed from the earth, the radiation coming from the sun appears to be essentially equivalent to that coming from a back surface at 5762°K .

“The rate at which solar energy arrives at the top of the atmosphere is called solar constant I_{sc} ”. This is the amount of energy received in unit time on a unit area perpendicular to the sun's direction at the mean distance of the earth from the sun. Because of the sun's distance and activity vary throughout the year, the rate of arrival of solar constant is thus an average from which the actual values vary up to 3 percent in either direction. The National Aeronautics and Space Administration's (NASA) standard value the solar constant, expressed in three common units, is as follows:

- (i) 1.353 kilowatts per square meter
- (ii) 116.5 Langleys per hour (1 Langley being equal to $1\text{cal}/\text{cm}^2$ of solar radiation received in one day)
- (iii) 429.2 Btu per Sqr.ft. per hour.

The distance between the earth and the sun varies a little through the year. Because of this variation, the extra – terrestrial flux also varies. The earth is closest to the sun in the summer and farthest away in the winter. This variation in the Intensity of solar radiation (I) that reaches the earth. This can be approximated by the equation.

$$\frac{I}{I_{sc}} = 1 + 0.033 \cos \frac{360(n-2)}{365}$$
$$= 1 + 0.033 \cos \frac{360 \times n}{365}$$

where n is the day of the year.

2.3 Solar Radiation at the Earth's Surface

The solar radiation that penetrates the earth's atmosphere and reaches the surface differs in both amount and character from the radiation at the top of the atmosphere. In the first place. Part of the radiation is reflected back in to the space, especially by clouds. Further more, the radiation entering the atmosphere is partly absorbed by molecules in the air. Oxygen and Ozone (O_3), formed from oxygen, absorb nearly all the Ultraviolet radiation, and water vapour and carbon dioxide absorb some of the energy in the infrared range. In addition, part of the solar radiation is scattered (i.e. its direction has been changed) by droplets in clouds by atmosphere molecules, and by dust particles.

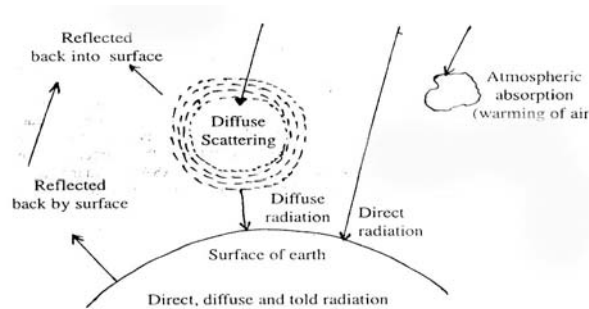


Fig 2.1 Solar Radiation at the Earth's surface

Solar Radiation that has not been absorbed or scattered and reaches the ground directly from the sun is called "Direct Radiation" or Beam Radiation. Diffuse radiation is that Solar Radiation received from the sun after its direction has been changed by reflection and scattering by the atmosphere. Because of the Solar Radiation is scattered in all directions in the atmosphere, diffuse radiation comes to the earth from all parts of the sky. The sum of the beam and diffuse radiation flux is referred to as total or global radiation.

2.4 Instruments for measuring solar radiation and sun shine

Solar Radiation flux is usually measured with the help of a pyranometer or a Pyrheliometer, sunshine recorder is used for measuring sunshine.

2.4.1. Pyranometer

A pyranometer is an instrument which measures either global or diffuse Radiation over a hemispherical field of view. A sketch of the Instrument as

Installed for the measurement of global radiation is shown in fig.2.1. Basically the pyranometer consists of a 'black' surface which heats up when exposed to solar radiation. Its temperature increases until its rate of heat gain by solar radiation equals its rate of heat loss by convection, conduction and radiation. The hot Junctions of a thermopile are attached to the black surface. While the cold Junctions are located in such a way that they do not receive the radiation. As a result, an e.m.f. is generated. This emf which is usually in the range of 0 to 10MV can be read, recorded or Integrated over a period of time and is a measure of the global radiation.

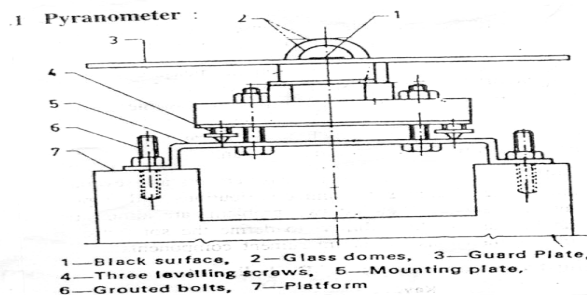


Fig 2.2. Pyranometer

The pyranometer shown in fig 2.2 is used commonly in India it has its hot Junctions arranged in the form of a circular disc of diameter 25MM and is coated with a special black lacquer having a very high absorptivity in the solar wave length region. Two concentric hemispheres, 30 and 50MM in diameter respectively made of optical glass having excellent transmission characteristics are used to protect the disc surface from the weather. An accuracy of about ± 2 percent can be obtained with the Instrument.

2.4.2 Sun Shine Recorder

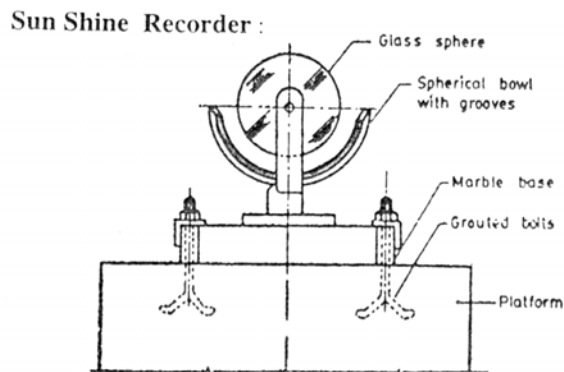


Fig 2.3 Sun Shine Recorder

The duration of bright sun shine in a day is measured by means of a sunshine recorder shown in fig2.3 the sun's Rays are focussed by a glass sphere to a point on a card strip held in a groove in a spherical bowl mounted concentrically with the sphere. Whenever there is bright sunshine, the image formed is intense enough to burn a spot on the cord strip. Though the day as the sun moves across the sky, the image moves along the strip. Thus, a burnt trace whose length is proportional to the duration of sunshine is obtained on the strip. A photograph of the instrument is shown in plate 3 (bottom).

2.5 Solar Energy Utilization – Basic ideas about the pre-historic way of using solar energy

Energy is a common Man's daily commodity: The world energy consumption in 1975 was 8002 million tons of coal equivalent and is expected to shoot up to 27,400 million tons of coal equivalent in the year 2000. It is becoming scarce day by day even then its demand is on the increase. The increased population has led to depletion of energy. The process of mankind has influenced the subsequent exploitation of new sources of energy from time to time. The utilization of coal, the development of hydro electricity, the discovery of oil and gas and the advents of nuclear energy are significantly mile stones in human history. Each new source brought about a preformed change in the life style of the people. Each new source supplemented the other.

The size of the balance of fossil fuels will be over within a hundred years. Hence it is essential to tap the other sources of energy to supplement the existing energy demands of all non-conventional energy source, solar energy holds the greatest promise as it is abundant, renewable and pollution free. Its collection, storage on conversion is also easy. Hence worldwide attention is now focused on various methods of utilization of solar energy.

All life on the earth depends on solar energy. Green plants make food by means of photosynthesis. Light is essential from in this process to take

place. This light usually comes from sun. Animals get their food from plants are store houses of solar energy.

The solar energy that falls on India in one minute, is enough to supply the energy needs of our country for one day. Man has made very little use of this enormous amount of solar energy. That reaches the earth he has used solar energy indirectly, for many thousands of years. Wind mills which are driven by wind that results from infrared solar energy.

2.6 Solar Energy applications

1. Heating and cooling of residential building.
2. Solar water heating.
3. Solar drying of agricultural and animal products.
4. Salt production by evaporation of seawater.
5. Solar cookers.
6. Solar engines for water pumping.
7. Solar Refrigeration.
8. Solar electric power generation.
9. Solar photo voltaic cells, which can be used for electricity.
10. Solar furnaces.

2.6.1 Solar Collectors

A solar collector is a device for collecting solar radiation and transfer the energy to fluid passing in contact with it. Utilization of solar energy requires solar collectors. These are generally of two types.

- (i) Non- concentrating (or) flat plate solar collector.
- (ii) Concentrating (focusing) type solar collector.

The solar energy collector, with its associated absorber, is the essential component of any system for the conversion of solar radiation in to more usable form (e.g heat or electricity). In the non-concentrating type, the collector area is the same as the absorber area. On the other hand, in concentrating collectors, the area intercepting the solar radiation is greater. By means or concentrating collectors, much higher temperatures can be

obtained than with the non-concentrating type. Concentrating collectors may be used to generate medium pressure steam. They use many different arrangements of mirrors and lenses to concentrate the sun's rays on the boiler. This type shows better efficiency than the flat plate type. For best efficiency, collectors should be mounted to face the sun as it moves through the sky.

(i) Flat plate collectors (non-concentrating)

Where temperatures below about 90°C are adequate as they are for space and service water heating flat plate collectors, which are of the non-concentrating type, are particularly convenient. They are made in rectangular panels from about 1.7 to 2.9 sq.m, in area, and are relatively simple to construct and erect. Flat plates can collect and absorb both direct and diffuse solar radiation, they are consequently partially effective even on cloudy days when there is no direct radiation.

Flat plate solar collectors may be divided into two main classifications based on the type of heat transfer fluid used.

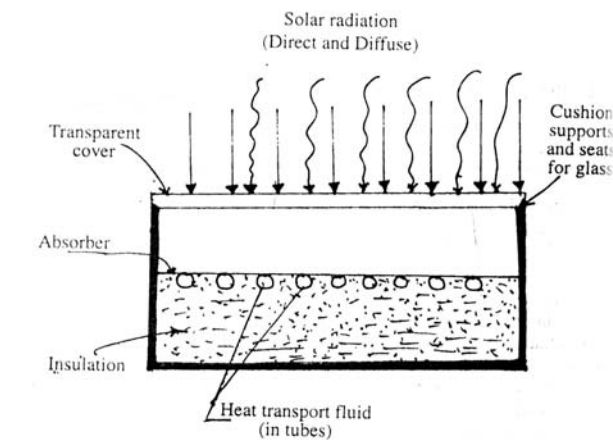


Fig. 2.4. Liquid heating flat plate collector

A. Liquid heating Flat Plate Collector

Liquid heating flat plate collectors are used for heating water and non-freezing aqueous solutions.

There are many flat-plate collector designs, but most are based on the principle shown in fig 2.4. It is the plate and tube type collector. It basically

consists of a flat surface with high absorptivity for solar radiation called the absorbing surface. Typically a metal plate, usually of copper, steel or aluminum material with tubing of copper in thermal contact with the plates are the most commonly used materials. The absorber plate is usually made from a metal sheet 1 to 2 mm in thickness, while the tubes, which are also of metal, range in diameter from 1 to 1.5cm. They are soldered, brazed or clamped to the bottom of the absorber plate with the pitch ranging from 5 to 15 Cm, In some designs, the tubes are also in line and integral with the absorber plate.

The primary function of the absorber is to absorb maximum radiation reaching it through the glazing, to lose maximum heat upward to the atmosphere and down ward through the back of the container and to transfer the retained heat to the working fluid. Black painted absorbers are preferred because they are considerably cheaper and good absorbers of radiation.

Heat is transferred from the absorber plate to a point of use by circulation of fluid (usually water) across the solar heated surface. Thermal insulation of 5 to 10cm. Thickness is usually placed behind the absorber plate to prevent the heat losses from the rear surface. Insulation materials is generally mineral wool or glass wool or fiber glass.

The front covers are generally glass that is transparent to incoming solar radiation and opaque to the infra-red re-radiation from the absorber. The glass covers act as a convection shield to reduce the losses from the absorber plate beneath. The glass thickness of 3 and 4 mm are commonly used. The usual practice is to have 2 covers with specific ranging from 1.5 to 3cm.

Advantages of second glass which is added above the first one are

- (i) Losses due to air convection are further reduced. This is important in windy areas.
- (ii) Radiation losses in the infra-red spectrum are reduced by a further 25%, because half of the 50% which is emitted out wards from the first glass plate is back radiated.

(ii) Concentrating (focusing) type solar collector

Focusing collector or concentrating type solar collector is a device to collect solar energy with high intensity of solar radiation on the energy absorbing surface. Such collectors generally use optical system in the form of reflectors or refractors. A focusing collector is a special form of flat-plate collector modified by introducing a reflecting or refracting surface between the Solar Radiation and the absorber. In these collectors radiation falling on a relatively large area is focused on to a receiver or absorber of considerably smaller area. As a result of the energy concentration, fluids can be heated to temperatures of 500°C or more.

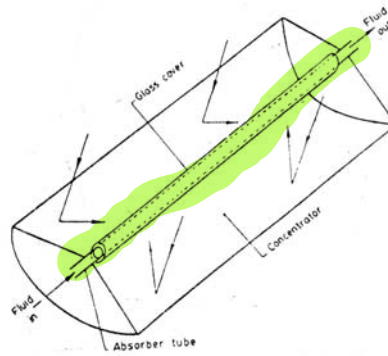


Fig. 2.5 Parabolic concentrating collector

A Schematic diagram of a typical concentrating collector is shown in fig 2.5 the collector consists of a concentrator and a receiver. The concentrator shown is a mirror reflector having the shape of a cylindrical parabola. It focuses the sunlight on to its axis where it is absorbed on the surface of the absorber tube and transferred to the fluid flow through it. A concentric glass cover around the absorber tube helps in reducing the convective and radiative losses to the surroundings. In order that the sun's rays should always be focussed on to the absorber tube, the concentrator has to be rotated. This movement is called tracking in the case of cylindrical parabolic concentrators, rotation about a single axis is generally required. Fluid temperatures up to around 300°C can be achieved in cylindrical parabolic focussing collector system.

The generation of still higher working temperatures is possible by using parabolic reflectors as shown in fig 2.6 which have a point focus.

A paraboloidal dish collector brings solar radiation to a focus at a point actually a small central volume. A dish 6.6m diameter has been made from about 200 curved mirror segments forming a paraboloidal surface. The absorber, located at the focus, is a cavity made of zirconium-copper alloy with a black chrome selective coating. The heat-transport fluid flows into and out of the absorber cavity through pipes bonded to the interior. The dish can be turned automatically about two axes so that the sun is always kept in a line with the focus and the base of the paraboloidal dish. Thus, the can be fully tracked at essentially all times.

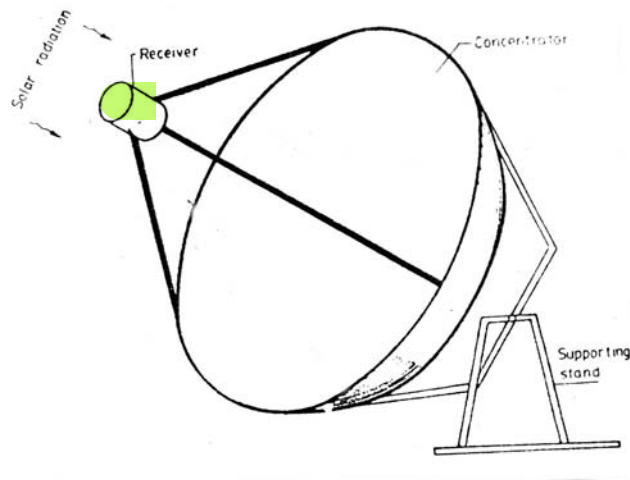


Fig 2.6 Dish Collector (Point Focus)

2.6.3. Solar Cooker

In our country energy consumed for cooking shares a major portion of the total energy consumed in a year. In villages 95% of the consumption goes only to cooking. Variety of fuel like coal, kerosene, cooking gas, firewood, dung cakes and agricultural waste are used the energy crisis is affecting everyone. It is affecting the fuel bills for those who use it for heating the houses and cooking their food. The poor of the developing countries who have been using dry wood, picked up from the fields and forests as domestic fuel, have been affected in their own way, due to scarcity of domestic fuel in the rural areas. At present, fire wood and cow dung too precious to allowed to be used for burning and cooking. It is very useful to improve the fertility of the soil, it should be used in proper way. The supply of wood is also fast depleting because of the indiscriminate felling of trees in the rural areas and the denudation of forests. There is a rapid deterioration in the supply of these

fossil fuels like coal, kerosene or cooking gas. The solution for the above problem is the harnessing of solar energy for cooking purpose.

The most important is that the solar cooker is a great fuel saver. The department of new conventional energy source has calculated that a family using a solar cooker 275 days a year would save 800kgs of fire wood or 65 liters of kerosene. Similarly an industrial Canteen or a Hostel mess using the larger community solar cooker which can cook for 20 to 25 people could save 400kgs of fire wood or 335 liters of kerosene per year.

Types of Solar Cooker

Basically there are three designs of solar cooker;

- (i) Flat plate box type solar cooker with or without Reflector.
- (ii) Multi Reflector type solar cooker.
- (iii) Parabolic disc concentrator type solar cooker.

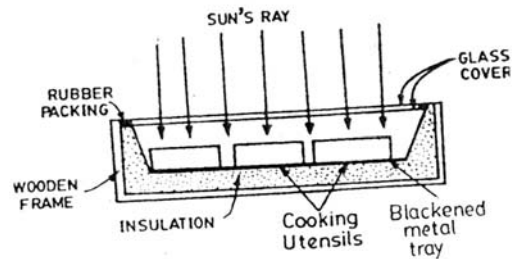


Fig 2.7 Flat Plate box type solar cooker

Flat plate box type design is the simplest of all the designs. This cooker allows solar radiation to enter through a double walled glass cover placed inside a blackened box which is well insulated and made airtight. Maximum no load temperature with a single reflector reaches up to 160°C.

In Multi Reflector type, four square or triangular or rectangular reflectors are mounted on the oven body. They all reflect the solar radiation into the cooking zone in which cooking utensils are placed. Temperature obtained is of the order of 200°C. The maximum temperature can reach to 250°C if the compound cone reflector system is used.

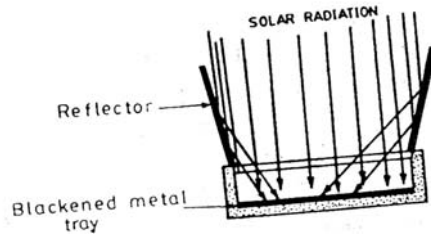


Fig 2.8 (a)

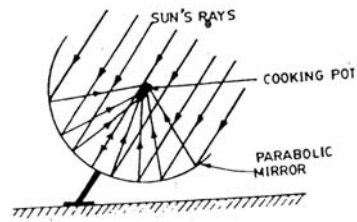


Fig 2.8 (b)

In parabolic type cooker, parallel sun's rays are made to reflect on a parabolic surface and concentrated on a focus on which the Utensils for cooking are placed. The temperature of the order of 450°C can be obtained in which solar radiation are concentrated on to a focal point. Principle of operations of solar cookers is shown in fig 2.8 (b).

Design principle and constructional details of a box type Solar Cooker

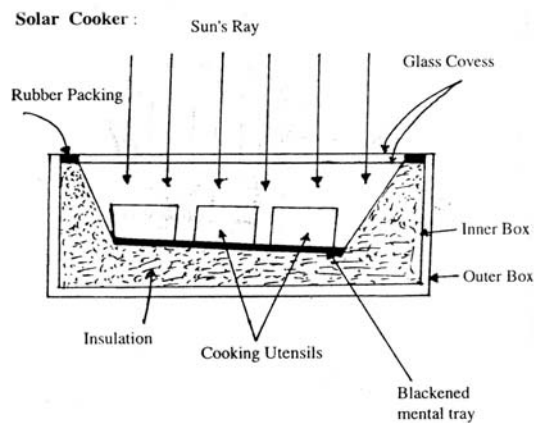
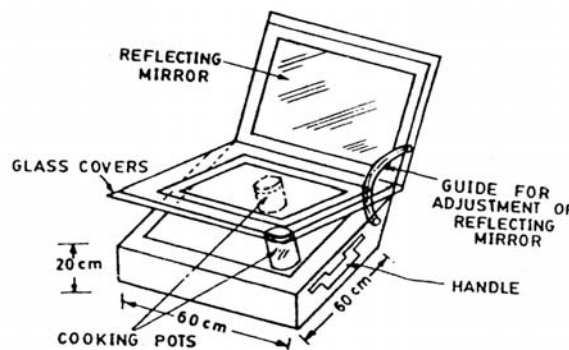


Fig 2.9 Box Type Solar Cooker

The principal of operation of box type solar cooker is illustrated in fig2.9. The solar rays penetrate through the glass covers and absorbed by a blackened metal tray kept inside the solar box. The solar radiation entering the box are of short wave length. Two glass covers are provided to minimize the heat loss. The loss due to convection is minimized by making the box air tight by providing a rubber strip all around between the upper lid and the box. Insulating materials like glass wool, saw dust or any other material is filled in the space between blackened tray and outer cover of the box. This minimize heat loss due to conduction with type of cooker is placed in the sun the

blackened surface starts absorbing sun rays and temperature inside the box starts rising. The cooking pots, which are also blackened are placed inside with food material get heat energy and food will be cooked in a certain period of time depending upon the intensity of solar radiation and material of insulation provided. The amount of solar radiation intensity can be increased by provided mirror or mirrors. The solar cooker is made up of inner and outer metal or wooden box with double glass sheet on it. Absorber tray or blackened tray is painted black with suitable black point like black board point. This point is dull in colour and can with stand the maximum temperature attained inside the cooker as well as water vapour coming out of the cooking utensils. The top cover contains 3mm thick two plain glasses fixed on wooden or metal frame, keeping about 25mm distance between the two. The entire top cover can be made tight with padlock hasp. Neoprene rubber sealing is provided around the contact surfaces of the glass cover and hinged on one side of the glass frame. A mechanism (guide for adjusting mirror) is provided to adjust the reflector at different angles with the cooker box when the reflector is adjusted to shine in the cooker box, 115°C to 125°C. Temperature is achieved inside the cooker box. Addition of the reflector is useful in cooking earlier particularly in winter. The solar cooker is able to cook about 1.25kg dry food materials, which is enough for a family of 5 to 7 persons. The total weight of the cooker is about 22kgs. Overall dimensions of a typical model are 60x60x20cm height.



Merits and limitations of a Solar Cooker

Following are the some merits of a solar cooker;

- (1) No attention is needed during cooking as in other devices.
- (2) No fuel is required.

- (3) Negligible maintenance cost.
- (4) No pollution.
- (5) Vitamins of the food are not destroyed and food cooked is nutrition and delicious with natural taste.
- (6) No problem of over flowing of food.

Limitations of a Solar Cooker are

- (i) One has to cook according to the sunshine, the menu has to preplanned.
- (ii) One cannot cook at short notice and food can not be cooked night or during cloudy days.
- (iii) It takes comparatively more time.
- (iv) Chapatias are not cooked because high temperature for baking is required.

2.6.4 Solar Water heater

It is a device to heat water using solar energy. Solar water heaters are one of the best options to be adapted in the developing country. Solar water heating systems are commercially produced in the country. Most of the systems available in India are designed to give water temperature from 60 to 90°C. These are suitable for pre heating feed water to boiler and processing industries and hot water application in Hotels, Bakeries, Industries etc.

The term solar water heater includes conventional flat plate collector with either thermosiphon or forced circulation flow system. A solar water heater normally consists of the following components:

- (a) A flat plate collector to absorb solar radiation and convert it into thermal energy.
- (b) Storage tank to hold water for use and cold water feeding the flat plate collector.
- (c) Connecting pipes inlet and out let, for feeding cold water from the storage tank and taking hot water to the storage tank or point of use.

A diagram of a simple, small capacity, natural circulation system suitable for domestic purpose is shown in fig 2.11. The two main components

of the storage tank, the tank being located above the level of the collector. As water in the collector is heated by solar energy, it flows automatically to the top of the water tank and its place is taken by colder water from top of the tank. Whenever this is done, cold water automatically enters at the bottom. An auxiliary heating system is provided for use on cloudy or rainy days.

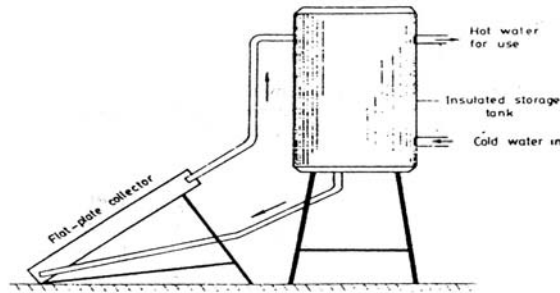


Fig 2.11 Solar Water Heater

Typically, such system have capacities ranging from 100 to 200 liters and adequately supply the needs of a family of four or five persons. The temperature of the hot water delivered ranges from 50 to 70°C. Solar water heating is a good example to illustrate one of the assets of the direct use of solar. This is the possibility of matching the temperature achieved in the heating device with the temperature required for end use. As a result of this matching, the thermodynamic efficiency based on considerations of availability of energy can be shown to be higher in the case of solar water heating system than a water heating system using natural gas or electricity.

Solar water heaters of the natural circulation type were used fairly widely from the beginning of the twentieth century till about 1940 until cheap oil and natural gas became available. Now they are a being installed again. They are in wide spread use in countries like Israel, Australia and Japan.

2.6.5. Solar distillation

Fresh water is a necessity for the sustenance of life and also the key to Man's prosperity. It is generally observed that arid, semi arid and coastal areas which are thinly populated and scattered, one or two family members are always busy in bringing fresh water from a long distance. In these areas

solar energy is plentiful and can be used for converting saline water into distilled water. The pure can be obtained by distillation in the simplest solar still, generally known as the “basin type solar still”.

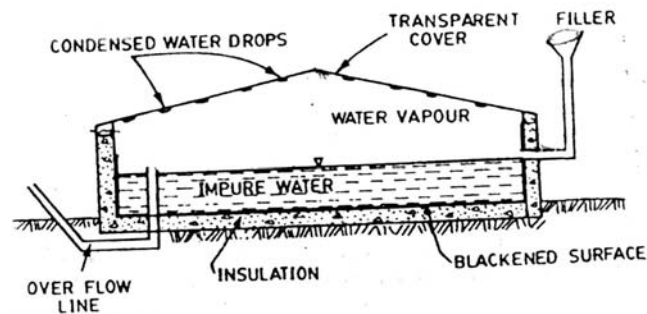


Fig 2.12 Solar Distillation

Solar water still is shown schematically in fig 3.1, it consists of a blackened basin containing saline water at a shallow depth, over which is a transparent air tight cover that encloses completely the space above the basin. It has a roof-like shape. The cover which is usually glass may be of plastic, is sloped towards a collection through solar radiation passes through the cover and is absorbed and converted into heat in the black surface. Improve water in the basin or tray is heated and the vapor produced is condensed to purified water on the cooler interior of the roof. The transparent roof material transmits nearly all radiation falling on it and absorbs very little; hence it remain cool enough to condense the water vapor. The condensed water flows down the sloping roof and is collected in through at the bottom. Saline water can be replaced in the operation by either continuous operations or by batches.

2.6.6 Solar Pumping

Solar pumping consists in utilizing the power generated by solar energy for water pumping useful for irrigation.

Solar energy offers several features that make its utilization for irrigation pumping quite attractive, first, the greatest need for pumping occurs during the summer months when solar radiation is greatest second, pumping can be intermittent to an extent, during periods of low solar radiation when

pumping decreases, evaporation losses from crops are also low. Finally relatively in expensive pumped storage can be provided in the forms of bonds.

A number of recently constructed solar irrigation pump installations are now operational. The major obstacle to increase use of solar irrigation system at this time is their Relatively high capital cost. If the costs of solar pumps can be substantially reduced and assuming that conventional fuel costs continue to rise, solar; irrigation could become economical, and increased use of such system might be anticipated in future.

The basic system consists of the following components:

- (1) The solar collector.
- (2) The heat transport system.
- (3) Boiler or Heat exchanges.
- (4) Heat engine.
- (5) Condenser.
- (6) Pump

The solar pump is not much different from a solar heat engine working in a low temperature cycle. The sources of heat is the solar collector, and sink is the water to be pumped. A typical solar powered water pumping system is shown in fig2.3.

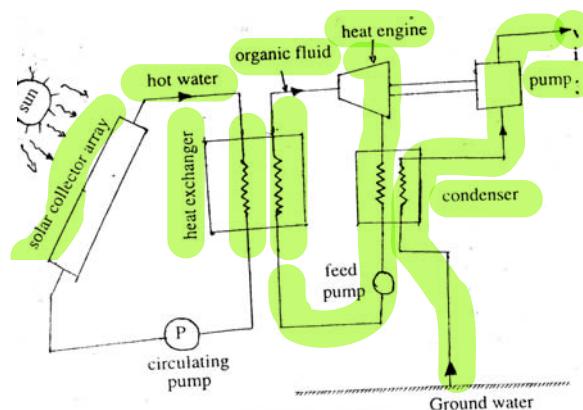


Fig 2.13 Solar Water Pumping

The primary components of the system are an array of flat-plate collectors and an Rankine engine with an organic fluid as the working

substance. During operation a heat transfer fluid (Pressurized water) flows through the collector arrays. Depending upon the collector configuration, solar flux and the operating conditions of the engine, the fluid will be heated in the collector to a higher temperature, the solar energy which is thus converted to the thermal energy. The fluid (water) flows into a heat exchanger (boiler), due to temperature, gradient, and comes back to the collector. This water yields its heat to an intermediate fluid in the boiler. This fluid evaporates and expands in the engine before reaching the condenser, where it condenses at low pressure. The condenser is cooled by the water to be pumped. The fluid is then reinjected in the boiler to close the cycle. The expansion engine or Rankine engine is coupled to the pump and it could of course be coupled to an electric generation.

2.6.7 Electricity from Solar Energy

Electricity energy is the most convenient form of energy. It is easy to use, transport, control and transform into other forms of energy.

Modern society has an insatiable hunger for energy. This need for energy will continue to increase as the newly developing countries become more industrialized and the mature nations increase their scope of Mechanization. To satisfy this need, vast quantities of coal and petroleum products are required. More recently, the advent of nuclear energy has added vast quantities for future needs.

Although sun is the ultimate source of all the power which man has at his disposal, the conversion of solar radiation directly into electrical power by some cheap and efficient means has been sought for several decades. Many different methods have been tried for this purpose, but none of these could compete with conventional fossil fuel or hydro electric power plants.

2.6.8 Solar Photo Voltaics

The direct conversion of solar energy into electrical energy by means of the photo voltaic effect, that is the conversion of light (or other electromagnetic radiation) into electricity. The photo voltaic effect is defined

as the generation of the electromotive force as a result of the absorption of ionizing radiation energy conversion devices which are used to convert sun light to electricity by the use of the photo voltaic effects are called solar cells. A single converter cell is called a solar cell or more generally, a photo voltaic cell, and combination of such cells, designed to increase the electric power out put is called a solar module or solar array.

Photo voltaic cells are made of semi conductors that generate electricity when they absorb light. As photons are received, free electrical changes are generated that can be collected on contacts applied to the surface of the semi conductors. Because solar cells are not heat engines, and therefore do not need to operate at high temperatures, they are adopted to the weak energy flux of solar radiation, operating at some temperature. These devices have theoretical efficiencies of the order of 25 percent. Actual operating efficiencies are less than this value, and decrease fairly rapidly with increasing temperature.

The best known applications of photo voltaic cells for electrical power generation has been is spacecraft, for which the Silicon cell is the most highly developed type. The Silicon cell consists of a single crystal of silicon into which a doping material is diffused to form a semi conductor. Since the early day of solar cell development, many improvements have been manufactured with areas $2 \times 2 \text{ cm}$, efficiencies approaching 10 percent, and operating at 28°C . The efficiency is the power developed per unit area of array divided by the solar energy flux in the free space (1.353 KW/m^2).

For terrestrial applications, silicon solar cells have shown operating efficiencies of about 12 to 15 percent. Though Silicon is one of the earth's most abundant materials, it is expensive to extract (from sand, where it occurs mostly in the form SiO_2) and refine to the purity required for solar cells. The greater barrier to solar cell application lies in the costs of the cells themselves. Reducing the cost of Silicon Cells is difficult because of the cost of making single crystal. One very promising method is being developed to produce continuous thin ribbons of single-crystal Silicon to reduce fabrication costs.

Cells made from the ribbon have so far shown efficiencies of around 8 percent. Several other kinds of photo cells are in the laboratory stage of development. Cadmium Sulfide cells are other possibilities. So far, efficiencies have been in the range of 3 to 8 percent and these cells have been less durable than Silicon cells owing to degradation with exposure to Oxygen, water vapor and sunlight, especially at elevated temperatures. The active part of the Cadmium Sulfide cds cell is a thin polycrystalline layer of cds, about $10\mu\text{m}$. Thick on which a layer of Cu_2S compound perhaps $0.1\mu\text{m}$ thick is grown. These cells can be made by deposition on long sheets of substrates, a process that might be adaptable to expensive mass production.

Photo voltaic cells could be applicable to either small or large power plants, since they function well on a small scale, and may be adaptable to local energy generation on building roof tops. The cost of the energy storage and power conditioning equipment might, however, make generation in large stations the most economical method; solar cells have also been used to operate irrigation pumps, navigational signals high way emergency call system, rail road crossing warnings, automatic meteorological stations, etc; in location where access to utility power lines is difficult.

2.6.9 Applications of Solar Photo-Voltaic System in Rural Areas

A variety of PV system configurations have been developed and deployed for rural applications such as drinking water supply, street lighting irrigation water pumping and for operation of electronic equipment's. Government of India is sponsoring, a program for Popularizing solar lighting, solar water pumping etc; by providing capital subsidies and concessional interest on borrowed capital.

Solar lighting

Electricity for lighting during night is one of the most convenient and preferred form of energy. However in our Country out of 6 Lacks villages, 1 lakh villages are still to be electrified. Even in electrified villages, only a quarter of house-holds have proper connection. Owing to power shortage in

many stages, electricity supply situation in villages is precarious and mostly it is not available even in electrified villages for lighting when it is most needed.

The bulk of rural house-holds in India, normally use kerosene lanterns lanterns for meeting their lighting requirements. It is estimated that around 100 million kerosene lanterns are used in India. These lanterns provided insufficient and poor quality of light.

For village lighting, three major system Configuration are available;

- (1) Domestic lighting system or solar lantern.
- (2) Pole mounted stand alone street lighting system.
- (3) Non-grid interactive centralized lighting system.

(i) Solar lantern

Solar photo-voltaic powered lights called lanterns fig 2.14(a) are considered to be alternative solution to village lighting needs. A typical solar lantern consists of small photo-voltaic module, alighting device, a high frequency investor, battery charge controller and appropriate housing. During day time, module is placed under the sun and is connected to lantern through cable for charging a typical lantern uses a 10 watt lamp. The expected life of the lamp is 3 to 5 years.

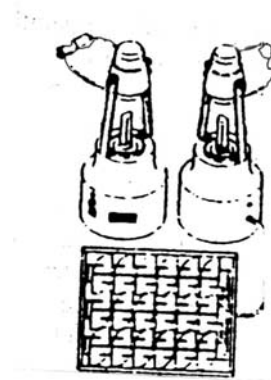


Fig 2.14(a)

Storage battery is one crucial component in lantern, Recombinant maintenance free absorbed electrotpe batteries are being used. The battery has a life of 3 to 5 years. Sealed nickel Cadmium battery is a good option considering their deep discharge characteristics.

It is important to have reliable electronics to operate the lamp and provide suitable protection. A high freequency investor is being used to excite compact fluorescent lamp and a charge controller which protect battery from over charing.

(ii) Street lighting system

It consists of two photo-voltaic modules, mounting frame, 4m long pole, battery box, tubular type lead-acid battery, charge controller, inverter and day light senses fig 2.14(b) Time module sensing is used to switch on lights on the evening. It works for one fluorescent tube lights of 20 watts for whole night.

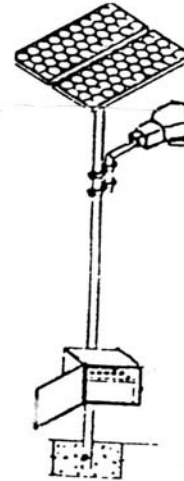


Fig. 2.14(b)

QUESTIONS

Short Answer type questions

- (1) Define solar constant?
- (2) Define solar energy?
- (3) What is diffuse radiation?
- (4) What is global radiation?
- (5) What are the instruments used for measuring solar radiation and sunshine?
- (6) What are the applications of solar energy?
- (7) What are the different types of solar cooker?
- (8) Explain the maintenance of a solar cooker?
- (9) What are the components of solar water heater?
- (10) What are the basic components of solar water pumping?
- (11) Define photo voltaic effect?
- (12) What are the different applications of solar photo voltaic system in Rural area?

Essay type questions

- (1) Explain Pyrometer with the help of neat sketch?
- (2) Explain basic ideas about the Pre-historic ways of using solar energy?
- (3) Explain any one of the solar collectors with the aid of neat sketch?
- (4) What is the importance of solar cooker?

- (5) Explain the design principle of constructional details of a box type solar cooker?
- (6) Explain briefly about solar water pumping with the help of a neat sketch?
- (7) Describe briefly about photo voltaic system?
- (8) Explain briefly about the applications of solar photo voltaic system in Rural Areas?
- (9) Explain briefly about solar water heaters with the help of a neat sketch?

CHAPTER – 3 WIND ENERGY

Wind energy

Winds are caused because of two factors.

- (1) The absorption of solar energy on the earth's surface and in the atmosphere.
- (2) The rotation of the earth about its axis and its motion around the sun.

Because of these factors, alternate heating and cooling cycles occur, differences in pressure are obtained, and the air is caused to move. The potential of Wind energy as a source of power is large. This can be judged from the fact that energy available in the wind over the earth's surface is estimated to be 1.6×10^7 K.W Besides the energy available is free and clean.

The problems associated with Utilizing wind energy are that:

- (i) The energy is available in dilute form, because of this conversion machines have to be necessarily large.
- (ii) The availability of the energy varies considerably over a day and with the seasons.

For this reason some Means of storage have to be devised if a continuous supply of power is required.

A wind mill converts the kinetic energy of moving air into mechanical energy that can be either used directly to run the machine or to run the generator to Produce electricity.

Classification of wind mills



(a) Horizontal Wind mills

i. Horizontal Axis single blade Wind mills

If extremely long blades are mounted on rigid hub. Large blade root bending moments can occur due to tower shadow, gravity and sudden shifts in wind directions on a 200ft long blade. Fatigue load may be enough to cause blade root failure.

To reduce rotor cost, use of single long blade centrifugally balanced by a low cost counter Weight as shown in fig3.1. The relatively simple rotor hub consists of a Universal Joint between the rotor shaft and blade allowing for blade. This type of hub design contains fewer parts and costs less.

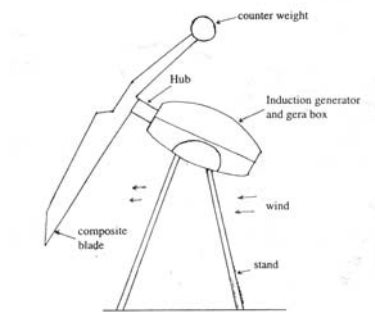


Fig 3.1 Single blade wind mill

(ii) Horizontal axis – two bladed wind mills

In this arrangement rotor drives generator through a step-up gear box. The components are mounted on a bed plate which is mounted on a pintle at the Top of the tower. The two blade rotor is Usually designed to be oriented down wind of the tower. The arrangement of all the Components used in horizontal axis wind mill is shown in fig3.2.

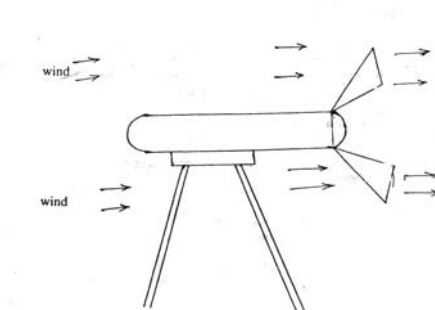


Fig 3.2 Double blade wind mill

When the machine is operating its rotor blades are continuously flexed by Unsteady aerodynamic, gravitational and inertial loads. If the blades are metal, flexing reduces their fatigue life. The tower is also subjected to unsteady load and dynamic interactions between the components of the machine-tower system can cause serious damage.

(iii) Horizontal Axis – Multi bladed Wind Mills

This type of wheel have narrow rims and Wire spokes. The wire spokes support light weight aluminum blades. The rotors of this design have high strength to weight ratios and have been known to survive hours of free wheeling operation in 100kmph winds. They have good power Co-efficient, high starting torque and added advantage of simplicity and low cost.

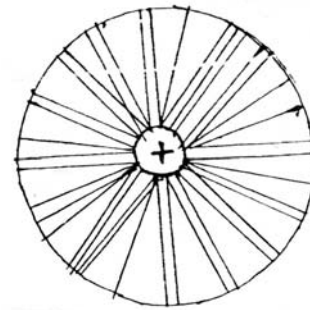


Fig. 3.4 Multi bladed wind mill

(b) Vertical Wind Mills

Wind turbines mounted with the axis of rotation in a vertical Position have advantage that they are omni-directional that is, they need not to be turned to Force the wind. The Vertical mounted Wind Machines eliminates the need for some of the complex mechanical devices and control systems necessary for horizontal mounted wind Machines.

Two types of vertical axis Wind Machines have received attention. The Darrieus rotor consists of two or three convex metal blades with an air foil cross section, mounted on a Central shaft which is supported by bearings at the top and bottom. The rotor assembly is held in position by guy wires running from the top of the rotor to the ground.

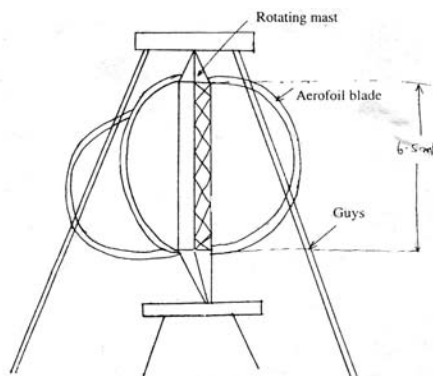


Fig 3.5 Vertical wind mill (Darrieus rotor)

The savonius rotor consists of a long solid's- shaped surface mounted to turn at the center of 's' the savonius rotor is self starting and has an efficiency of about 31% while the Darrieus rotor has a slightly higher efficiency of 35% but is not self starting.

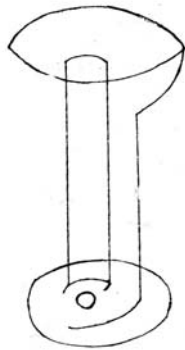


Fig 3.6 (a) Savonius rotor

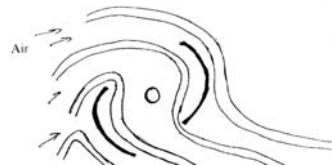


Fig 3.6 (b) Air flow through a savonius rotor

Advantages of Wind energy

- (1) The wind energy is free, inexhaustible and does not need transportation.
- (2) Wind mills will be highly desirable and economical to the rural areas which are far from existing grids.
- (3) Wind power can be used in combination with hydroelectric plants. Such that the water level in the reservoir can be maintained for longer periods.

Disadvantage of Wind energy

- (1) Wind power is not consistent and steady, which makes the complications in designing the whole plant.
- (2) The wind is a very hazard one. Special and costly designs and controls are always required.
- (3) The cost factor, which has restricted the development of wind power in large scale for feeding to the existing grid .
- (4) It has low power coefficient.
- (5) Careful survey is necessary for plant location.

Wind Energy

Questions

(a) Short Answer Questions

- (1) How the Wind mills are classified?
- (2) What are the advantages of Wind Power?
- (3) What are the disadvantages of Wind Power?

(b) Essay type Questions

- (1) Explain briefly about the horizontal Wind mills with neat sketches?
- (2) Explain briefly about Vertical Wind mills with neat sketches?

CHAPTER – 4 BIO ENERGY

4.1 Introduction

Bio gas is generated through a process of anerobic digestion of Bio Mass. Bio Mass is organic matter produced by plants, both terrestrial (those grown on land) and aquatic (those grown in water) and their derivatives. It includes forest crops and residues, crops grown especially for their energy content on “energy farms” and animal manure. Unlike coal, oil and natural gas, which takes Millions of years to form, bio mass can be considered as a renewable energy source because plant life renews and adds to it self energy year. It can also be considered a from of solar energy as the latter is used indirectly to grow these plants by photosynthesis.

Bio Mass means organic matter and Photo Chemical approach to harness solar energy means harnessing of solar energy by photo synthesis. Solar energy is stored in the from of chemical energy. Hence solar energy → Photosynthesis → Bio Mass->energy generation.

Out of several sources of renewable energy like solar, wind, tidal, wave energy, Geothermal energy, nuclear energy, energy through bio mass are important features in our Country. Biogas Production Technology contributes in following ways.

Advantages of Bio Gas technology

- (1) It provides a better and cheaper fuel cooking, lighting and for power generation.
- (2) It produces good quality, enriched manure to improve soil fertility.
- (3) It proves an effective and convenient way for sanitary disposal of human excreta, improving the hygienic conditions.
- (4) It generates social benefits such as reducing burden on forest for meeting cooking fuel by cutting of tree for fuel wood, reduction in the drudgery of women and children etc.

- (5) As a smokeless domestic fuel, it reduces the incidence of eye and lung diseases.
- (6) It also helps in generation of productive employment

Bio Gas and its Compositions

Bio gas contains 55-70% methane and 30-45% carbon dioxide as well as small quantities of (N_2 , H_2 , H_2S) some gases. It is lighter than the air and has an ignition temperature of approximately $700^\circ C$. The temperature of the flame is $870^\circ C$. its calorific value is approximately 4713 kcal/m^3 . The methane content of bio gas produced from different feed stock is given in Table .1.

Content of Methane in Bio-gas produced from different feed stocks

S.No.	Feed Stock	Content of Methane in Bio gas in Percentage
1.	Cattle Manure	54-56
2.	Pig Manure	57
3.	Poultry Manure	55
4.	Farm yard Manure	55
5.	Straw	55
6.	Grass	60
7.	Leaves	58
8.	Kitchen Waste	50-52
9.	Human excreta	60

The bio gas system is most suitable technology to solve the energy problems in rural areas, as it Produces Manure, clean fuel and improves rural sanitation. It's thermal energy Per Unit volume is sufficient to meet domestic energy needs the comparison of bio gas with other fuels is given in table II.

Comparison of Bio-gas with other fuels

S.No.	Name of fuel and Unit	Calorific value (K Cal)	Mode of Burning
1.	Gobar (M^3)	4713	Standard burner
2.	Kerosene (lit)	9122	Pressure stove
3.	Fire wood (Kg)	4700	Open chulla
4.	Cow-dung cake (Kg)	2092	Open chulla
5.	Charcoal (Kg)	6930	Open chulla
6.	Soft cake (Kg)	6292	Open chulla

S.No.	Name of fuel and Unit	Calorific value (K Cal)	Mode of Burning
7.	Butane (LPG) Kg	10882	Standard burner
8.	Coal gas (M ³)	4004	Standard burner
9.	Electricity (Kwn)	860	Hot plate

4.2.2 History of Bio-gas

In the field of an aerobic digestion of waste Material India and China are recognized world leaders. This practice is based upon an age-old tradition of composting human, animal and plant wastes to produce and organic fertilizer. In fact bio gas programme has been recognized for making available a clean and efficient fuel for cooking, lighting, engine and the liquid slurry as a fertilizer and soil conditioner all over the World. Since the 1920's there have been sporadic attempts made to recover bio gas from sewage wastes and animal dung. A few bio gas plants for sewage disposal and bio gas recovery were installed in Europe and the U.S.A. in 1920's and 1930's respectively.

Although the Chinese have been experimenting with bio gas since the 1950's and China had its first bio gas Plant in 1936, yet it was reported only in the 70's in Sichvan. Mainly, there was a movement to extend the practice and reproduce the digester in a large way throughout the country side.

The history of bio gas from cattle dung in India goes back to 1939 when Indian Agricultural Research Institute (ARI) New Delhi, was able to ferment cattle dung to produce Methane.

Subsequently Prof N.V.Joshi; at Poona patented a model of bio gas plant in 1945. Later on Jashbhai, J.Patel evolved a simple model called "Gramalaxmi" and patented it in the year 1951. Mean while, J.J.Patel continued his efforts to simplify his eariler patented. Model, which resulted in the development of two chamber digester with central guide for free up and down movement of steel gas holder. This design was accepted by the Khadi and Villages Industries Commission (KVIC) Bombay, in India during 1962. Subsequently, many Scientists, Engineers, Government, Semi-Government

and different social organisations contributed to the development of bio-gas technology.

The Planning Research and Action Division (PRAD) of State Planning Institute, Government of Uttar Pradesh, Lucknow developed a cheap and convenient model of bio gas in 1957, and called it Janata bio gas plant. After independence the Ministry of Agriculture, Government of India, was looking the bio-gas programme, related development and extension activities. K.V.I.C and state Government's continued the programme in 1970-1980. The national Project on Bio-gas development was launched in 1980-81 under Ministry of Agriculture. Currently Department of non-conventional energy source(DNES), the Ministry of Power and Non-Conventional energy is also actively involved in supporting research and diffusion of bio gas technology and many institutions and State Agricultural Universities are supporting research programme on bio gas technology.

4.2 Process of Bio gas, generation – Wet Process, dry Process

(i) Wet process

(a) Anaerobic digestion

Bio gas is produced by the bacterial decomposition of wet sewage sludge, animal dung or green plants in the absence of oxygen. Feed stocks like – wood shavings, straw, and refuse maybe used, but digestion takes much longer. The natural decay process 'anaerobic decomposition' can be speeded up by using a thermally insulated, air-tight tank with a stirrer Unit and heating system. The gas collect in the digester tank above the slurry and can be piped off continuously. At the optimum temperature (35°C) complete decomposition of animal or human farces takes around 10 days. Gas yields depend critically on the nature of the waste-Pig manure, for example, is better than cow dung or house hold refuse. The residue left after digestion is valuable fertilizer. It is also rich in protein and could be dried and used as animal feed-supplement.

(b) Fermentation

As stated, ethanol or ethyl alcohol is produced by the Fermentation of sugar solution by natural yeast's. After 30 hours of fermentation the brew or

'beer' contains 6-10% alcohol and this can readily be removed by distillation. Traditionally, the fibrous residues from plant crops like sugar cane bagasse have been burnt to provide the heat. Suitable feed stocks include crushed sugar cane and beet, fruit etc sugar can also be manufactured from vegetable starches and Cellulose, Maize, Wheat grain, or Potatoes. For they must be ground or pulped and then cooked with enzymes to release the starch and convert it to fermentable sugars. Cellulose materials like wood, paper waste or straw, require harsher pre-treatment typically milling and hydrolysis with hot acid. One tonne of sugar will produce up to 520 liters of Alcohol, a tonne of grain, 350 liters and a tonne of wood, an estimated 260 to 540 liters. After fermentation, the residue from grains and other feed stocks contains high protein content and is a useful cattle-feed supplement.

The hydrolysis and distillation step require a high energy input; for woody feed stocks direct combustion or pyrolysis is probably more productive at present, although stem treatment and new low-energy enzymatic hydrolysis techniques are under development. The energy requirement for distillation is also likely to be cut drastically. Alcohol can be separated from the beer by many methods which are now under intensive development. These include solvent extraction, reverse osmosis, molecular sieves and use of new desiccants for alcohol drying. It may soon be possible to have the energy required for alcohol production to produce a greater net energy gain.

(iii) Dry Process

Pyrolysis

A wide range of energy-rich fuels can be produced by roasting dry woody matter like straw and wood-chips. The material is fed into a reactor vessel or retort in a pulverized or shredded form and heated in the absence of air. As the temperature rises the cellulose and lignin break down to simpler substances which are driven off leaving a char residue behind. This process has been used for centuries to produce charcoal.

The end products of the reaction depend critically on the conditions employed, at lower temperatures around 500°C, organic liquid predominate, whilst at temperatures nearer 1000°C combustible mixture of gases results.

4.3 Raw Materials available for Bio gas Fermentation

(i) Manure: This kind of Material is available from animal waste and poultry waste. Their compositions vary with the feeding stuff. In Chinese rural areas, the fresh manure is used to be fermented with shorter retention time (about two months). The manure from human, cattle, chicken and pigs are subject to inhabitation when fermented without other Carbon source. Manures can be used as Bio mass.

(ii) Plants: Most of the Plants of the grass family can be used for fermentation. Both aquatic and terrestrial plants can be used as Bio mass. As such they are difficult to be biodegraded. In order to ferment them more easily, pretreatment and a longer retention time are generally needed. Adequate amount of nitrogen rich elements urea etc; should be added for the fermentation.

(iii) Industrial Organic Waste Water: Industrial effluents from food processing, biochemical pharmacy, paper making etc; can be treated by anaerobic fermentation. Most organic matters of these liquid are soluble and their composition are more stable than that of agricultural wastes, while the water quantities may fluctuate. Some effluents may be nutrient deficient or even toxic. Thus pretreatment, such as the elimination of toxicity, adding nutritional additives and so on, are necessary for anaerobic digestion. In food processing Industries plenty of organic waste will be available. By utilizing this organic waste we can produce Bio gas by anaerobic digestion.

(iv) Organic Matter in Municipal Wastes: They are in solid and inorganic contents need separation. Organic matter is available in Municipal wastes. By utilizing this organic matter. Bio-gas can be produced.

4.4 Selection of site for Installation of a Bio gas plant

Careful selection of the best site for installation of a bio gas unit should be done for every beneficiary. The following points should be taken into consideration.

- (1) The distance between the plant and site of gas consumption or kitchen should be less to Minimize cost on gas pipe line and gas leakage. For a plant of capacity 2m^3 , the optimum distance is 10 Cm.
- (2) It should be near the cattle-shed to minimize the distance for carrying cattle dung and transportation cost.
- (3) There should be enough space for storage of digested slurry or construction of compost pit.
- (4) It should be 10 to 15 meters away from any drinking water well to prevent contamination of water.
- (5) The area should be free from roots of trees which are likely to creep into the digester and cause damage.
- (6) It should be open to receive the Sun's rays for most part of the day and to keep the plant in warm. The sunlight should fall on the plant as temperature between 15°C to 30°C is essential for gas generation at good rate.
- (7) It should be on an elevated area so that the plant does not get submerged during normal rains.
- (8) Sufficient space must be available for day to day Operation and Maintenance. As a guide line 10 to 12 M^2 area is needed per M^3 of the gas.
- (9) Plenty of water must be available as the Cow dung slurry with a solid concentration of 7% to 9% is used.

4.5 Materials required for the Construction of bio gas plant

The Materials required for the construction of bio gas plant are listed in table1, however, the quantity of material will depended on type and size of the bio gas plant. The quality of the construction material should be ensured before use.

Construction material for bio gas plant

S.No.	Material	Quality needed
1.	Bricks	Good quality bricks preferably Machine made first class bricks with uniform size and shape.
2.	Cement	Pure port and land with no impurities packed in Polythene bags.
3.	Sand	Fine and Coarse with good quality.

S.No.	Material	Quality needed
4.	Steel for Bigger size plant.	Standard
5.	Concrete	Hard with out any impurities
6.	Reinforced concrete for larger tank	Having good quality composition of cement, sand and water
7.	Asbestos - Cement pipes for inlet and out let.	Good qualities with no leakage and should have uniform diameter and length as required.
8.	Ferro – Cement	Having few layers of iron wire Mesh plastered with diameter and length as required.
9.	Bricks Ferro – Cement.	Outer wall of brick, inner surface plastered with Ferro-Cement.
10.	Plastic	Fiber-glass reinforced plastics PVC polythene, etc.

4.6 Constructional Details of Biogas Plant

Floating gas holder type

The construction techniques used in Bio gas plant should be simple with low demand of Materials, low in Cost and it should be easily known to Mason. The standard requirement for bio gas plant Construction is specially trained skilled Mason to make it leak proof and air tight. Otherwise, the chances of failure of the plant would be more.

1. Layout

The first step is to Mark the lay out of the plant on selected and leveled site by marking the out – line of the plan on the ground according to required dimensions using accord stretched from the center.

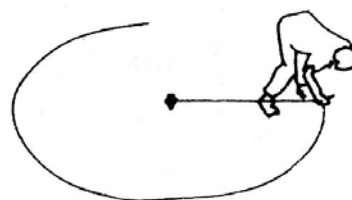


Fig 4.1 Marking the out line of the plant

2. Excavation

Excavate with care so that the sides do not collapse. The deeper the Pit, the more dangerous the excavation work becomes. Especially in sandy soil walls, the pit should be supported with a frame of bamboo poles. The excavated soil should be thrown at least half a meter away from the site so that it does not fall in when the construction work is in progress. The diameter of the Pit to be excavated, should be equal to the base diameter.

3. Floor

The bottom should be consolidated and levelled. If the bottom is muddy or soft and sandy, a layer of broken bricks or stones must be pounded in until the ground is firm. Bricks for the floor, including wall foundations are set on their edge. This method gives sufficient strength. Where the ground condition is very bad concert or reinforced concrete should be used.

4.Side Wall

A circular well should be constructed. It is essential to back fill between the wall and the sides after every 30Cm height has been added to the walls. Back filling should be done thoroughly with a piece of wood and water added to help compact the soil shown in fig4.2 failure to properly ram the back fill will cause cracking in the walls.



Fig 4.2 Back filling

5. Partition Wall

A Partition wall is constructed to divide the circular well into two equal halves in floating gas holder plants of $3M^3$. It controls the flow of slurry. A half brick wall is enough. It is built up to the level of the top surface of the central guide frame or ledge.

6. Inlet and Outlet Pipes

These should be fixed slanting and inserted at an appropriate height in the walls. There should not be any bends in the pipe as these cause blockage. The lower end of the inlet is placed in about the center of the compartment. This position is not vital, but what is important bottom and dead patches of unmoving slurry are not above to form the mouth of the pipe is about 30-35Cm above the floor of the digester. This prevents any blockage of the inlet pipe and also provides some volume in case stones, sand etc; get inside the plant and collect over a period of time. The lower end of the out let pipe is constructed in a similar way. The only difference is that the mouth of

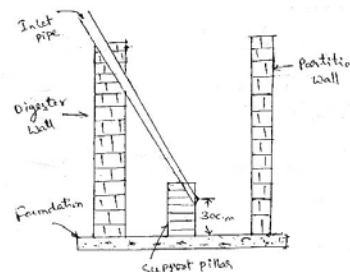


Fig 4.3 Inlet and outlet Pipes

the pipe is set about 25-30Cm above the floor as it is less likely that stones, sand etc; will come down this pipe.

The difference in the top level of the inlet and out let pipe should be at least 0.45M to allow digested slurry to come out automatically when fresh slurry is added.

7. Plastering

The digester should be plastered from inside with 1:3 Mortar of cement sand and cured properly.

8. Slurry Mixing Tank

In order to mix dung water thoroughly before feeding into the digester, a Masonry Tank is provided in the bio gas plant.

It may be provided with a Mechanical or hand operated stirrer for preparation of homogenous Mixture. The bottom of the tank is given a slope opposite to the direction of inlet chamber in order to prevent entry of sand or in-organic materials in to the digester.

9. Compost Pit

These can be dug in the ground or lined with Masonry. The volume should be sufficient to receive the amount of Slurry Put in per day Multiplied by the number of days required for emptying the pits.

10.Placing of the gas holder

The guide frame is greased generally and then the gas holder is mounted carefully without causing any tilting of the guide pipe and damage to the digester wall.

11. Testing and Operation

The digester Portion should also be tested for water leaks. When the plant is found leak proof, it is ready to fed with dung slurry and for the production of gas.

4.7 Utilization and benefits of Biogas Technology

- (i) It provides simultaneous dual benefits in terms of fuel for cooking and lighting as well as fertilizer.
- (ii) Bio gas burns efficiently without smoke or smell and eliminates the possibility of eye and lung diseases caused by smoke.
- (iii) It reduces the drudgery of women and children in terms of providing relief from the collection of fire wood and preparation of dung cakes.
- (iv) The gas can also be used in dual-fuel engines, where up to 80% of the diesel can be replaced with biogas.
- (v) Besides gas, the digested slurry can be used as natural fertilizer which improves the soil fertility.
- (vi) Utilization of waste materials in the plant not only improves the sanitary condition of your village but also prevents deforestation and assures e-ecollegial and environmental balance.

Bio gas is a cheep and clean fuel. It burns with a blue flame. Which is smoke-free. When it is burnt in silk mantle lamps, it illuminates better than kerosene in petromax lanterns. Bio gas can replace Petrol and diesel in engines.

4.8 Common Operational Problems

The common problems encountered with bio gas units and their remedial measures are given in table 4.1.

Defect	Cause	Remedy
1.Installation		
(i) Cracking of digester wall.	(i) Sinking of foundation or improper back filling.	(i) Repair and do proper back-filling.
(ii) Falling of partition wall.	(ii) Unequal Pressure of slurry in the inlet and out let chambers.	(ii) Fill the slurry initally from both inlet and out let sides. If the wall has fallen, then empty the digester and rebuild it.
(iii) Gas leakage.	(iii) Improper welding of iron gas holder or improper construction of dome.	(iii) Check and repair

(iv) Accumulation of water in pipe line.	(iv) Improper installation of water trap.	(iv) Check levels and set the water trap at correct position,
2.Operation		
(i) No gas after first filling of plant	(i) Lack of time	(i) It may take 3-4 weeks.
(ii) Gas holder does not rise or slurry level does not rise in inlet and out let chambers, even though gas is being produced.	(ii)(a) Insufficient addition of slurry (b)Leak in gas holder and pipe (c) Hard scum	(ii)(a) Add More slurry (b)Check and Correct (c) Rotate drum or agitate slurry bamboo pole in fixed dome type plant.
(iii) No gas at stove but plenty in the plant.	(iii)(a) Gas pipe blocked by (b)Insufficient pressure (c) Gas out let blocked by scum.	(iii) (a) Remove water condensate from Moisture trap. (b)Increase weight on gas holder. (c) Disconnect the out let valve from the hose pipe and clean it by pouring water.
(iv) Gas does not burn	(iv) Wrong kind of gas	(iv) Add properly mixed slurry
(v) Flame far from burner.	(v) Pressure too high or deposition of Carbon on the nozzle.	(v) Adjust gas out let valve and clean nozzle.
(vi) Flame dies quickly.	(vi) Insufficient pressure.	(vi) Check quantity of gas. Increase pressure by putting weight on gas holder and break the scum by rotating the holder.
(vii) Corrosion of Iron gas holder and pipe line.	(vii) Improper Maintenance.	(vii) Paint periodically and replace corroded material.
(viii) Clogging of inlet and out let pipes.	(viii) Addition of materials other than cleaning dung.	(viii) Use cattle dung only and clean pipes by pushing a bamboo pole in and out of them.
(ix) Insanitary condition around bio gas plant.	(ix) (a) Improper digestion. (b) Improper disposal of slurry.	(ix)(a) Add correct quantity of slurry (b) Use slurry for composting of crop residue.

4.9. Economical, social environmental and health benefits of bio gas utilization

Economical benefits of bio gas utilization

From the national economic point of view the bio gas yields following economic benefits:

- (1) Bio gas technology, which is based on recycling of readily available resources in rural areas, gives comparatively cheaper and better fuel for cooking lighting and power generation.
- (2) An individual can reduce the consumption of commercial energy sources such as fire wood, coal, kerosene, etc. by adopting waste recycling technology which vigorously help in reducing the family fuel budget.
- (3) The problem of uncertainty of availability of commercial energy can be resolved by use of bio gas technology.
- (4) The rural population of the country uses fire wood for meeting their cooking requirements. This reduces the national forest wealth. Our forest area can be conserved by using bio gas.
- (5) The dependency on chemical fertilizer for better-agricultural production has increased to a great extent after independence in India. Bio gas slurry can be proved a best organic fertilizer which helps in improving soil fertility and crop production.
- (6) Presently country is facing the problem of foreign exchange and balance of payment. Bio gas technology reduces the import of chemical fertilizers by using home made organic fertilizer and also petroproducts.
- (7) Bio gas technology utilizes effectively, the man power and resources, resulting in self-sufficiency and self reliance in the society.

Social Benefits

Bio gas is one of the best options in rural areas, which provides self-sufficiency in energy and helps in increasing standard of living of rural people. The social benefits from bio gas utilization are as follows:

- (1) Bio gas burns giving shootless flame and smokeless cooking, as such it provides cleanliness in the houses.

- (2) The cooking on bio gas is faster and also women is not required to waste their time to collect fuel from forest, as such it reduces the drudgery of women who can use her free time for other developmental activities.
- (3) The bio gas provides lighting in the rural areas, which are far away from electrical supply lines. Thus it helps the children to use their time in study of employment.
- (4) It helps in generation of employment opportunities to village artisans. This also stops the Migration of people from rural to urban areas in search of employment.
- (5) Generally, rural women and children spend their energy and time for collection of fuel, chopping wood, collecting wood and crop stalks or buying coal or kerosene for cooking from distant places, energy and time saved due to sue of bio gas can be used for constructive work or going to school for education.

Environmental and Health Benefits

Bio gas system contributes to maintain clean and healthier environment by processing human growth. More transportation means have increased the pressure on environment. The scientists are worried about environmental protection and looking for appropriate means for the same, which are locally available. In present circumstances bio gas utilization would be a solution for environmental protection for healthy and prosperous society. Environmental benefits from bio gas are enormous. Some of them are as under:

- (1) lungs and eye diseases are very common among village women and children due to smoky kitchen. Bio gas utilization reduces the disease spread. Thus reducing health problem rush in hospitals and waste of national wealth.
- (2) Sanitation problems in villages through systematic collection and through proper processing of animal dung and human excreta will be solved.
- (3) It helps to prevent deforestation. consequently it controls soil erosion, and floods.

4.10. KVIC Bio gas Plant

The Floating gas holder digester developed in India is of Masonry construction with gas holder made of M.S. Plates. The drum in the KVIC Model is the costliest component and its life is comparatively less.

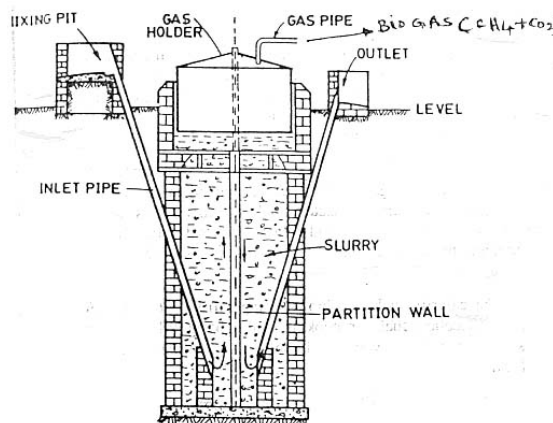


Fig. 4.4 KVIC Bio-gas plant

The design of KVIC plant was developed and perfected in India in the year 1945. This was taken up propagation in the villages in the year 1962, by Khadi and village industries commission, Bombay. Therefore, it is known as KVIC design. The design is available in sizes of 1 cum to 140 cum gas per day. In KVIC plant the gas is stored in mild steel drum of storage capacity of 30-40 percent of plant size at a pressure of about 10cm of water column. Which is sufficient to carry it up to a length of 20 meters to 100 meters, depending on the size of the plant.

The plant consists of two parts. The digester, which is well containing the animal waste in the form of a slurry, and the dome which floats on the slurry and serves as the gas holder.

The digester is normally below ground level and two pipe lines lead to its bottom. One for feeding the animal waste slurry and the other for spent slurry called sludge to come out after it has under gone fermentation. It is worth noting that the sludge to come out retains all the nitrogen, phosphorous and potassium and in an excellent fertilizer. A vertical partition wall divides it into two equal parts and serves to direct the flow of the slurry.

The gas generation process occurs in two stages. In the first stage, the complex organic substance contained in the waste are acted upon by a certain kind of bacteria called acid formers and are broken up into small chain simple acids. In the second stage, these acids are acted upon by another kind of bacteria which produces Methane and Carbon dioxide.

The calorific value of bio gas ranges from 1600 to 2500 KJ/m³. It is an excellent fuel for cooking and lighting. When compared with diesel it is also be a very good fuel for compression, ignition engines and can save 70 to 80 percent of diesel.

4.10.1 Pragathi Design Bio gas Plant

The design has been developed by United Socio-Economic Development and Research Programme (UNDARP) Pune, in order to have a cheaper floating drum bio gas plant. In this design the depth of pit is less than K.V.I.C plant so that it can be constructed in hilly and high water tabel areas. The cost of Pragathi plant is 20% less than KVIC plant.

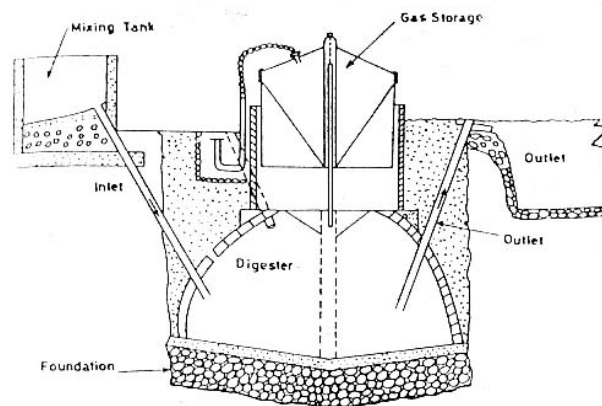


Fig 4.5 Pragathi Design Bio gas Plant

The design shown in fig 4.5, indicates its different parts. The foundation of this plant is of conical shape, with difference of one feet between outer periphery and its center so as to reduce the earth and digester wall work. It is constructed at the base of the pit with cement, sand and concrete, keeping the site conditions in view so it can bear the load due to weight of slurry in the digester.

The digester of Pragathi design plant start from the foundation in dome shape there by reducing the constructional area, for same digester volume, thus reducing the cost of construction of the plant. The wall thickness of digester is kept 75mm only. Dome shape construction takes place up to a collar base, where a central guide frame is provided. the digester wall above guide-frame is constructed in cylindrical shape,

Partition wall is constructed in the digester for 4cum. and bigger sizes so as to control the flow of slurry inside the digester. It divides the digester into two parts separating inlet and outlet.

The inlet is through a pipe, placed while constructing digester wall. It is used for feeding daily slurry into the digester and is generally of 100mm diameter. The outlet pipe is also 100mm in diameter, and fixed while constructing digester wall. The asbestos cement pipe can be used for inlet and outlet.

The guide frame is made of angle iron and steel pipe, is embedded in the digester wall at top of spherical portion of digester. The central guide pipe holds gas holder which is also made of M.S sheet and angle iron. It floats up and down along pipe depending on the quantity of gas in the drum.

4.10.2 Janata Bio gas Plant

This was first developed by the planning, Research and Action division, Lucknow, in 1978. It is an improved version of the Chinese fixed-dome bio gas plant. The plant is shown in fig 4.6 with different parts. The foundation of Janata Bio gas plant is laid at the base of the underground pit on a leveled ground to bear the load of the slurry as well as digester walls.

Digester is cylindrical in shape constructed with bricks and cements to retain dung slurry for a retention time, so that the bio gas is produced from the slurry in the digester. It should be noted that the diameter and height ratio of the digester is kept 1.75:1.

The gas is stored in gas portion which is an integral part of plant, between dome and digester. Where the usable gas is stored, the height of the gas portion is above the inlet and outlet openings to the beginning of dome, and is equal to maximum volume of the gas to be stored (30-40) (Percent of plant Capacity) and equal to volume of slurry to be displaced in inlet and outlet.

Dome is constructed over the gas portion, with volume of 60 percent of the plant capacity. It must be constructed very carefully integrated it with digester and gas portion so that no leakage of gas can take place. The gas out let pipe is fixed at the top of dome for laying the line.

Inlet and outlet portions are constructed for putting the fresh slurry in side the plant and to take the digested slurry out. The inlet and outlet are of larger sizes. Provided on each side of the digester, facing each other. The opening to the digester for feeding the waste material and effluent out let from it are also of large sizes. The discharge of slurry out of the plant is due to pressure of the gas in the plant. Over the inlet portion an inlet Mixing tank is also constructed to mix the dung and water.

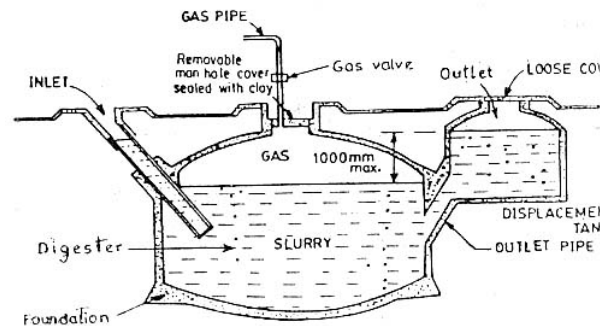


Fig 4.6 Jnanatha Bio gas Plant

4.10.3 Deenbandhu Bio Gas Plant

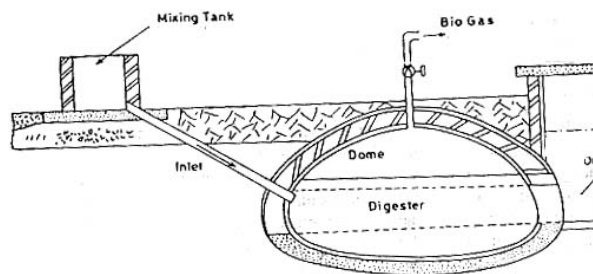


Fig 4.7 Deenbandhu Bio gas Plant

This is also a fixed dome plant development by action for food production, New Delhi, which is a allow cost bio gas plant. The principle of working of this plant is same as that of Janata model, except configuration of inlet entrance and digester.

The foundation of the plant is constructed in the segment of spherical shape as shown fig4.7 on the outer periphery of this foundations the dome shaped digester is constructed with same base diameter. In this way the digester, gas portion and dome look as a single unit. The surface area of bio gas plant is reduced with same digester volume, reducing the earth work and cost of construction with out sacrificing the efficiency. The higher compressive strength of the brick masonry and concrete makes it a safe structure as the plant is always under compression. A spherical, Structure loaded from the convex side will be under compression and therefore, the internal load will not have any residual effect on the structure.

At the top of the foundation a window opening is kept (outlet portion) for the out ward movement of the digested slurry. The asbestos cement pipe of 15Cm diameter is used for inlet instead of separate opening. The pipe is embedded in the digester wall at a fixed position ,just opposite to out let opening, to avoid short circuiting of fresh material and digested slurry.

The volume of the out let is increased to produce requisite gas pressure through the weight of the displaced slurry. At the top of the dome a gas outlet pipe is fixed as incase of Janata plant.

Bio gas energy – Questions

Short Answer Questions

1. Define Biogas energy?
2. What are the raw materials required for fermentation?
3. What are the benefits of Bio gas technology?
4. What is Bio Mass?
5. Write the Percentage compositions of Bio gas?
6. What is aerobic and anaerobic?

Essay type Questions

1. Write the brief History of Bio gas?
2. Explain briefly about the process of Bio gas?
3. Explain briefly about the selection site for installation of a Bio gas plant?

4. Explain briefly about the material requirement for construction of Bio gas plant?
5. Explain briefly about the construction techniques of bio gas plant?
6. Explain briefly about the common problems with bio gas units and their remedies?
7. What are the economic benefits of Bio gas Utilization?
8. What are the social benefits of Bio gas Utilization?
9. What are the environmental and Health Benefits of bio gas Utilization?
10. Explain KVIC Bio gas plant with the help of neat sketch?
11. Explain Pragathi design Bio gas plant with the help of neat sketch?
12. Explain Janata Bio gas plant with the help of neat sketch?
13. Explain Deenbandhu Bio gas plant with the help of neat sketch?

CHAPTER – 5 TIDAL POWER PLANTS

5.0 Introduction to Tidal Power Plants

The development of a nation is estimated from the total amount of energy it produces and consumes in relation to its size and Population. Human-Progress has been judged from the ways in which man has been able to develop and harness energy.

The present shortage of the energy production sources due to depletable nature of fossil fuels has awakened the world human community to search for new unconventional and replenishable energy sources. According to the energy experts, exhaustible resources extracted from the ground (coal, oil, gas, and uranium), in the face increasing demand, results into a cycle. This cycle starts at zero production increases exponentially than decreases exponentially and finally comes to zero. It is estimated that Petroleum reserves of the world will be nearly exhausted by 2020. Under such circumstances, man has to find out some source of energy for his survival. In this context, three sources of energy, tidal, wind and solar have been brought to be most promising.

What is tidal power? Tide is periodic rise and fall of the water level of the sea. Tides occur due to the attraction of seawater by the moon. These tides can be used to produce electrical power which is known as tidal power.

When the water is above the mean sea level, it is called flood tide and when the level is below the mean level, it is called ebb tide. A dam is constructed in such a way that a basin gets separated from the sea and a difference in the water level is obtained between the basin and sea. The constructed basin is filled during high tide and emptied during low tide passing through sluices and turbine respectively. The Potential energy of the water stored in the basin is used to drive the turbine which in turn generates electricity as it is directly coupled to an alternator.

Though the idea of utilizing tides for human service relates to eleventh century when tidal mills were used in England but the use of tidal power for electric power generation is hardly a decade old as the world's first Rance tidal power plant of 240 MW capacity in France was commissioned by President de Gaulle in 1965 who described it a magnificent achievement in the human life.

Tidal power has been a dream for engineers for many years and it remained dream because of large capital cost involved in its development. But after the inauguration of Rance Tidal Project, a new chapter in the history is now opened.

5.1 Factors affecting the suitability of the site for tidal power plant

The feasibility and economic vulnerability of a tidal power depends upon the following factors.

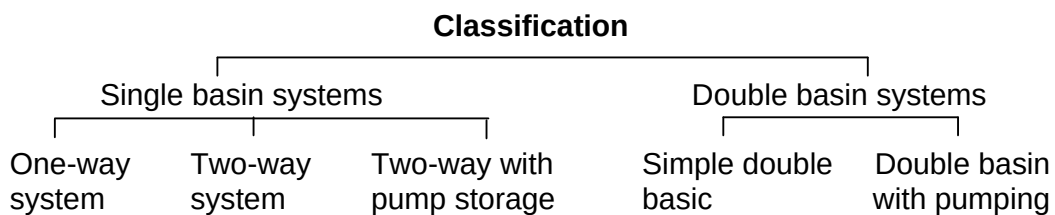
1. The power produced by a tidal plant depends mainly on the range of tide and the cubature of the tidal flow occurring in the estuary during a tidal cycle which can be stored and utilized for power generation. The cubature of the tidal flow not only depends on the tidal range but on the width of estuary mouth.
2. The minimum average tide range required for economical power production is more.
3. The site should be such that with a minimum cost of barrage it should be possible to create maximum storage volume. In addition to this, the site selected should be well protected from waves action.
4. The site should not create interruption to the shipping traffic running through the estuary other wise the cost of the plant will increase as locks are to be provided.
5. Silt index of the water of the estuary should be as small as possible to avoid the siltation troubles. The siltation leads to reduction of the range of tides and reduces the power potential of the plant.
6. The fresh water prism that falls into the reservoir of the tidal plant (due to the surface flows in the streams having out fall in the estuary) eats away the valuable storage created for storing the tidal prism. Therefore, the

ratio of fresh water prism to tidal water prism becomes an important index in determining the economic feasibility of a tidal scheme. The effective and cheaper will be the power production with decreasing the ratio mentioned above.

5.2 Classification of tidal Power Plants

The tidal power plants are generally classified on the basis of the number of basins used for the power generation. They are further subdivided as one-way or two-way system as per the cycle of operation for power generation.

The classification is represented with the help of a line diagram as given below



5.3 Working of Different Tidal Power Plants

5.3.1 Single basin-One-way cycle

This is the simplest form of tidal power plant. In this system, a basin is allowed to get filled during flood tide and during the ebb tide. The water flows from the basin to the sea passing through the turbine and generates power. The power is available for a short duration during ebb tide.

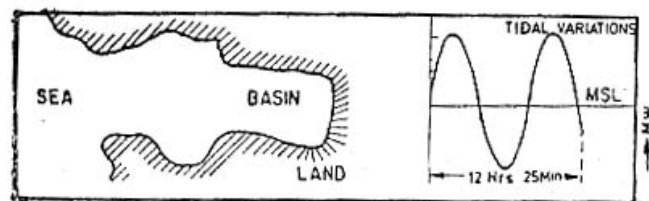
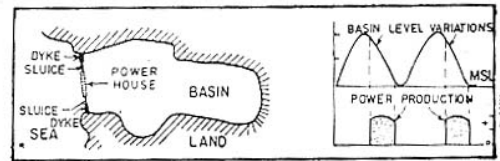


Fig 5.1 Single basin Tidal Power Plant

Fig 5.1 shows a single tide basin before the construction of dam and Fig5.2 shows the diagrammatic representation of a dam at the mouth of the basin and power generation during the falling tide.

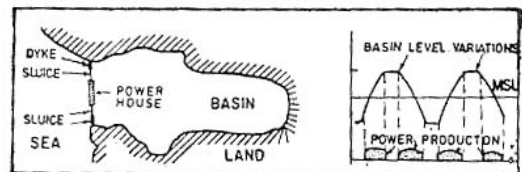


(b) Single basin, one-way tidal power plant.

Fig 5.2

5.3.2 Single-basin two-way cycle

In this arrangement power is generated both during flood tide as well as ebb tide also. The power generation is also intermittent but generation period is increased compared with one-way cycle. However the peak power obtained is less than the one-way cycle. The arrangement of the basin and the power cycle is shown in Fig 5.3.



Single-basin two-way tidal power plant.

Fig 5.3

The main difficulty with this arrangement, the same turbine must be used as Prime mover as ebb and tide flows pass through the turbine in opposite directions. Variable pitch turbine and dual rotation generator are used for such schemes.

5.3.3. Single-basin two-way cycle with pump storage

The Rance tidal power plant in France uses this type of arrangement. In this system, power is generated both during flood and ebb tides. Complex machines capable of generation Power and Pumping the water in either directions are used. A part of the energy produced is used for introducing the difference in the water levels between the basin and the sea at any time of the tide and this is done by pumping water into the basin up or down. The period of power production with this system is much longer than the other two described earlier. The cycle of operation is shown in Fig 5.4.

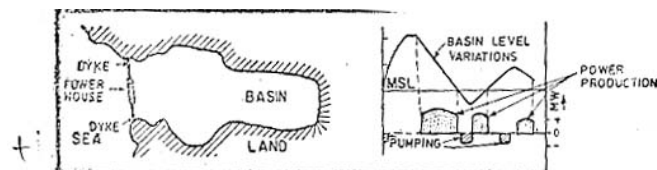


Fig 5.4

Single-basin, two-way tidal plant coupled with pump storage system.

5.3.4 Double basin type

In this arrangement, the turbine is set up between the two basins as shown in Fig 5.5. one basin is intermittently filled by the flood tide and other is intermittently drained by the ebb tide. Therefore a small capacity but continuous power is made available with this system as shown in Fig 5.5. The main disadvantage of this system is that 50% of the Potential energy is sacrificed in introducing the variation in the water levels of the two basins.

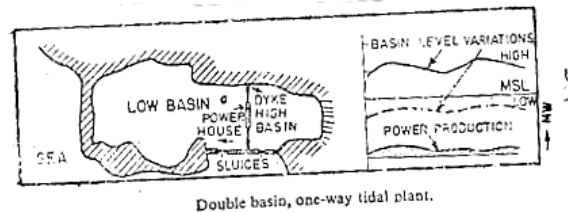


Fig 5.5

5.3.5 Double basin with Pumping

In this case, off peak power from the base load plant in a interconnected transmission system is used either to pump the water up the high basin. Net energy gain is possible with such a system if the pumping head is lower than the basin-to-basin turbine generating head.

5.4 Advantages and disadvantages of Tidal Power Plants

Advantages

1. Exploitation of tidal energy will in no case make demand for large area of valuable land because they are on bays.
2. It is free from pollution as it does not use any fuel.
3. It is much superior to hydro-power plant as it is totally independent of rain which always fluctuates year to year. Therefore, there is certainty of power supply as the tide cycle is very definite.
4. As in every form of water power, this will also not produce any unhealthy waste like gases, ash, atomic refuse which entails heavy removal costs.
5. Tidal Power is superior to conventional hydro power as the hydro plants are known for their large seasonal and yearly fluctuations in the output of energy because they are entirely dependent upon the nature's cycle of rainfall, which is not the case with tidal as monthly certain power is assured. The tides are totally independent on nature's cycle of rainfall.

6. Another notable advantage of tidal power is that it has a unique capacity to meet the peak power demand effectively when it works in combination with thermal or hydroelectric system.
7. It can provide better recreational facilities to visitors and holiday makers, in addition to the possibility of fish farming in the tidal basins.

Disadvantages

1. These Power plants can be developed only if natural sites are available.
2. As the sites are available on the bay which will be always far away from the load centers. The power generated must be transported to long distances. This increases the transportation cost.
3. The supply of power is not continuous as it depends upon the timing of tides. Therefore some arrangements (double basin or double basin with pump storage) must be made to supply the continuous power. This also further increases the capital cost of the plant.
4. The capital cost of the plant (Rs.5000/kw) is considerably large compared with conventional-power plants (hydro, thermal)
5. Sedimentation and siltration of the basins are some of the added problems with tidal power plants.
6. The navigation is obstructed.
7. It is interesting to note that the output of power from tidal power plant varies with lunar cycle, because the moon largely influences the tidal rhythm, where as our daily power requirement is directly related to solar cycle.

In addition to all the above mentioned (imitations of tidal power, the utilization of tidal energy on small scale has not yet proved economical.

Components of Tidal Power plants

There are three main Components of a tidal Power plant. i.e,

- (i) The Power house
- (ii) The dam or barrage
- (iii) Sluice-ways from the basins to the sea and vice versa.

The turbines, electric generators and other auxiliary equipment's are the main equipments of a power house. The function of dam to form a barrier between the sea and the basin or between one basin and the other in case of multiple basins.

The sluice ways are used either to fill the basin during the high tide or empty the basin during the low tide, as per operational requirement. These are gate controlled devices.

It is generally convenient to have the power house as well as the sluice-ways in alignment with the dam.

The design cycle may also provide for pumping between the basin and the sea in either direction. If reversible pump turbines are provided, the pumping operation can be taken over at any time by the same machine. The modern tubular turbines are so versatile that they can be used either as turbines or as pumps in either direction of flow. In addition, the tubular passages can also be used as sluice-ways by locking the machine in to a stand still. As compared to conventional plants, this, however, imposes a great number of operations in tidal power plants. For instance, the periodic opening and closing of the sluice-way of a tidal plant are about 730 times in a year.

Dam (Barrage)

Dam and barrage are synonymous terms. Barrage has been suggested as a more accurate term for tidal power scheme, because it has only to withstand heads a fraction of the structure's height, and stability problems are far more modest. However, the literature does not always make the distinction, even though heads are small with tidal power cutoffs.

Tidal power barrages have to resist waves whose shock can be severe and where pressure changes sides continuously.

The barrage needs to provide channels for the turbines in reinforced concrete. To build these channels a temporary coffer dam is necessary, but it is now possible to build them on land, float them to the site, and sink them into place.

Tidal barrages require sites where there is a sufficiently high tidal range to give a good head of water – the minimum useful range is around three meters. The best sites are bays and estuaries, but water, can also be impounded behind bounded reservoir built between two points on the same shore line.

The location of the barrage is important, because the energy available is related to the size of trapped basin and to the square of the tidal range. The nearer it is built to the mouth of bay, the larger the basin, but the smaller the tidal range. A balance must also be struck between increased out put and increased material requirements and construction costs.

Gates and Locks

Tidal power basins have to be filled and emptied. Gates are opened regularly and frequently but heads vary in height and on the side where they occur, which is not the case with conventional river projects. The gates must be opened and closed rapidly and this operation should use a minimum of power. Leakage, is tolerable for gates and barrages. Since we are dealing with seawater, corrosion problems are actuate, they have been very successfully solved by the cathodic protection and where not possible by paint. Gate structures can be floated as modular units into place.

Though, in existing plants, vertical lift gates have been used. The technology is about ready to substitute a series of flap gates that operates by water pressure. Flap gates are gates that are positioned so as to allow water in to the holding basin and require no mechanical means of operation. The flap gates allow only in the direction of the sea to basin. Hence, the basin level rises well above to sea level as ebb flow area is far less than flood flow area.

Power house

Because small heads only are available, large size turbines are needed; hence, the power house is also a large structure. Both the French and Soviet operating plants use the bulb type of turbine of the propeller type, with revisable blades, bulbs have horizontal shafts coupled to a single generator. The cost per installed kilowatt drops with turbine size, and perhaps larger turbines might be installed in a future major tidal power plant.

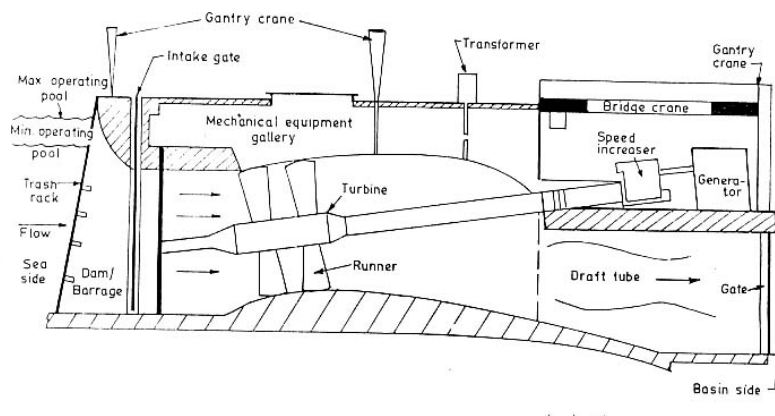


Fig 5.6 Schematic Lay out of Tidal Power House

Questions

Short answer Questions

- 1) What are the components of Tidal Power plant?
- 2) What are the advantages and disadvantages of Tidal Power plant?
- 3) Define Tidal energy?

Essay type Question

- 1) Explain briefly about Tidal Power plant and Mention the advantages and disadvantages of tidal power plant?

CHAPTER – 6 FUEL CELLS

6.0 Introduction

Fuel Cells are efficient and quiet, operate on a variety of hydrocarbon fuels, and produces almost no objectionable emissions. The concept has been proven in numerous small scale applications. The recent infusion of power cost from fossil fuels may convert this promising device into a major source of electric power generation.

The electric Power Research Institute and United Technologies Corporation recently agreed to construct a 4.8MW capacity plant, with an investment of 25 million dollars. It is predicted that if the fuel cell clears a number of technical hurdles and proves financially viable, it could contribute 20,000MW power to the US grid, by the end of 1985. This would save \$1 billion in power generating costs and 100 million barrels of fuel oil annually.

6.1 Working of Fuel Cell

The fuel cell represents one of the successful ways by passing the heat cycle and converting the chemical energy of fuels directly into electricity. It may be defined as an electrochemical device for the continuous conversion of the portion of the free energy change in a chemical reaction to electrical energy. It is distinguished from a battery in that it operates with continuous replenishment of the fuel and oxidant at active electrode areas and does not require recharging.

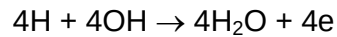
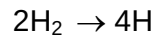
The arrangement of fuel cell components is shown in fig 6.1

The working of the fuel cell is explained here with reference to the Hydrogen-Oxygen fuel cell using aqueous electrolyte. The fuel cell consists of an anode, a cathode and an electrolyte. Hydrogen fuel is fed into the anode side of the cell. Positive H_2 ions move from the anode-side and enter the electrolyte through porous cell walls. The anode is left with a negative charge. Air is fed into the cathode side. O_2 ions enter the electrolyte leaving

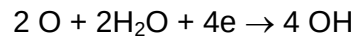
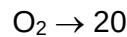
the cathode side with a positive charge. Excess anode electrons flow to the cathode creating a current flow H_2 and O_2 ions combine in the electrolyte to form water which leaves the cell as steam.

The reaction taking place at the electrodes are given below

(1) Hydrogen electrode (anode)



(2) Oxygen electrode (cathode)



(3) Overall cell reaction

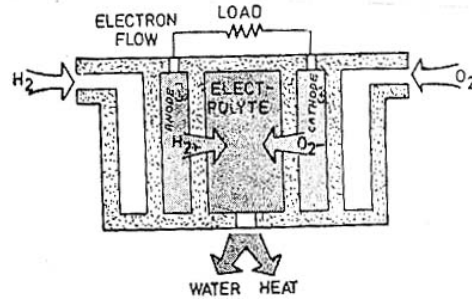
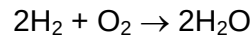


Fig 6.2 Fuel Cell

The above reactions indicate H_2 molecules break up into H Atoms at the anode and they combine with OH ion to form water and free electron at anode. The formed free electrons travel to the cathode through the external circuit as shown in fig. At the cathode, O_2 molecules break up into two O atoms and these atoms combine with the four electrons arriving by the external circuit and two molecules of water (out of 4 molecules produced at the anode) to form 4OH ions. The OH ions migrate towards the anode and are consumed there. The electrolyte remains invariant. It is prime requirement that the composition of electrolyte should not change as the cell operates. The major difficulty experienced in the design of fuel cell is to obtain sufficient fuel-electrode-electrolyte reaction sites in a given volume. There are many other types of cells as ion exchange membrane cell, direct and indirect oxidation fuel cells, molten carbonate fuel cells and many others.

The present research work done in this area is to increase the efficiencies or modify the design of the three major components of fuel cell power plant to increase the overall efficiency of the system. The arrangement of the fuel cell power plant which will be used in future is shown in fig6.3.

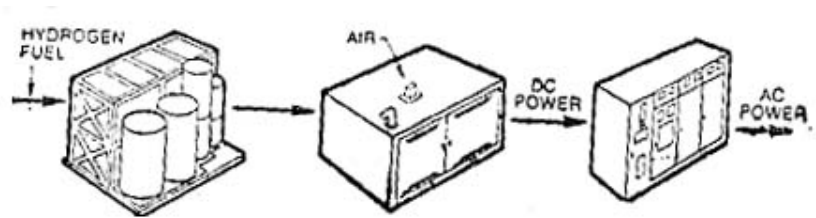


Fig 6.3

(a) Processor

It converts a hydrocarbon fuel into a hydrogen rich gas that can be accepted by the cell. One of the main problems regarding fuel-cell development is finding away to convert low grade fuel into H₂ rich gas that has a minimum of impurities. Material such as sulphur in fuel can adversely affect the fuel processor. Presently the research is directed to find a cell electrolyte that is more tolerant of fuel impurities. It is predicted that the developments in both processor and electrolyte will allow the use of standard fuel oil as fuel in the 1990s and ultimately residual oil and coal gas.

Hydrogen generation processes from the petrochemical industry are being adopted to the fuel cell's needs steam reforming is an other successful fuel processing technique now commonly used with phosphoric-acid electrolyte fuel cells. The life of such a processor is presently limited to about 15,000 hrs.

(b) Electrolyte

Initial work has been done chiefly with phosphoric acid which has an operating temperature of 160°C to 200°C. This electrolyte can give fuel cells efficiency of 50%.

Next generation of fuel cells will use molten carbonate electrolyte. Operating at a temperature of 500 to 750°C, this material is more tolerant of fuel impurities and gives a fuel cell efficiency of 40 to 50%. Life for existing molten carbonate fuel cells is about 10,000 hrs. A 50,000hr, life is expected to be achieved by 1990.

Solid oxide may be electrolyte of the third generation of fuel cells, which are probably 20 years away. This electrolyte will give a fuel efficiency of 60% or better and will be compatible with many of the coal gasification processes.

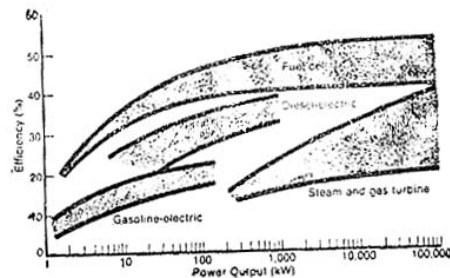
(c) Inverter

The last element in the fuel cell power generation system which converts DC to AC does not yet exist commercially in the sizes needed for large scale power conversion. To date inverters with a capacity of 20 to 40KW are in limited use and some capable of 1.8MW have been tested. Major research work is presently directed to develop the inverters of high capacity.

The predicted performance of fuel cell power plant is shown in fig relative to three other power generation methods.

6.2 Advantages of Fuel Cells

- 1) The fuel cell converts its fuel directly to electric power. Pollutant levels range from 1/10 to 1/50,000 of those produced by a fossil fuel power plant as there it no combustion
- 2) No cooling water is needed so it can be located at any desired place.
- 3) As it does not make noise. It can be readily accepted in residential areas.
- 4) The fuel cell takes little time to go into operation.
- 5) It would be an ideal reserve power source with in large conventional power plants to handle peak or emergency loads.
- 6) There is no efficiency penalty for part load operation. Efficiency remains constant from 100% to 25 % of rated load.
- 7) There is no maximum or minimum size for a fuel cell power plant. Individual fuel cells are joined to form stacks. The stacks are joined to



form power modules. The number of modules can be tailored to power plant requirements.

- 8) The land requirement is considerably less compared with conventional power plants.
- 9) Possibly the greatest advantage of the fuel cell is its high operating efficiency. Present-day fuel cell efficiency is 38% and it is expected to reach to 60% before the end of this century.
- 10) Fuel cell power plants may further cut generation costs by eliminating or reducing line losses. Fuel cell power plant in rural areas or highly congested residential areas would eliminate the need for long lines to bring in power from remote generating stations.
- 11) A wide variety of fuels can be used with the fuel cell. Although presently limited to using substances that produce pure H_2 rich gas, the cell may one day be able to operate on fuels derived from low grade shale oils or highly sulfur coals.
- 12) The maintenance charges are low as there are no moving parts and outages are also less.
- 13) Fuel cells have an overload capacity of 50 to 100% for a short duration.
- 14) The weight and volume of the fuel cell is considerably low compared to other energy sources.
- 15) In H_2 - O_2 cell, the reaction product is water which is portable.

QUESTIONS

Essay type Questions

1. Explain briefly about the working of Fuel Cell?
2. What are the advantages of Fuel Cells?

12. APPLICATION OF NON-CONVENTIONAL & RENEWABLE ENERGY SOURCES

12.1 Concept of Renewable Energy

Renewable energy sources also called non-conventional energy, are sources that are continuously replenished by natural processes. For example, solar energy, wind energy, bio-energy - bio-fuels grown sustain ably), hydropower etc., are some of the examples of renewable energy sources

A renewable energy system converts the energy found in sunlight, wind, falling-water, sea-waves, geothermal heat, or biomass into a form, we can use such as heat or electricity. Most of the renewable energy comes either directly or indirectly from sun and wind and can never be exhausted, and therefore they are called renewable.

However, most of the world's energy sources are derived from conventional sources-fossil fuels such as coal, oil, and natural gases. These fuels are often termed **non-renewable** energy sources. Although, the available quantity of these fuels are extremely large, they are nevertheless finite and so will in principle 'run out' at some time in the future

Renewable energy sources are essentially *flows* of energy, whereas the fossil and nuclear fuels are, in essence, *stocks* of energy

Various forms of renewable energy

- Solar energy
- Wind energy
- Bio energy
- Hydro energy
- Geothermal energy
- Wave and tidal energy

This chapter focuses on application potential of commercially viable renewable energy sources such as solar, wind, bio and hydro energy in India.

12.2 Solar Energy

Solar energy is the most readily available and free source of energy since prehistoric times. It is estimated that solar energy equivalent to over 15,000 times the world's annual commercial energy consumption reaches the earth every year.



India receives solar energy in the region of 5 to 7 kWh/m² for 300 to 330 days in a year. This energy is sufficient to set up 20 MW solar power plant per square kilometre land area.

Solar energy can be utilised through two different routes, as solar thermal route and solar electric (solar photovoltaic) routes. Solar thermal route uses the sun's heat to produce hot water or air, cook food, drying materials etc. Solar photovoltaic uses sun's heat to produce electricity for lighting home and building, running motors, pumps, electric appliances, and lighting.

Solar Thermal Energy Application

In solar thermal route, solar energy can be converted into thermal energy with the help of solar collectors and receivers known as solar thermal devices. The Solar-Thermal devices can be classified into three categories:

- Low-Grade Heating Devices - up to the temperature of 100°C.
- Medium-Grade Heating Devices -up to the temperature of 100°-300°C
- High-Grade Heating Devices -above temperature of 300°C

Low-grade solar thermal devices are used in solar water heaters, air-heaters, solar cookers and solar dryers for domestic and industrial applications.

Solar water heaters

Most solar water heating systems have two main parts: a solar collector and a storage tank. The most common collector is called a *flat-plate collector* (see Figure 12.1). It consists of a thin, flat, rectangular box with a transparent cover that faces the sun, mounted on the roof of building or home. Small tubes run through the box and carry the fluid – either water or other fluid, such as an antifreeze solution – to be heated. The tubes are attached to an absorber plate, which is painted with special coatings to absorb the heat. The heat builds up in the collector, which is passed to the fluid passing through the tubes.



Figure 12.1 Solar Flat plate collector

An insulated storage tank holds the hot water. It is similar to water heater, but larger in size. In case of systems that use fluids, heat is passed from hot fluid to the water stored in the tank through a coil of tubes.

Solar water heating systems can be either active or passive systems. The active system, which are most common, rely on pumps to move the liquid between the collector and the storage tank. The passive systems rely on gravity and the tendency for water to naturally circulate as it is heated. A few industrial application of solar water heaters are listed below:

- ❑ *Hotels*: Bathing, kitchen, washing, laundry applications
- ❑ *Dairies*: Ghee (clarified butter) production, cleaning and sterilizing, pasteurization
- ❑ *Textiles*: Bleaching, boiling, printing, dyeing, curing, ageing and finishing
- ❑ *Breweries & Distilleries*: Bottle washing, wort preparation, boiler feed heating
- ❑ *Chemical /Bulk drugs units*: Fermentation of mixes, boiler feed applications
- ❑ *Electroplating/galvanizing units*: Heating of plating baths, cleaning, degreasing applications
- ❑ *Pulp and paper industries*: Boiler feed applications, soaking of pulp.

Solar Cooker

Solar cooker is a device, which uses solar energy for cooking, and thus saving fossil fuels, fuel wood and electrical energy to a large extent. However, it can only supplement the cooking fuel, and not replace it totally. It is a simple cooking unit, ideal for domestic cooking during most of the year except during the monsoon season, cloudy days and winter months

Box type solar cookers: The box type solar cookers with a single reflecting mirror are the most popular in India. These cookers have proved immensely popular in rural areas where women spend considerable time for collecting firewood. A family size solar cooker is sufficient for 4 to 5 members and saves about 3 to 4 cylinders of LPG every year. The life of this cooker is upto 15 years. This cooker costs around Rs.1000 after allowing for subsidy. Solar cookers.(Figure 12.2) are widely available in the market.



Figure 12.2 Box Type Solar

Parabolic concentrating solar cooker:

A parabolic solar concentrator comprises of sturdy Fibre Reinforced Plastic (FRP) shell lined with Stainless Steel (SS) reflector foil or aluminised polyester film. It can accommodate a cooking vessel at its focal point. This cooker is designed to direct the solar heat to a secondary reflector inside the kitchen, which focuses the heat to the bottom of a cooking pot. It is also possible to actually fry, bake and roast food. This system generates 500 kg of steam, which is enough to cook two meals for 500 people (see Figure 12.3). This cooker costs upward of Rs.50,000.

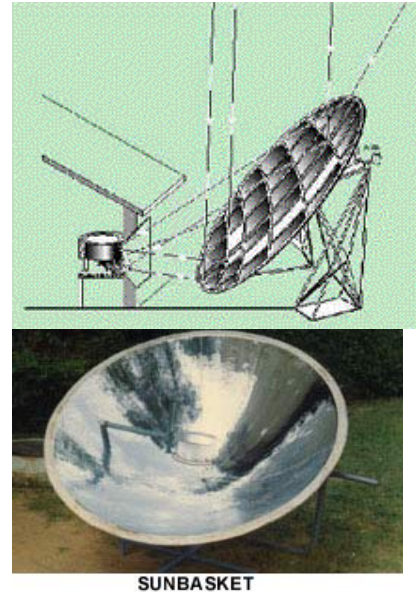


Figure 12.3 Parabolic Collector

Positioning of solar panels or collectors can greatly influence the system output, efficiency and payback. Tilting mechanisms provided to the collectors need to be adjusted according to seasons (summer and winter) to maximise the collector efficiency.

The period four to five hours in late morning and early afternoon (between 9 am to 3pm) is commonly called the "Solar Window". During this time, 80% of the total collectable energy for the day falls on a solar collector. Therefore, the collector should be free from shade during this solar window throughout the year - Shading, may arise from buildings or trees to the south of the location.

Solar Electricity Generation

Solar Photovoltaic (PV): Photovoltaic is the technical term for *solar electric*. Photo means "light" and voltaic means "electric". PV cells are usually made of silicon, an element that naturally releases electrons when exposed to light. Amount of electrons released from silicon cells depend upon intensity of light incident on it. The silicon cell is covered with a grid of metal that directs the electrons to flow in a path to create an electric current. This current is

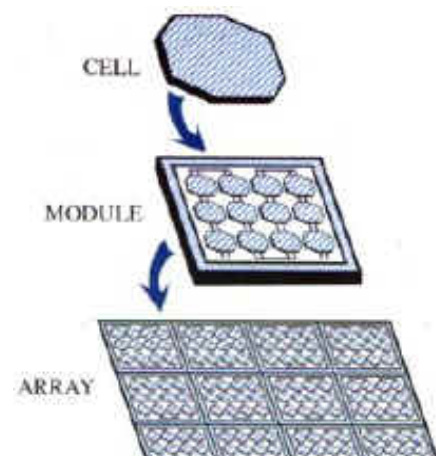


Figure 12.4 Solar Photovoltaic Array

guided into a wire that is connected to a battery or DC appliance. Typically, one cell produces about 1.5 watts of power. Individual cells are connected together to form a solar *panel* or *module*, capable of producing 3 to 110 Watts power. Panels can be connected together in series and parallel to make a solar *array* (see Figure 12.4), which can produce any amount of Wattage as space will allow. Modules are usually designed to supply electricity at 12 Volts. PV modules are rated by their peak Watt output at solar noon on a clear day.

Some applications for PV systems are lighting for commercial buildings, outdoor (street) lighting (see Figure 12.5), rural and village lighting etc. Solar electric power systems can offer

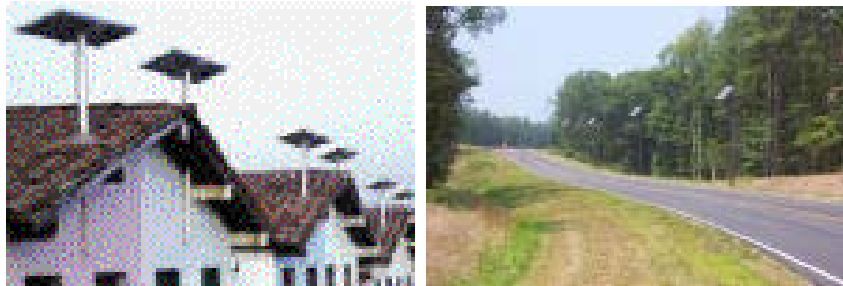


Figure 12.5 Photovoltaic Domestic and Streetlights

independence from the utility grid and offer protection during extended power failures. Solar PV systems are found to be economical especially in the hilly and far flung areas where conventional grid power supply will be expensive to reach.

PV tracking systems is an alternative to the fixed, stationary PV panels. PV tracking systems are mounted and provided with tracking mechanisms to follow the sun as it moves through the sky. These tracking systems run entirely on their own power and can increase output by 40%.

Back-up systems are necessary since PV systems only generate electricity when the sun is shining. The two most common methods of backing up solar electric systems are connecting the system to the utility grid or storing excess electricity in batteries for use at night or on cloudy days.

Performance

The performance of a solar cell is measured in terms of its efficiency at converting sunlight into electricity. Only sunlight of certain energy will work efficiently to create electricity, and much of it is reflected or absorbed by the material that make up the cell. Because of this, a typical commercial solar cell has an efficiency of 15%—only about one-sixth of the sunlight striking the cell generates electricity. Low efficiencies mean that larger arrays are needed, and higher investment costs. It should be noted that the first solar cells, built in the 1950s, had efficiencies of less than 4%.

Solar Water Pumps

In solar water pumping system, the pump is driven by motor run by solar electricity instead of conventional electricity drawn from utility grid. A SPV water pumping system consists of a photovoltaic array mounted on a stand and a motor-pump set compatible with the photovoltaic array. It converts the solar energy into electricity, which is used for running the motor pump set. The pumping system draws water from the open well, bore well, stream, pond, canal etc

Case Example:

Under the Solar Photovoltaic Water Pumping Programme of the Ministry of Non-conventional Energy Sources during 2000-01 the Punjab Energy Development Agency (PEDA) has completed installation of 500 solar pumps in Punjab for agricultural uses.



Figure 12.6 Photovoltaic Water Pumping

Under this project, 1800 watt PV array was coupled with a 2 HP DC motor pump set. The system is capable of delivering about 140,000 litres water every day from a depth of about 6 – 7 metres. This quantity of water is considered adequate for irrigating about 5 –8 acres land holding for most of the crops. Refer Figure 12.6.

12.3 Wind Energy

Wind energy is basically harnessing of wind power to produce electricity. The kinetic energy of the wind is converted to electrical energy. When solar radiation enters the earth's atmosphere, different regions of the atmosphere are heated to different degrees because of earth curvature. This heating is higher at the equator and lowest at the poles. Since air tends to flow from warmer to cooler regions, this causes what we call winds, and it is these airflows that are harnessed in windmills and wind turbines to produce power.

Wind power is not a new development as this power, in the form of traditional windmills -for grinding corn, pumping water, sailing ships – have been used for centuries. Now wind power is harnessed to generate electricity in a larger scale with better technology.



Wind Energy Technology

The basic wind energy conversion device is the wind turbine. Although various designs and configurations exist, these turbines are generally grouped into two types:

1. **Vertical-axis wind turbines**, in which the axis of rotation is vertical with respect to the ground (and roughly perpendicular to the wind stream),
2. **Horizontal-axis turbines**, in which the axis of rotation is horizontal with respect to the ground (and roughly parallel to the wind stream.)

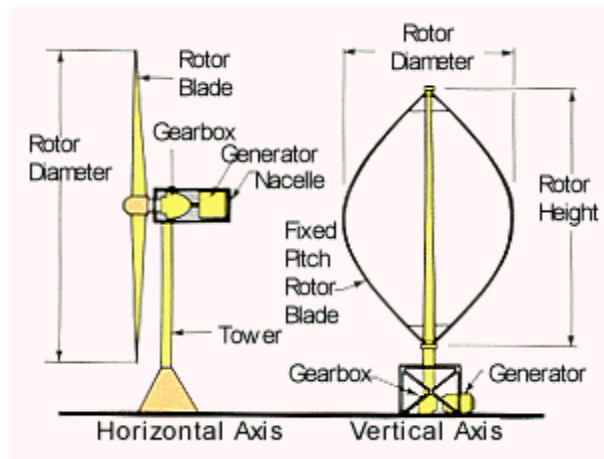


Figure 12.7 Wind Turbine Configuration

The Figure 12.7 illustrates the two types of turbines and typical subsystems for an electricity generation application. The subsystems include a blade or rotor, which converts the energy in the wind to rotational shaft energy; a drive train, usually including a gearbox and a generator, a tower that supports the rotor and drive train, and other equipment, including controls, electrical cables, ground support equipment, and interconnection equipment.

Wind electric generators (WEG)

Wind electric generator converts kinetic energy available in wind to electrical energy by using rotor, gear box and generator. There are a large number of manufacturers for wind electric generators in India who have foreign collaboration with different manufacturers of Denmark, Germany, Netherlands, Belgium, USA, Austria, Sweden, Spain, and U.K. etc. At present, WEGs of rating ranging from 225 kW to 1000 kW are being installed in our country.

Evaluating Wind Mill Performance

Wind turbines are rated at a certain wind speed and annual energy output

Annual Energy Output = Power x Time

Example: For a 100 kW turbine producing 20 kW at an average wind speed of 25 km/h, the calculation would be:

$$100 \text{ kW} \times 0.20 \text{ (CF)} = 20 \text{ kW} \times 8760 \text{ hours} = 175,200 \text{ kWh}$$

The Capacity Factor (CF) is simply the wind turbine's actual energy output for the year divided by the energy output if the machine operated at its rated power output for the entire year. A reasonable capacity factor would be 0.25 to 0.30 and a very good capacity factor would be around 0.40. It is important to select a site with good capacity factor, as economic viability of wind power projects is extremely sensitive to the capacity factor.

Wind Potential

In order for a wind energy system to be feasible there must be an adequate wind supply. A wind energy system usually requires an average annual wind speed of at least 15 km/h. The following table represents a guideline of different wind speeds and their potential in producing electricity.

Average Wind Speed km/h (mph)	Suitability
Up to 15 (9.5)	No good
18 (11.25)	Poor
22 (13.75)	Moderate
25 (15.5)	Good
29 (18)	Excellent

A wind generator will produce lesser power in summer than in winter at the same wind speed as air has lower density in summer than in winter.

Similarly, a wind generator will produce lesser power in higher altitudes - as air pressure as well as density is lower -than at lower altitudes.

The wind speed is the most important factor influencing the amount of energy a wind turbine can produce. Increasing wind velocity increases the amount of air passing the rotor, which increases the output of the wind system.

In order for a wind system to be effective, a relatively consistent wind flow is required. Obstructions such as trees or hills can interfere with the wind supply to the rotors. To avoid this, rotors are placed on top of towers to take advantage of the strong winds available high above the ground. The towers are generally placed 100 metres away from the nearest obstacle. The middle of the rotor is placed 10 metres above any obstacle that is within 100 metres.

Wind Energy in India

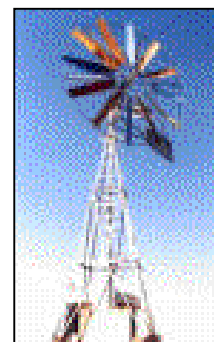
India has been rated as one of the most promising countries for wind power development, with an estimated potential of 20,000 MW. Total installed capacity of wind electric generators in the world as on Sept. 2001 is 23270 MW. Germany 8100 MW, Spain- 3175 MW, USA 4240 MW, Denmark 2417 MW, and India - 1426 MW top the list of countries. Thus, India ranks fifth in the world in Wind power generation.

There are 39 wind potential stations in Tamil Nadu, 36 in Gujarat, 30 in Andhra Pradesh, 27 in Maharashtra, 26 in Karnataka, 16 in Kerala, 8 in Lakshadweep, 8 Rajasthan, 7 in Madhya Pradesh, 7 in Orissa, 2 in West Bengal, 1 in Andaman Nicobar and 1 in Uttar Pradesh. Out of 208 suitable stations 7 stations have shown wind power density more than 500 Watts/ m².

Central Govt. Assistance and Incentives

The following financial and technical assistance are provided to promote, support and accelerate the development of wind energy in India:

- Five years tax holiday
- 100% depreciation in the first year
- Facilities by SEB's for grid connection
- Energy banking and wheeling and energy buy back
- Industry status and capital subsidy
- Electricity tax exemption
- Sales tax exemption



Applications

- Utility interconnected wind turbines generate power which is synchronous with the grid and are used to reduce utility bills by displacing the utility power used in the household and by selling the excess power back to the electric company.
- Wind turbines for remote homes (off the grid) generate DC current for battery charging.
- Wind turbines for remote water pumping generate 3 phase AC current suitable for driving an electrical submersible pump directly. Wind turbines suitable for residential or village scale wind power range from 500 Watts to 50 kilowatts.

12.4 Bio Energy

Biomass is a renewable energy resource derived from the carbonaceous waste of various human and natural activities. It is derived from numerous sources, including the by-products from the wood industry, agricultural crops, raw material from the forest, household wastes etc.



Biomass does not add carbon dioxide to the atmosphere as it absorbs the same amount of carbon in growing as it releases when consumed as a fuel. Its advantage is that it can be used to generate electricity with the same equipment that is now being used for burning fossil fuels. Biomass is an important source of energy and the most important fuel worldwide after coal, oil and natural gas. Bio-energy, in the form of biogas, which is derived from biomass, is expected to become one of the key energy resources for global sustainable development. Biomass offers higher energy efficiency through form of Biogas than by direct burning (see chart below).

Application

Bio energy is being used for:
Cooking, mechanical applications, pumping, power generation
Some of the devices : Biogas plant/ gasifier/burner, gasifier engine pump sets, stirling engine pump sets, producer gas/ biogas based engine generator sets

25 kg Fresh Dung		
Direct Burning		Biogas
5 kg dry dung	Product	1 cu m biogas
10460 kCal	Gross energy	4713 kCal
10%	Device efficiency	55%
1046 kCal	Useful energy	2592 kCal
None	Manure	10kg air dried manure

Biogas Plants



Biogas is a clean and efficient fuel, generated from cow-dung, human waste or any kind of biological materials derived through anaerobic fermentation process. The biogas consists of 60% methane with rest mainly carbon-di-oxide. Biogas is a safe fuel for cooking and lighting. By-product is usable as high-grade manure.

A typical biogas plant has the following components: A digester in which the slurry (dung mixed with water) is fermented, an inlet tank - for mixing the feed and letting it into the digester, gas holder/dome in which the generated gas is collected, outlet tank to remove the

spent slurry, distribution pipeline(s) to transport the gas into the kitchen, and a manure pit, where the spent slurry is stored.

Biomass fuels account for about one-third of the total fuel used in the country. It is the most important fuel used in over 90% of the rural households and about 15% of the urban households. Using only local resources, namely cattle waste and other organic wastes, energy and manure are derived. Thus the biogas plants are the cheap sources of energy in rural areas. The types of biogas plant designs popular are: floating drum type, fixed dome-type and bag-type portable digester.

Biomass Briquetting

The process of densifying loose agro-waste into a solidified biomass of high density, which can be conveniently used as a fuel, is called Biomass Briquetting (see Figure 12.8). Briquette is also termed as "Bio-coal". It is pollution free and eco-friendly. Some of the agricultural and forestry residues can be briquetted after suitable pre-treatment. A list of commonly used biomass materials that can be briquetted are given below:

CornCob, JuteStick, Sawdust, PineNeedle, Bagasse, CoffeeSpent, Tamarind, CoffeeHusk, AlmondShell, Groundnutshells, CoirPith, BagaseePith, Barleystraw, Tobaccodust, RiceHusk, Deoiled Bran



Figure 12.8 Biomass Briquetting

Advantages

Some of advantages of biomass briquetting are high calorific value with low ash content, absence of polluting gases like sulphur, phosphorus fumes and fly ash- which eliminate the need for pollution control equipment, complete combustion, ease of handling, transportation & storage - because of uniform size and convenient lengths.

Application

Biomass briquettes can replace almost all conventional fuels like coal, firewood and lignite in almost all general applications like heating, steam generation etc. It can be used directly as fuel instead of coal in the traditional chulhas and furnaces or in the gasifier. Gasifier converts solid fuel into a more convenient-to-use gaseous form of fuel called producer gas.

Biomass Gasifiers

Biomass gasifiers (see Figure 12.9) convert the solid biomass (basically wood waste, agricultural residues etc.) into a combustible gas mixture normally called as producer gas. The conversion efficiency of the gasification process is in the range of 60%–70%. The producer gas consists of mainly carbon-monoxide, hydrogen, nitrogen gas and methane, and has a lower calorific value (1000–1200 kcal/Nm³).



Figure 12.9 Biomass Gasifiers

Gasification of biomass and using it in place of conventional direct burning devices will result in savings of at least 50% in fuel consumption. The gas has been found suitable for combustion in the internal combustion engines for the production of power.

Applications:

Water pumping and Electricity generation: Using biomass gas, it is possible to operate a diesel engine on dual fuel mode—part diesel and part biomass gas. Diesel substitution of the order of 75 to 80% can be obtained at nominal loads. The mechanical energy thus derived can be used either for energizing a water pump set for irrigation purpose or for coupling with an alternator for electrical power generation - 3.5 KW - 10 MW

Heat generation: A few of the devices, to which gasifier could be retrofitted, are dryers— for drying tea, flower, spices, kilns for baking tiles or potteries, furnaces for melting non-ferrous metals, boilers for process steam, etc.

Direct combustion of biomass has been recognized as an important route for generation of power by utilization of vast amounts of agricultural residues, agro-industrial residues and forest wastes. Gasifiers can be used for power generation and available up to a capacity 500 kW. The Government of India through MNES and IREDA is implementing power-generating system based on biomass combustion as well as biomass gasification

High Efficiency Wood Burning Stoves

These stoves save more than 50% fuel wood consumption. They reduce drudgery of women saving time in cooking and fuel collection and consequent health hazards. They also help in saving firewood leading to conservation of forests. They also create employment opportunities for people in the rural areas.

Bio fuels

Unlike other renewable energy sources, biomass can be converted directly into liquid fuels— biofuels— for our transportation needs (cars, trucks, buses, airplanes, and trains). The two most common types of biofuels are *ethanol* and *biodiesel*. See Figure 12.10.

Ethanol is an alcohol, similar to that used in beer and wine. It is made by fermenting any biomass high in carbohydrates (starches, sugars, or celluloses) through a process similar to brewing beer. Ethanol is mostly used as a fuel additive to cut down a vehicle's carbon monoxide and other smog-causing emissions. Flexible-fuel vehicles, which run on mixtures of gasoline and up to 85% ethanol, are now available.

Biodiesel, produced by plants such as rapeseed (canola), sunflowers and soybeans, can be extracted and refined into fuel, which can be burned in diesel engines and buses. Biodiesel can also be made by combining alcohol with vegetable oil, or recycled cooking greases. It can be used as an additive to reduce vehicle emissions (typically 20%) or in its pure form as a renewable alternative fuel for diesel engines.



Figure 12.10 Biodiesel Driven Bus

Biopower

Biopower, or biomass power, is the use of biomass to generate electricity. There are six major types of biopower systems: *direct-fired*, *cofiring*, *gasification*, *anaerobic digestion*, *pyrolysis*, and *small - modular*.

Most of the biopower plants in the world use direct-fired systems. They burn bioenergy feedstocks directly in boiler to produce steam. This steam drives the turbo-generator. In some industries, the steam is also used in manufacturing processes or to heat buildings. These are known as combined heat and power facilities. For example, wood waste is often used to produce both electricity and steam at paper mills.

Many coal-fired power plants use cofiring systems to significantly reduce emissions, especially sulfur dioxide emissions. Cofiring involves using bio energy feedstock as a supplementary fuel source in high efficiency boilers.

Gasification systems use high temperatures and an oxygen-starved environment to convert biomass into a gas (a mixture of hydrogen, carbon monoxide, and methane). The gas fuels a gas turbine, which runs an electric generator for producing power.

The decay of biomass produces methane gas, which can be used as an energy source. Methane can be produced from biomass through a process called anaerobic digestion. Anaerobic digestion involves using bacteria to decompose organic matter in the absence of oxygen. In landfills –scientific waste disposal site - wells can be drilled to release the methane from the decaying organic matter. The pipes from each well carry the gas to a central point where it is filtered and cleaned before burning. Methane can be used as an energy source in many ways. Most facilities burn it in a boiler to produce steam for electricity generation or for industrial processes. Two new ways include the use of microturbines and fuel cells. Microturbines have outputs of 25 to 500 kilowatts. About the size of a refrigerator, they can be used where there are space limitations for power production. Methane can also be used as the "fuel" in a fuel cell. Fuel cells work much like batteries, but never need recharging, producing electricity as long as there is fuel.

In addition to gas, liquid fuels can be produced from biomass through a process called pyrolysis. Pyrolysis occurs when biomass is heated in the absence of oxygen. The biomass then turns into liquid called pyrolysis oil, which can be burned like petroleum to generate electricity. A biopower system that uses pyrolysis oil is being commercialized.

Several biopower technologies can be used in small, modular systems. A small, modular system generates electricity at a capacity of 5 megawatts or less. This system is designed for use at the small town level or even at the consumer level. For example, some farmers use the waste from their livestock to provide their farms with electricity. Not only do these systems provide renewable energy, they also help farmers meet environmental regulations.

Biomass Cogeneration

Cogeneration improves viability and profitability of sugar industries. Indian sugar mills are rapidly turning to bagasse, the leftover of cane after it is crushed and its juice extracted, to generate electricity. This is mainly being done to clean up the environment, cut down power costs and earn additional revenue. According to current estimates, about 3500 MW of power can be generated from bagasse in the existing 430 sugar mills in the country. Around 270 MW of power has already been commissioned and more is under construction.

12.5 Hydro Energy

The potential energy of falling water, captured and converted to mechanical energy by waterwheels, powered the start of the industrial revolution.

Wherever sufficient head, or change in elevation, could be found, rivers and streams were dammed and mills were built. Water under pressure flows through a turbine causing it to spin.

The Turbine is connected to a generator, which produces electricity (see Figure 12.11). In order to produce enough electricity, a hydroelectric system requires a location with the following features:

Change in elevation or head: 20 feet @ 100 gal/min = 200 Watts.
100 feet head @ 20 gal/min gives the same output.

In India the potential of small hydro power is estimated about 10,000 MW. A total of 183.45 MW small Hydro project have been installed in India by the end of March 1999. Small Hydro Power projects of 3 MW capacity have been also installed individually and 148 MW project is under construction.

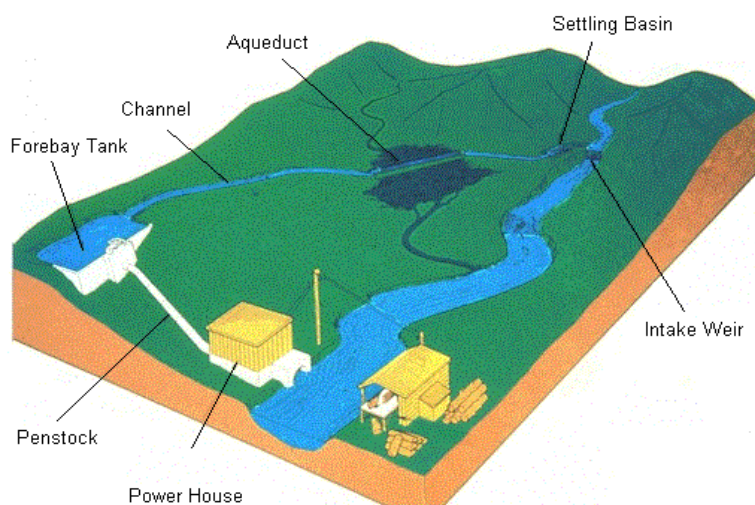
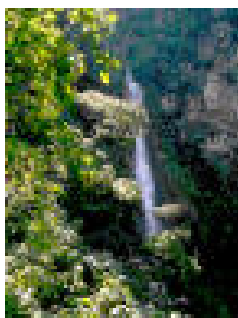


Figure 12.11 Hydro Power Plant

Small Hydro

Small Hydro Power is a reliable, mature and proven technology. It is non-polluting, and does not involve setting up of large dams or problems of deforestation, submergence and rehabilitation. India has an estimated potential of 10,000 MW



Micro Hydel

Hilly regions of India, particularly the Himalayan belts, are endowed with rich hydel resources with tremendous potential. The MNES has launched a promotional scheme for portable micro hydel sets for these areas. These sets are small, compact and light weight. They have almost zero maintenance cost and can provide electricity/power to small cluster of villages. They are ideal substitutes for diesel sets run in those areas at high generation cost.

Micro (upto 100kW) mini hydro (101-1000 kW) schemes can provide power for farms, hotels, schools and rural communities, and help create local industry.

12.6 Tidal and Ocean Energy

Tidal Energy

Tidal electricity generation involves the construction of a barrage across an estuary to block the incoming and outgoing tide. The head of water is then used to drive turbines to generate electricity from the elevated water in the basin as in hydroelectric dams.



Barrages can be designed to generate electricity on the ebb side, or flood side, or both. Tidal range may vary over a wide range (4.5-12.4 m) from site to site. A tidal range of at least 7 m is required for economical operation and for sufficient head of water for the turbines.

Ocean Energy

Oceans cover more than 70% of Earth's surface, making them the world's largest solar collectors. Ocean energy draws on the energy of ocean waves, tides, or on the thermal energy (heat) stored in the ocean. The sun warms the surface water a lot more than the deep ocean water, and this temperature difference stores thermal energy.

The ocean contains two types of energy: thermal energy from the sun's heat, and mechanical energy from the tides and waves.

Ocean thermal energy is used for many applications, including electricity generation. There are three types of electricity conversion systems: closed-cycle, open cycle, and hybrid. Closed cycle systems use the ocean's warm surface water to vaporize a working fluid, which has a low boiling point, such as ammonia. The vapour expands and turns a turbine. The turbine then activates a generator to produce electricity. Open-cycle systems actually boil the seawater by operating at low pressures. This produces steam that passes through a turbine / generator. The hybrid systems combine both closed-cycle and open-cycle systems.

Ocean mechanical energy is quite different from ocean thermal energy. Even though the sun affects all ocean activity, tides are driven primarily by the gravitational pull of the moon, and waves are driven primarily by the winds. A barrage (dam) is typically used to convert tidal energy into electricity by forcing the water through turbines, activating a generator.

India has the World's largest programmes for renewable energy. Several renewable energy technologies have been developed and deployed in villages and cities of India. A Ministry of Non-Conventional Energy Sources (MNES) created in 1992 for all matters relating to Non-Conventional / Renewable Energy. Government of India also created Renewable Energy Development Agency Limited (IREDA) to assist and

provide financial assistance in the form of subsidy and low interest loan for renewable energy projects.

IREDA covers a wide spectrum of financing activities including those that are connected to energy conservation and energy efficiency. At present, IREDA's lending is mainly in the following areas: -

- Solar energy technologies, utilization of solar thermal and solar photo voltaic systems
- Wind energy setting up grid connected Wind farm projects
- Small hydro setting up small, mini and micro hydel projects
- Bio-energy technologies, biomass based co-generation projects, biomass gasification, energy from waste and briquetting projects
- Hybrid systems
- Energy efficiency and conservation

The estimated potential of various Renewable Energy technologies in India by IREDA are given below.

Energy source estimated potential

Solar Energy	20 MW / sq. km
Wind Energy	20,000 MW
Small Hydro	10,000 MW
Ocean Thermal Power	50,000 MW
Sea Wave Power	20,000 MW
Tidal Power	10,000 MW
Bio energy	17,000 MW
Draught Animal Power	30,000 MW
Energy from MSW	1,000 MW
Biogas Plants	12 Million Plants
Improved Wood Burning Stoves	120 Million Stoves
Bagasse-based cogeneration	3500 MW

Cumulative achievements in renewable energy sector (As on 31.03.2000)

<u>Sources / Technologies</u>	<u>Unit Upto 31.03.2000</u>
Wind Power	MW 1167
Small Hydro	MW 217
Biomass Power & Co-generation	MW 222
Solar PV Power	MW / Sq. km 42
Urban & MSW	MW 15.21
Solar Heater	m ² . Area 480000
Solar Cookers	No. 481112
Biogas Plants	Nos. in Million 2.95
Biomass Gasifier	MW 34
Improved Chulhas	Nos. in Million 31.9

Summary. — The contribution is conceived for non-nuclear experts, intended as a synthetic and simplified overview of the technology related to energy by nuclear fission. At the end of the paper, the Reader will find a minimal set of references, several of them on internet, useful to start deepening the knowledge on this challenging, complex, debated albeit engaging energy source.

1. – Why nuclear energy is still an option

As an introductory reflection, we should recognise that energy represents a very complex equation, where no easy or ultimate solutions are yet available.

Today, notwithstanding a renewed criticism arisen after the Fukushima event, nuclear energy by fission is still an option in several Countries.

The main reasons are

- the cost of electricity produced is usually cheap or at least competitive with other energy sources, provided that some boundary conditions are confirmed (*e.g.* construction time schedule kept);
- nuclear is an almost CO₂-free energy on the whole life-cycle, together with renewables [1,2], see fig. 1, hence global warming and related environmental concerns are substantially reduced;
- nuclear is a high-quality industry hence is usually used to improve and develop country's economy, since the largest part of the investment is on the construction phase and not on the fuel cost as for the fossil fuels (likely to be covered with a

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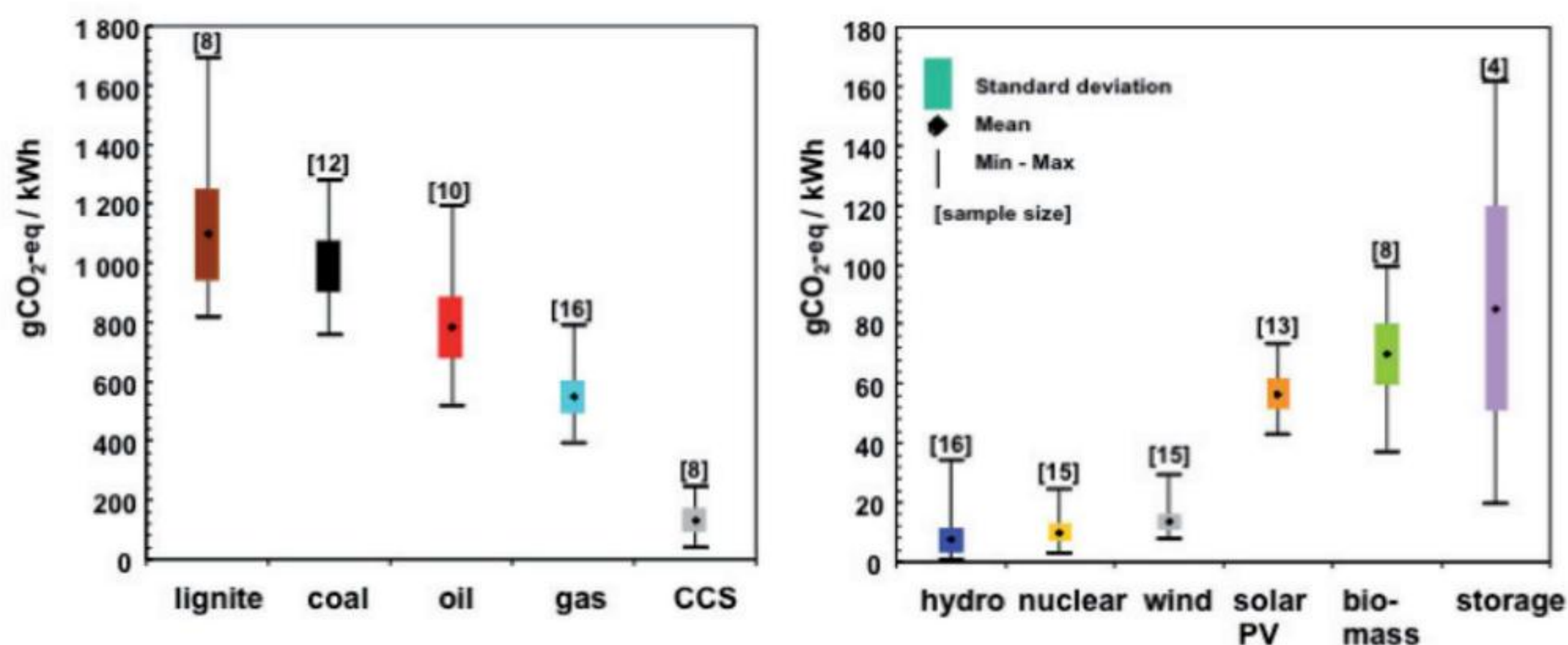
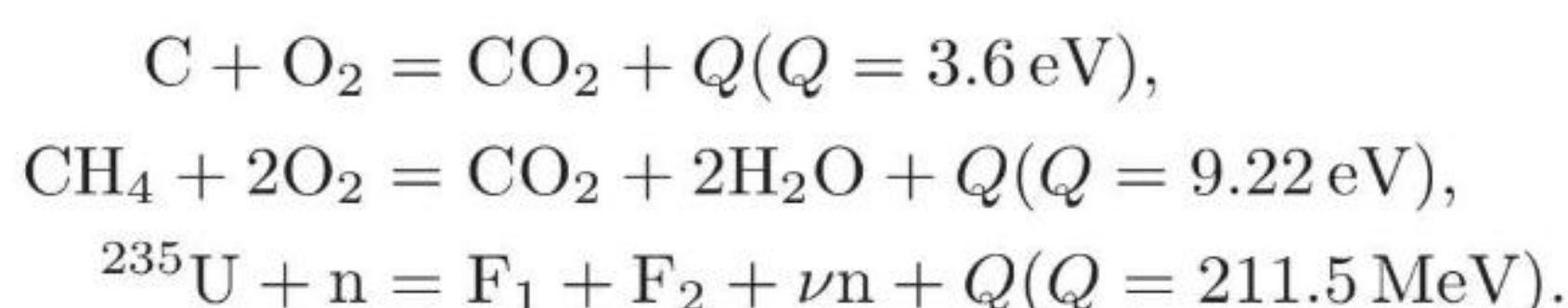


Fig. 1. – Estimation of life-cycle CO₂-equivalent emissions per electricity produced with different fuels.

large disbursement outside the country), moreover that phase could be carried out with local industries for more than 50%.

The main interest in this source of energy is physically based on the so-called “mega factor”. When comparing the energy released by a chemical reaction, as in the combustion phase of a coal (C) atom or a methane (CH₄) molecule, with a nuclear reaction on a nucleus of uranium (²³⁵U isotope) by a neutron (n), the amount of energy (Q, in ElectronVolt) differs from six orders of magnitude:



That means, on a larger scale, 5 grams of nuclear fuel (UO₂) are energetically equivalent to 640 kg of wood, or 360 m³ of methane, or 400 kg of coal, or 350 kg of oil, moreover with no CO₂ emissions.

Today, only a small amount of natural Uranium (which isotopic composition is: ²³⁴U, 0.006%; ²³⁵U, 0.712%; ²³⁸U, 99.282%) is exploited in the current, thermal nuclear reactors, since only the ²³⁵U isotope undergoes fission with thermal neutrons. In the future, 70% of the natural content of this element could be used, by transmutation of the largest isotope (²³⁸U) into a fissile isotope with fast neutrons, in fast nuclear reactors.

The high energy density of this fuel reflects also in the land use and in the infrastructure requirements of a power station.

For a 3000 MWe nuclear power station, roughly 150 hectares are needed, a size that doubles for a coal or oil fired power plant of the same installed power, due to the fuel storage on-site. Renewables are more demanding: for one third of the installed power, some square km are needed for hydro-power (a large dam), dozens of square km for solar panels, hundreds for a wind farm, thousands for a biomass plantation.

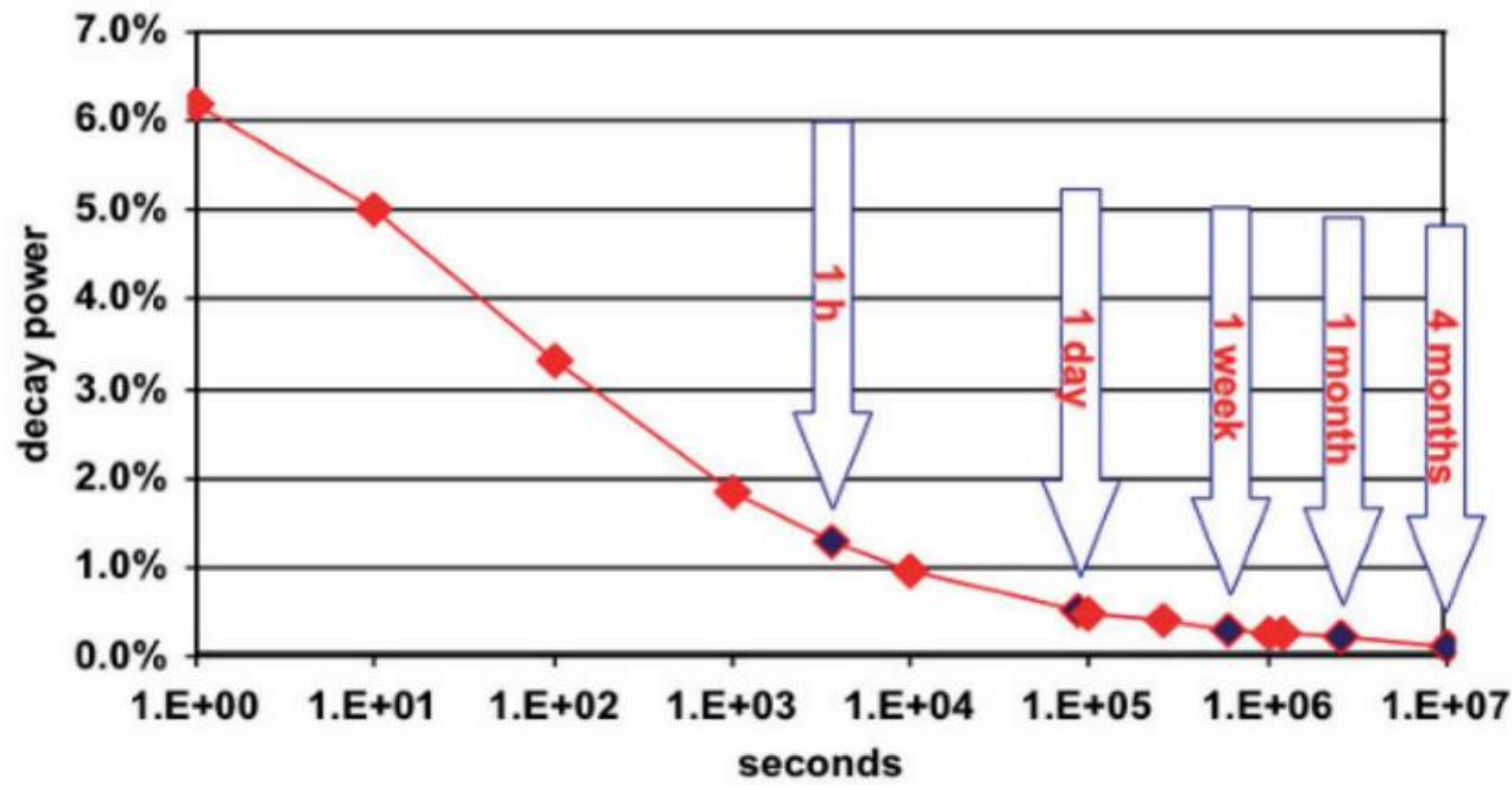


Fig. 2. – Nuclear decay power after shutdown.

While one single truck, loading the new fuel is required to annually operate a nuclear reactor, one oil tanker a week is needed for an oil-fired power station, or one trainload per day for the coal-fired power station, or the umbilical dependence from a pipeline for the gas-fired power station.

2. – Why a nuclear reactor is different from any other power plant

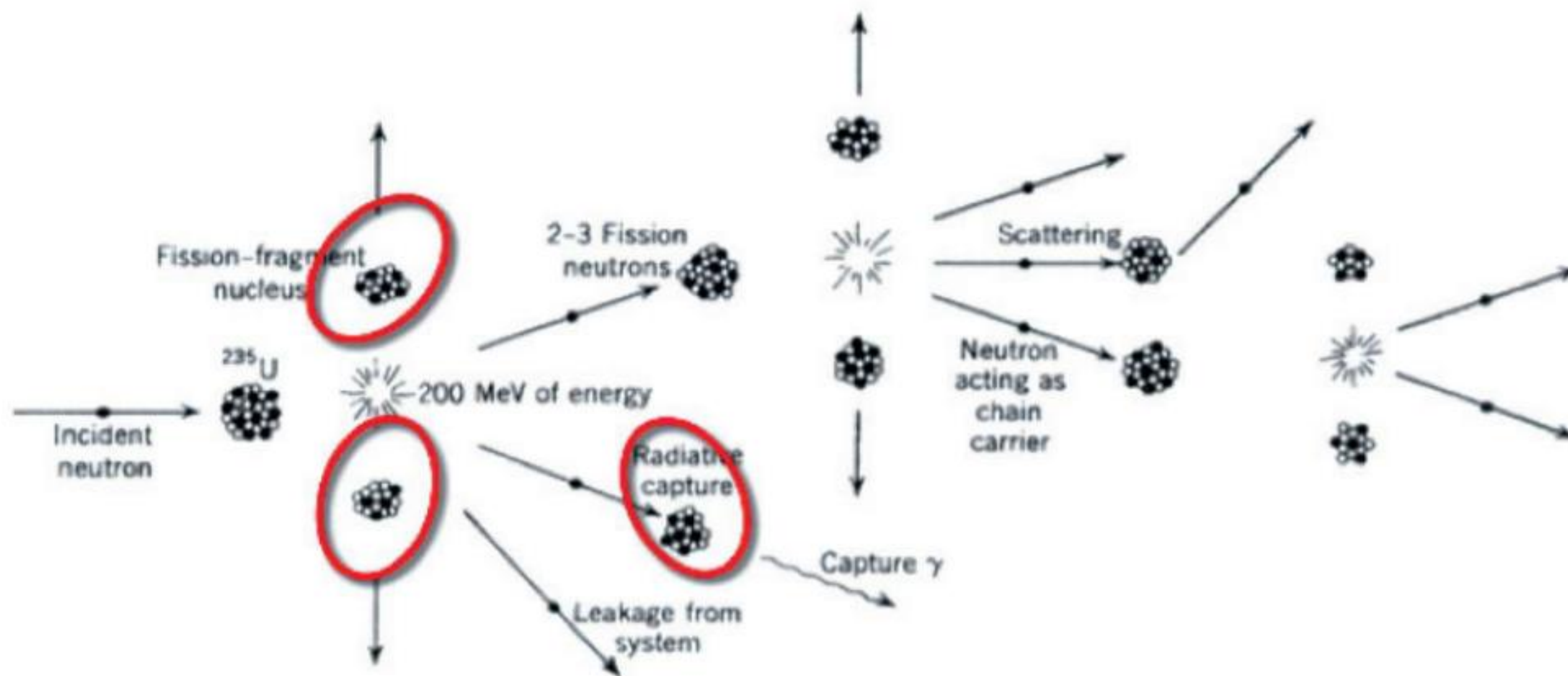
Together with the pros, come the cons.

Nuclear energy and nuclear reactors have unique, critical features that must be duly addressed, in order to exploit the above-mentioned positive characteristics while reducing the corresponding risks.

The main issues are on safety and nuclear waste, topics that are typical in any debate on nuclear. Those issues have their founding phenomenon in the fission event.

During that event, caused by a slow, thermal neutron interacting with a nucleus of ^{235}U , more than 200 MeV of energy are released, the largest part as kinetic energy of the couple of fission fragments that are generated. Those isotopes are rich in neutrons hence instable and decay to more energetically stable configurations by emitting energy, in radiation form. The fission fragments release that decay energy with delay after the fission event, with characteristics time ranging from seconds to years.

That radioactivity represents the first critical and unique aspect for a nuclear reactor: even if the neutrons are absorbed hence the fissions stopped, the fuel continues emitting radiation. The resulting decay power is equal to almost 6% of the full thermal power in the first second after the reactor shutdown, then decays with time but remains at sensible levels for long time (*e.g.*, 0.2% after 1 month, see fig. 2). That means safety systems to reject the decay heat are required to enter and remain in operation for each nuclear reactor, in case of accident. Otherwise, as occurred in Fukushima, the fuel could overheat and fail, releasing the radioactivity into the safety containment building and possibly to the environment, if the safety containment is subject to a further failure.



the “nuclear waste”: Fission Products and Transuranic elements

Fig. 3. – The nuclear fission chain reaction: energy released and nuclear waste generated.

A second unique feature is the possibility to increase the power of the nuclear reactor beyond the designed nominal power, behaviour impossible in the other power plants. The fission events and the neutron population are balanced during a normal, stable operation of the reactor: a constant production of neutrons generated by the previous fission events, generates in turn a constant and stable number of new fissions. A balanced chain reaction is reached when one neutron, among the 2 or 3 produced for each fission, is able to generate a further fission in the fuel. The remaining 1 or 2 neutrons are absorbed into the fuel, the structural materials of the reactor, the fluid acting as moderator and coolant, or leak from the reactor. If more than one neutron continues the fission process, the chain reaction could diverge, generating an exponential increase of the corresponding power. That means further safety systems are required for the nuclear reactor, able to stop the chain reaction as soon as the control is lost or whatsoever accident occurs. The shutdown control rods are neutron absorbers entering in operation in few seconds to stop the nuclear reaction.

A third unique feature, related to the nuclear waste issue, is related to the absorption of neutrons by the fuel, as mentioned before. The large amount of ^{238}U present in the fuel, usually enriched only at 3–4% in the fissile isotope ^{235}U , leads to the transmutation of the heaviest Uranium isotope into further, heavier nuclides by neutron capture. Thus transuranic actinides like plutonium (Pu), americium (Am), curium (Cm) are created by sequential neutron absorption into the fuel. These actinides are rich in neutrons hence instable, radioactive alfa-emitters and several of them are long-lived (*e.g.*, Pu half life is equal to 23 000 years).

These transuranic elements generated by neutron capture, together with the fission fragments produced by neutron fission, are the real, dangerous nuclear waste, (see fig. 3) bringing in a small amount of matter (10% of the nuclear waste volume) 90% of the whole radioactivity produced by a nuclear reactor.

A fourth feature is the possibility to trigger exothermic chemical reactions during severe accident scenarios, leading to hydrogen production. The cladding of the nuclear fuel is usually made of zirconium alloy, since that type of steel is a lower neutron absorption material. But in case of severe accidents, typically with a loss of fuel cooling capabilities, if the cladding temperature rises beyond 1200 K the water steam reacts with the zirconium, which generates a fast oxidation that produces hydrogen, as occurred in Fukushima. The related safety systems to avoid such a scenario adopt hydrogen burners, catalytic recombiners or inertised containment buildings.

A fifth uniqueness of the nuclear reactors refers to economics. The production cost of electricity (€/kWh_e) is largely based (> 50%) on capital investment cost, *i.e.* the overnight cost of construction of the nuclear power plant plus the financial interests during the construction period, while the fuel costs are around 25%–30% and operation and maintenance costs the remaining part. Exactly the opposite than any other fossil fuelled power plant, where more than 70% of the cost of electricity production is the fuel cost. This cost structure implies that nuclear is competitive when fossil fuels cost is high and the cost of money is low.

3. – Basics of nuclear reactors

From a simplified, technical point of view, a nuclear reactor is a sort of nuclear boiler producing steam, which is sent to a turbine that moves a generator, hence producing electricity.

More than 80% of the nuclear power plants in operation nowadays belong to the pressurised water reactor (PWR) or to the boiling water reactor (BWR) type (see fig. 4).

Both of them use water as moderator to slow down (thermalize) the neutrons to increase the fission probability of ²³⁵U, as well as fluid to cool the fuel. The main difference is that in PWRs the water is kept in liquid phase by high pressure (155 bar), to enhance the moderation feature, hence a secondary circuit is needed to produce the steam, while in BWR the steam is generated directly into the primary circuit and sent to the turbine.

The nuclear fuel is usually in the form of UO₂ pellets, 8 mm diameter and 12 mm height, piled up into zircalloy cladding cylinders 3.5 m length. A square matrix 8 × 8 (BWR) or 17 × 17 (PWR) of those fuel rods forms one single fuel assembly. According to the size of the reactor, hundreds of fuel assemblies form the nuclear core, to be cooled by the water.

The shutdown control rods, the water cooling and the water injecting systems are the main safety systems connected to the primary and secondary circuits (only for PWRs) of the reactor.

The last barrier to avoid radioactive release towards the environment is the safety containment system. Usually in PWRs a steel or a concrete containment is provided, able to withstand the maximum pressure and temperature created by the steam released by the primary cooling system into the building during a loss-of-coolant accident. In BWRs a different strategy is adopted: the steam is released into a dry-well chamber,

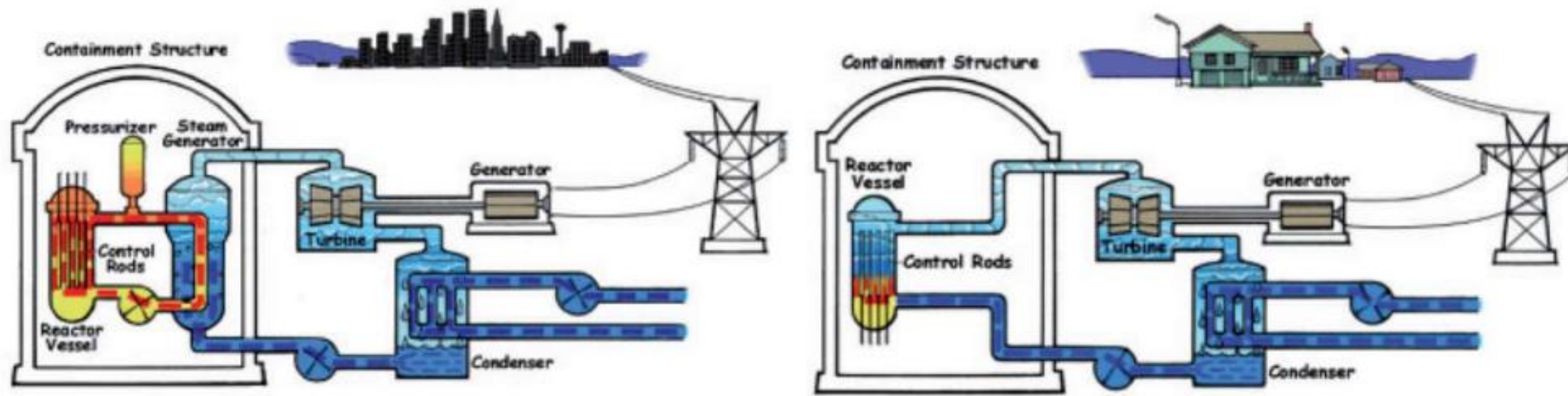


Fig. 4. – Left: PWR type; right: BWR type.

then directed into a wet-well chamber by means of large piping that guide the steam into a suppression water pool, where it is condensed. Both the containments have to withstand also to external accidents, ranging from natural (tornadoes, floods, earthquakes) to man-made (airplane crash) events.

A more general classification for reactors adopting thermal, moderated neutrons is reported in table I.

TABLE I. – *Main thermal reactors, classified by type of moderator-coolant-fuel.*

Moderator	Coolant	Fuel type, enrichment	Reactor type
H ₂ O	H ₂ O	UO ₂ , ~ 3%	PWR, BWR
D ₂ O	D ₂ O	UO ₂ , nat.	CANDU
C	H ₂ O	UO ₂ , nat.	RBMK
C	CO ₂	U metallic, nat.	MAGNOX
C	CO ₂ He	UO ₂ or UC ₂ , 1-2% up to 93%	AGR HTGR

4. – Basics of nuclear fuel cycle

A long and complex journey is required to produce and manage the nuclear fuel (see fig. 5).

The ore of the natural uranium is mined with the same classical methods adopted in the mining industry, in open or underground mines, or is extracted by leaching.

The ore is then concentrated and purified, to eliminate all the rare earths and other chemical elements than UO₂, that could represent a poison for the neutrons in the final nuclear fuel.

Since the content in ²³⁵U in the natural isotopic mixture is only 0.7%, the fuel is usually enriched up to 3–4% in order to optimise its use in the nuclear reactor. The enrichment process requires the conversion from UO₂ to UF₆, a fluorinated compound that can be transported in solid state and easily transformed into a gas by heating it at low temperature (60 °C). The gaseous state is needed to mechanically enrich the fuel,

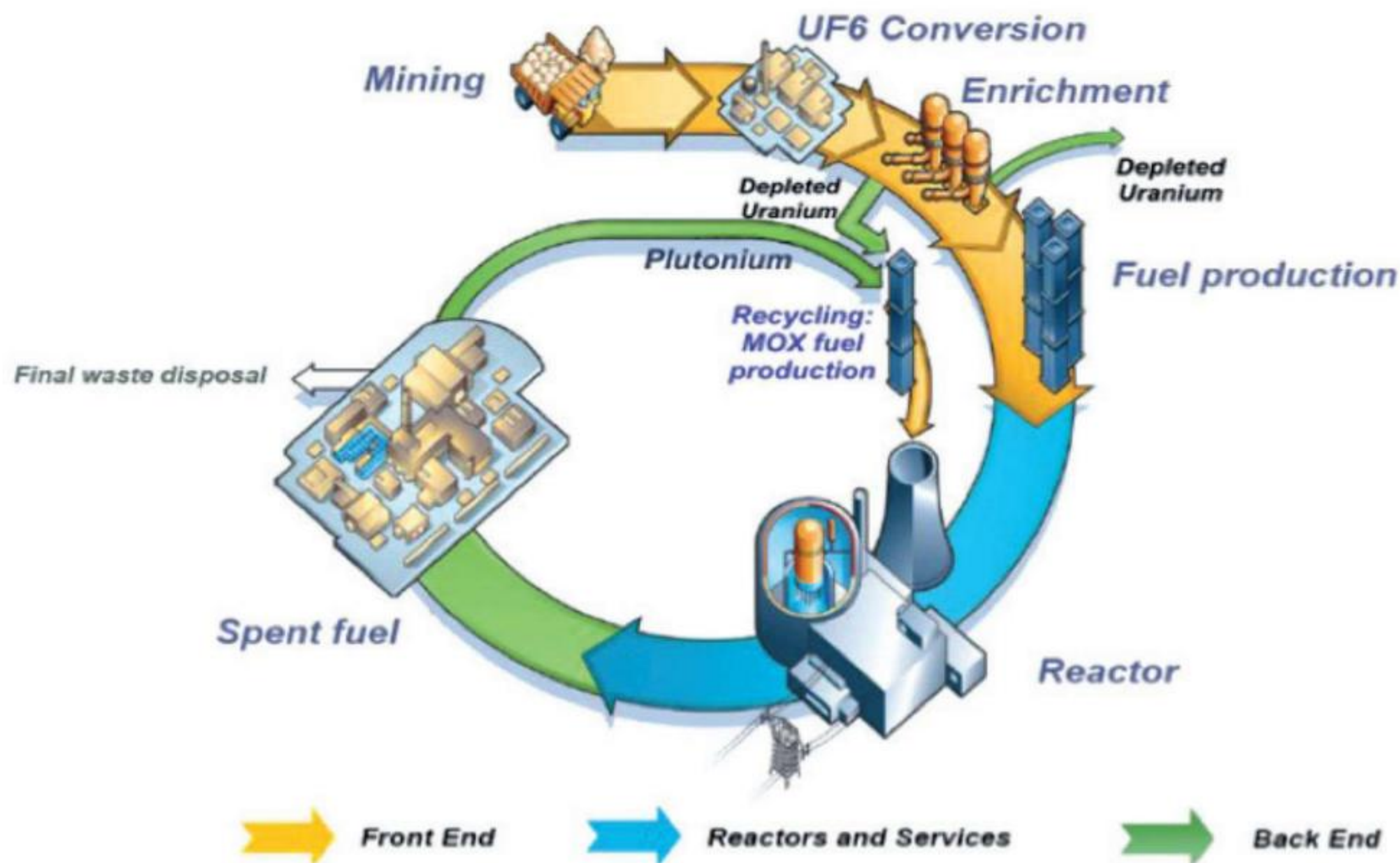


Fig. 5. – The main steps of the Nuclear Fuel Cycle.

i.e. increasing the ^{235}U content with respect to ^{238}U . The available enrichment processes are based on the gaseous diffusion across a porous membrane and on the ultracentrifuge technology. Both the processes exploit the different velocity or the different centrifugal force acting on the $^{235}\text{UF}_6$ molecules with respect to the $^{238}\text{UF}_6$ ones. But the ultracentrifugation method is more effective for the separation capacity (1 order of magnitude better than the gaseous diffusion) and for the energy consumption (50 times less than the electricity consumed by the gaseous diffusion).

Once enriched, the UF_6 must be re-converted into UO_2 to create the fuel assemblies. The UO_2 powder is synthesised to form stable, ceramic pellets, which fill in the zircalloy fuel rods, than grouped in square matrix to create a fuel assembly.

The fresh fuel assemblies replace the spent fuel ones into the reactor, once the power plant is shutdown for refuelling and maintenance operations every 12–18 months.

The spent fuel contains mainly ^{238}U (still $> 95\%$), the fission fragments and the transuranic elements accumulated during the fuel burnup period, as well as some ^{235}U the reactor was not able to burn. The transuranic elements include ^{239}Pu , a new fuel for the reactors but also a strategic element, used to produce the nuclear warheads.

At this point in the cycle, the spent fuel assemblies could be sent to the temporary or, in the near future, to the final waste repository, creating an “open cycle”. Otherwise if a “closed cycle” is selected, the spent fuel assemblies are sent to a recycling facility, to retrieve the fissile isotopes ^{235}U and ^{239}Pu (to produce further fuel, the so-called MOX, Mixed OXide fuel, $\text{UO}_2 + \text{PuO}_2$) and to separate and concentrate the fission products and the transuranic elements, which are conglomerated into a special glass matrix able

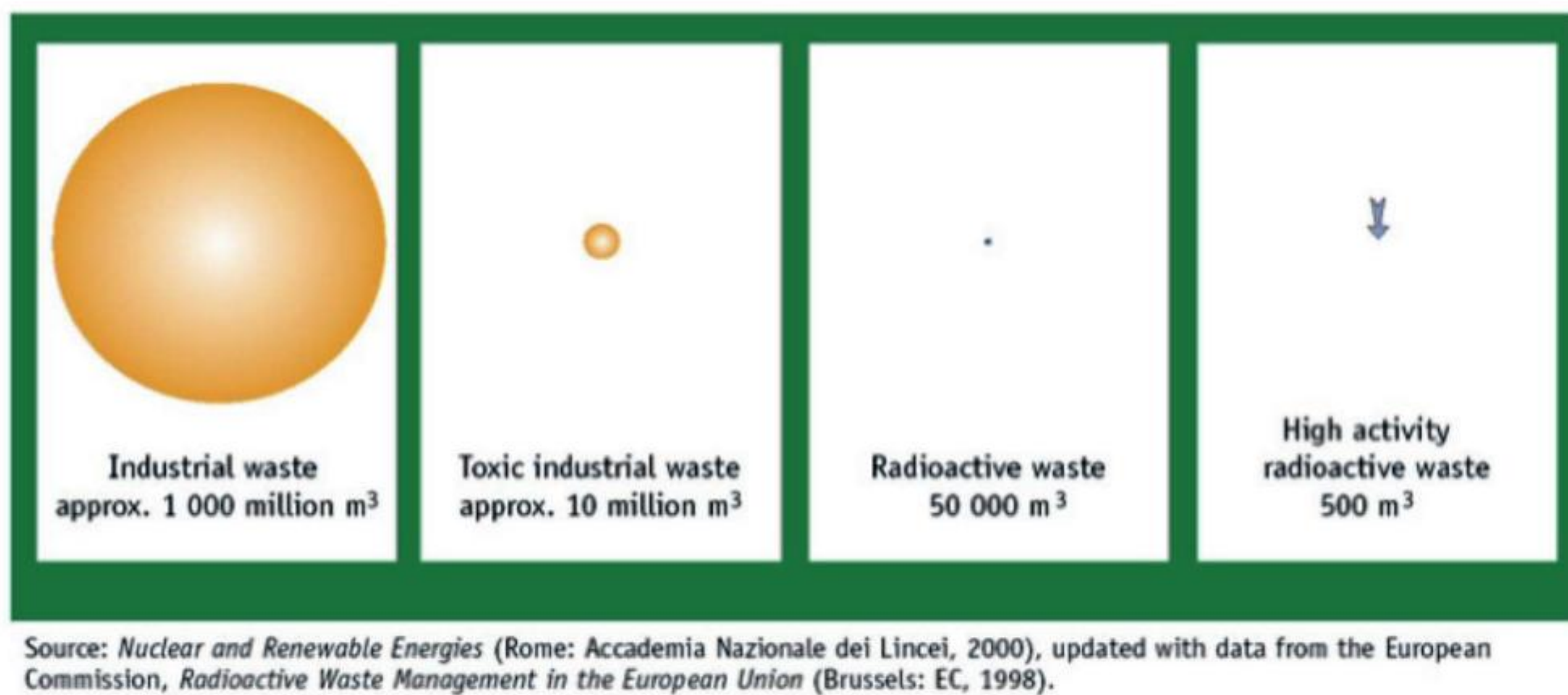


Fig. 6. – Waste generation comparison: industrial and nuclear waste yearly produced in the European Union.

to efficiently reject the decay heat and to avoid any chemical or water attack in the millennia. The largest mass of the spent nuclear fuel is the depleted uranium ^{238}U , stored in canisters on site.

A couple of options are envisaged as final solution for the high radioactive nuclear waste coming from the spent fuel: the final, geological repository and the waste burning.

Nowadays some countries like Finland, Sweden and France are preparing the geological repository underground (around 500 m depth), in stable layers of rock or rock salt or clay, to place the whole spent fuel assemblies coming from the open cycle or the separated and concentrated nuclear waste coming from the closed cycle.

The alternative will be to transmute or “burn” the high radioactive and long-lived isotopes from the fission products and the transuranic elements into fast neutron reactors, usually liquid metal cooled. Some new generation reactors (or “Generation IV” reactors) are under development also for this purpose.

As a final annotation, the real burden given by the nuclear waste by comparison with the industrial waste annually produced should be considered. A self-explanatory picture is offered in fig. 6, referred to the EU production.

5. – Basics of nuclear safety

Since the beginning of the development of the nuclear energy, the nuclear safety is founded on the concept of “defence-in-depth”. A consecutive set of safety barriers, both of technological and operational type, is created around the source of the hazard, *i.e.* the nuclear fuel containing the highly radioactive isotopes.

The first barrier is the fuel matrix itself: the synthesised, ceramic UO_2 is selected to accommodate a large part of the gaseous fission products into its porosity, and to resist to the temperature reached by the fuel during the normal operation and to the neutron flux, which has the capability to change the material structure.

The second barrier is represented by the fuel cladding, the third is the primary system boundary and the fourth is the containment building.

A further level of safety is offered by the set of safety systems connected to the primary, the secondary and the containment systems. Redundancy, separation and diversification are the key requirements, to reduce the risk of failure. Even if one single safety system is enough to cope with a specific accident, a multiplicity of two, three or even four is usually envisaged (redundancy). Moreover, different types of safety systems are adopted to obtain the same function (diversification).

The first safety system is the shutdown control rods system, to stop the reaction chain. A second is the decay heat removal system, to avoid fuel overheating. A third is the safety injection system, to keep the core cooled even in case of a break in the primary circuit, leading to a loss of coolant. A fourth is the containment spray system, to condensate the steam inside the pressure containment building. The same system is envisaged also inside the reactor pressure vessel for the BWRs.

The last operational barrier is represented by the emergency planning procedures, adopted to manage or evacuate the surroundings of the nuclear power plant in case all the previous safety barriers suffer some faults.

Both a deterministic and a probabilistic approach are followed in the design and the evaluation of the safety level of a nuclear reactor. The main goal is to minimise the risk of accident and to provide a suitable set of safety systems to cope with all the different and possible accident scenarios that may occur, in case of internal or external initiating events, natural or man-made.

But the real, estimated risk is usually different from the risk perception. So it is not surprising if the nuclear energy is perceived as less safe than the other energy sources or even other, daily-life human activities. Statistical data are anyway available in the literature, so a scientifically and technically based comparison can be performed. Again it would not be surprising, after that analysis, to see that to live around a nuclear power station is less risky than to be hit by lightning having a tour on the mountains, or to die at home due to a domestic accident or to die due to a collision while driving our own car.

6. – Nuclear reactor generations

Several “generations” of nuclear reactors (see fig. 7) have been developed since the first demonstration of the possibility to ignite and control the fission nuclear chain reaction, by Enrico Fermi the 2nd of December 1942 under the Stadium of the University of Chicago.

In the 50s and 60s, the first generation of nuclear reactors went into operation in the Western World as well as in Russia and in Japan: gas-graphite, PWRs and BWRs were the main technologies adopted, with sizes of the plant in the range of few dozens up to few hundreds of MWe.

Then the second generation came, up to the 90s, generally an evolution of the previous reactors, mainly driven by the economy of scale law, demonstrating the commercial strength of the nuclear technology. The size grew up to 1200–1300 MWe.

Generations of Nuclear Energy

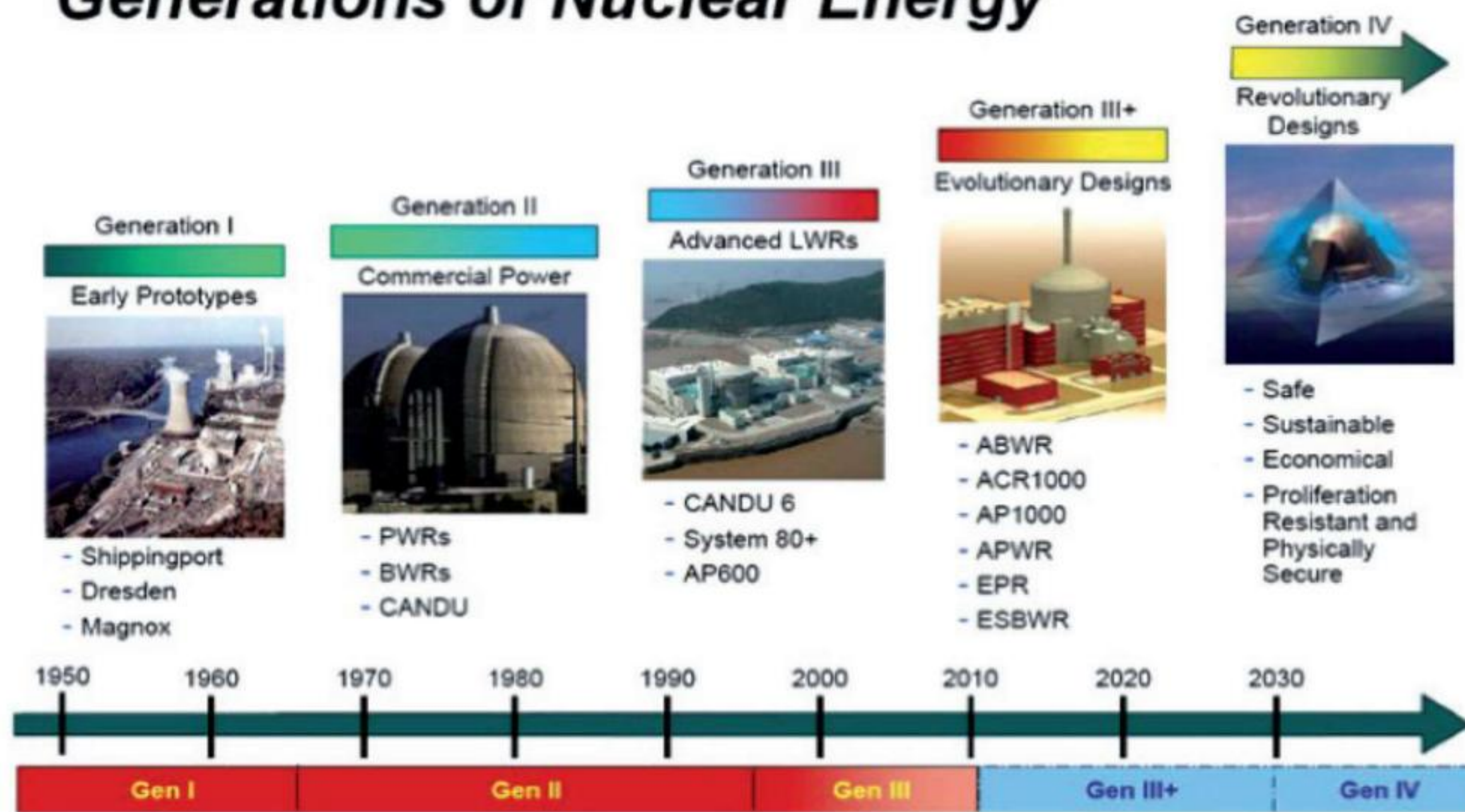


Fig. 7. – Nuclear reactor generations: from the pioneering age to the next decades.

From the 90s up to the early period of the new millennium, another generation of nuclear reactors were developed, the so called Generation III or III+: almost in any country where the nuclear energy was well developed, advanced and innovative projects were launched.

The economy of scale was in some cases still the driver, but in other designs the system simplification including a size reduction was addressed. In all the new projects, higher safety level was the main goal, after the Three Mile Island and the Chernobyl events.

Two different safety strategies have been adopted: some designs implement a further enhancement of the “classic” active safety, with redundant and diversified systems relying on energy sources (mainly electricity) external or internal to the site; other designs implement the “innovative” passive safety concept, where no or less need of electricity, mechanical components to be activated and human intervention are required, since the driving forces are usually based on “natural” physical principles, like the natural circulation of fluids at different temperatures hence densities.

Some examples of that different approach to reach a new, high level of safety are offered by the French EPR and the US-Japanese ABWR (see fig. 8), based on active systems, and by the US AP1000 and the US-Japanese ESBWR (see fig. 9), based on passive systems.

The first is a PWR with very large size (1600 MWe) and presents a quadruple redundancy of the safety systems (highlighted with four different colours in the figure), to substantially reduce the probability of unavailability of at least one system, which is



Fig. 8. – GenIII reactors based on Active Safety Systems. Left: EPR; right: ABWR.

enough to cope with the accident. Moreover, the containment system is a double concrete building, the first conceived to address internal accidents and the related increase of pressure, the second to withstand external events including airplane crashes. Again, to minimise the unavailability of the safety systems even in case of external accidents, the four safety systems are distributed around the bunkered reactor containment. Also the four emergency diesel generators are placed in two bunkered buildings.

EPRs are now under construction in Finland, France and China.

The ABWR is an advanced boiling reactor, already built and in operation in Japan, large size (1350 MWe), with ten internal pumps placed at the bottom of the reactor pressure vessel, to avoid piping penetrations hence eliminating the risk of large primary piping breaks. It implements the better knowledge and active safety systems so far developed on BWRs.

Based on the passive safety philosophy are the AP1000 and the ESBWR.

The first is a PWR, 1100 MWe size, with two instead of four primary loops and corresponding steam generators and active primary pumps, which adopts passive safety both for the primary and the containment systems. A large pool inside the containment is used as heat sink to reject the decay heat and as water supplier to inject coolant inside the reactor: both functions are performed in natural circulation hence without pumps and electricity. Moreover, also the containment building exploits external, natural circulation of air or evaporation of water for its cooling in case of accident, since it is made by steel instead of concrete. A second concrete external building protects the internal steel one from aircraft crash, tornadoes and similar events.

AP1000 reactors are under construction in US and China.

The second is a fully passive BWR, since not only the safety systems but even the

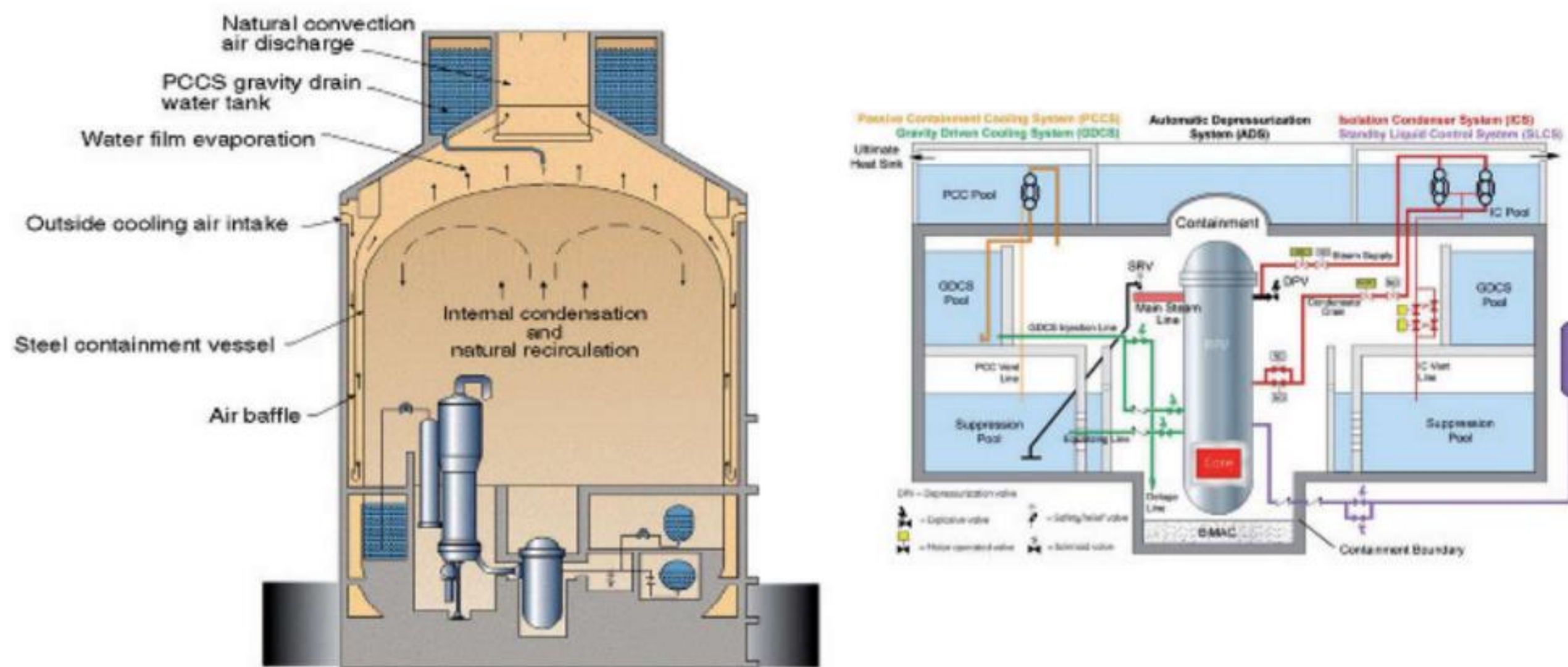


Fig. 9. – GenIII reactors based on Passive Safety Systems. Left: AP1000; right: ESBWR.

primary coolant flows in natural circulation, in normal operation. Several pools provide a large amount of water both for gravity injection into the core and for steam condensation hence heat rejection, in case of accident. Notwithstanding this fully natural behaviour, the reactor size is large: 1520 MWe.

Even severe accident scenarios are addressed in those designs. Some of them, like the EPR, apply the “ex-vessel” core retention strategy, with the melted corium assumed as released outside the primary reactor pressure vessel and then distributed in a suitable cavity under the reactor and there cooled. Other, like the AP1000, adopt the “in-vessel” core retention strategy, with the melted fuel collected at the bottom of the reactor pressure vessel and there cooled, both by the internal (water injection) and the external (cavity flooding) side.

In the path of development of nuclear reactor generations, the last, future step in the nuclear fission technology is the so-called Generation IV [3]. In 2000, ten Countries launched an international committee of nuclear scientists and technical experts, to identify and select the nuclear technologies for the future, able to satisfy the main requirements of

- sustainability, with more effective nuclear fuel exploitation, less waste production and less long-time burden,
- economics,
- enhanced safety, with no need for emergency response,
- non-proliferation and physical protection.

The result was a set of six technologies: the Gas-cooled Fast Reactor, the Molten Salt Reactor, the Supercritical Water-cooled Reactor, the Very-High Temperature Gas Reactor, the Lead-cooled fast Reactor and the Sodium-cooled Fast Reactor.

Those types of reactor should be developed through an international cooperation and become available in the years 2030–2040. Mainly due to the challenging environments and operating conditions required, new materials and innovative solutions are needed. The Sodium-cooled and the Lead-cooled options seem the most promising in the expected time, given the experience gained in the past on those technologies.

The GenIV fast reactors are expected also to offer a ultimate contribution in solving the problem of the high level nuclear waste, since they could reduce the radiotoxicity of the transuranic elements and the long-lived fission products to acceptable levels in a reduced amount of time, from 100 000 to 300 years.

7. – A future after Fukushima?

Are all the new features and developments of nuclear reactors, described in the previous sections, enough to ensure a future to the nuclear energy?

After Fukushima (see fig. 10), it seems hard to answer. But first of all, some items should be pointed out referring to the Japanese events. Some rough and brutal data may help: the earthquake and the resulting tsunami were cause of more than 14 000 deaths and 11 000 missing, hundreds of thousands evacuees, more than 100 000 buildings fully destroyed, 1 700 roads damaged, 50 bridges and one dam collapsed. The same brutal data referred only to the Fukushima-Daiichi nuclear power station, report 3 deaths due to non-radiation causes, 21 workers contaminated, almost all under the dose limits prescribed by the Japanese Safety Authority, 190 000 evacuees, soil and water contaminated and still under monitoring.

A recent preliminary report issued by the World Health Organisation on the health risk [4] assessment of the radiation doses released by the Fukushima reactors, concludes that a very small increase (4–7%) in the normal risk of generating leukaemia or a solid cancer in the whole lifetime could occur to the most affected populations (2 zones in the Fukushima Prefecture).

Moreover, the investigation committees that operated and operate in analysing the accident and the possible causes, at national as well as at international level, are con-



Fig. 10. – Effects of the earthquake and corresponding tsunami in Japan on March 11, 2011.

1. INTRODUCTION

Geothermal power plants can be divided into two main groups, steam cycles and binary cycles. Typically the steam cycles are used at higher well enthalpies, and binary cycles for lower enthalpies. The steam cycles allow the fluid to boil, and then the steam is separated from the brine and expanded in a turbine. Usually the brine is rejected to the environment (re-injected), or it is flashed again at a lower pressure. Here the Single Flash (SF) and Double Flash (DF) cycles will be presented.

A binary cycle uses a secondary working fluid in a closed power generation cycle. A heat exchanger is used to transfer heat from the geothermal fluid to the working fluid, and the cooled brine is then rejected to the environment or re-injected. The Organic Rankine Cycle (ORC) and Kalina cycle will be presented.

2. SINGLE FLASH

A flow sheet for the SF cycle is shown in Figure 1. The geothermal fluid enters the well at the source inlet temperature, station 1. Due to the well pressure loss the fluid has started to boil at station 2, when it enters the separator. The brine from the separator is at station 3, and is re-injected at station 4, the geothermal fluid return condition. The steam from the separator is at station 5, where the steam enters the turbine. The steam is then expanded through the turbine down to station 6, where the condenser pressure prevails.

The condenser shown here is air cooled, with the cooling air entering the condenser at station *c1* and leaving at station *c2*. The condenser hot well is at station 7. The fluid is re-injected at station 4.

Typically, such a process is displayed on a thermodynamic T-s diagram, where the the temperature in the cycle is plotted against the entropy (Figure 2). The condition at station 1 is usually compressed liquid. In vapour dominated fields, such as Lardarello in Italy, the inflow is in the wet region close to the vapour saturation line.

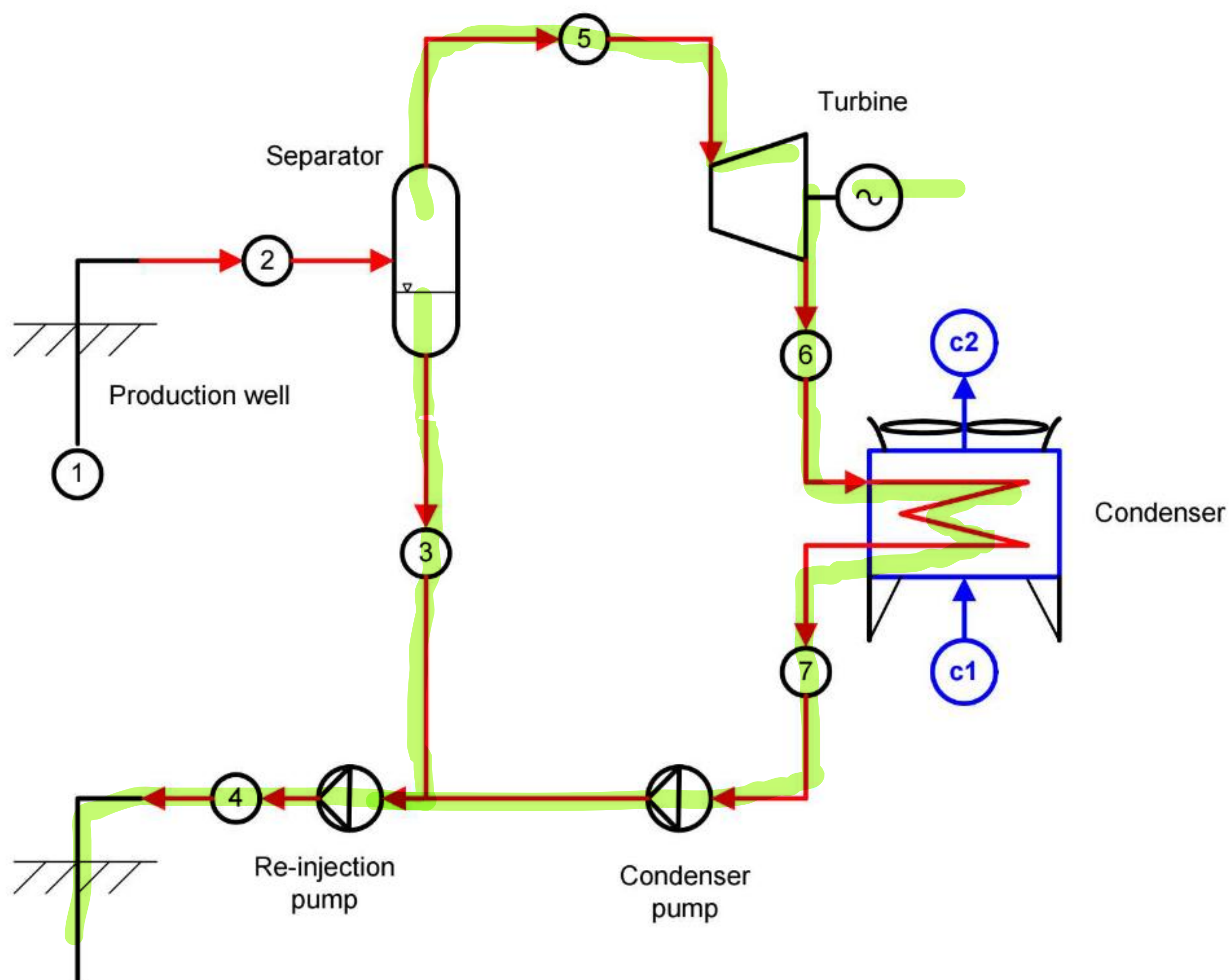


FIGURE 1: Single flash cycle schematic

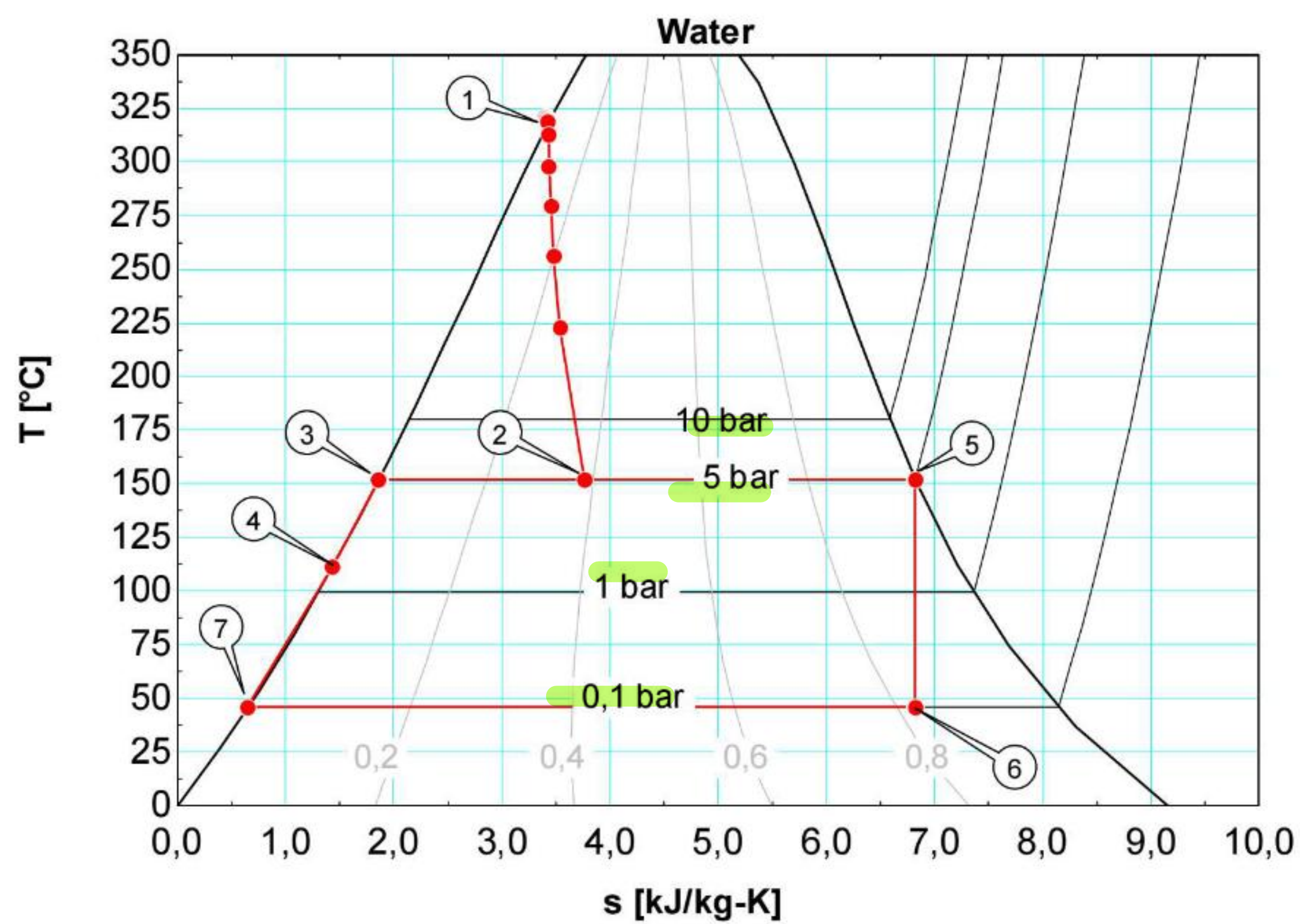


FIGURE 2: T-s diagram of a single flash cycle

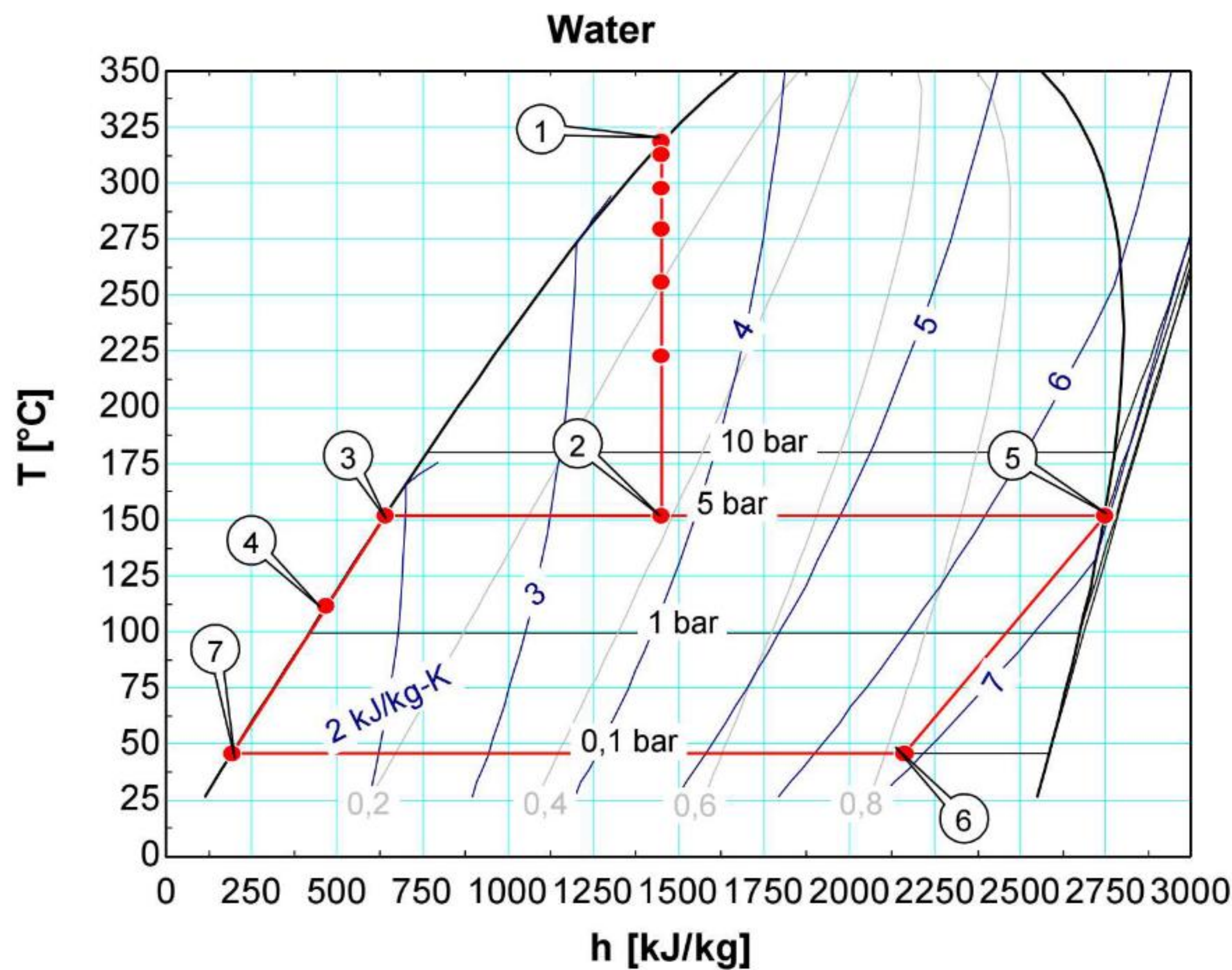


FIGURE 3: T-h diagram of a single flash cycle

3. DOUBLE FLASH

A flow sheet for the DF cycle is shown in Figure 4. The geothermal fluid enters the well at the source inlet temperature, station 1. Due to the well pressure loss the fluid has started to boil at station 2, when it enters the separator. The brine from the separator is at station 3, and is throttled down to a lower pressure level at station 8. The partly boiled brine is then led to a low pressure separator, where the steam is led to the turbine at station 9. The turbine is designed in such a way, that the pressure difference over the first stages is the same as the pressure difference between the high and low pressure separators. The mass flow in the lower pressure stages of the turbine is then higher than in the high pressure stages, just the opposite of what happens in a traditional fuel fired power plant with a bleed for the feedwater heaters from the turbine.

The brine from the low pressure separator is at station 10, and is then re-injected at station 4, the geothermal fluid return condition.

The steam from the high pressure separator is at station 5, where the steam enters the turbine. The low pressure steam enters the turbine a few stages later, at station 9. The steam is then expanded through the turbine down to station 6, where the condenser pressure prevails.

The condenser shown here is air cooled, with the cooling air entering the condenser at station *c1* and leaving at station *c2*.

The condenser hot well is at station 7. The fluid is re-injected at station 4.

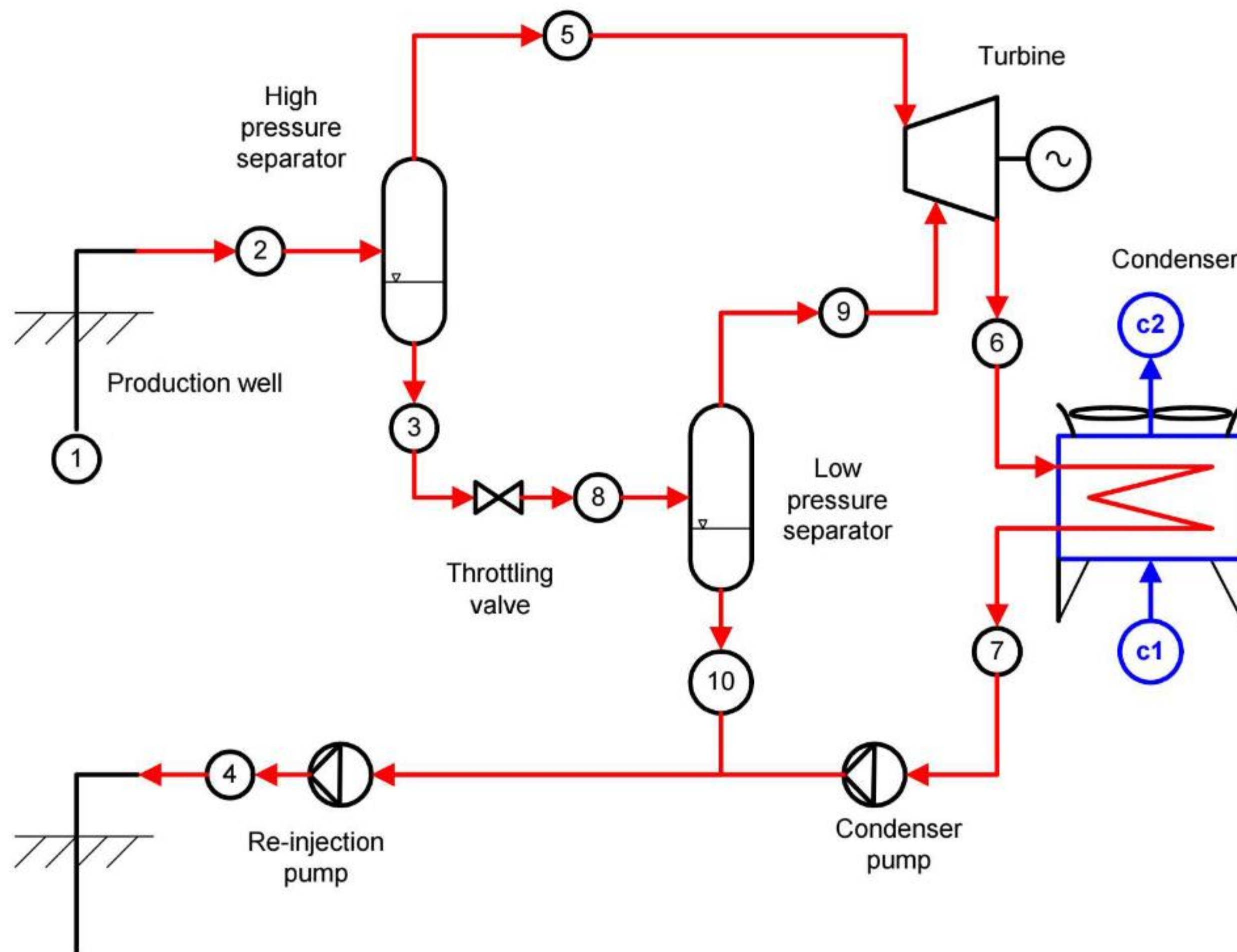


FIGURE 4: Double flash cycle schematic

The double flash cycle is presented in Figure 5 on a T-s diagram and on a T-h diagram in Figure 6.

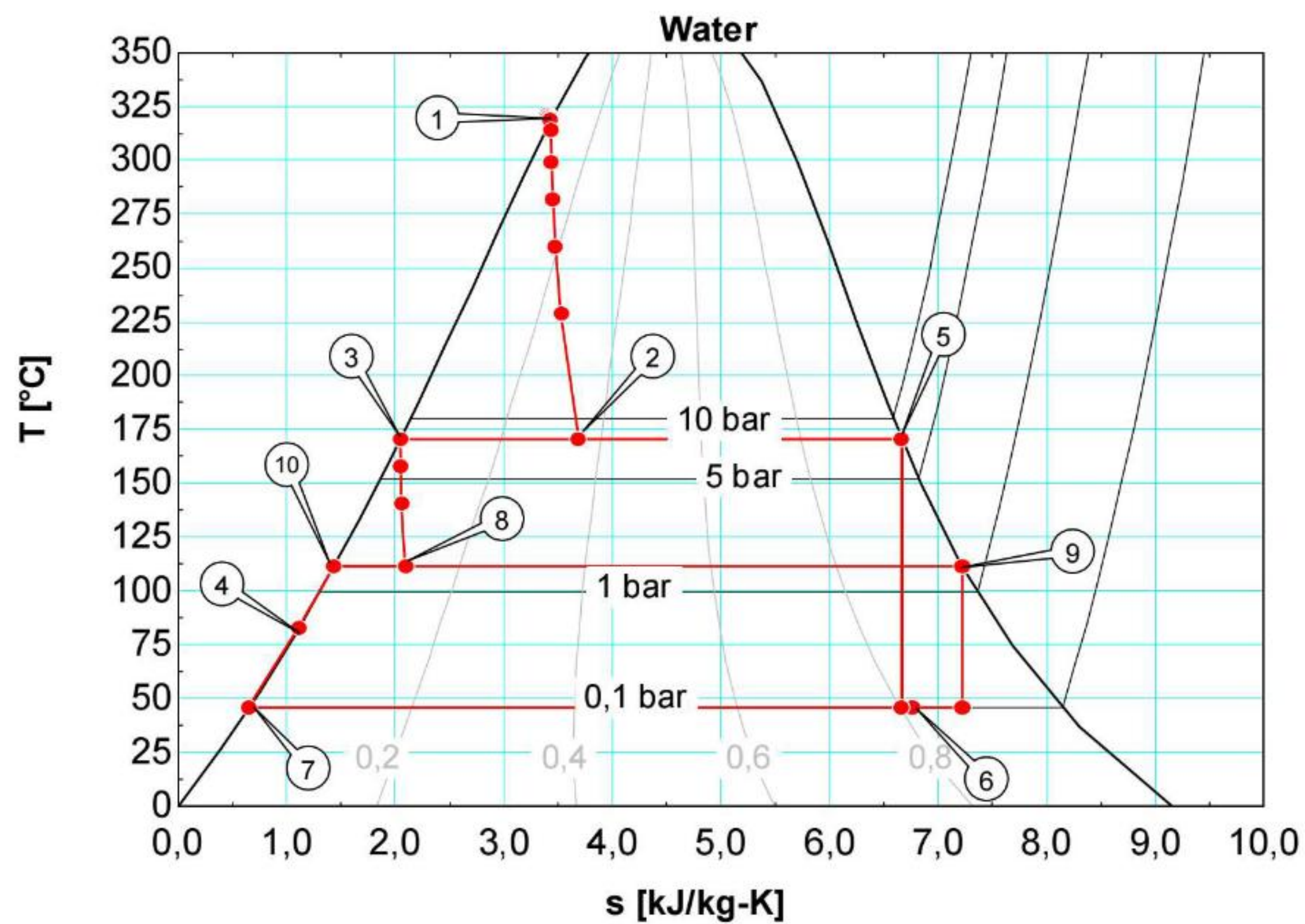


FIGURE 5: T-s diagram of a double flash cycle

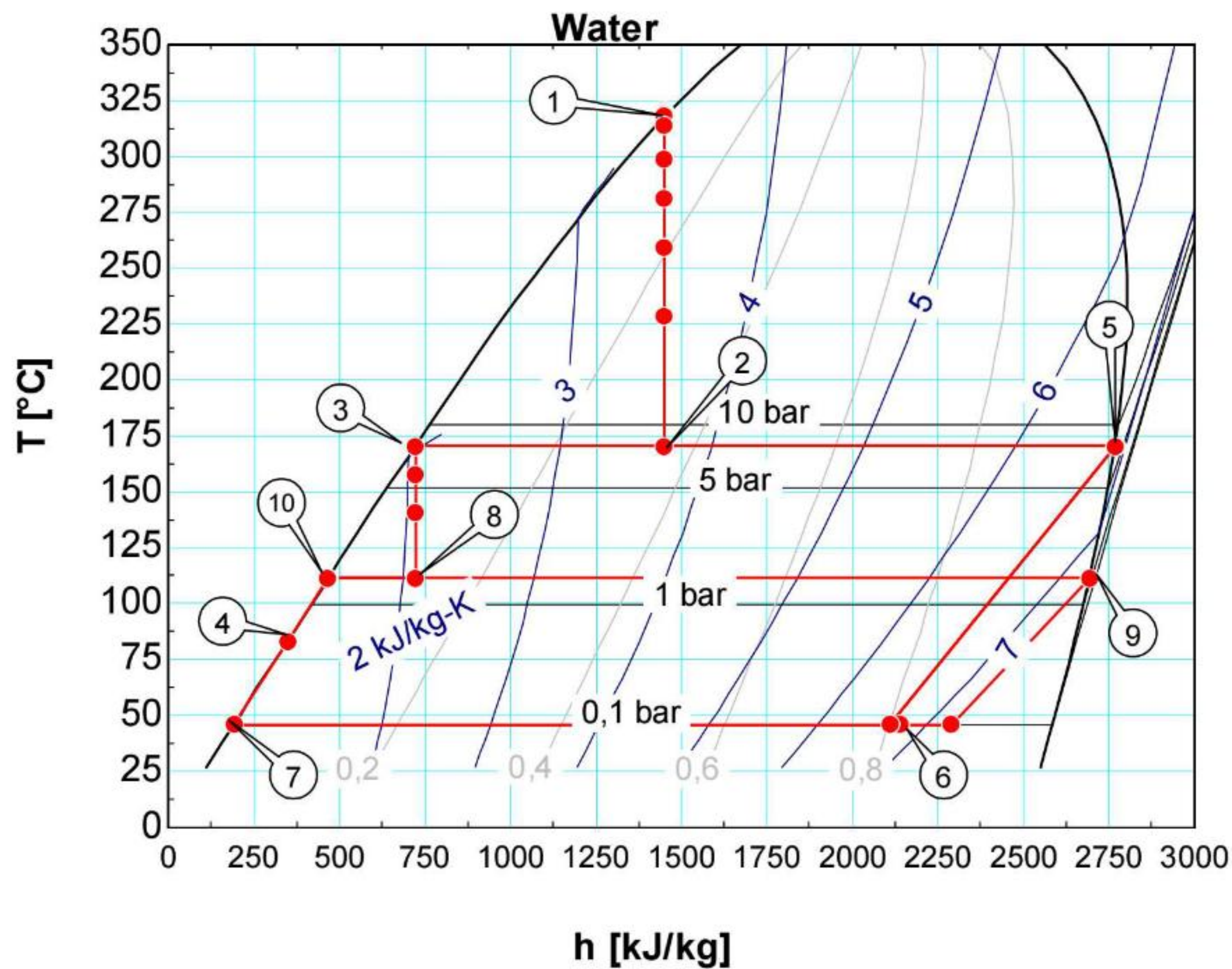


FIGURE 6: T-h diagram of a double flash cycle

4. ORGANIC RANKINE CYCLE (ORC)

A flow sheet for the ORC cycle is shown in Figure 7. The geothermal fluid enters the well at the source inlet temperature, station $s1$. The fluid is frequently liquid water. If the pressure is kept sufficiently high, no non-condensable gases will be separated from the liquid, and a gas extraction system is not necessary. The fluid is then cooled down in the vaporizer, and sent to re-injection at station $s2$.

The ORC with regeneration is presented in Figure 8 on a T-s diagram and on a T-h diagram in Figure 9.

Pre-heated (in the regenerator) ORC fluid enters the vaporizer at station 2. The fluid is heated to saturation in the vaporizer, or even with superheat in some cases. The vapour leaves the vaporizer at station 3, and enters the turbine.

The exit vapour from the turbine enters the regenerator at station 4, where the superheat in the steam can be used to pre-heat the condensed fluid prior to vaporizer entry. The now cooled vapour enters the condenser at station 5, where it is condensed down to saturated liquid at station 6.

A circulation pump raises the pressure from the condenser pressure up to the high pressure level in station 1. There the fluid enters the regenerator for pre-heat before vaporizer entry.

The condenser shown here is air cooled, with the cooling air entering the condenser at station $c1$ and leaving at station $c2$.

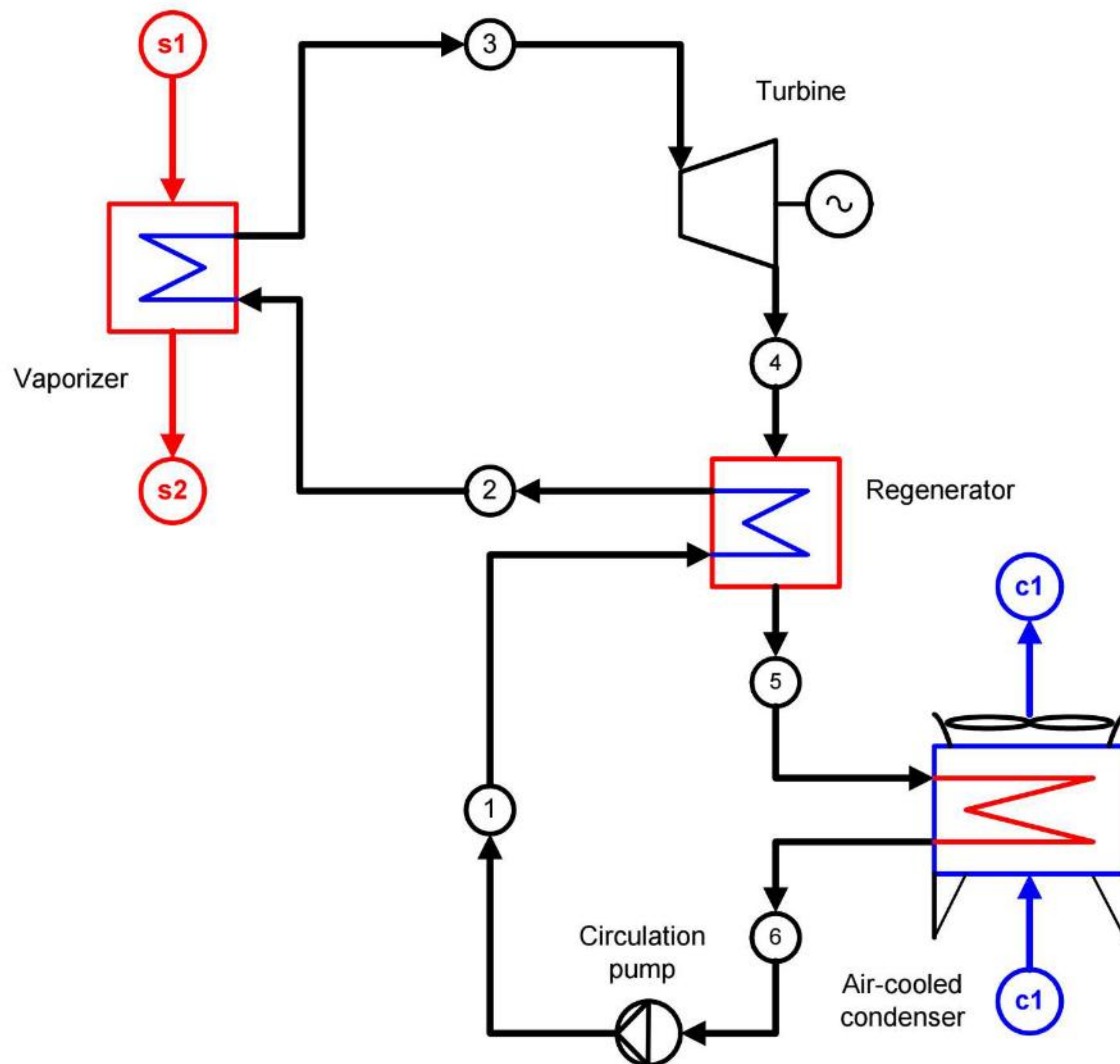


FIGURE 7: Flow diagram for an ORC cycle with regeneration

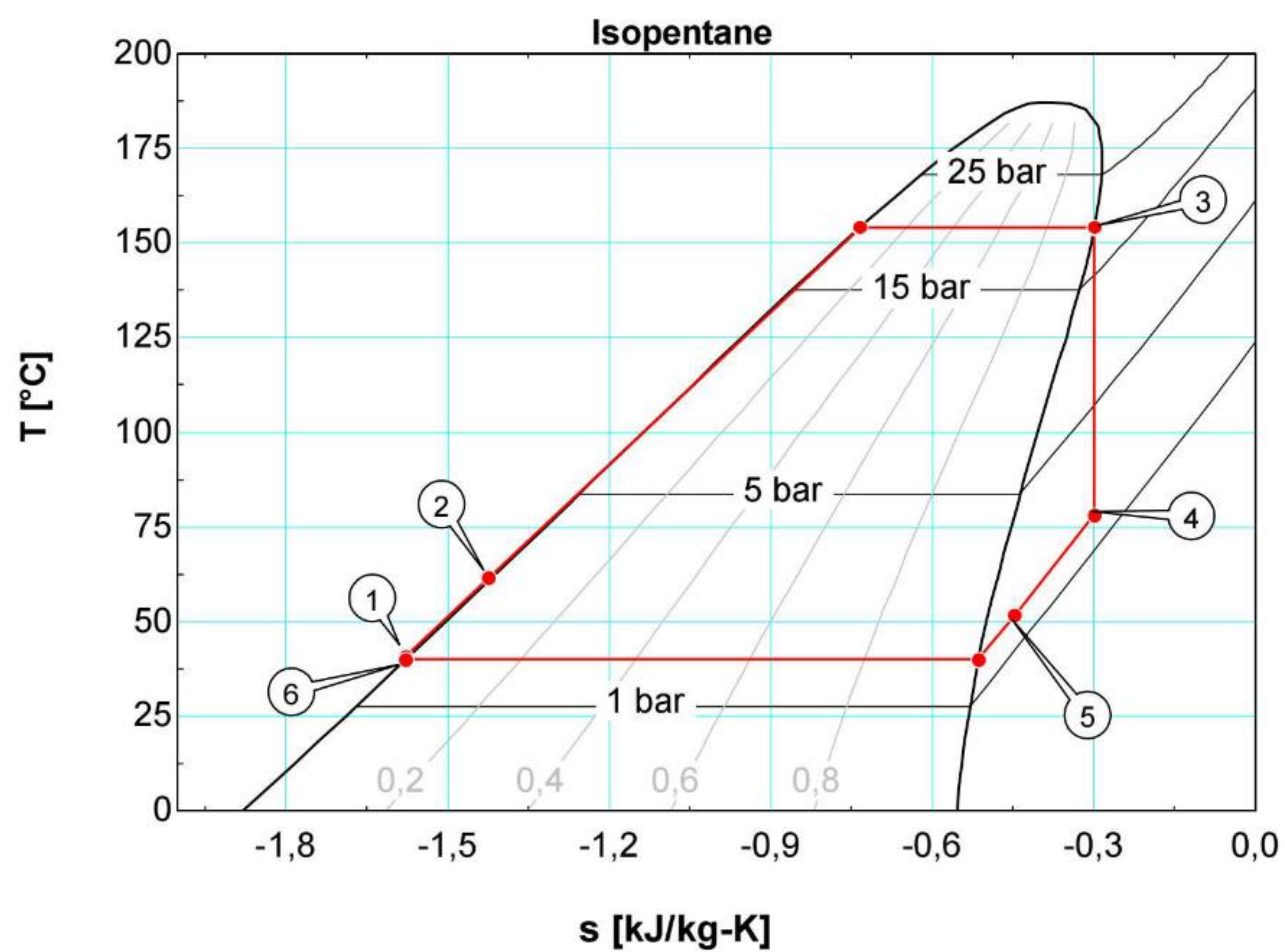


FIGURE 8: T-s diagram of an ORC cycle with regeneration

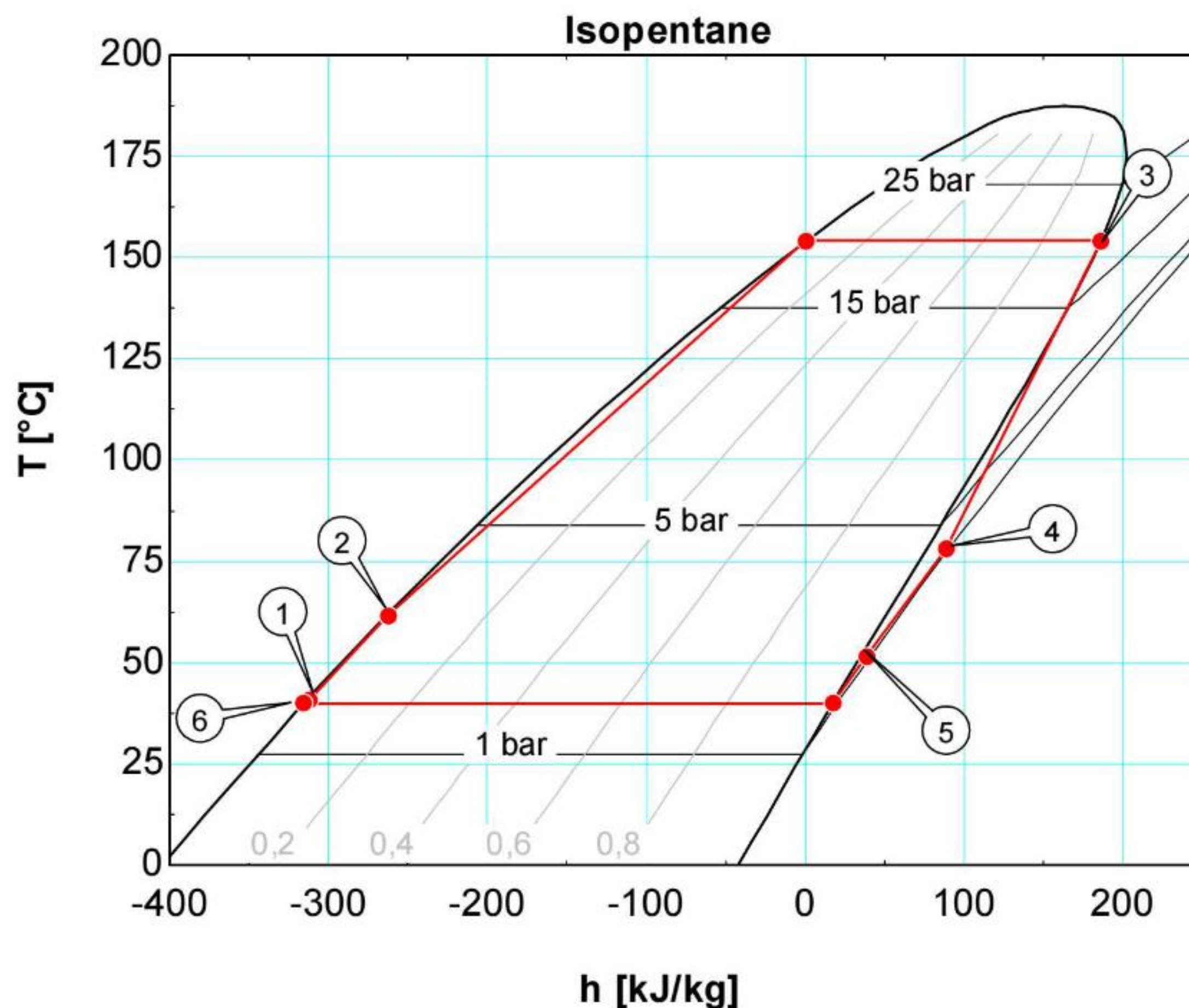


FIGURE 9: T-h diagram of an ORC cycle with regeneration

5. REGENERATION

Regeneration will increase the power plant efficiency. Then a part of the rejected heat is recovered for input to the power plant. If the plant were run on fuel, this would lead to direct fuel savings. This is not the case in geothermal power production. There the wells have certain maximum flow rate, and the well cost is usually entirely fixed, has very little if any relation to the flow from the well. The more the fluid from the well can be cooled, the more heat can be input to the power plant.

Regeneration increases the temperature of the working fluid at the vaporizer entry, and leads thus to higher geothermal fluid exit temperature from the vaporizer. The heat removal from the geothermal fluid is thus partly replaced by the recovered heat.

There is frequently a lower temperature limit on the geothermal fluid temperature. This limit may be imposed by chemistry (danger of scaling) or the requirements of a secondary process, such as district heating. If this is the case, regeneration can help.

Figure 10 shows a calculation of an isopentane ORC cycle, with geothermal fluid temperature of 200°C. It is assumed that the well flow is 1 kg/s, condensation temperature is 40°C. Three curves are calculated, no regeneration at all, if 50% of the heat available is used for regeneration and finally if all the available heat is used. The available heat is the heat which can be removed from the turbine exit vapour until the vapour reaches dew point. After that the vapour temperature is the same as the condensation temperature and no regeneration can occur. Note that the following calculation results are based on an ideal ORC cycle.

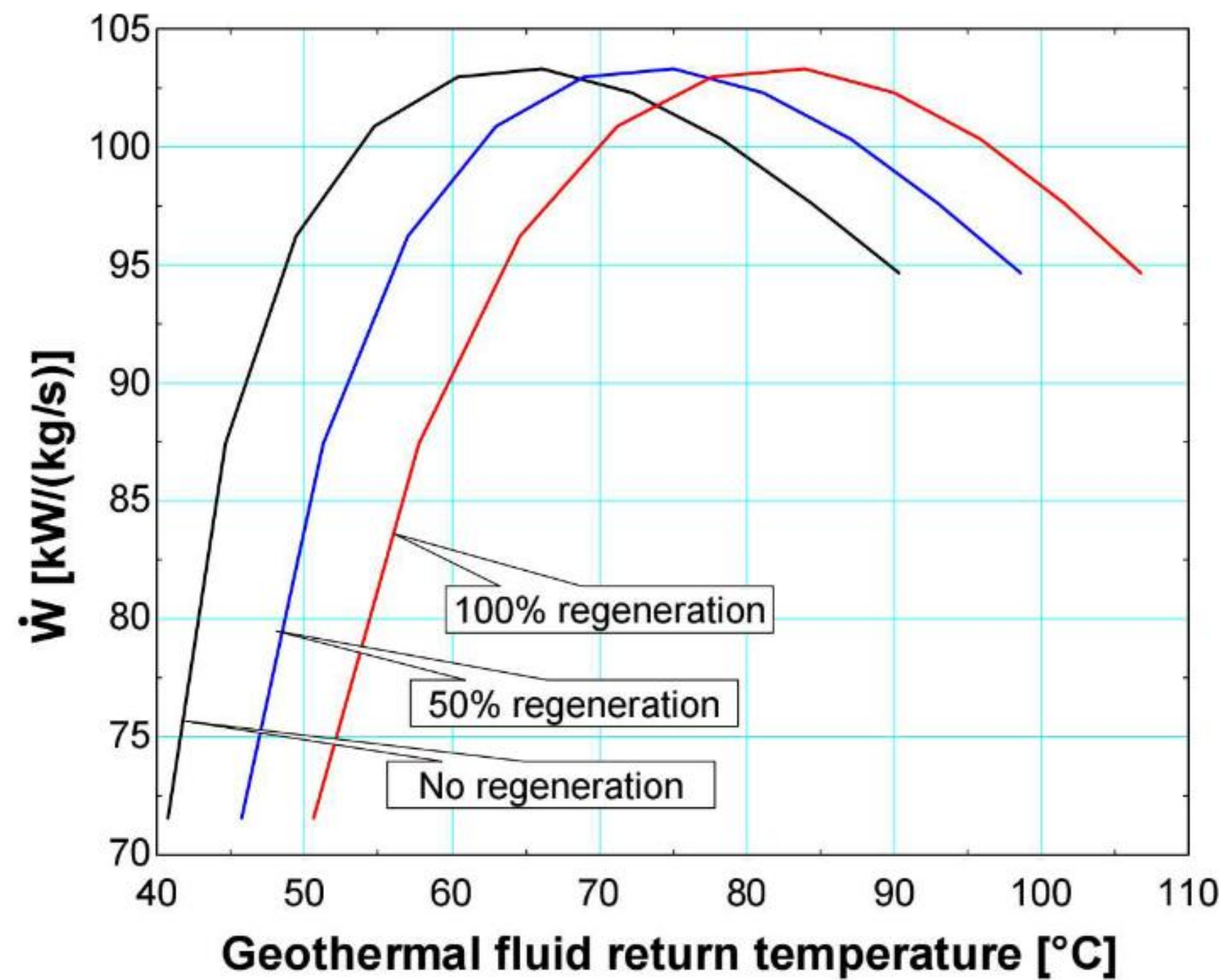


FIGURE 10: Power obtained from 1 kg/s of 200°C geothermal fluid (ideal cycle)

It can be seen from Figure 10, that the point of highest power is moved upwards by ca 20°C by the effect of the regeneration. Regeneration can thus keep the maximum power for geothermal fluid temperatures from ca 65°C to ca 85°C. It has to be kept in mind that a regenerator will be large and expensive, as well as causing pressure drop and associated losses. A cycle without regeneration will be more economical, if the geothermal fluid does not have any temperature limitation. Regeneration will not, repeat not, increase the produced power, even if it increases the efficiency. The increase of efficiency results only from less input of heat from the geothermal fluid. And this heat is normally free of charge.

The thermal efficiency increases when the geothermal return temperature increases (Figure 11). This is in accordance with the second law of thermodynamics, as the average input temperature of the heat increases, and thereby the efficiency.

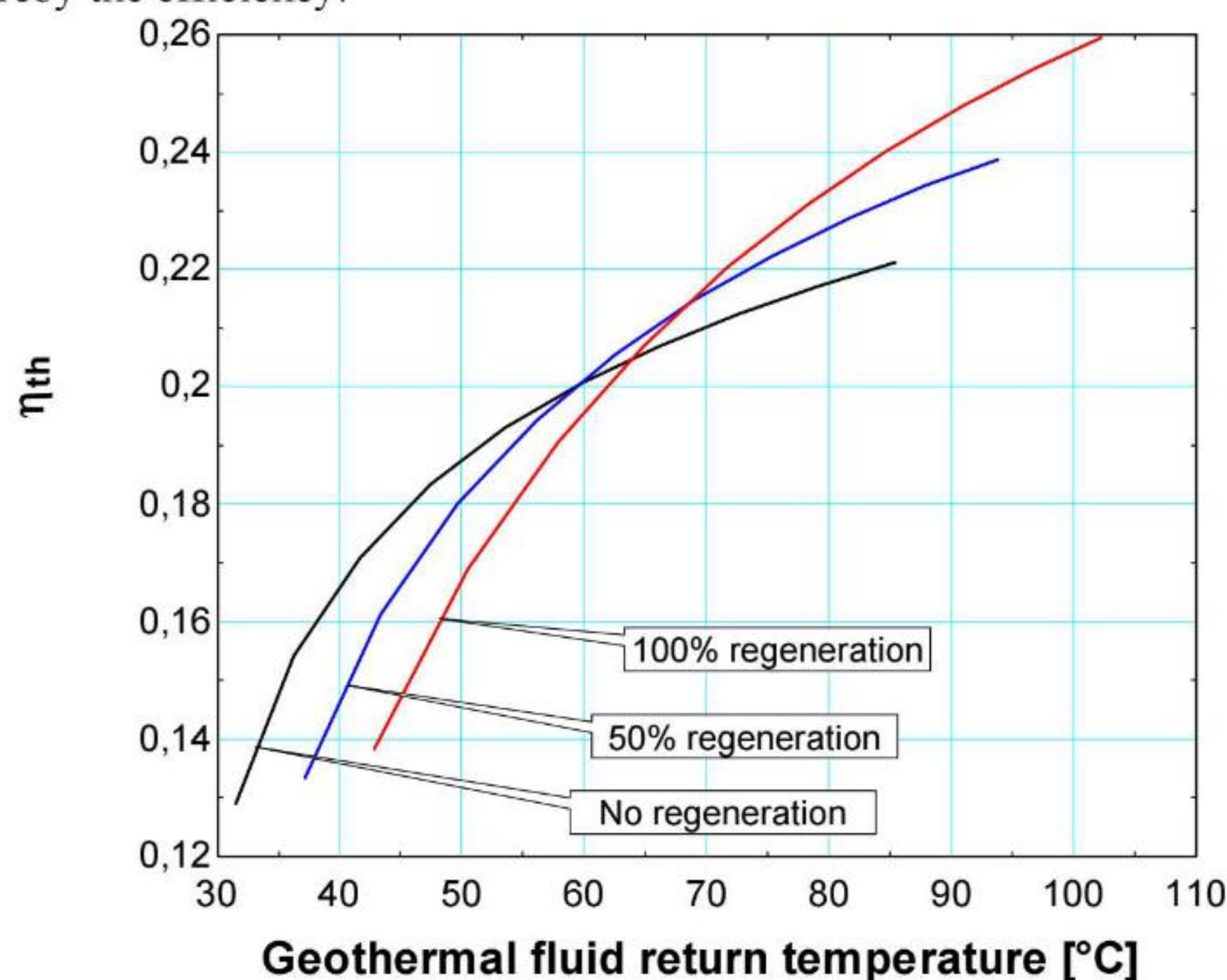


FIGURE 11: Thermal power plant efficiency (ideal cycle)

The power plant effectiveness is independent of the regeneration ratio, if the effectiveness is drawn as a function of the geothermal return temperature (Figure 12). And obviously this is a linear relation of the return temperature.

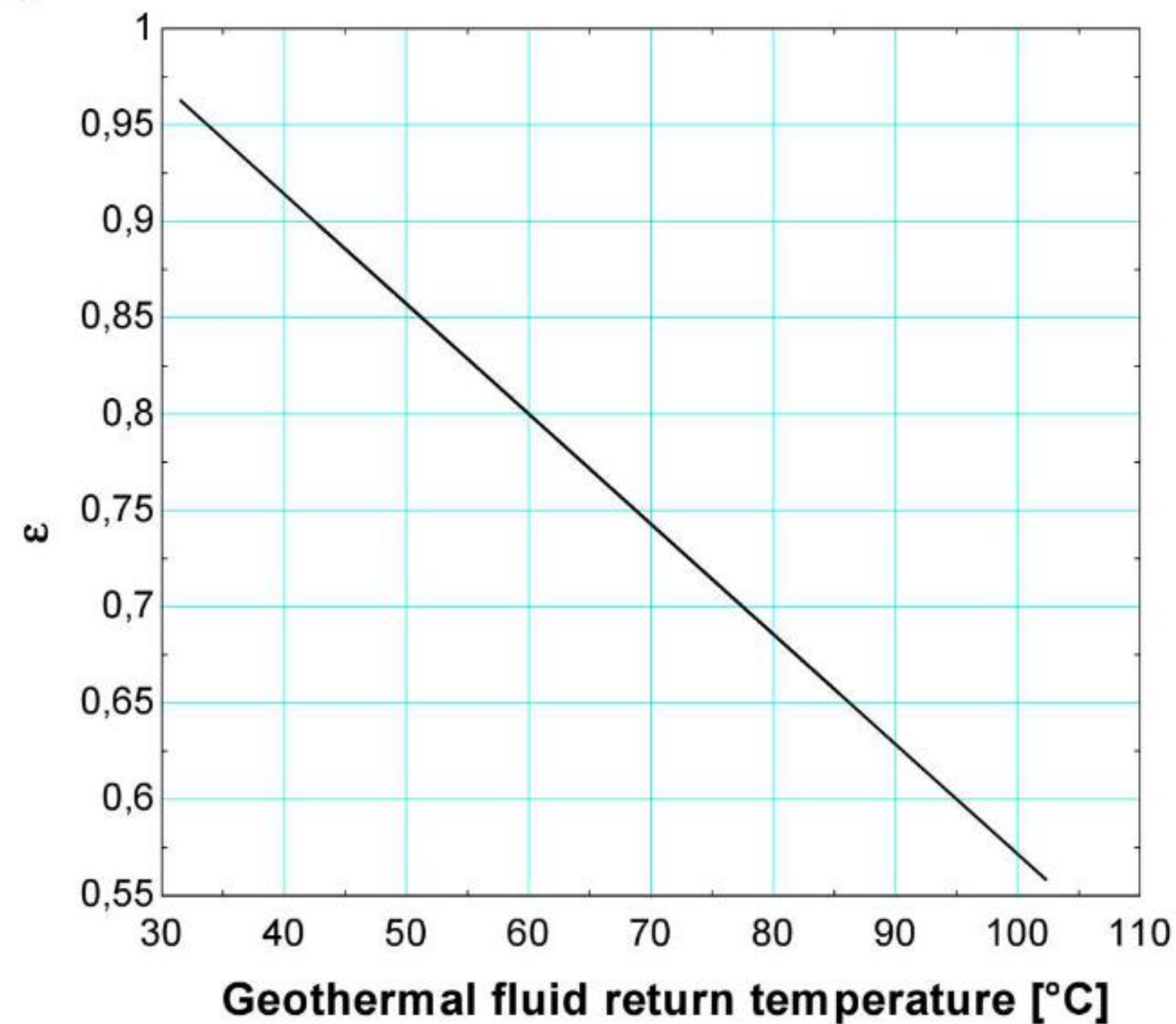


FIGURE 12: Power plant effectiveness (ideal cycle)

The total efficiency is found by multiplying the thermal efficiency by the effectiveness (Figure 13), and logically this is the same set of curves as the curves for the power obtained from the geothermal flow at the very beginning.

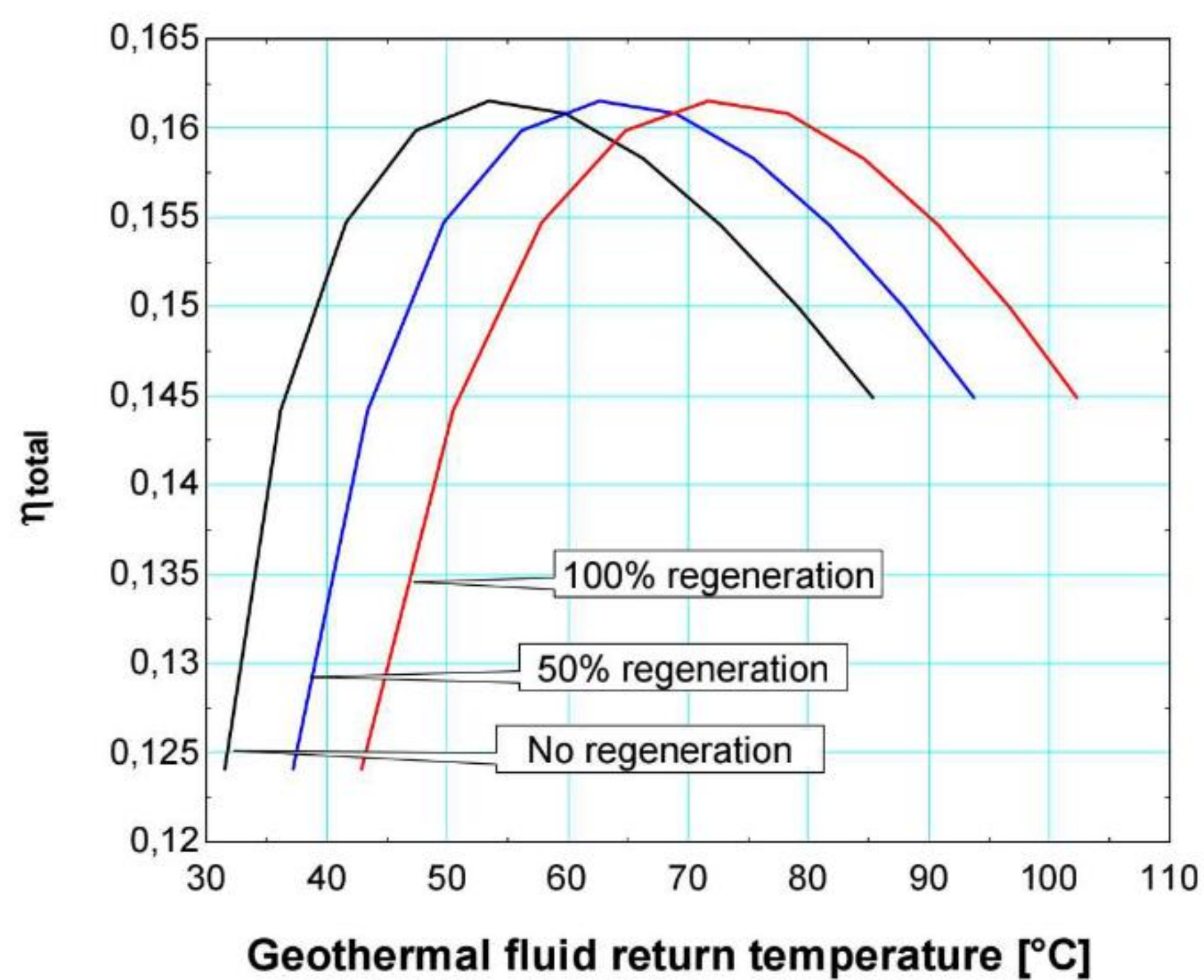


FIGURE 13: Thermal power plant total efficiency (ideal cycle)

The conclusion is simply that regeneration serves only to move the highest power production towards higher geothermal return temperature.

A final note is that a real cycle will show lower efficiency for higher regeneration, so regeneration will always reduce the maximum power available from a given geothermal flow stream. Regeneration will as well increase the plant cost, and has thus to be seen as a measure to preserve power, if a secondary process or geothermal fluid chemistry limits the return fluid temperature.

6. KALINA

The Kalina cycle is patented by the inventor, Mr Alexander Kalina. There are quite a few variations of the cycle. The Kalina power generation cycle is a modified Clausius-Rankine cycle. The cycle is using a mixture of ammonia and water as a working fluid. The benefit of this mixture is mainly that both vaporization and condensation of the mixture happens at a variable temperature. There is no simple boiling or condensation temperature, rather a boiling temperature range as well as condensation range. This is due to the fact, that the phase change process is a combined process, both the phase change of the substance and absorption/separation of ammonia from water.

6.1 The fluid

A phase diagram for ammonia – water mixture at 30 bar pressure is shown in Figure 14. The lower curve is the so-called bubble curve, when the first vapour bubble is created. This bubble has higher ammonia content than the boiling liquid. As the bubble ammonia content is higher than that of the liquid, the ammonia content in the liquid phase will be reduced. The upper curve is the so-called dew curve, when the last liquid drop evaporates. This drop has considerably lower ammonia content than the vapour.

The boiling process for 50% mixture is indicated in Figure 14.

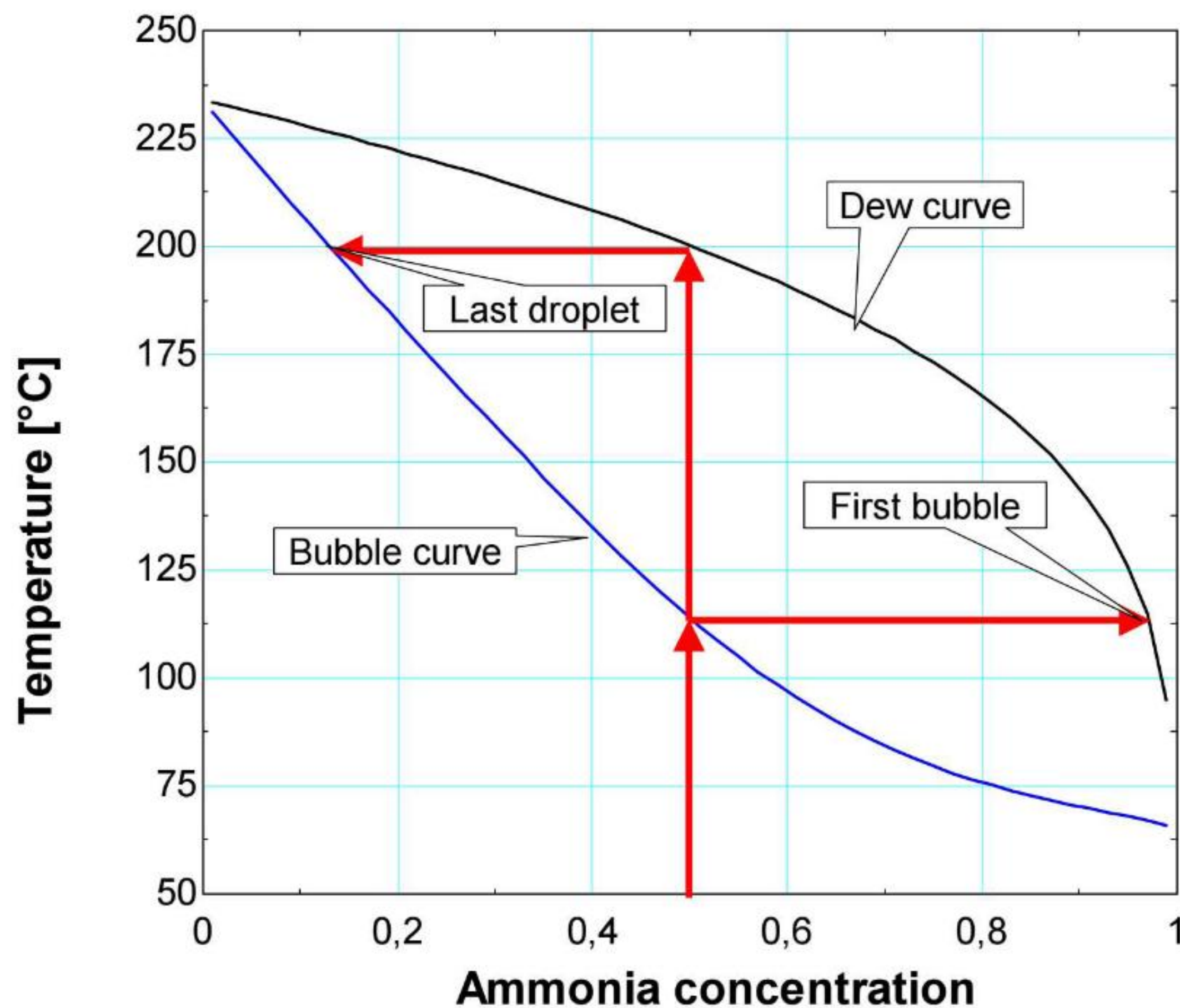


FIGURE 14: Ammonia-water phase diagram at 30 bar pressure

The temperature range for the boiling of the mixture at 30 bar is shown in Figure 15. The temperature range from bubble to dew is largest at approximately 67% ammonia concentration, and is then close to 95°C.

The mixture has thus a finite heat capacity, which is beneficial if the heat source is a liquid with constant or close to constant heat capacity.

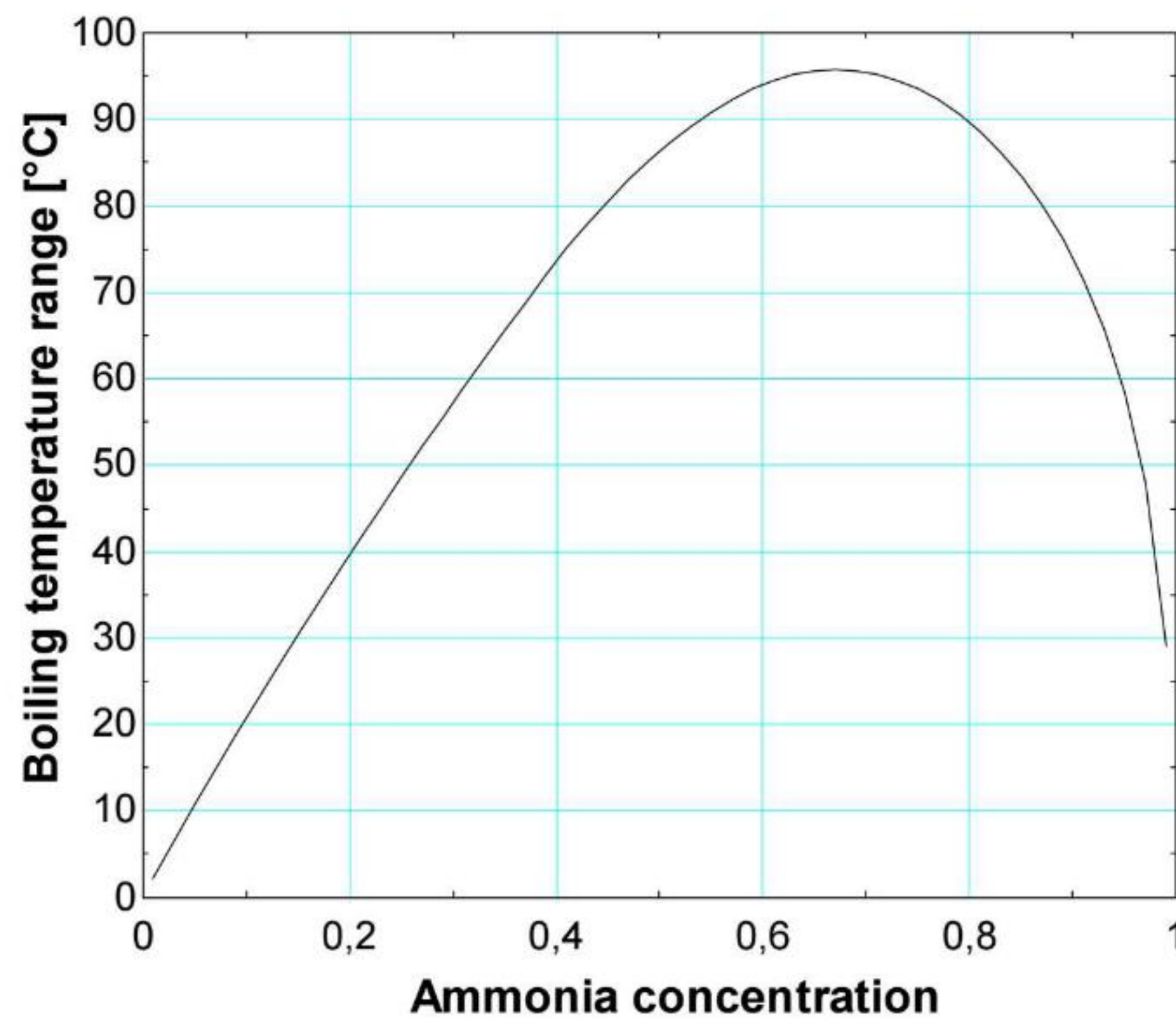


FIGURE 15: Ammonia-water boiling range at 30 bar pressure

The enthalpy of vaporization is as well dependent on the ammonia concentration (Figure 16).

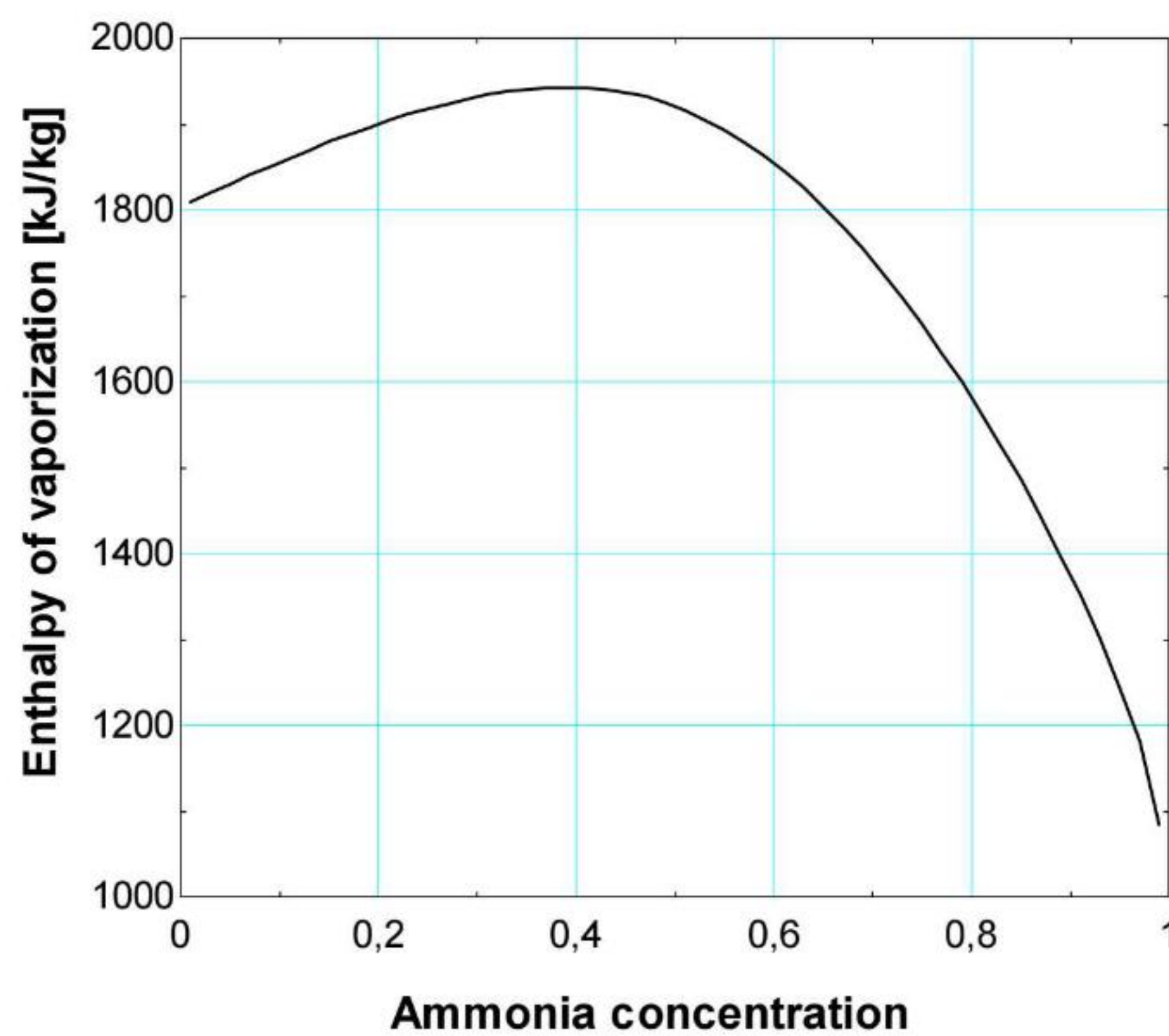


FIGURE 16: Ammonia-water enthalpy of vaporization at 30 bar pressure

Ammonia-water mixture is technically well known and widely used as a working fluid. Ammonia-water mixtures have been used in absorption refrigeration systems for decades. And ammonia is no newcomer to the technical field, it has been used in chemical and refrigeration processes for very long time. Ammonia is toxic, but the safeguards are well established.

Following are temperature - enthalpy diagrams for the mixture, at 25%, 75% and 95% ammonia concentration (Figures 16-18). The change in the curve form for boiling at constant pressure is to be noted. For low ammonia concentration, the largest temperature increase is at the beginning of the boiling, for high concentration at the end of the boiling. Intermediate concentration has S-formed boiling curve, and is therefore best suitable for power generation.

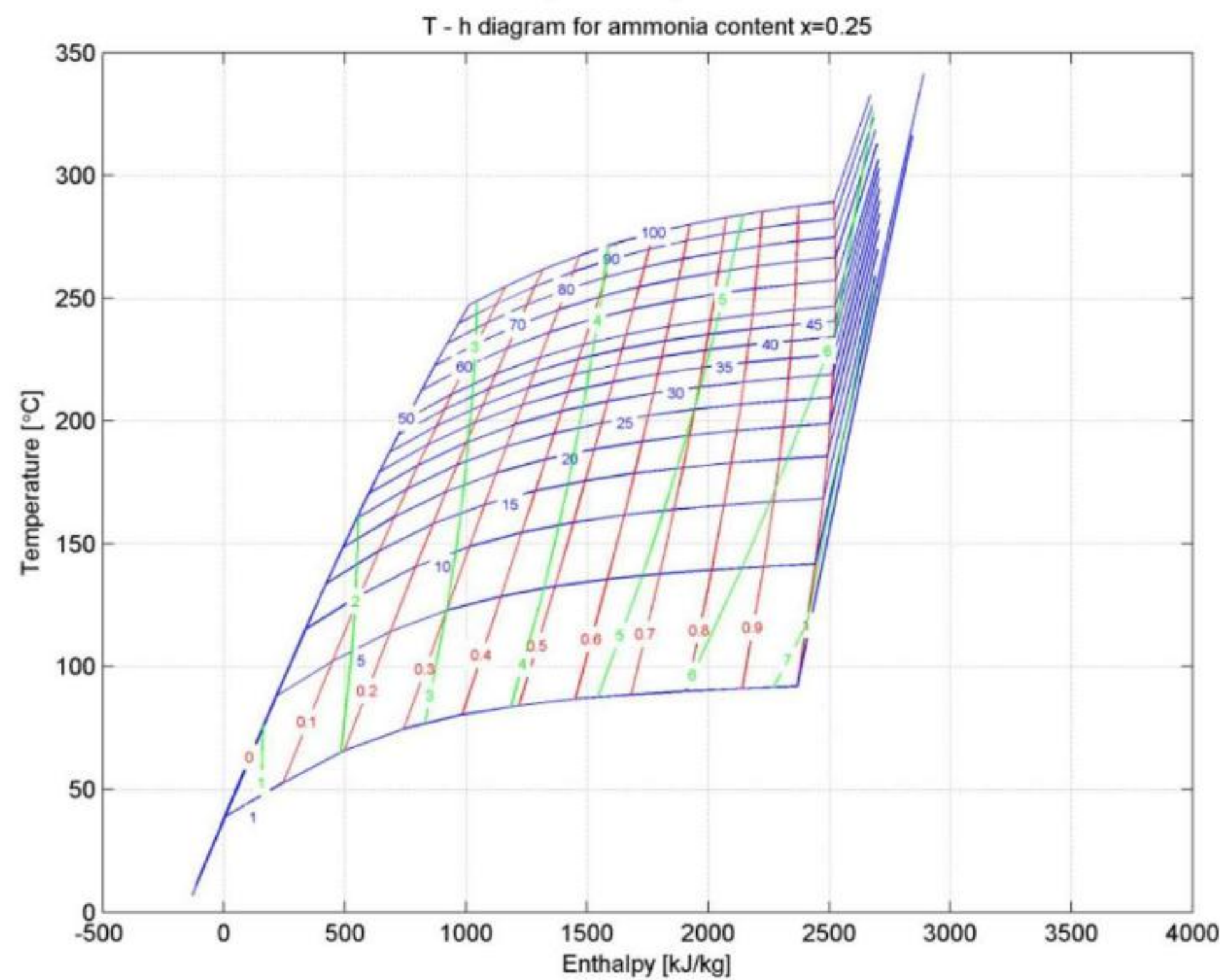


FIGURE 17: T-h diagram for 25% ammonia-water mixture

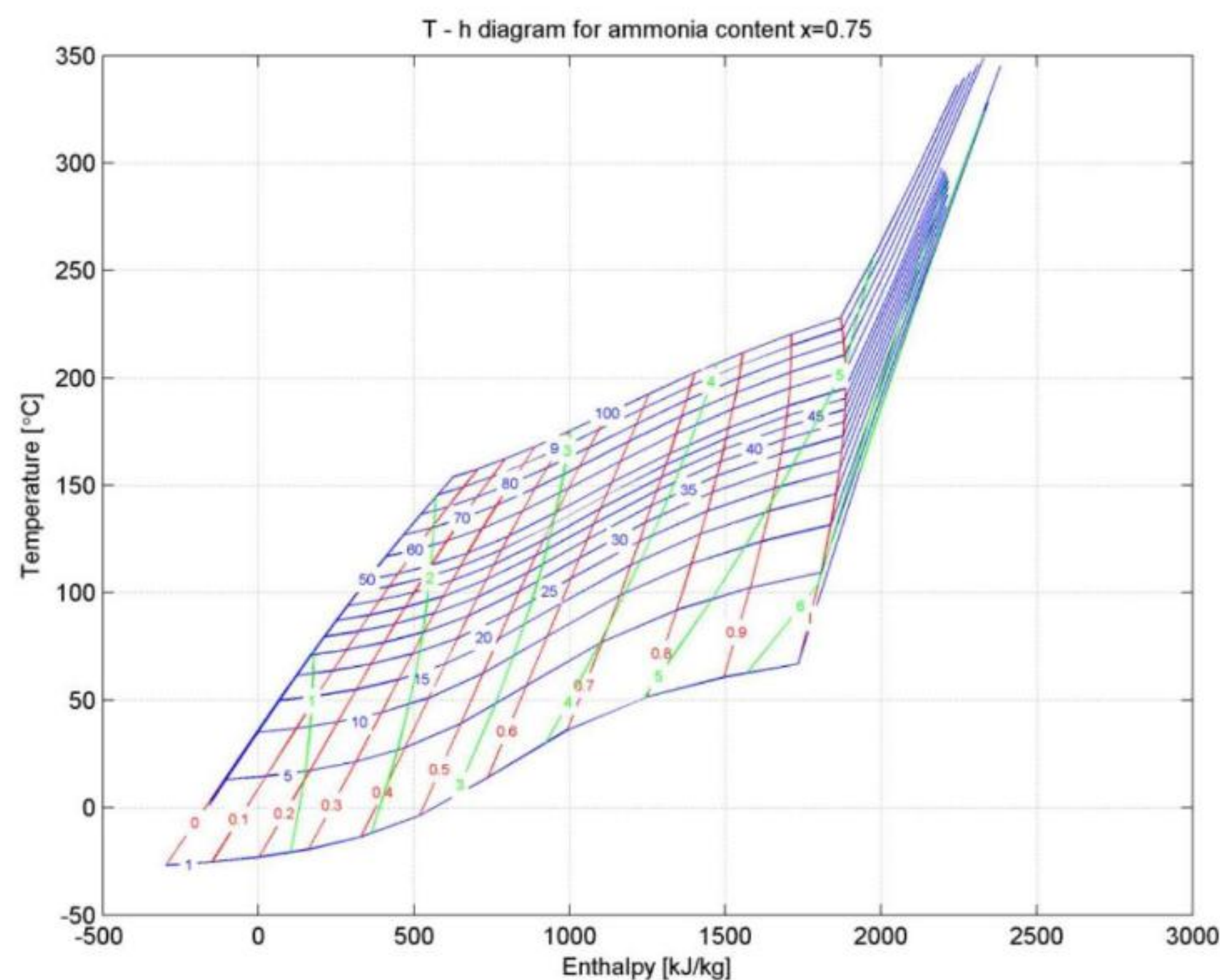


FIGURE 18: T-h diagram for 75% ammonia-water mixture

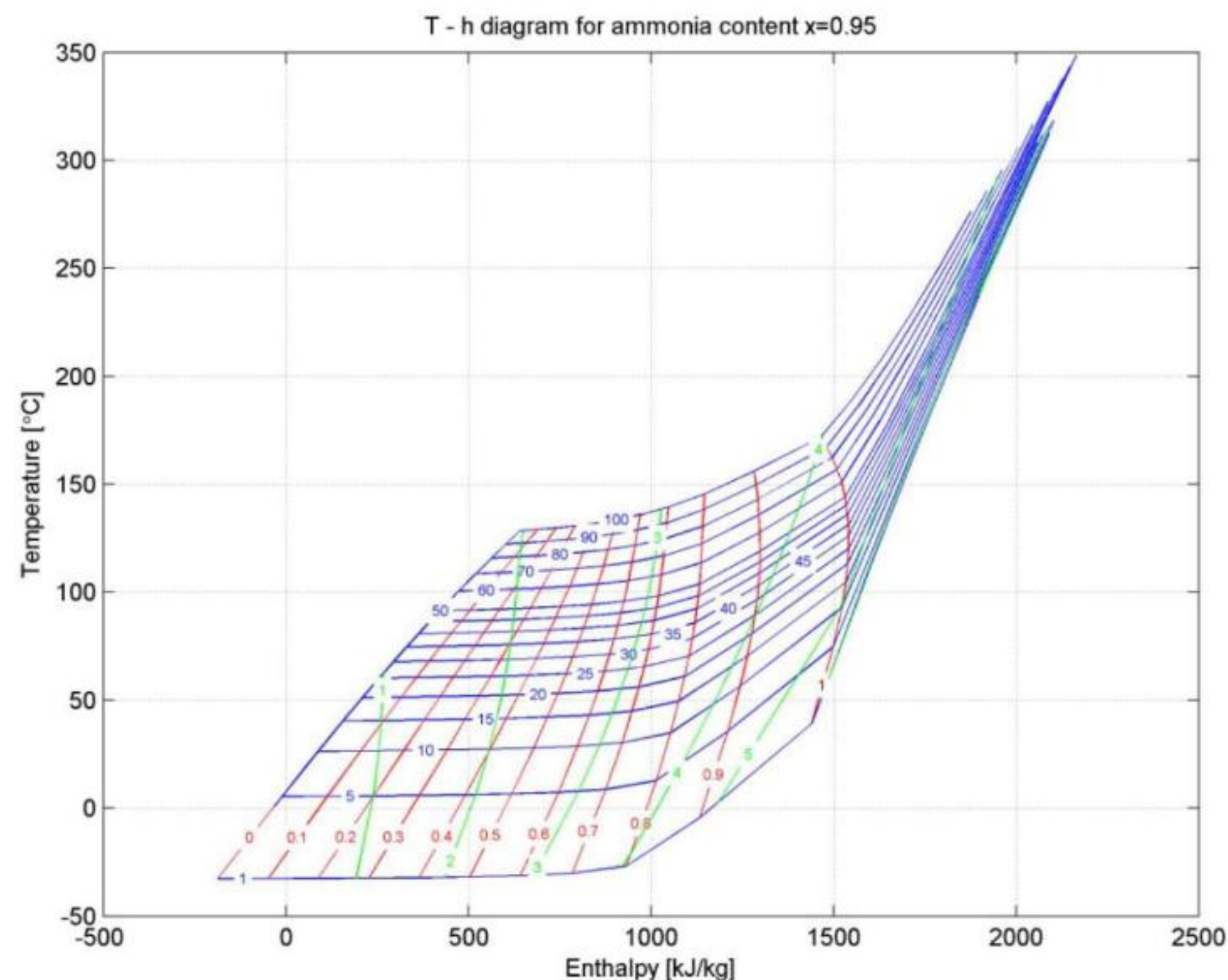


FIGURE 19: T-h diagram for 95% ammonia-water mixture

6.2 The cycle

A flow sheet for the Kalina saturated cycle is shown in Figure 20. The cycle is “saturated” because there is no superheat in the cycle. The fluid is not boiled entirely in the vaporizer, and the vapour-liquid mixture is then separated afterwards. This is done in order to maximise the vapour temperature at the vaporizer outlet.

The geothermal fluid enters the well at the source inlet temperature, station $s1$. The fluid is frequently liquid water. If the pressure is kept sufficiently high, no non-condensable gases will be separated from the liquid, and a gas extraction system is not necessary. The fluid is then cooled down in the vaporizer, and sent to re-injection at station $s2$.

Pre-heated (in the regenerators) liquid ammonia-water mixture enters the vaporizer at station 3. The fluid is boiled partly in the vaporizer. The liquid-vapour mixture leaves the vaporizer at station 4, and enters the separator.

The separated liquid leaves the separator and enters the high temperature regenerator at station 7. After the high temperature regenerator the liquid is throttled down to the condenser pressure in station 8, and mixed with the turbine exit vapour from station 6.

The ammonia-rich vapour enters the turbine at station 5, and is expanded to the condenser pressure at station 6. The exit vapour mixed with the throttled liquid (now at the average ammonia concentration) from the high temperature regenerator enters the low temperature regenerator at station 9.

The cooled fluid from the low temperature regenerator enters the condenser at station 10. The fluid gas now started to condense, and the ammonia concentration is not the same in the liquid or vapour phase. An absorption process is going on, where the ammonia rich vapour is absorbed into the leaner liquid, in addition to condensation due to lowering of the mixture temperature. The kinetics of the absorption process determines the rate of absorption, whereas heat transfer and heat capacity controls the condensation process.

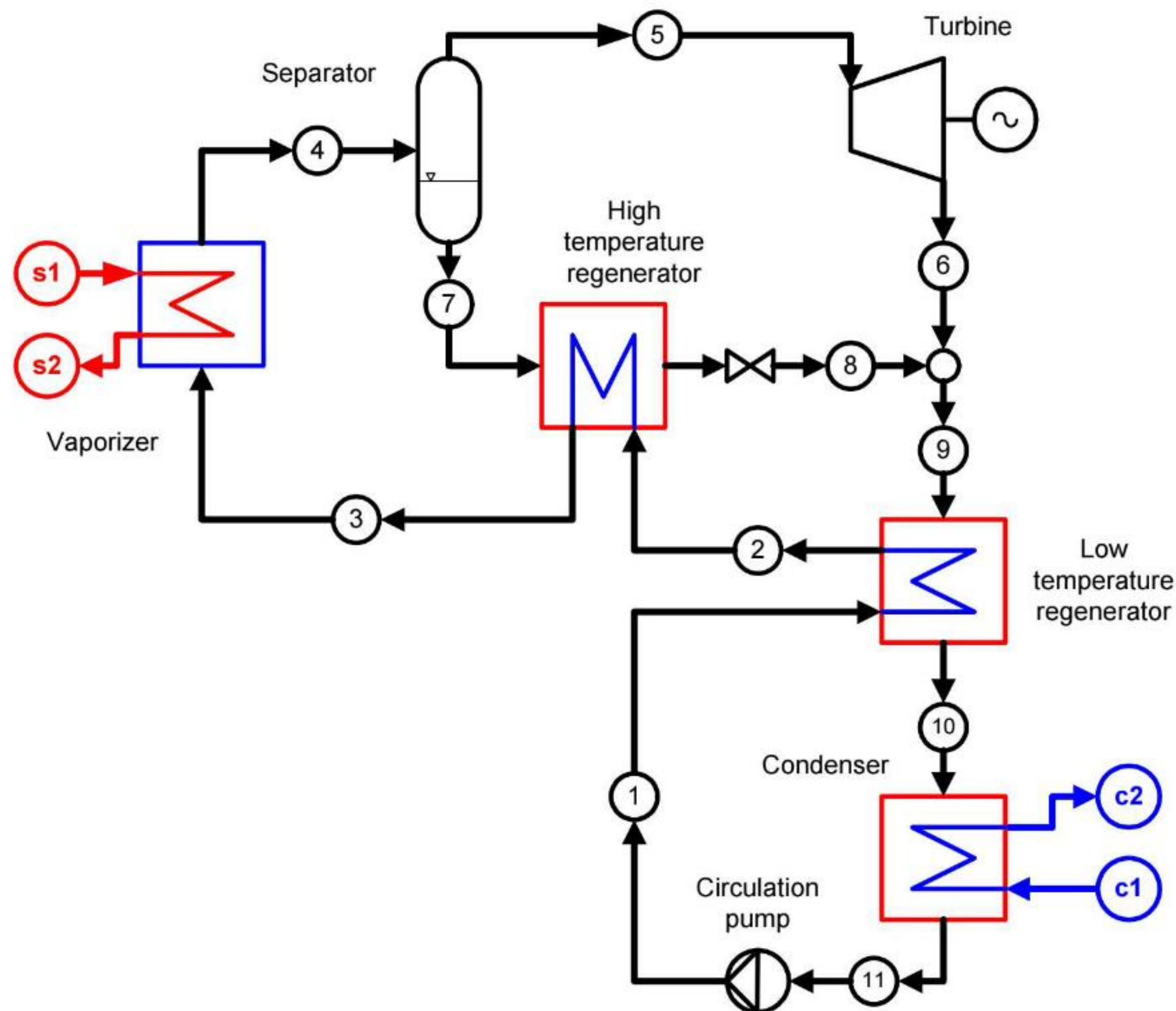


FIGURE 20: Flow diagram of a saturated Kalina cycle

Finally all the mixture is in saturated liquid phase in the hot well of the condenser at station 11. The circulation pump raises the fluid pressure up to the higher system pressure level, and the liquid is then preheated in the regenerators in stations 1 through 3.

The condenser shown here is water cooled, with the cooling water entering the condenser at station *c1* and leaving at station *c2*.

6.3 External heat exchange

A mixture of ammonia and water will not boil cleanly, but as well change the chemical composition. The vapour will be more ammonia – rich, whereas the liquid will be leaner for the partially boiled mixture. This can be seen from the phase diagram of ammonia –water mixture presented earlier. Similar variation of the chemical composition will be encountered in the condenser for the partially condensed mixture. This results in a variable temperature during the heat exchange process both in the vaporizer and the condenser. Following is a heat exchanger diagram for a vaporizer, where typical curves have been drawn both for isopentane and 80% ammonia – water mixture (Figure 21).

The temperature difference between the primary and the secondary fluid in the Kalina vaporizer is small compared for the isopentane vaporizer, even for similar or same pinch temperature difference. Entropy is generated whenever heat is transferred over a finite temperature difference, thus is the entropy generation in the Kalina vaporizer less, and thereby the destruction of exergy less. On the other hand the Kalina vaporizer will need larger heat exchange area due to the smaller temperature difference. And the diagram shows well that the logarithmic temperature difference approach for the sizing cannot be used, as the fluid heat capacity is far from being constant.

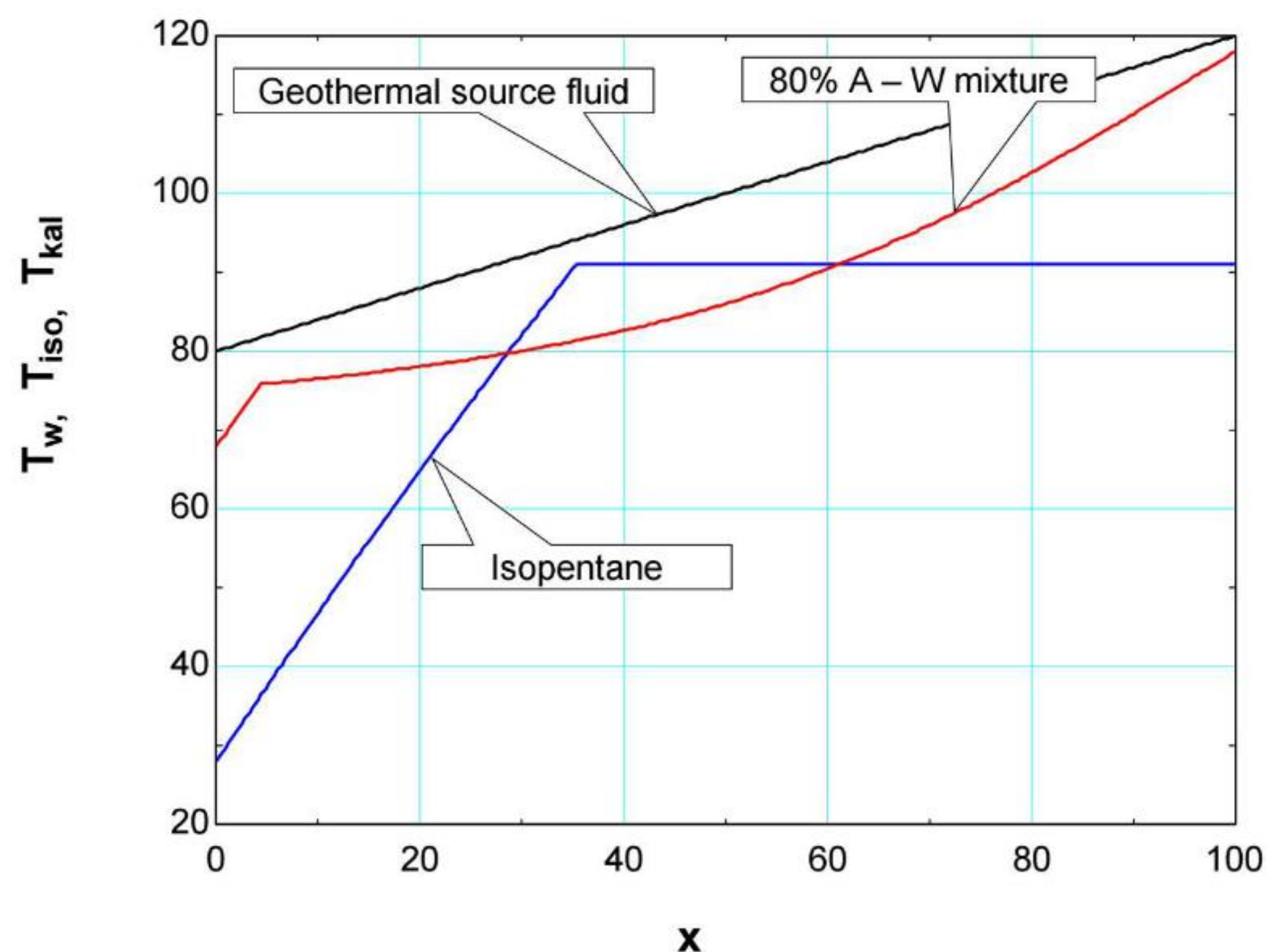


FIGURE 21: Heat exchanger diagram for a vaporizer in a binary power plant
 $x=100$ is at geothermal fluid entry, and $x=0$ at the outlet

A similar situation is in the condenser. There ammonia rich vapour is absorbed and condensed, with the associated changes in chemical composition of both liquid and vapour. Following is a heat exchanger diagram of both isopentane and ammonia – water mixture in a water cooled condenser (Figure 22).

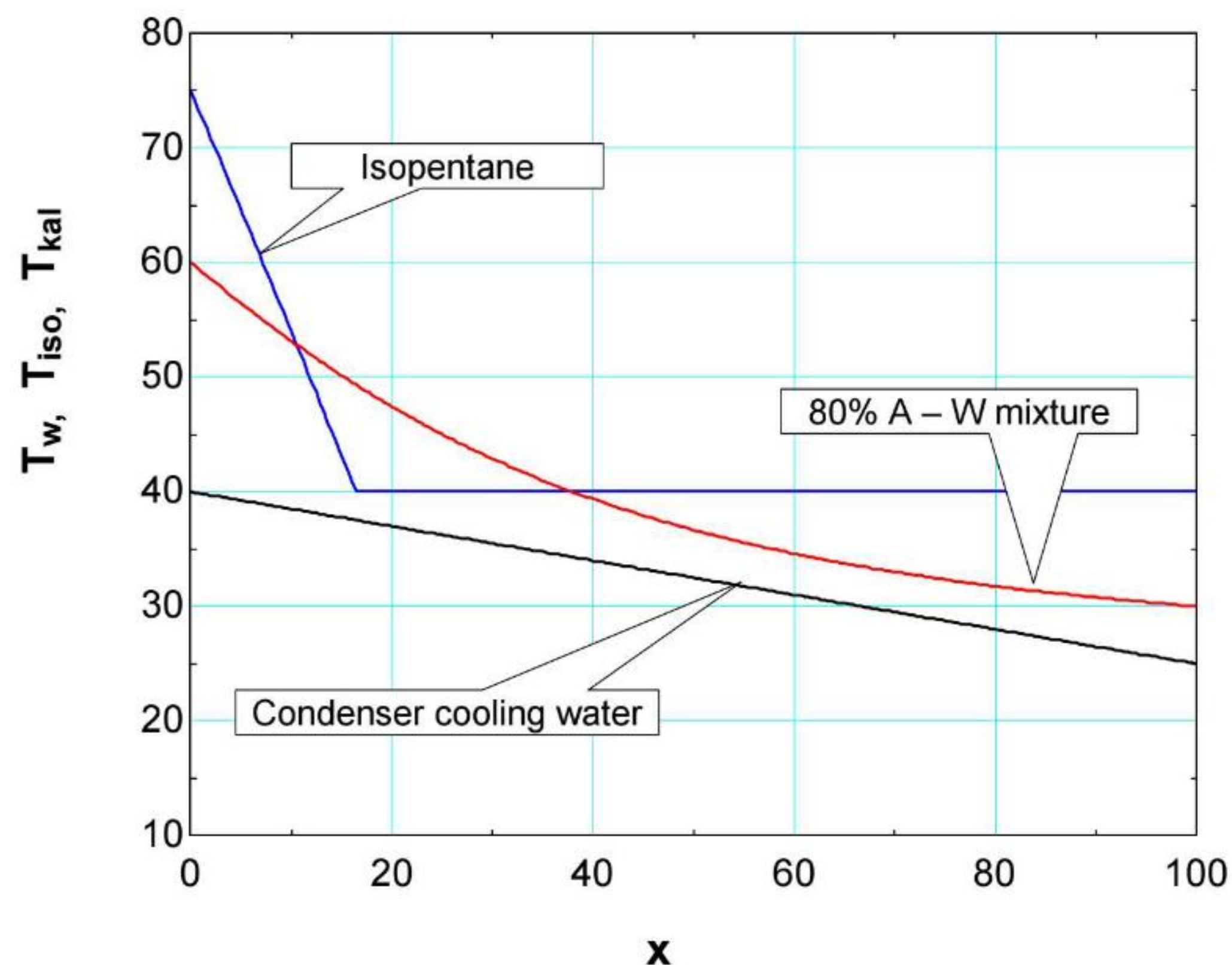


FIGURE 22: Heat exchanger diagram for a condenser in a binary power plant
 $x=100$ is at cooling water entry, and $x=0$ at the outlet

Both fluids will have the pinch point internally in the condenser, and the ammonia – water mixture will even have the pinch at an unknown point. The isopentane will obviously have the pinch at the fluid dew point, but it cannot be known beforehand at which vapor ratio the pinch for the ammonia – water mixture will be.

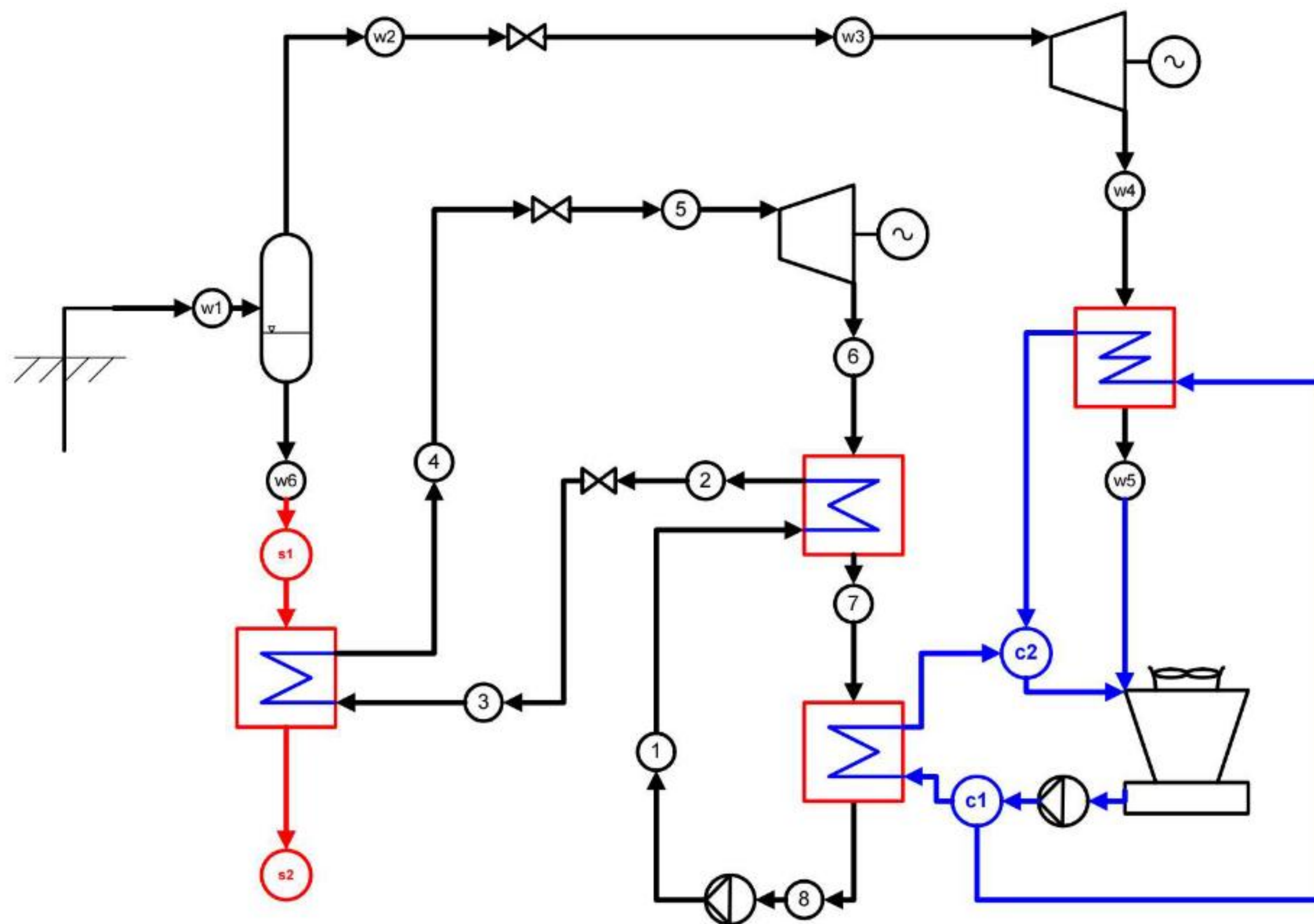


FIGURE 24: A single flash condensing cycle combined with an ORC cycle

By appropriate selection of working fluid, the geothermal fluid can be economically cooled further down than what is possible with the flash cycles. This will increase the power plant effectiveness at the cost of efficiency, as previously said. But at the end an optimum value for the plant return temperature of the geothermal fluid emerge, and this temperature will give the highest power plant output for a given flow stream from the wells.

The binary cycles have the disadvantage of having a secondary working fluid, often expensive, toxic and flammable. This leads to expensive safety measures required for the power plant.

When the geothermal fluid temperature is medium to low, the ORC cycle becomes more economical than the flash cycles. If the fluid temperature is below say 180°C it is likely that an ORC cycle will be more economical than a flash cycle. This is as well valid for higher temperatures if the gas content in the fluid is high.

The ORC cycle gives normally high power plant effectiveness. The cycle can be modified by adjusting the level of regeneration to suit the secondary process requirements (such as bottoming district heating system) or chemical limitations regarding the plant geothermal fluid return temperature. When regeneration is used, the plant efficiency increases and the plant effectiveness is reduced, as discussed before.

Another advantage of the ORC cycle is that it can be easily adapted to fluid with partial steam. The constant vaporizer boiling temperature has then to fit with the condensation temperature of the partial steam in the geothermal fluid.

When the geothermal fluid temperatures get lower than say $150\text{--}160^{\circ}$ the Kalina cycle seems normally to be superior to the ORC cycle. Kalina is better fit for situations where the geothermal fluid is only liquid water, due to the variable temperature of the vaporizer boiling and separation process in the ammonia – water mixture.

Other technical differences between these two binary cycles are that the pressure level of the Kalina cycle are higher than for a corresponding ORC cycle. The turbine cost in the ORC cycle is thus higher, due to high volume flow in the turbine at lower pressure. On the other hand, then all equipment in the Kalina cycle will have a higher pressure class than in the ORC cycle. The piping dimensions will be larger in the ORC cycle due to higher volume flow.

There does not seem to be any major difference in requirements of the piping/equipment material for these two cycles, with the exception of the turbine. Turbine corrosion has been encountered in the Kalina cycle, leading to the use of titanium as material for the turbine rotor.

Fluid safety measures are similar, as the ORC cycle uses commonly flammable working fluids. The precautions needed due to the flammability seem to be at the same order of magnitude as the requirements due to the toxicity of ammonia.

The technical complexity of the two cycles is at the same order of magnitude. The complexity is highly dependent on the level of regeneration used in the cycle, and therefore a comparison of the cycle complexity has to be made with caution. Obviously a non-regenerated ORC cycle is a lot less complex than a highly regenerated Kalina cycle with two temperature levels of regeneration. But this is not a fair comparison.

At the time of writing this text, the only geothermal Kalina plant is at Husavik, Iceland. A second plant is being built at Unterhaching in Bavaria, Germany. Presently the limited use of the Kalina cycle may be its biggest disadvantage, but this will most probably change during the coming years. ORC cycles are widespread and have been in use for decades.

9. POWER PLANT COMPONENTS

This chapter treats the main equations and short discussion of the main power plant components. The most relevant items related to geothermal engineering are as well discussed shortly.

9.1 Well and separator

A simplified model of the well and separator is presented in Figure 25. Station 1 is the undisturbed geothermal reservoir. The main thermal parameter for the reservoir with regard to the power plant

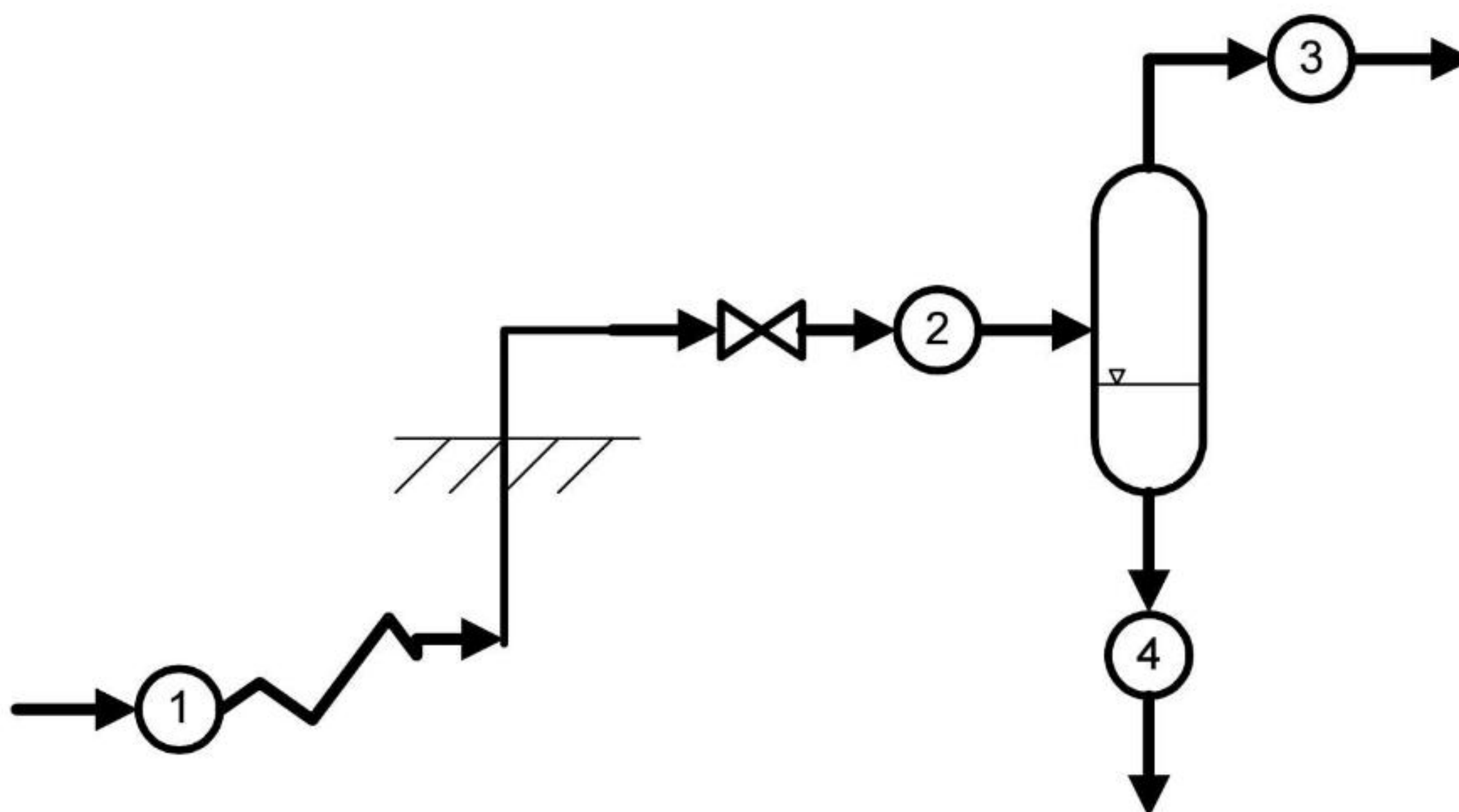


FIGURE 25: A simple model of well and separator

design is the field enthalpy, or energy content of the fluid. Station 2 is the entry of the steam – water separator, station 3 is the steam outlet from the separator and station 4 is the brine outlet of the separator.

The wells have certain productivity, i.e. there is a relation between the wellhead pressure and the flow from the well. The productivity is individual from well to well, and this relation is further complicated by the fact that the well may not be artesian, that is a well pump is required to harvest fluid from the well. Generally this relation can be presented as:

$$\dot{m}_1 = f(p_2) \quad (1)$$

where the function takes the presence of a well pump into account, as well the field characteristic.

The flow up the well and in the geothermal primary system can usually be treated as isenthalpic, that is that the heat loss in the well and the piping is neglected. No fluid loss is assumed, leading to:

$$\dot{m}_2 = \dot{m}_1 \quad (2)$$

$$h_2 = h_1 \quad (3)$$

The throttling in the well and primary system results most frequently in that the fluid starts to boil, which results in that the temperature is a direct function of the separator pressure (Station 2). If the well is non-artesian and a well pump is used, the pressure may be kept sufficiently high to avoid boiling, in which case a separator is not needed at all and the source fluid is liquid in the sub-cooled region at all times.

If boiling occurs and a separator is employed, the relation between temperature and pressure is:

$$T_2 = T_{sat}(p_2) \quad (4)$$

defined by the thermodynamic properties of steam and water.

The steam fraction is then defined by the energy balance over the separator. The heat flow in the incoming mixture of steam and water (from the well) equals the sum of the energy flows in the steam and the brine from the separator. The mass flow of steam from the separator will thus be:

$$\dot{m}_3 = \dot{m}_2 \frac{h_2 - h_4}{h_3 - h_4} \quad (5)$$

The separator is working in the (thermodynamic) wet area, containing a mixture of steam and water in equilibrium. All temperatures in the separator will thus be equal, assuming that there are no significant pressure losses or pressure differences within the separator.

$$T_3 = T_2 = T_4 = T_{sat}(p_2) \quad (6)$$

The enthalpy of the steam outgoing stream from the separator is thus the enthalpy of saturated steam at the separator pressure.

$$h_3 = h_g(p_2) \quad (7)$$

Mass balance holds over the separator, the sum of steam mass flow and brine mass flow equals the mass flow of the mixture from the wells towards the separator.

$$\dot{m}_4 = \dot{m}_2 - \dot{m}_3 \quad (8)$$

The selection of separator pressure is very critical for the power plant. If the wellhead pressure is low, boiling may occur in the formation around the well, which may lead to scaling within the cracks and narrow flow passages in the formation. This will lead to short well life.

Higher separator pressure means that better steam is available for the turbine (higher enthalpy), but the amount will be less, dictated by the separator energy balance as well as less well productivity due to higher wellhead pressure. This may as well influence the separation of non-condensable gases from the geothermal fluid.

The selection of the separator pressure is thus an optimization process, economical, thermodynamic and geothermal.

9.2 Vaporizer

The vaporizer is the first component of an ORC or a Kalina power plant (Figure 26). Station s1 is the entry of the geothermal fluid to the vaporizer, and station s2 is the outlet. Station 1 is the entry of the power plant working fluid (liquid) to the vaporizer, and station 2 is the outlet of the working fluid vapour or mixture towards the turbine.

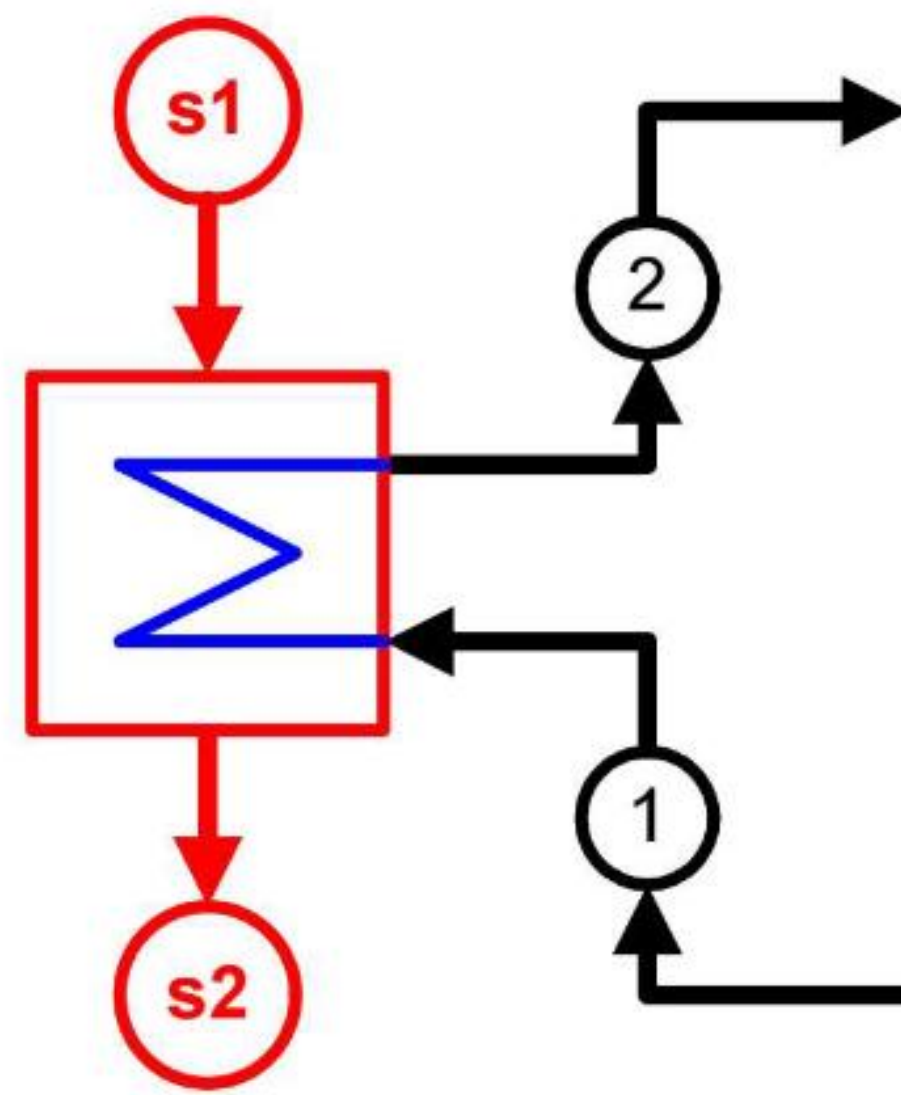


FIGURE 26: ORC vaporizer schematic

Obviously the heat removed from the source fluid has to equal the heat added to the working fluid.

$$\dot{m}_s (h_{s1} - h_{s2}) = \dot{m}_{\text{working fluid}} (h_2 - h_1) \quad (9)$$

The fluid condition at station 2 is determined by the cycle and the turbine requirements, for an ORC cycle this would be saturated or slightly superheated, for most Kalina cycles it would be in the wet region, with vapor fraction at 50-100%.

The vaporizer is nothing but a heat exchanger between the hot source fluid and the cold working fluid of the cycle. It has to be observed that the temperature of the hot fluid is higher than the one of the cold fluid throughout the vaporizer. As well it must be kept in mind that the relation between the

enthalpy and the temperature is highly non-linear, requiring that the vaporizer is divided into appropriate sections for the calculation.

The source fluid outlet temperature is critical as regards scaling. This temperature must be kept sufficiently high to avoid scaling on the source fluid side of the exchanger. Cleaning of the source fluid side may be necessary, so the vaporizer design must take this into account. Any geothermal fluid may be corrosive, so an appropriate material has to be used for the vaporizer.

9.3 Turbine

The turbine converts a part of the vapour enthalpy to shaft work, and then electricity in the generator (Figure 27). Station 1 is the vapour inlet to the turbine, and station 2 is the turbine exit.

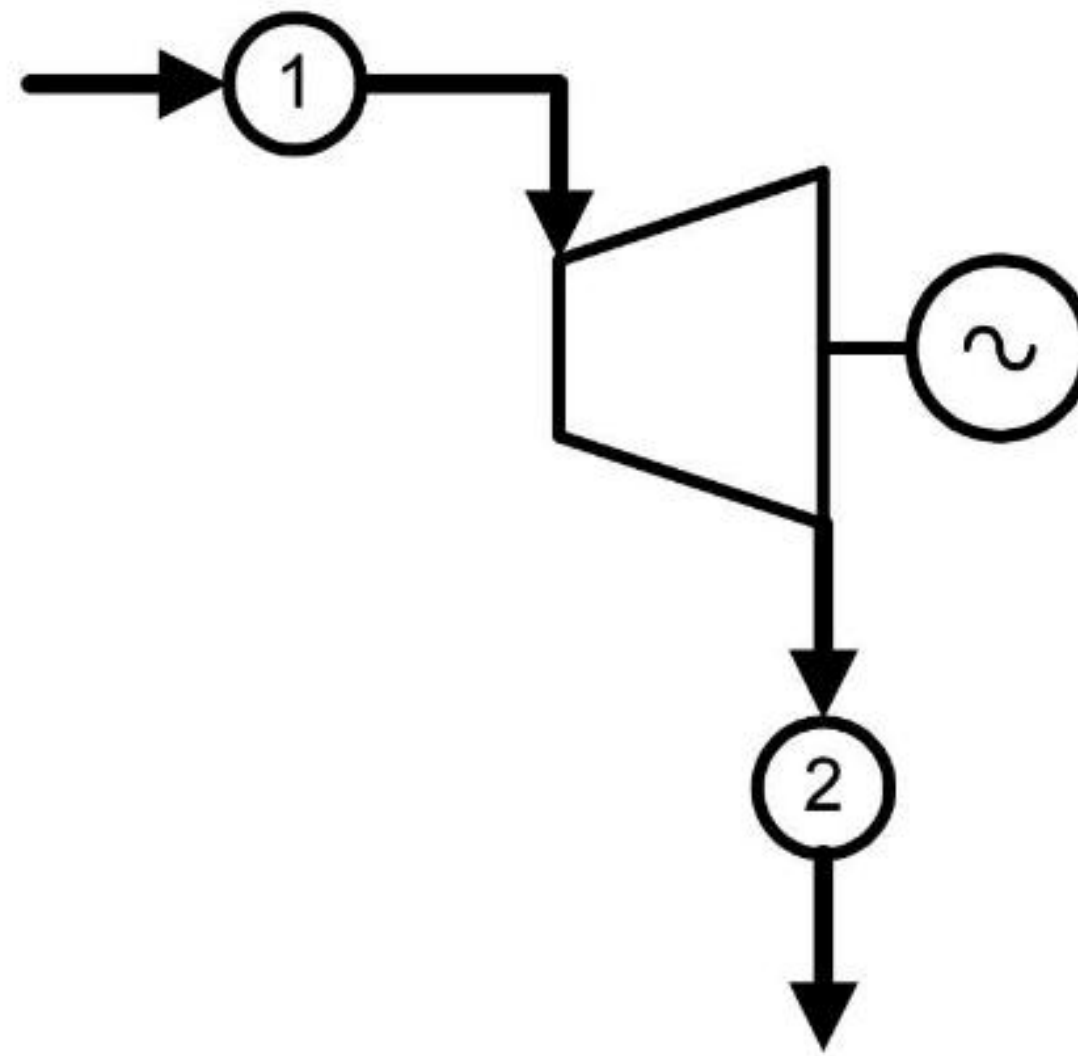


FIGURE 27: Turbine schematic

The ideal turbine is isentropic, having no second law losses. In this case the entropy of the incoming vapor equals the entropy in the exhaust steam. The corresponding enthalpy change (reduction) of the vapor is the largest enthalpy change possible. The isentropic exit enthalpy is then the enthalpy at the same entropy as in the inlet and at the exit pressure, which is roughly the same as prevails in the condenser.

$$h_{2,s} = h(s_1, p_2) \quad (10)$$

The turbine isentropic efficiency is given by the turbine manufacturer. This efficiency is the ratio between the real enthalpy change through the turbine to the largest possible (isentropic) enthalpy change. The real turbine exit enthalpy can then be calculated:

$$h_2 = h_1 - \eta_{is}(h_1 - h_{2,s}) \quad (11)$$

The work output of the turbine is then the real enthalpy change multiplied by the working fluid mass flow through the turbine.

$$\dot{W} = \dot{m}(h_1 - h_2) \quad (12)$$

The expansion through the turbine may result in that the exit vapor is in the wet region, or that a fraction of the mass flow is liquid. This can be very harmful for the turbine, resulting in erosion and blade damage.

The Kalina cycle uses a mixture of ammonia and water, so that the droplets created in the turbine are electrically conductive. It is the meaning of the writer that this conductivity is the reason for the corrosion encountered in the turbine in Husavik, Iceland. This corrosion has been avoided by the usage of non-magnetic titanium for the turbine rotor.

Many of the working fluids for the ORC cycle are retrograde, which means in this context that the exit vapor is superheated. The heat removal in the condenser will then partly be “de-superheat”, heat transfer out of the vapor at temperature higher than the final condensing temperature.

Ammonia-water mixture is not retrograde, but as the condensation will occur at a variable temperature, the heat removal process is very similar to that of the retrograde fluids.

9.4 Recuperator

The regenerator is a heat exchanger between the hot exit vapour from the turbine and the condenser (Figure 28). It is a de-superheater in the ORC cycle, transferring heat from the turbine exit vapour to the condensate from the condenser.

Station 1 is the turbine exit vapour, station 2 is the regenerator outlet towards the condenser, station 3 is the inlet of the condensate from the condenser, and station 4 is the pre-heated feed to the vaporizer.

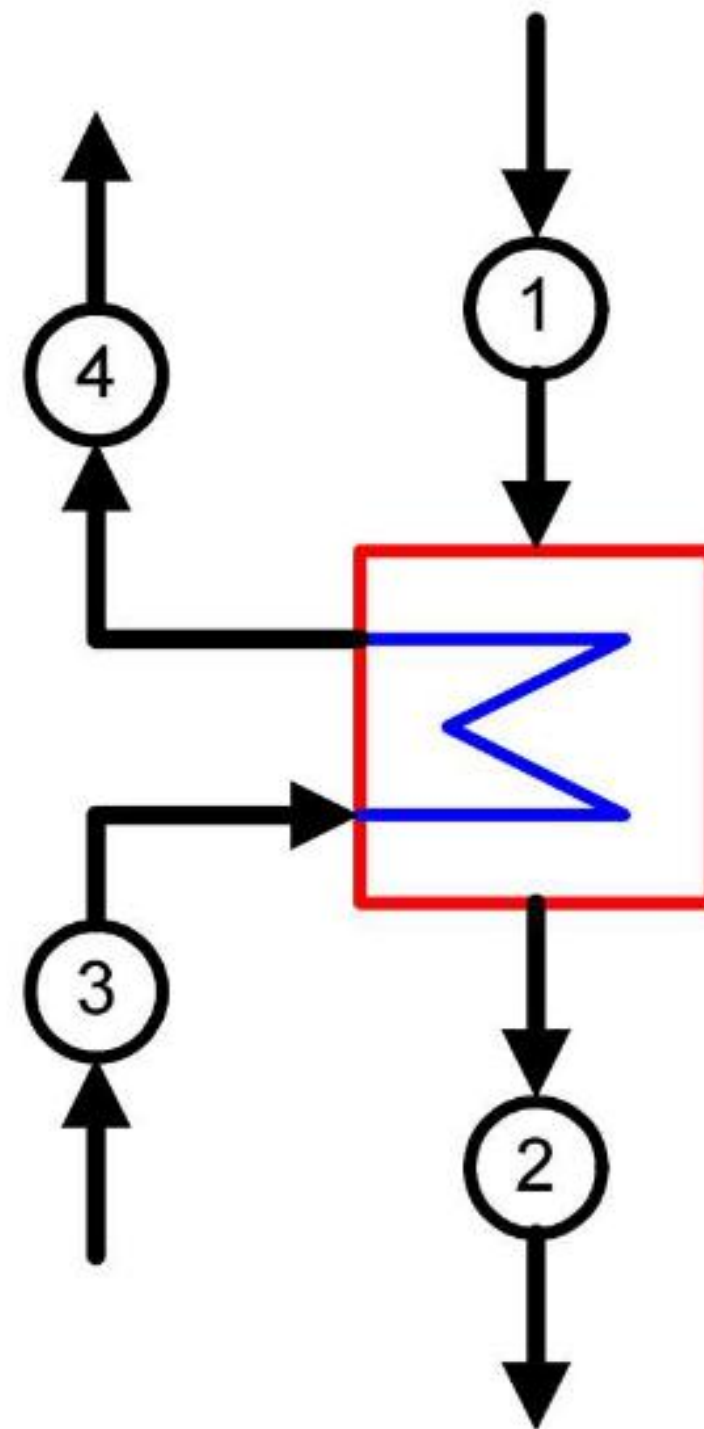


FIGURE 28: Recuperator schematic

The heat removed from the turbine exhaust vapor is equal the heat added to condensate:

$$\dot{m}(h_1 - h_2) = \dot{m}(h_4 - h_3) \quad (13)$$

The mass flow is the same on both sides of the regenerator. The hot fluid from the turbine is on the hot side, will be condensed in the condenser and then pumped right away through the cold side of the regenerator towards the vaporizer.

It has to be observed that the temperature of the hot fluid is higher than the one of the cold fluid throughout the regenerator. The fluid behavior is usually close to linear, so it is normally not necessary to divide the regenerator into sections.

The effect of regeneration on the cycle has been treated earlier in this text.

9.5 Condenser

The condenser may be either water or air cooled (Figure 29). The calculations for the condenser are roughly the same in both cases, as the cooling fluid (air or water) is very close to linear. Station 1 is the working fluid coming from the regenerator (or turbine in the case of a non-regenerated cycle). Station 2 is the condensed fluid, normally saturated liquid with little or no sub-cooling. Station c1 is the entry of the cooling fluid, station c2 the outlet.

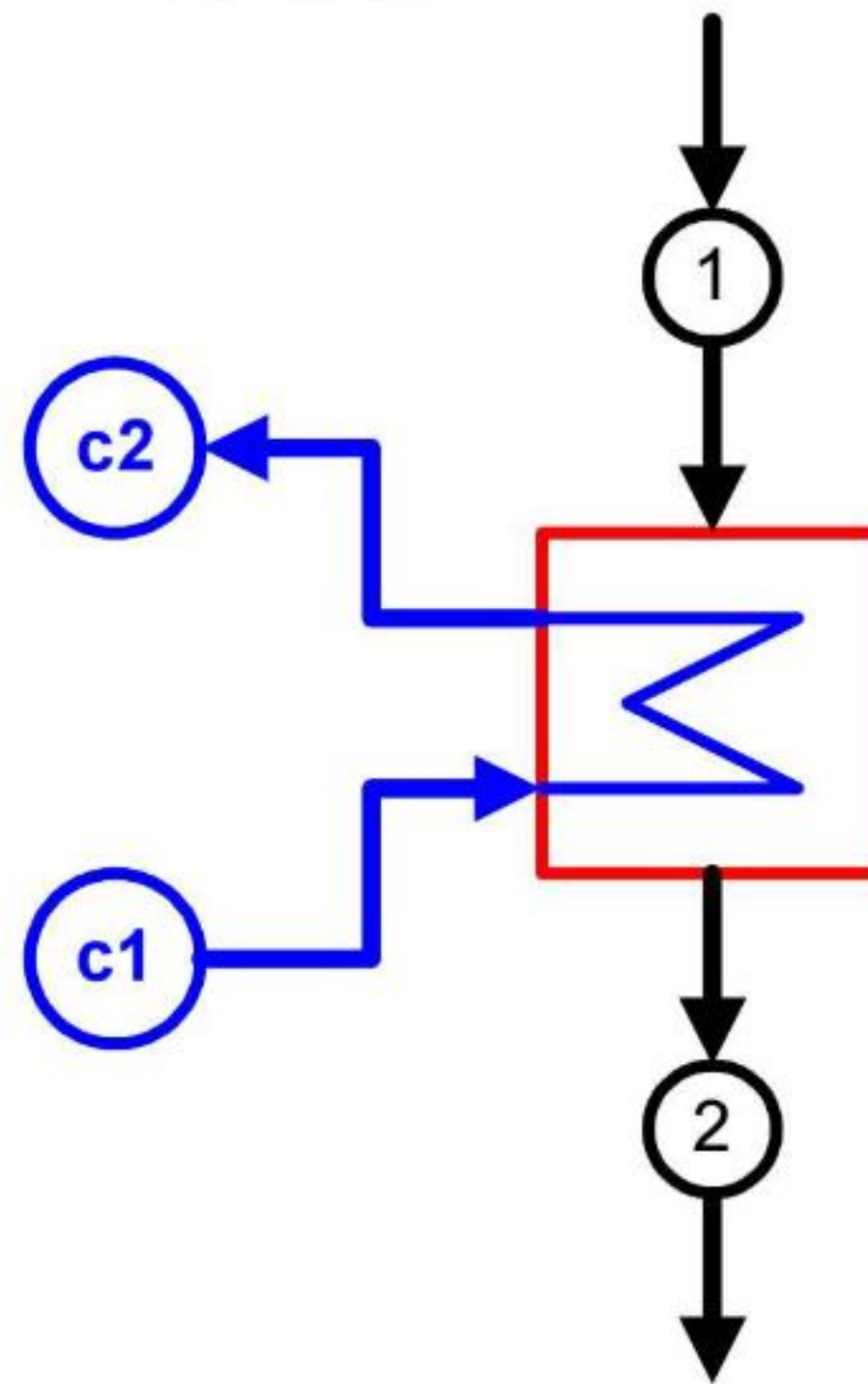


FIGURE 29: Condenser schematic

The condenser is nothing but a heat exchanger between the hot vapor from the regenerator/turbine and the cooling working fluid of the cycle. It has to be observed that the temperature of the hot fluid is higher than the one of the cold fluid throughout the condenser. As well it must be kept in mind that the relation between the enthalpy and the temperature is non-linear, requiring that the vaporizer is divided into appropriate sections for the calculation. This is especially valid for Kalina cycles, in the ORC cycle there is only a property change at the dew point, where de-superheat ends and condensation begins.

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The phenomenon of rise and fall in the ocean waters, called tides, is due to the attractive forces between the celestial bodies; Sun, Earth and the Moon. When the ocean water rises to a maximum extent, it is called spring tide and when they fall off to the lowest possible extent, it is called neap tide. With progress in technology, the usage of electric and electronic devices is exponentially increasing and there is a need to produce extra power other than the existing, in order to meet the future demands. Tidal energy can be considered as one of the best existing source of renewable energies. Unlike the wind, solar, thermal energy etc., tidal energy is something that has a long term perspective and it can be forecasted more accurately. Tidal energy is clean and not depleting. Because of these features it is unique and suitable to use it as a power generating source in the future. There are various types of tidal power plants across the world with varying tidal elevation. Also, the method of conversion of tidal energy into electrical energy is site specific. But generally, the method followed for extracting energy from tides is similar to the conventional hydroelectric power plants. In this paper, the tides at some locations across the world and along the Indian coast, tidal power plants across the world, resource allocation of tidal power plants, advantages and disadvantages of tidal power will be reviewed from the literature.

Keywords: *Tidal energy, tidal barrage, tidal stream, electrical energy.*

1. INTRODUCTION

There are numerous diverse forms of ocean energy that are being explored as potential sources for energy extraction. Some of them are ocean current energy, tidal energy, wave energy, offshore wind energy and thermal energy (Nicholls and Turnock, 2008). Even though the tidal power is still an immature concept, it is definitely a major contributor for electricity generation from renewable sources in the near future (Shaikh and Shaikh, 2011).

The occurrence of tides was witnessed from Roman times and this energy was used on rivers such as the joint estuary of the Tigris, the Tibet River in Rome and Euphrates Rivers even much earlier (Charlier and Finkl, 2009). The flow of water in the form of tides, induced due to the relative positions of the planets Sun-Earth-Moon can surely be considered as one of the reliable sources of energy if suitable systems are designed with an economic plan (Nicholls and Turnock, 2008). In the recent past, use of renewable energy resources for the extraction of energy has been the point of discussion (Cave and Evans, 1984; Bryden et al., 1995; Sodin, 2001).

Researchers working on renewable energy are mainly interested to extract the energy from tides because of its advantages over other forms of renewable energies as the forecasting of tides is easier and accurate from time and magnitude point of view, the density of sea water is much denser than wind, lack of extreme flow speeds and negligible aesthetic damage. (Shaikh and Shaikh, 2011).

1.1 Tides

Tides are the periodic motion of the waters of the sea due to the inter-attractive forces between the celestial bodies. Tides are very long-period waves that move through the oceans in response to the

forces exerted by the moon and sun. Tide and current are not the same. Tide is the vertical rise and fall of the water and tidal current is the horizontal flow. In simple words, the tide rises and falls, the tidal current floods and ebbs. The principal of tidal forces are generated by the Moon and Sun. The Moon is the main tide-generating body. Due to its greater distance, the Sun's effect is only 46 per cent of the Moon's.

1.2 Energy from the tides

There are three types of tides: diurnal, semidiurnal and mixed. (Hagerman G., Polagye B, 2006). Tidal Energy is one of the new and evolving technologies, which is commercially not viable and still in Research & Development (R&D) stage. Tidal energy is inexhaustible and can be considered as a renewable energy source (Shaikh and Shaikh, 2011). It is an advantage because it is less vulnerable to climate change; while the other sources are all vulnerable to the random changes in climate (Nicholls and Turnock, 2008). The review given by the Energy Technology Support Unit (ETSU) on the Tidal Stream Energy was the initial attempt to estimate the energy from tidal stream resources in the UK (ETSU, 1993). The points marked by the ESTU were later studied and modified in 2001 in a document submitted to the UK Department of Trade and Industry (DTI) by Binnie, Black and Veatch. Most of the existing technology used for tidal energy conversion is from the wind power industry (Bahaj et al., 2007; Batten et al, 2007; Fraenkel, 2002). Researchers have predicted that UK has is capable to produce over 20% of its electrical needs from its tidal resources (Callaghan, 2006). It is also a fact that the studies carried out so far in predicting the energy that can be extracted from tides, has only focused on the past and present availability of the energy. But it is also important to consider and address the effects of exploiting the renewable energy sources for energy extraction. There has to be an understanding among the developers as to when and where to stop the energy extraction so that there is minimum or no disturbance caused to the regular natural phenomenon.

1.3 Tidal energy in India

The tides that are generated along some parts of the Indian coastline have the potential to extract energy from the turbines. The tidal elevation in India is as high as 8.5 m at Bhavnagar, Gujarat and as low as 0.5 m at the Southern part of India. Survey of India predicts tide levels at some locations along the Indian coastline and Tide Tables are published for every year (Sanil Kumar V. et al, 2006).

As per the studies carried out by Central Water and Power Commission (CWPC) in 1975, the Gulf of Kutch and Gulf of Khambhat in Gujarat and Sunderbans area in West Bengal are the only suitable sites in India for the production of Tidal Energy. In 1980s, Central Electricity Authority (CEA) took up a study for the assessment of tidal energy potential in India. CEA listed few places of Potential Tidal Energy extraction in India shown in Table 1.

No tidal power generation plant has been installed in India due to its high cost of generation of electricity and lack of techno-economic viability. However, there are proposals for setting up of tidal power stations at Gujarat.

Table 1. Tidal Energy Potential in India

Region	State	Tidal potential (MW)
Gulf of Khambhat	Gujarat	7000
Gulf of Kutch	Gujarat	1200
Gangatic Delta, Sunderbans	West Bengal	100

1.3.1 Demonstration Project at Sunderbans

A report was submitted by West Bengal Renewable Energy Development Agency (WBREDA) in 2001 for setting up a 3.65 megawatt capacity tidal power station at Durgaduani Creek in Sunderbans Island of West Bengal. These details were submitted to the Ministry on June 2006. Also, WBREDA

entered into a MoU with the National Hydroelectric Power Corporation Limited (NHPC), Faridabad for updating of the Project Report and its execution. The updated Report prepared by NHPC was received by the Ministry in November, 2007. The NHPC Limited was given responsibility to complete the project. However, the project has been discontinued due to very high tender cost.

1.3.2 Tidal Power Projects in Gulf of Kutch, Gujarat

A committee was established under the Central Electricity Authority (CEA) on the 900MW Kutch Tidal Power Project for estimating the cost of the project. The project was not found to be commercially viable due to high capital cost as well as high cost of generation of electricity.

In January 2011, Gujarat signed a MoU with Gujarat Power Corporation Ltd. (GPCL) for establishing a 250 MW tidal power project at Mandavi district in Gulf of Kutch. GPCL has initially started a 50MW tidal power project in Kutch. GPCL has made a request for grant for the tidal power plant to Ministry of New and Renewable Energy (MNRE).

The experience gained in the above project will decide the future course of action for the advancement of tidal energy in India.

1.4 Tidal energy around the world

The necessity to reduce CO₂ emissions and gradual increase in cost of fossil fuel has resulted in a significantly increased use of tidal energy (Nicholls and Turnock, 2008). Today, tidal energy around the world is increasingly being considered as a potential source of renewable energy (Bryden and Scott, 2007). Extreme tides are found in many locations across the globe. Some of them are: the Pentland Firth, Scotland; the Severn estuary; the Aleutians; the fjords of Norway; the Philippines; the Straits of Messina, Italy; the Bosphorus, Turkey; the English Channel; Indonesia, and the straits of Alaska and British Columbia.

The first major hydroelectric plant was put to operation in 1967 that used the energy of the tides to generate electricity. It produced about 540,000 kW of electricity (Charlier and Finkl, 2009). Studies have shown that the European territorial waters have 106 locations for extracting tidal energy that would provide electricity of 48 TW per year. It is estimated around 50,000 MW of installed capacity being achievable along the coasts of British Columbia alone. There are greater predictions of extracting energy of about 90,000 MW off the North West coast of Russia and about 20,000 MW at the inlet or Mezen river and White Sea. There are also estimations along the West coast of India having potential to generate 8,000MW.

Table 2 gives the highest available tidal levels in some of the regions that have the potential to establish tidal power stations. Tidal power plants have already been set up at some of these places and some are still in the planning phase. The main characteristics of four large-scale tidal power plants that were constructed after World War II and currently exist are given in Table 3.

Table 2. Highest tides of the global ocean (Gorlov A. M., 2001)

Site	Country	Tidal elevation (m)
Bay of Fundy	Canada	16.2
Severn Estuary	England	14.5
Port of Ganville	France	14.7
La Rance	France	13.5
Puerto Rio Gallegos	Argentina	13.3
Bay of Mezen (White Sea)	Russia	10
Penzhinskaya Guba (Sea of Okhotsk)	Russia	13.4

Table 3.Existing large tidal power plants (Gorlov A. M., 2001)

Site	Country	Bay area (km ²)	Avg. tide (m)	Installed Power (MW)
La Rance	France	22	8.55	240
Kislaya Guba	Russia	1.1	2.3	0.4
Annapolis	Canada	15	6.4	18
Jiangxia	China	1.4	5.08	3.9

2. METHODS OF TIDAL ENERGY EXTRACTION

Different methods have been suggested by authors for the extraction of tidal energy. However the basic principle behind the methods remains same. However, there are two primary methods to extract energy from the tides.

- a. Estuaries into which large amounts of ocean water flows in due to high tidal range, are captured behind barrages and the turbines are rotated by utilizing the potential energy of the stored water.
- b. The kinetic energy of moving water can be used to extract energy similar to the principle of extraction of wind energy.

Both methods that are mentioned above have been suggested and followed and each has its own advantages and disadvantages (Bryden and Melville, 2004). It may also be possible to employ pumping strategies for barrages to obtain better efficiency and to match electricity demand better (MacKay, 2007).

The devices that are used in the energy generation vary in size, shape and specifications. ISSC (2006) has classified the devices into three types:

- a. Tidal barrages that store tidal flow and generate power through discharge.
- b. Tidal fences which block a passage and extract energy in either or both directions of tidal flow.
- c. Tidal current devices which are fixed or moored within a tidal stream.

2.1 Tidal barrage

Tidal barrage is a structure generally built across the mouth of the estuary through which the water flows in and out of the basin. The tidal barrage has sluice gates that allows the flow of water in and out of the basin. The water flows into the bay during high tide and the water is retained by closing the sluice gates at the beginning of low tide. The barrage gates are controlled by knowing the tidal range of the location and operating it at right times of the tidal cycle. There are turbines located at the sluice gates which produce electricity when the gates are opened during the low tide. Using this principle, authors have mentioned different ways of extraction of energy like ebb generation, flood generation, ebb and flood generation, pumping, two basin schemes etc. Figure 1 shows the Plan view of a hypothetical tidal barrage. Even though the barrage method has high theoretical efficiency, only one large scale tidal barrage has been constructed at La Rance, France (Blunden and Bahaj, 2006).

The advantage of using barrage to method to generate electricity in comparison with fossil fuels is that it reduces the greenhouse effects, to provide a better environment. La Rance tidal power plant, France is an example for barrage method. On the top of the barrage there is a four-lane highway that cuts 35 km of distance between the towns of Saint Malo and Dinard representing.



Figure 1.Top view of La Rance tidal power plant barrage (750m in length)

2.2 Tidal stream energy

In the early 1990s, tidal power was mainly focused on harnessing the tidal flow and generating the energy by means of potential storage rather than through tidal stream. Tidal stream technologies have made massive progress towards commercialisation in the last decade. Extensive research is being carried out in UK waters related to tidal stream energy. UK has a target to achieve 20% of its electricity requirement through ocean resources by 2020. About 40 energy converting machines are being developed and prototypes are being tested in the labs and in waters of UK (Irena, 2014). Since the tidal stream energy is still an emerging technology, it has no standardizations, but variety of devices are being developed to make use of the water flow to extract electricity. However, the efficiency of each of the devices has to be flawlessly examined by extensive testing to choose the appropriate device for a particular location.

2.3 Calculation of tidal energy

The total tidal energy is the energy due to the tidal stream (kinetic energy) and the energy due to release of the stored water in the basin (potential energy). It is also a fact that the increase in tidal variation or the tidal stream energy results in increase of energy extraction to a large extent (Shaikh and Shaikh, 2011).

2.3.1 Kinetic energy

The kinetic energy of the stream flow flowing across the cross section with a velocity is given by (Bryden et al., 2004).

$$P = \frac{1}{2} \rho A V^3 \quad (1)$$

ρ is the density of sea water (kg/m^3)

C_p is the power coefficient

A is the area of cross section of the channel (m^2)

V is current velocity (m/s)

The power output or the efficiency of the turbine " ξ " depends on the design of the turbine. The power output for a turbine from these kinetic systems can be obtained by the following equation (Shaikh and Shaikh, 2011).

$$P = \frac{\xi}{2} \rho A V^3 \quad (2)$$

ξ is turbine efficiency

P is power generated (in watts)

ρ is density of the water (seawater is 1025 kg/m³)

A is sweep area of the turbine (in m²)

V is velocity of the flow

2.3.2 Potential energy

The potential energy is mainly dependent on the tidal prism of the basin. Potential energy obtained due to the stored water can be calculated as (Gorlov, 2001; Shaikh and Shaikh, 2011).

$$E = \frac{1}{2} A \rho g h^2 \quad (3)$$

h is the vertical tidal range,

A is the horizontal area of the barrage basin,

ρ is the density of water = 1025 kg per cubic meter (seawater varies between 1021 and 1030 kg/m³)

g is the acceleration due to the Earth's gravity = 9.81 m/s²

From equation 3, it can be seen that the potential energy varies with square of tidal range. So, a barrage should be placed in such a location where it is possible to achieve maximum storage head. Black and Veatch (2003) suggest that the ideal water depths to achieve the best possible power output at few potential sites around the UK range between 25 and 40 m and the recommend diameter of the rotor to range between 10 m and 20 m. (Frost et. al, 2015). The inlets that are between islands having large basin area are considered to have a greater amplification effect because of the reduction in the throat area and the water depth relative to the surroundings, producing a venturi effect. This accelerates the water as it is forced through a channel with a smaller cross-sectional area (Bryden I G and Melville G T, 2004).

2.4 Tidal energy devices

There are wide range of tidal energy devices used for energy extraction based on site conditions. The details of some of them is given in Table 4.

Table 4. Tidal energy extraction devices

Turbine	Power Output	Description	Image
Seagen, Marine Current Turbines Ltd. UK	1 MW	Twin two bladed rotor, sheath mounted, horizontal axis tidal turbine	
The Blue Concept, Hammerfest Strom, Norway	1 MW	Three bladed pile mounted horizontal axis tidal turbine	

THG, Tidal Hydraulic Generator	3 MW	Array of four bladed horizontal axis turbines attached to a gravity mounted frame	
TidEL SMD Hydrovision, UK	1 MW	Twin two bladed tethered rotor, able to yaw into the current	
Stingray, Engineering Business Ltd., UK	500 kW	Oscillating Hydroplane	
Blue energy, Blue energy Inc., Canada	500 kW	Four bladed, moored, surface piercing, vertical axis turbine	

3. RECOMMENDED SITES FOR INSTALLATION OF TIDAL POWER PLANTS

Some preliminary standards are given by Couch and Bryden to identify sites that are suitable for the development of a tidal energy extraction. The most important variables generally considered are:

1. The local water depth: Existing device technology concepts are generally limited to operational water depths of 25–45 metres.
2. The location of the nearest exploitable grid connection: For an immature industry, the economics of tidal energy extraction require easy access to a nearby grid connection with spare capacity; otherwise the capital cost cannot be viably recouped across the life of the project.
3. An energetic and persistent resource: Large mean spring and neap tide velocities are highly desirable. Some sites have the added advantage of minimizing the low velocity periods of the tidal cycle as the local dynamics ensure that the tidal flow reverses through the slack period at an accelerated rate. The sites that the developers are interested to extract energy tend to have peak spring tidal velocities of 3+ m/s.

If these three primary criteria are met, a site is considered to have solid potential for future development. The majority of coastal locations can be rejected out of hand by consideration of just these three variables (Couch and Bryden, 2006).

The English Channel, the Arctic Ocean, The Gulf of Mexico, The Amazon, The Straits of Magellan and Taiwan are some of the possible locations for locating tidal devices. Table 6 shows some potential sites for tidal power installations. There are estimates that the energy that can be globally extracted is around 1800 TWh/year (Nicholls and Turnock, 2008). But, it has to be taken care that the effect on environment, economic and social constraints have to be addressed (Sun X et al., 2008). It is suggested that 10% of the extraction of energy can be considered as guideline for harnessing renewable energy resources (Bryden I.G. et al., 2004). The estimated construction costs for existing and proposed tidal barrages are given in Table 5.

Table 5.Estimated construction costs for existing and proposed tidal barrages

Barrage	Country	Capacity (MW)	Power Generation (GWh)	Construction costs (Million USD)	Construction costs per kW (USD/kW)
Operating					
La Rance	France	240	540	817	340
Sihwa Lake	Korea	254	552	298	117
Proposed/planned					
Gulf of Kutch	India	50	100	162	324
Wyre barrage	UK	61.4	131	328	534
Garorim Bay	Korea	520	950	800	154
Mersey barrage	UK	700	1340	5741	820
Incheon	Korea	1320	2410	3772	286
Dalupiri Blue	Philippines	2200	4000	3034	138
Severn barrage	UK	8640	15600	36085	418
Penzhina Bay	Russia	87000	200000	328066	377

Note: Cost equivalent for 2012

Based on Wyre Energy Ltd., 2013.

Table 6. Some potential sites for tidal power installations (Gorlov A. M., 2001)

Site	Country	Bay area (Km ²)	Avg. Tide	Potential Power (MW)
Passamaquoddy	USA	300	5.5	400
Cook Inlet	USA	3100	4.35	Up to 18000
Mezen	Russia	2640	5.66	15000
Tugur	Russia	1080	5.38	6790
Severn	UK	490	8.3	6000
Mersey	UK	60	8.4	700
San Jose	Argentina	780	6.0	7000
Carolim Bay	Korea	90	4.7	480
Secure	Australia	130	8.4	570
Walcott	Australia	260	8.4	1750

Technology and economics, however, dictate that only those areas where currents can exceed 2m/s might be suitable for exploitation (Bryden I G and Melville G T, 2004).

4. ADVANTAGES AND DISADVANTAGES OF TIDAL POWER

Advantages:

-) Tidal power is a renewable and sustainable energy resource. It reduces dependence upon fossil fuels.
-) It produces no liquid or solid pollution. It has little visual impact.
-) Tidal power exists on a worldwide scale from deep ocean waters.
-) Tidally driven coastal currents provide an energy density four times greater than air, which means that a 15m diameter turbine will generate as much energy generated by a 60m diameter windmill.
-) Tidal currents are both predictable and reliable, a feature which gives them an advantage over both wind and solar systems. Power outputs can be accurately calculated far in advance, allowing for easy integration with existing electricity grids.

INTRODUCTION

Energy is a critical factor in developing countries for economic growth as well as for social development and human welfare. Hydropower is a renewable source of energy, which is economical, non-polluting and environmentally benign among all renewable sources of energy. For efficient operation of hydropower plants, in order to meet the electricity demand, the hydro energy is stored either in reservoirs for dam based schemes or settling basins for run-of-river schemes. These reservoirs or settling basins are filled with sediments over a period of time. This problem must be taken care of by sediment settling systems in power plants. However, lot of unsettled sediment pass through the turbines every year and turbine parts are exposed to severe erosion. The erosion of hydro turbine components is a major problem for the efficient operation of hydropower plants. These problems are more prominent in power stations which are of run-of-river types. The problem is aggravated if the silt contains higher percentage of quartz, which is extremely hard.

HYDRO POTENTIAL

India is endowed with economically exploitable and viable hydro potential assessed to be about 84,000 MW at 60% load factor. In addition, 6,780 MW in terms of installed capacity from Small, Mini, and Micro Hydel schemes have been assessed. Also, 56 sites for pumped storage schemes with an aggregate installed capacity of 94,000 MW have been identified. It is the most widely used form of renewable energy. India is blessed with immense amount of hydro-electric potential and ranks 5th in terms of exploitable hydro-potential on global scenario.

The present installed capacity as on September 30, 2013 is approximately 39,788.40 MW which is 17.39% of total

electricity generation in India. The public sector has a predominant share of 97% in this sector. National Hydroelectric Power Corporation (NHPC), Northeast Electric Power Company (NEEPCO), Satluj Jal Vidyut Nigam (SJVN), THDC, NTPC-Hydro are a few public sector companies engaged in development of Hydroelectric Power in India.

Bhakra Beas Management Board (BBMB), an illustrative state owned enterprise in north India has an installed capacity of 2.9 GW and generates 12,000-14,000 million units per year. The cost of generation of energy after four decades of operation is about 20 paise/kWh. [citation needed] BBMB is a major source of peaking power and black start to the northern grid in India. Large reservoirs provide operational flexibility. BBMB reservoirs annually supply water for irrigation to 12.5 million acres (51,000 km²; 19,500 sq mi) of agricultural land of partner states, enabling northern India in its green revolution.

HISTORY OF HYDROELECTRICITY

Archaeologists confirm that the history of the storage dam goes back to 5000 years to 3000 BC. Vitruvius was the first architect to explain a water wheel which could generate power. Then Barbegal from France in the 4th century AD worked on water wheels and generated a system of sixteen water wheels which followed the principle of kinetic energy into mechanical energy.

STRENGTHS OF HYDRO POWER

- Environmental friendly, clean renewable
- High degree of flexibility
- Part of multipurpose project with additional benefits.
- Pumped storage for optimal integrate operation of grid

- Least operational and maintenance cost
- Additional benefits of Flood control, Tourism, fishery.
- Well recognized for obtaining financial support.

WEAKNESS OF HYDRO POWER

- Mainly depends on rainfall/snowmelt.
- Run of river not for peaking
- High capital intensive.
- Remotely located
- Gestation period is very large.
- Relatively smaller units.
- Non standard occurrence.

OPPORTUNITIES OF HYDRO POWER

- Vast potential untapped
- Requirement for power peaking
- Greater concern towards increasing pollutions on land, water and in air causes leading inclination towards hydro
- Depletion of fossil fuels.

THREATS OF HYDRO POWER

- Ambitious plan for thermal/nuclear programme for power.
- Growing concern of environment.
- Prone to natural calamities.
- Submergence of land and displacement of population
- Apprehension on seismic disturbance
- Over emphasis of other renewable energy sources development.

WORKING PRINCIPLE OF HYDRO- ELECTRIC POWER PLANT

In hydroelectric power plants the potential energy of water due to its high location is converted into electrical energy. The total power generation capacity of the hydroelectric power plants depends on the head of water and volume of water flowing towards the water turbine.

The water flowing in the river possesses two type of energy:

- The kinetic energy due to flow of water and
- Potential energy due to the height of water.

In hydroelectric power and potential energy of water is utilized to generate electricity.

The formula for total power that can be generated from water in hydroelectric power plant due to its height is given

$$P = q \cdot h \cdot g$$

Where “p” is the power produced in “watt”

“Q” is the rate of flow of water which in cubic meter/second

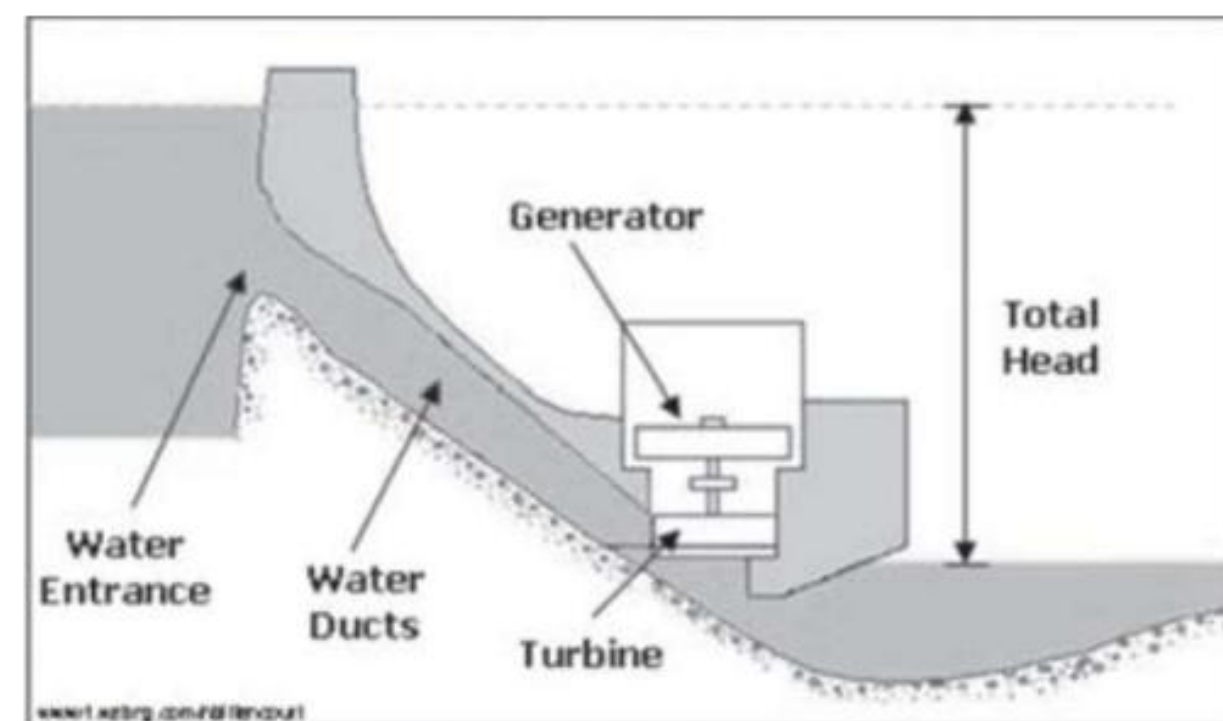


Fig. 1: Hydro Power Principle

“h”= height of water which is measured in “meter”

It's also head of water.

The difference between source of water (from where water is taken) and the water's outflow (where the water is used to generate electricity, it is the place near the turbines).

“g” is the gravity constant 9.81 m/second square

The formula clearly shows that the total power that can be generated from the hydroelectric power plants depends on two major factors

- The flow rate of water or volume of flow of water and
- Height or head of water.
- More the volume of water and
- More the head of water more is the power produced in the hydroelectric power plant.

To obtain the high head of water the reservoir of water should as high as possible and power generation unit should be as low as possible. The maximum height of reservoir of water is fixed by natural factors like the height of river bed, the amount of water and other environmental factors. The location of the power generation unit can be adjusted as per the total amount of power that is to be generated. Usually the power generation unit is constructed at levels lower than ground level so as to get the maximum head of water.

The total flow rate of water can be adjusted through the pen stock as per the requirements. If more power is to be generated more water can be allowed to flow through it.

COMPONENTS OF HYDROELECTRIC POWER PLANTS

Hydroelectric power plant requires various components for generating electrical power. Some of the major components in hydroelectric power plants are: Reservoirs, Dam, Trash Rack, Forebay, Surge Tank, Penstock, Spillway, Prime Mover and Generator, Draft Tube. The functions of all major components are discussed.

The basic requirement of a hydroelectric power station is a reservoir where large quantity of water is stored during

rainy season and used during the dry season. The reservoir is built by constructing a dam across the river. The water from the reservoir is drawn by the forebay through an open canal or tunnel. The water from the forebay is supplied to the water prime mover through the penstock which is located at the much lower level than the height of the water in the reservoir. Thus potential energy of water stored in reservoir is converted into kinetic energy and made to rotate the turbine. Turbine shaft is connected to synchronous generator or alternator for generating electricity. This generated power is stepped up using step-up transformer and delivered to load centers or grid. The regulation of water flow to the turbine depending on the electrical load demand is carried out by the governor system.

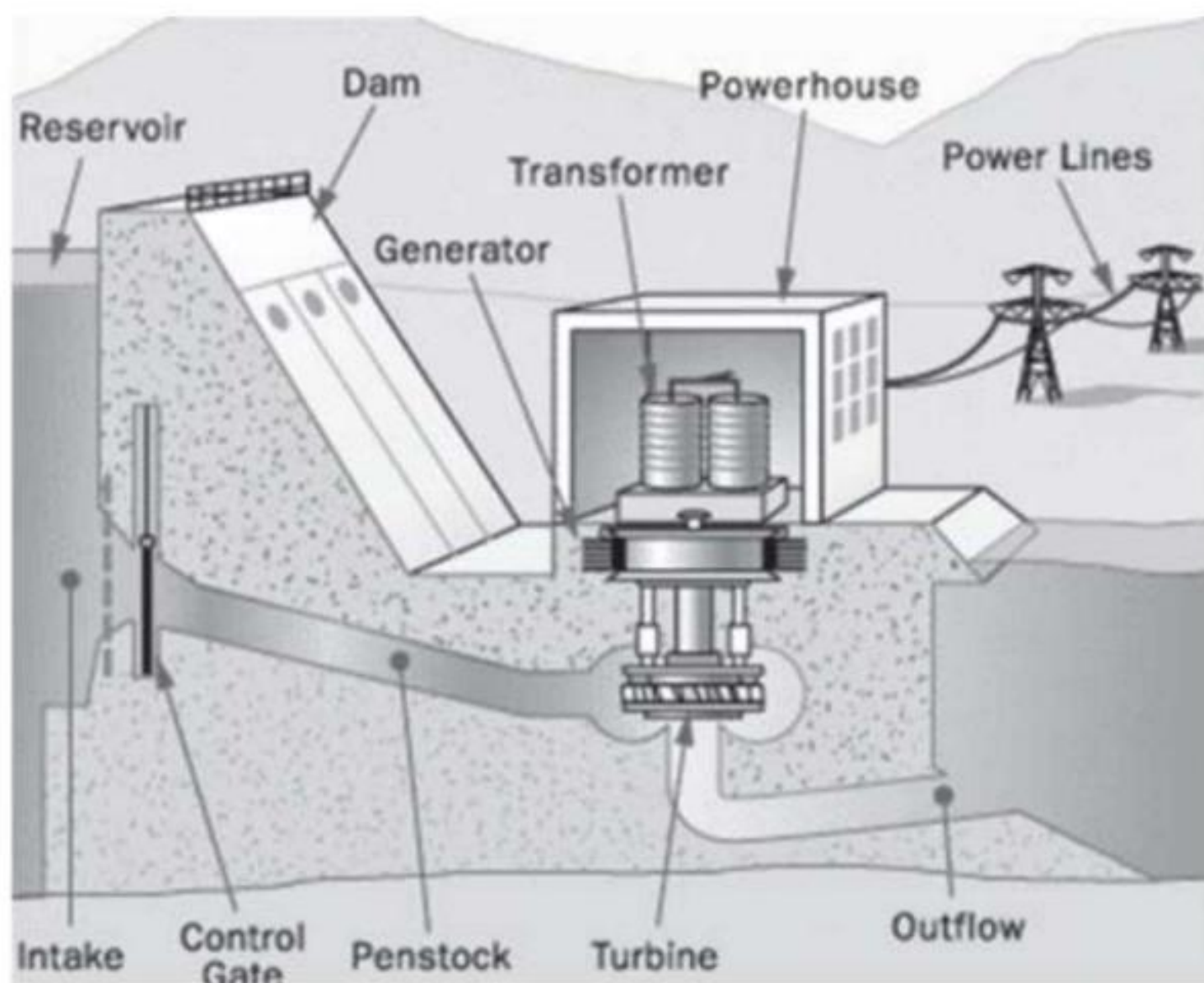


Fig. 2: Components of Hydro Electric Power Plant

Some of the components of hydroelectric power plants and their functions are given below:

WATER RESERVOIR

The function or purpose of reservoir is to store the water during rainy season and supply the same during dry season. This is in simple, water storage area. The water reservoir is the place behind the dam where water is stored. The water in the reservoir is located higher than the rest of the dam structure. The height of water in the reservoir decides how much potential energy the water possesses. The higher the height of water, the more its potential energy. The high position of water in the reservoir also enables it to move downwards effortlessly.

The height of water in the reservoir is higher than the natural height of water flowing in the river, so it is considered to have an altered equilibrium. This also helps to increase the overall potential energy of water, which helps ultimately produce more electricity in the power generation unit.

DAM

The function of dam is to increase the height of the water level (increase in the potential energy) behind it which ultimately increases the reservoir capacity. The dam also helps in increasing the working head of the power plant. Dams are generally built to provide necessary head to the power plant.

TRASH RACK

The water intake from the dam or from the forebay is provided with trash rack. The main function of trash rack is to prevent the entry of any debris which may damage the wicket gates and turbine runners or choke-up the nozzles of impulse turbine. During winter season when water forms ice, to prevent the ice from clinging to the trash racks, they are often heated electrically. Sometimes air bubbling system is provided in the vicinity of the trash racks which bring warmer water to the surface of the trash racks.

FOREBAY

The function of forebay is to act as regulating reservoir temporarily storing water when the load on the plant is reduced and to provide water for initial increment of an increasing load while water in the canal is being accelerated. In many cases, the canal itself is large enough to absorb the flow variations. In short, forebay is naturally provided for storage of water to absorb any flow variations if exist. This can be considered as naturally provided surge tank as it does the function of the surge tank. The forebay is always provided with some type of outlet structure to direct water to penstock depending upon the local conditions.

SURGE TANK

The main function of surge tank is to reduce the water hammering effect. When there is a sudden increase of pressure in the penstock which can be due sudden decrease in the load demand on the generator. When there is sudden decrease in the load, the turbine gates admitting water to the turbine closes suddenly owing to the action of the governor. This sudden rise in the pressure in the penstock will cause the positive water hammering effect. This may lead to burst of the penstock because of high pressures.

When there is sudden increase in the load, governor valves opens and accepts more water to the turbine. This results in creation of vacuum in the penstock resulting into the negative water hammering effect. Therefore the penstock should have to withstand both positive water hammering effect created due to close of governor valve and negative water hammering effect due to opening of governor valve. In order to protect the penstock from these water hammering effects, surge tank is used in hydroelectric power station.

A surge tank is introduced in the system between dam and the power house nearest. Surge tank is a tank provided to absorb any water surges caused in the penstock due to sudden loading and unloading of the generator. When the velocity of the water in the penstock decreases due to closing of turbine valves, the water level in the surge tank increases and fluctuating up and down till its motion is damped out by the friction. Similarly when the water accelerates in the penstock, water is provided by the surge tank for acceleration. Surge tank water level falls down and fluctuates up and down absorbing the surges.

INTAKE OR CONTROL GATES

These are the gates built on the inside of the dam. The water from reservoir is released and controlled through these gates. These are called inlet gates because water enters the power generation unit through these gates. When the control gates are opened the water flows due to gravity through the penstock and towards the turbines. The water flowing through the gates possesses potential as well as kinetic energy.

THE PENSTOCK

The penstock is the long pipe or the shaft that carries the water flowing from the reservoir towards the power generation unit, comprised of the turbines and generator. The water in the penstock possesses kinetic energy due to its motion and potential energy due to its height.

The total amount of power generated in the hydroelectric power plant depends on the height of the water reservoir and the amount of water flowing through the penstock. The amount of water flowing through the penstock is controlled by the control gates.

SPILLWAY

The function of spillway is to provide safety of the dam. Spillway should have the capacity to discharge major floods without damage to the dam and at the same time keeps the reservoir levels below some predetermined maximum level.

POWER HOUSE

A power house consists of two main parts, a sub-structure to support the hydraulic and electrical equipment and a superstructure to house and protect this equipment.

The superstructure of most power plants is the buildings that house all the operating equipment. The generating unit and the exciter is located in the ground floor. The turbines which rotate on vertical axis are placed below the floor level while those rotating on a horizontal axis are placed on the ground floor alongside of the generator.

WATER TURBINES

Water flowing from the penstock is allowed to enter the power generation unit, which houses the turbine and the generator. When water falls on the blades of the turbine the kinetic and potential energy of water is converted into the rotational motion of the blades of the turbine. The rotating blades cause the shaft of the turbine to also rotate. The turbine shaft is enclosed inside the generator. In most hydroelectric power plants there is more than one power generation unit.

There is large difference in height between the level of turbine and level of water in the reservoir. This difference in height, also known as the head of water, decides the total amount of power that can be generated in the hydroelectric power plant.

There are various types of water turbines such as Kaplan turbine, Francis turbine, Pelton wheels etc. The type of turbine used in the hydroelectric power plant depends on the height of the reservoir, quantity of water and the total power generation capacity.

GENERATORS

It is in the generator where the electricity is produced. The shaft of the water turbine rotates in the generator, which produces alternating current in the coils of the generator. It is the rotation of the shaft inside the generator that produces magnetic field which is converted into electricity by electromagnetic field induction. Hence the rotation of the shaft of the turbine is crucial for the production of electricity and this is achieved by the kinetic and potential energy of water. Thus in hydroelectricity power plants potential energy of water is converted into electricity.

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Role of Nuclear Engineering

- Before we can start understanding about nuclear power, it is essential to look at the motivation.
- We shall realise that if India has to meet its aspirations of providing decent living to its people, it is necessary that nuclear has to grow fast.
- If this is planned properly and executed, there are tremendous opportunities for nuclear engineers.
- Let us show this in a brief manner.

Population Growth

- The source for the information on facts and figures are taken from a recent publication by Grover and Chandra (*R.B. Grover and S. Chandra, "Scenario for growth of energy in India", Energy Policy, Vol. 34, 2006, pp 2834-2847*).
- Energy is generated for people to use it and hence we need to have a good picture of the population dynamics before we can project the energy requirements.
- The population statistics past, present and projected are given in the following Table:

Year	Population (Billions)	Growth Rate (per year)
2001	1.027	1.50
2011	1.19	1.02
2021	1.32	0.7
2031	1.41	0.4
2041	1.47	0.2
2051	1.50	0.0

- We expect to stabilise around 2050.
- We need to provide electricity for 1.5 billion people.

Electricity Production

Year	Power Production (Mwe)
1947	1363
1981	30214
1991	66086
2003	138730

- From the above table, we see that the average growth rate is 8.6% and is impressive.
- However, per capita power production is still small compared to global average.

Total Energy Scenario

- In 2001, total energy consumed by the world was 382 EJ (1 Exa Joule = 10^{18} J).
- Indians consumed 3.4% with a population share of 16.6%.
- USA consumed 24.5% with 4.6% population.
- Indian consumption was $1/5^{\text{th}}$ of the world average.
- Thus to improve the quality of life there has to be a growth in energy production and consumption.

Energy Split

- In 2003, Indians consumed 18.96 EJ.
- Domestic resources accounted for 15 EJ and imported 3.96 EJ.
- 71% of the above was for commercial purposes and 29% was for domestic purposes.
- Of the commercial, 92% was from fossil, 6% hydro, 1.7% nuclear and 0.2% wind.
- The import content was 8% for coal, 71% for oil and 13% for nuclear.

Energy Growth Rate Projection

Year	Total Energy (%)	Electrical Power (%)
2002-22	4.6	6.3
2022-32	4.5	4.9
2032-42	4.5	4.5
2042-52	3.9	3.9

- For a sustainable development a model has been evolved for electricity production. This is summarised in the table shown above.
- The contributions from coal, Hydrocarbon, Hydel, Renewable and Nuclear are 46%, 16%, 8%, 4% and 26% respectively. Thus nuclear has to grow fast if we have to meet our energy demands.

Nuclear Perspective

- For the projections given, if we have to succeed, several new technologies have to be demonstrated.
- The foremost among them is the Fast Breeder.
- Demonstration of Accelerated Driven Systems with thorium as fuel.
- It is important to note that India has to take the leadership role as the rest of the world does not have much interest in thorium based cycle.
- Having understood the need for accelerated growth of nuclear power, let us now learn the fundamentals of the subject.
- Before going into the details, a small overview is given to appreciate the details of the subject that will be taught.

Discovery of Fission

- Rutherford described as the father of nuclear physics, bombarded alpha particles on Nitrogen to eject a proton and thus started the business of splitting atom in 1917.
- James Chadwick is credited with discovering neutron in 1932. Being neutral, it can easily penetrate the Coulomb barrier and interact directly with the nucleus.
- Enrico Fermi, considered to be one of the most brilliant physicist, postulated and succeeded in producing transuranic elements by bombarding neutrons on Uranium. However, he did not realise that he had fissioned Uranium in 1934.
- Otto Hahn, Fritz Strassman and Lise Meitner in 1938 conducted experiments similar to Fermi.
- They noticed that instead of a heavy element formation they found lighter elements in the product.
- They were the first ones to postulate fission and pointed the large energy that will be released.
- We shall see later how this energy release is made possible.

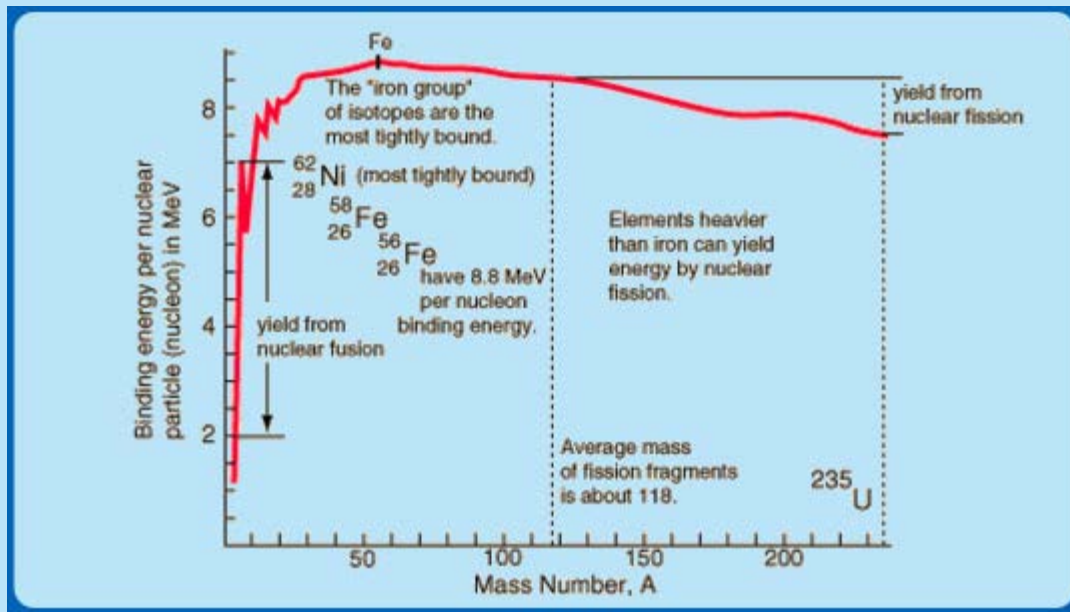
Rutherford's Model

- Rutherford, based on experimentation, had proposed a model for the atom.
- It has a central positively charged nucleus consisting of protons and neutrons.
- It has electrons revolving around to make the atom neutral.
- The charge and mass are summarised in the following table.

	Mass	Charge
Proton	$1.67261 \times 10^{-24} \text{ g}$	$+1.60219 \times 10^{-19} \text{ Coulombs}$
Electron	$9.10956 \times 10^{-28} \text{ g}$	$-1.60219 \times 10^{-19} \text{ Coulombs}$
Neutron	$1.67492 \times 10^{-24} \text{ g}$	0

Binding Energy

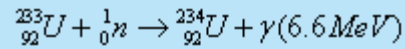
- Binding energy of a nucleus is defined as the energy required to break a nucleus into its constituents.
- It increases with the number of nucleons in a nucleus.
- It is often plotted in per nucleon basis as shown below



- We shall understand this figure in more detail in a later lecture.
- Higher the binding energy per nucleon, more stable is the nuclei.
- Iron is the most stable nuclei.
- We also note that as light elements fuse or heavy elements undergo fission, the products are more stable.

Fissile Elements

- Consider the following equation



- 6.6 MeV is released on account of neutron being absorbed or bound to U^{234} nucleus.
- Thus, Binding Energy of Last Neutron is 6.6 MeV.
- Energy necessary to be supplied to induce fission is called Critical Energy for Fission.
- The binding energy of last neutron and critical energy for fission for some isotopes are listed below:

Isotope	Critical Energy for Fission (MeV)	Binding Energy of Last Neutron (MeV)
Th^{232}	6.5	5.1
U^{234}	4.6	6.6
U^{236}	5.3	6.4
U^{239}	5.5	4.9
Pu^{240}	4.0	6.4

- Thus when neutron is absorbed in U^{233} the neutron deposits its binding energy in the compound nucleus (U^{234}), which is more than what is required to make U^{234} fission. Hence it splits spontaneously.
- Those elements which can be fissioned by zero energy neutrons are called fissile elements.
- Thus U^{233} , U^{235} and Pu^{239} are fissile elements.
- On the other hand, Th^{232} and U^{238} need neutrons with 1.4 MeV and 0.6 MeV respectively for inducing fission. Hence these are called fissionable nuclei.
- Natural Uranium has 99.3% of U^{238} and 0.7% of U^{235} .
- U^{235} is the only naturally occurring fissile element.

Prompt Neutron Spectrum

- Neutrons released from fission have an average Energy = 1.98 MeV.
- A large fraction of neutron is above the threshold energy of 0.6 MeV (refer table in previous page) needed to fission U^{238} .
- Thus some U^{238} is also directly fissioned in reactors.
- It has been noticed that as the energy of the colliding neutron is high, the probability of reaction is poor.
- This is attributed to the reduced time a neutron spends in the zone of interaction.
- Hence the probability of reaction is inversely proportional to the speed of the neutron.
- As the interaction decreases rapidly with increase in energy, and that the energy of the prompt neutrons are high, we need to slow them down.
- This is done by the use of moderator.

Moderators and Coolants

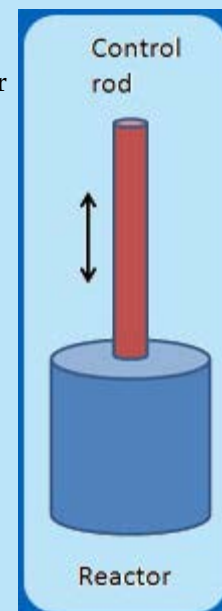
- Hydrogen in the form of water (H_2O) and heavy hydrogen in the form of heavy water (D_2O) are the most common moderators.
- In gas cooled reactors graphite (C) is used as the moderator.
- Coolant is employed to transport the heat generated in the core to the steam generator, where steam is generated.
- In water moderated reactors the same fluid also serves as coolant.
- There are some safety advantages if we use gas as a coolant. In such cases CO_2 or He is used.

Critical Size

- We shall see later in the course that all neutrons produced do not get absorbed in the reactor.
- Many neutrons leak out of the reactor.
- To keep the neutron population steady in a reactor we need to produce more neutrons than what is absorbed in fuel.
- As the number of neutrons produced depends on the volume and number of neutrons leaking is proportional to surface area, there exists a minimum size below which the reactor cannot operate (due to excess leakage).
- This is called the critical size.
- Thus, for a reactor to operate we need a critical size.
- This size depends on the concentration of the fissile elements.
- The size increases with the reduction in concentration.
- Thus conceptually, as fuel burns out we need to increase the size, which is impractical.
- To overcome this, the size of the reactor is increased to begin with and to compensate for the excess production, neutron absorbers are used.
- As fuel burns out the absorbers are gradually removed.

Control of Nuclear Reactors

- The process of power addition or reduction is done by insertion or removal of control rods.
- Originally when the fuel is fresh and the reactor is not operating, these rods are fully in.
- To start the reactor, these are slowly pulled out and the reactor begins operation when the rods are out by a small amount.
- As the fissile elements deplete, to compensate for them these are continuously pulled out.
- When they get pulled out completely, then reactor is stopped and fresh fuel added.



Fast Breeders

- We shall show later in the course that we get around 2 effective neutrons from fission induced by fully moderated neutrons.
- However, if we do not fully slow down and allow the neutron to induce fission at higher speeds, we get more effective neutrons per fission.
- While only one neutron is required to get the reactor operating steadily, the excess neutrons can be used to produce fresh fuel.
- This is called breeding.
- Since neutrons are not fully slowed down, it is called fast breeder.
- Such reactors are usually cooled by sodium.

Power Reactors

- Some of the characteristics of nuclear power reactors are summarised in the following Table:

Type	Moderator	Coolant	Pressure (bar)	Maximum Temp (C)	Eff. %
Pressurized Water Reactor	Light Water	Light Water	150	320	33
Boiling Water Reactor	Light Water	Light Water	70	250	33
Pressurised Heavy Water Reactor	Heavy Water	Heavy Water	80	320	32
Gas Cooled Reactor	Graphite	Carbondioxide	15	410	35
High Temperature Gas Reactor	Graphite	Helium	45	800	45
Liquid Metal Fast Reactor	-----	Sodium	1	580	42

Agenda for the Course

- Having realised that nuclear reactors have an important role to play in India's power generation scene, we shall now understand the principles of operation of a nuclear reactor.
- We shall understand that present and the reactors of immediate future shall operate on fission process.
- We shall understand how reactors are sized, moderated, cooled and controlled.
- We shall also understand the safety aspects that have to be kept on mind while designing these reactors.



Objectives

In this lecture you will learn the following

- This lecture brings out the features of nuclear power reactors.
- Description of BWR, PWR, PHWR and FBR's are included.

Objectives





Fig. 1 Nuclear Reactor with Containment

- This lecture gives some of the basic concepts used in power stations.
- A bird's eye view of a reactor site is shown in fig. 1.
- The two cylindrical buildings are the reactor building.
- The rectangular buildings behind the reactors are the turbine buildings.
- Two large hyperbola shaped towers are the cooling towers.

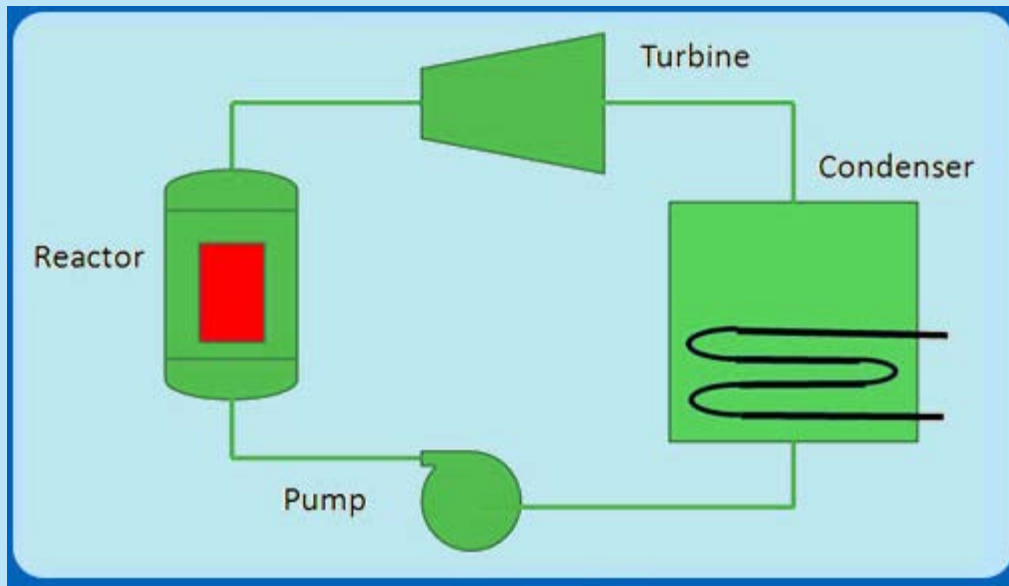


Fig. 2 Boiling Water Reactor (BWR)-I (BWR)

- The schematic view of Boiling Water Reactor (BWR) is shown in fig.2.
- In this reactor water boils in the reactor itself.
- The steam so produced is sent to turbine to produce work.
- The expanded steam is condensed and is then pumped back into the reactor.

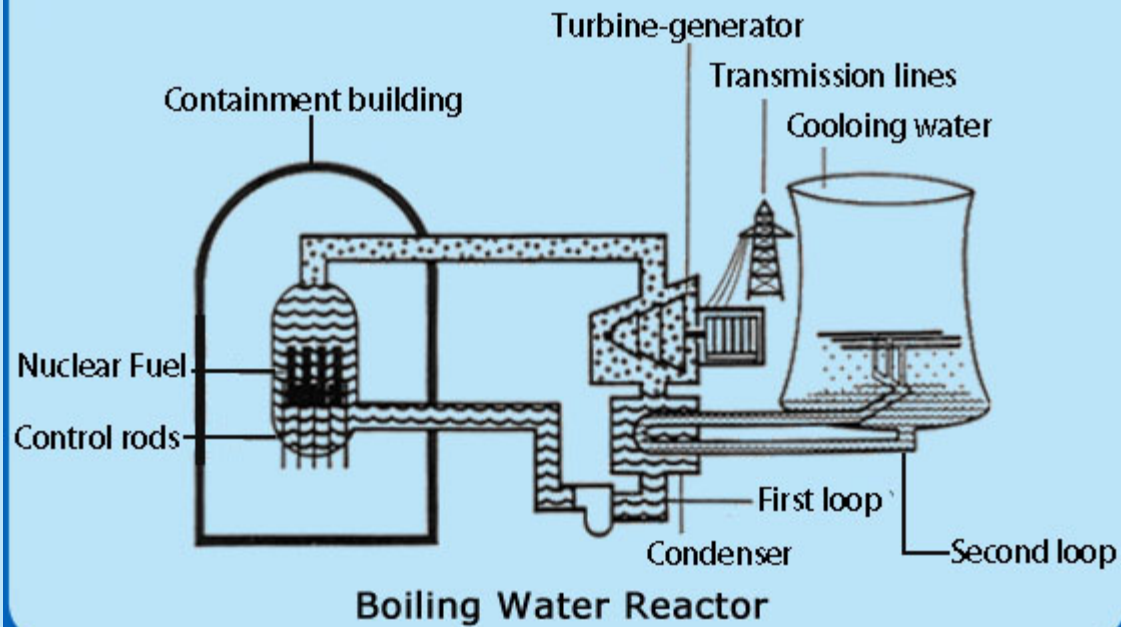


Fig.3 Typical Layout of BWR

- A more detailed schematic layout is shown in fig. 3.
- Note that control rods are inserted from the bottom of the vessel.
- Note that the reactor building is also called containment building.
- This is because in case of any accidents, the radioactive products are trapped (contained) in this building.
- Note that only reactor is in the containment building.
- The turbine and condenser are housed outside.

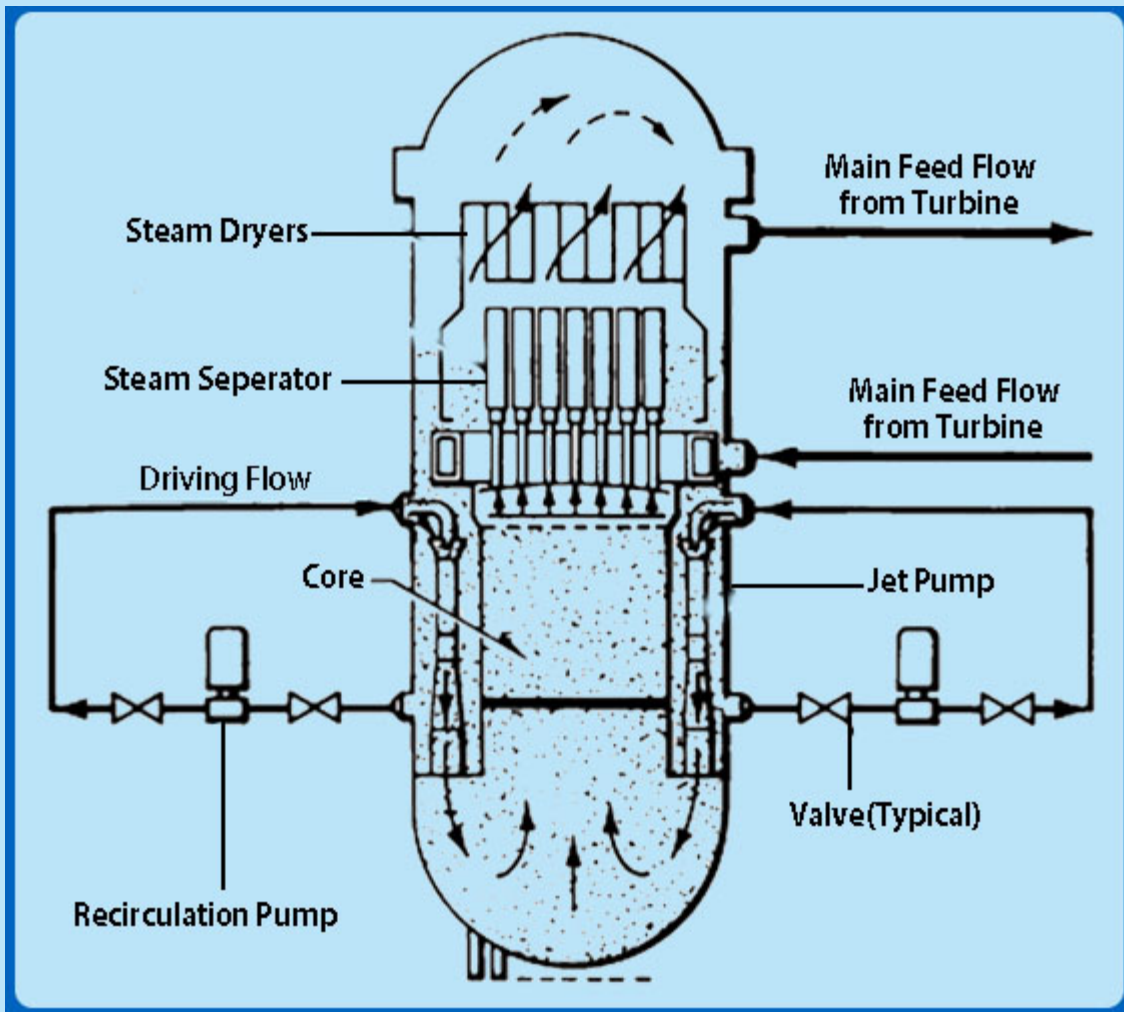


Fig.4 BWR Reactor Vessel

- The details of components housed inside the reactor vessel are shown.
- Jet pumps housed inside the reactor vessel helps in circulating fluid into the core.
- Steam separators at the top separate steam from water.
- Steam dryers are used to remove water droplets that are carried by steam.

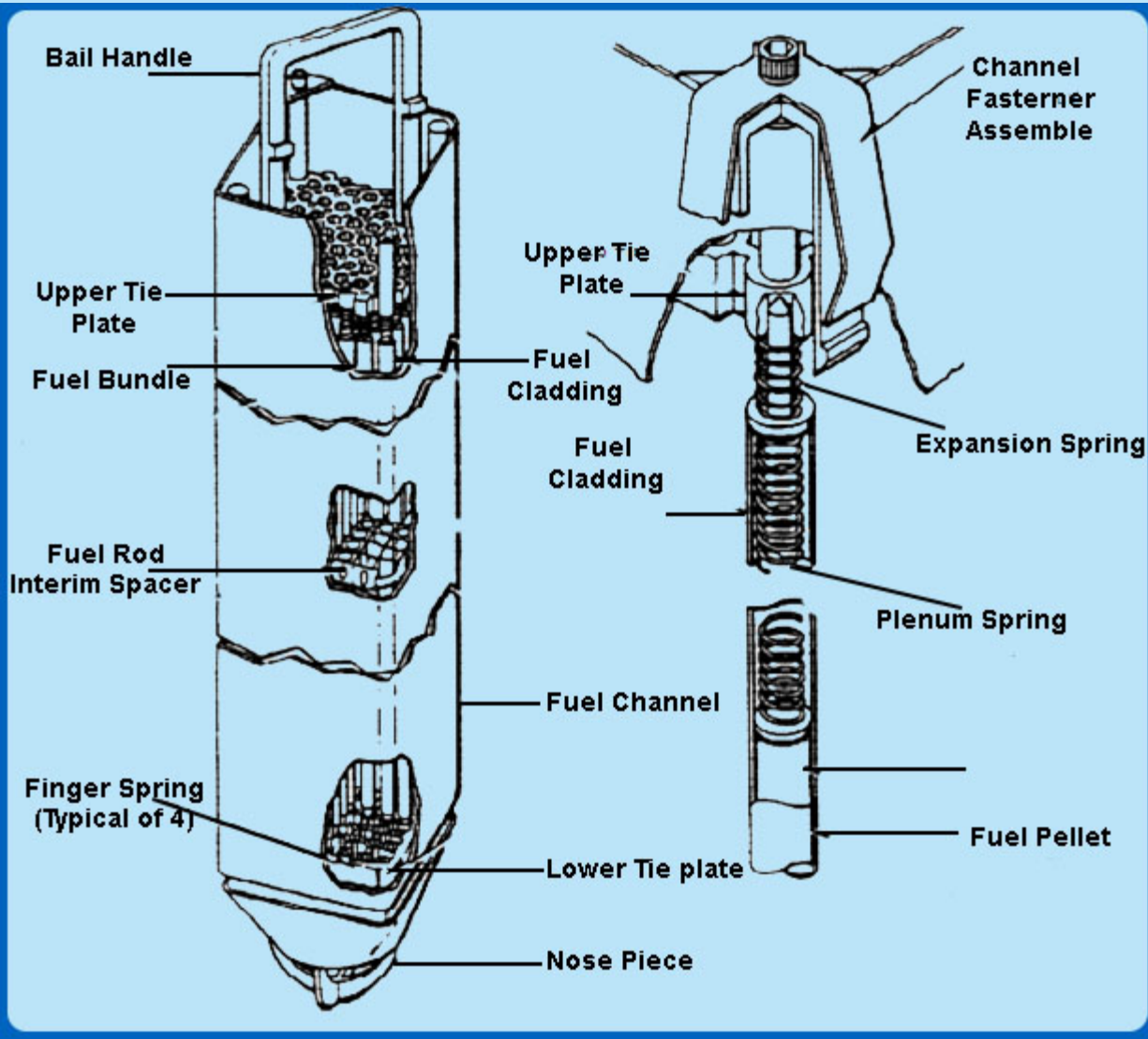


Fig.5 BWR Fuel Bundle

- The fuel bundle of a BWR is shown in fig. 5.
- It is an assembly of fuel rods arranged in a square array.
- Details inside each fuel rod is shown separately.
- The fuel is in the form of cylindrical pellets.
- A gas plenum(space) is provided at the top of fuel rod.
- This is to accommodate fission gases that diffuse out of the fuel.
- The bundle is wrapped with a metal sheet to prevent mixing of steam from adjacent bundles.

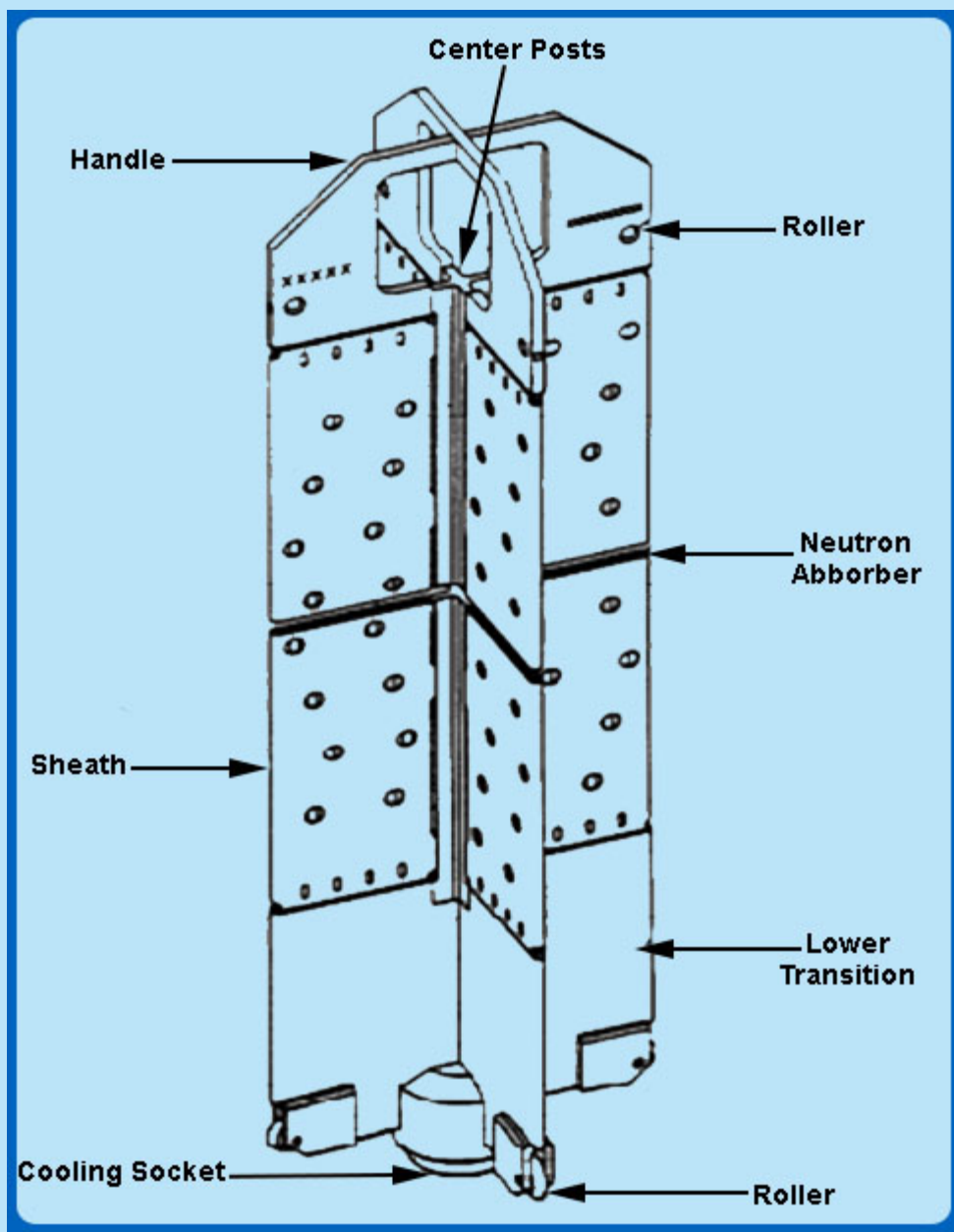


Fig.6 BWR Control Rod

- The schematic view of the control rod is shown in fig.6.
- It has a cross section in the form of a "cross" and is called cruciform control rod.
- Neutron absorbers are placed inside the cross.



Fig.7 BWR/6 Fuel assemblies & control rod module

- Figure 7 clearly shows the assembly of rod bundles.
- It also shows the position of the cruciform control rods.
- One cruciform control rod is positioned in between four fuel bundles.
- The handles on the rod bundles are used for lifting and positioning the rod bundles.

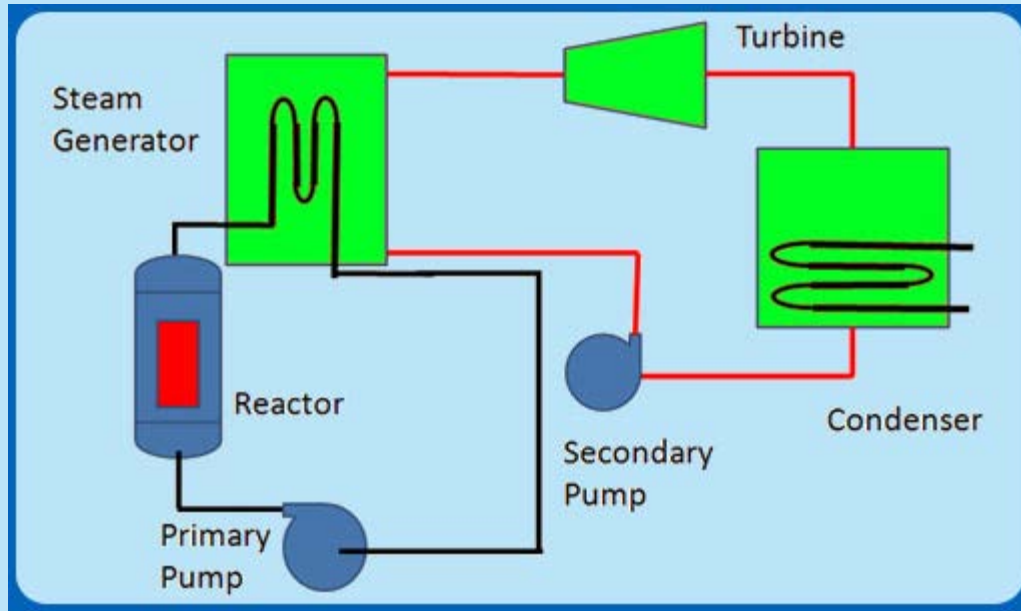


Fig.8 Pressurised Water Reactor

- The schematic view of a Pressurised Water Reactor (PWR) is shown in fig.8.
- In a PWR, the water is not allowed to boil in the reactor.
- Steam is produced in a steam generator.
- Thus there are two circuits, called primary and secondary.
- The primary coolant takes heat from fuel rods in the reactor and transports it to steam generator.
- In the steam generator, the primary coolant transfers heat to the secondary coolant.
- The primary coolant is pumped back using primary pump.
- In the secondary circuit, water is pumped into the steam generator by the secondary pump.
- The steam generated is expanded in the turbine, condensed in the condenser and pumped back.

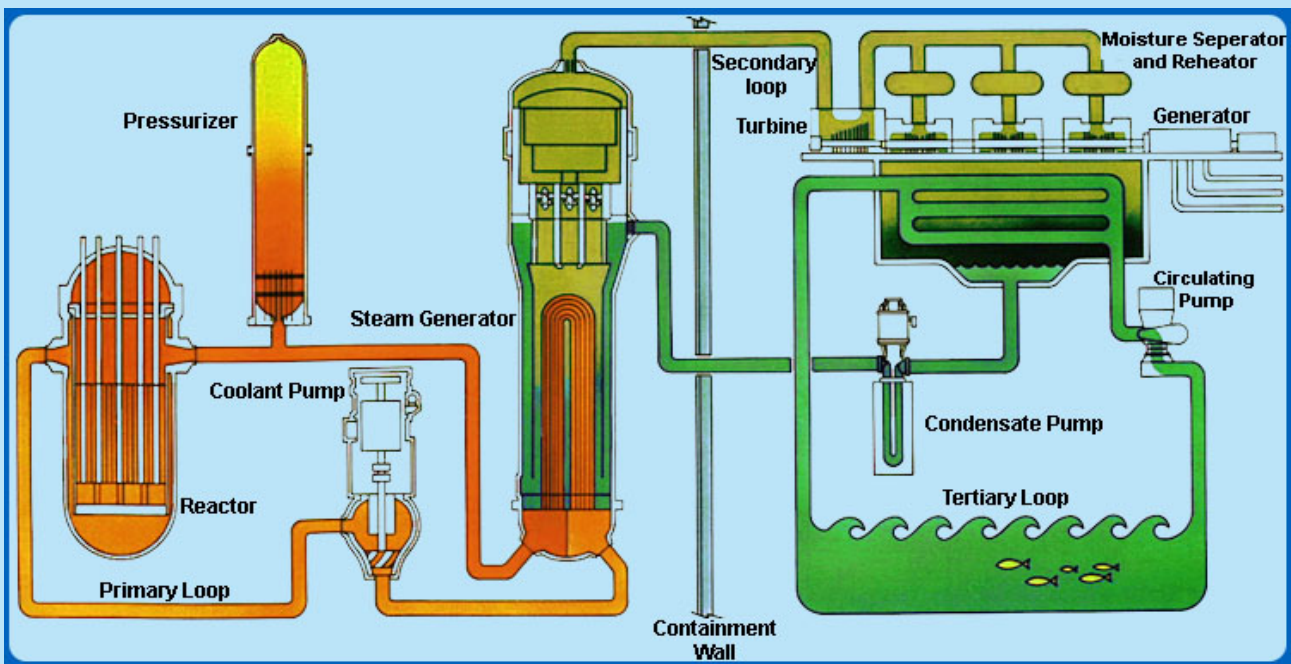


Fig.9 Westinghouse PWR

- The schematic view of the PWR manufactured by Westinghouse is shown in fig.9.
- An additional component called pressuriser is seen.
- It maintains the pressure in the system, should it try to rise or fall due a mismatch between heat generated in fuel and heat transmitted in the steam generator.
- It may be observed that the size of steam generator is larger than the reactor.
- More than one steam generator is used in a PWR.

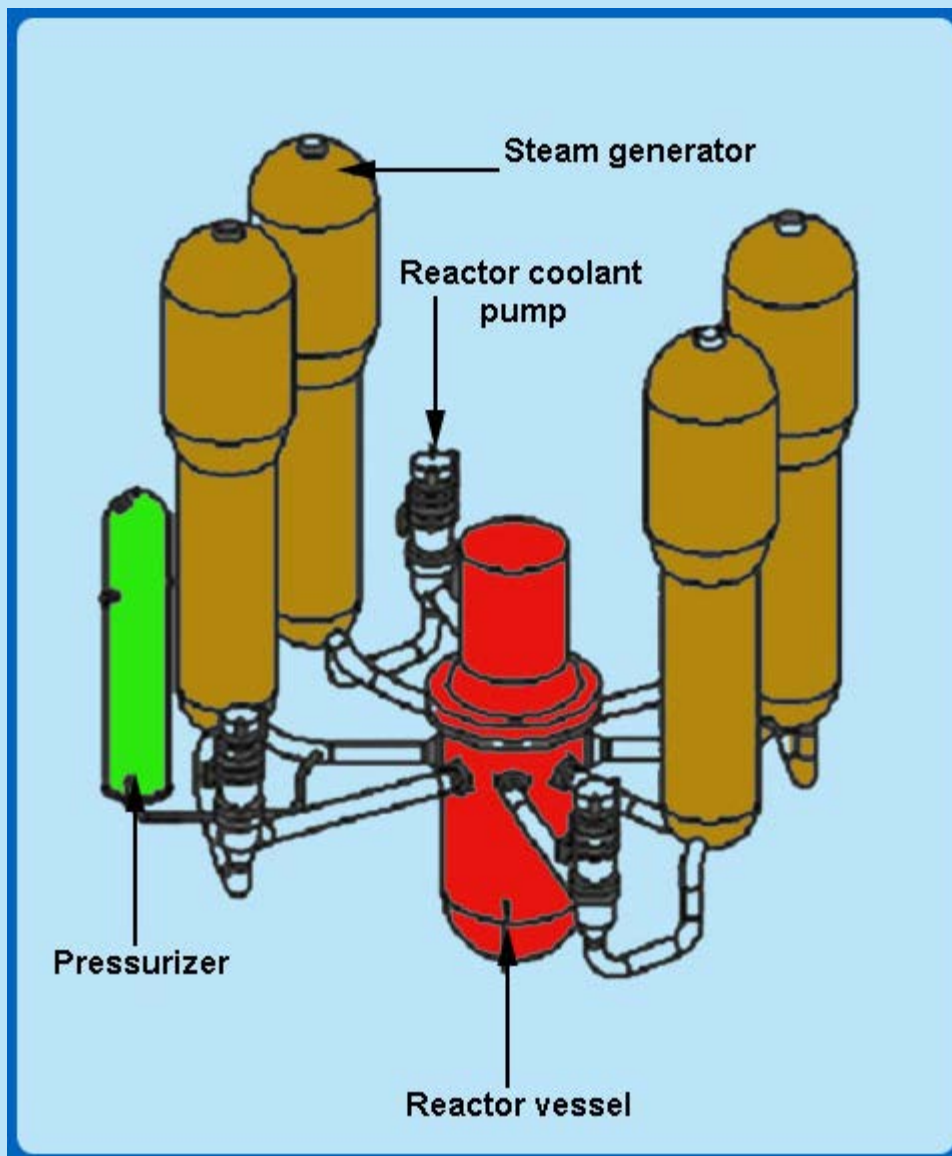


Fig.10 Four-Loop Plant

- A typical four loop plant is shown in fig. 10.
- It may be noted that one of the pump is hidden.

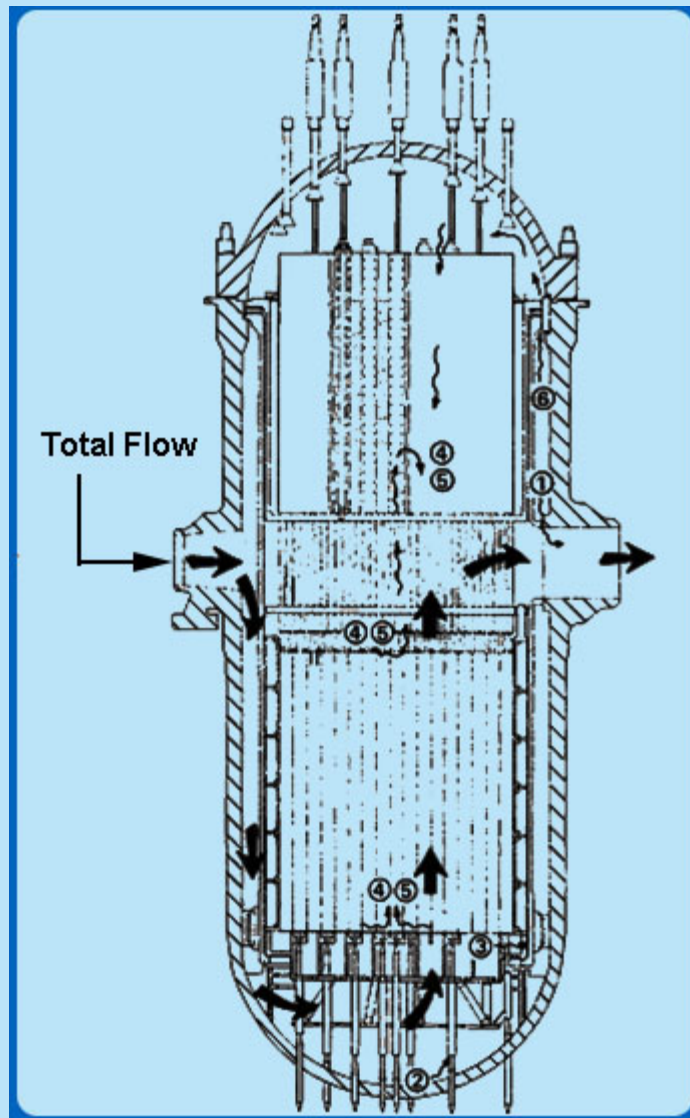


Fig.11 PWR Reactor Vessel

- Figure 11 shows the details of the PWR pressure vessel.
- It may be observed that the top half is used to house the control rods.
- The primary coolant fluid movement path is also shown.
- The fluid is made to move down by the presence of a core baffle.
- The fluid then rises from the bottom of the core.
- The heated water then flows out from the top of the core.

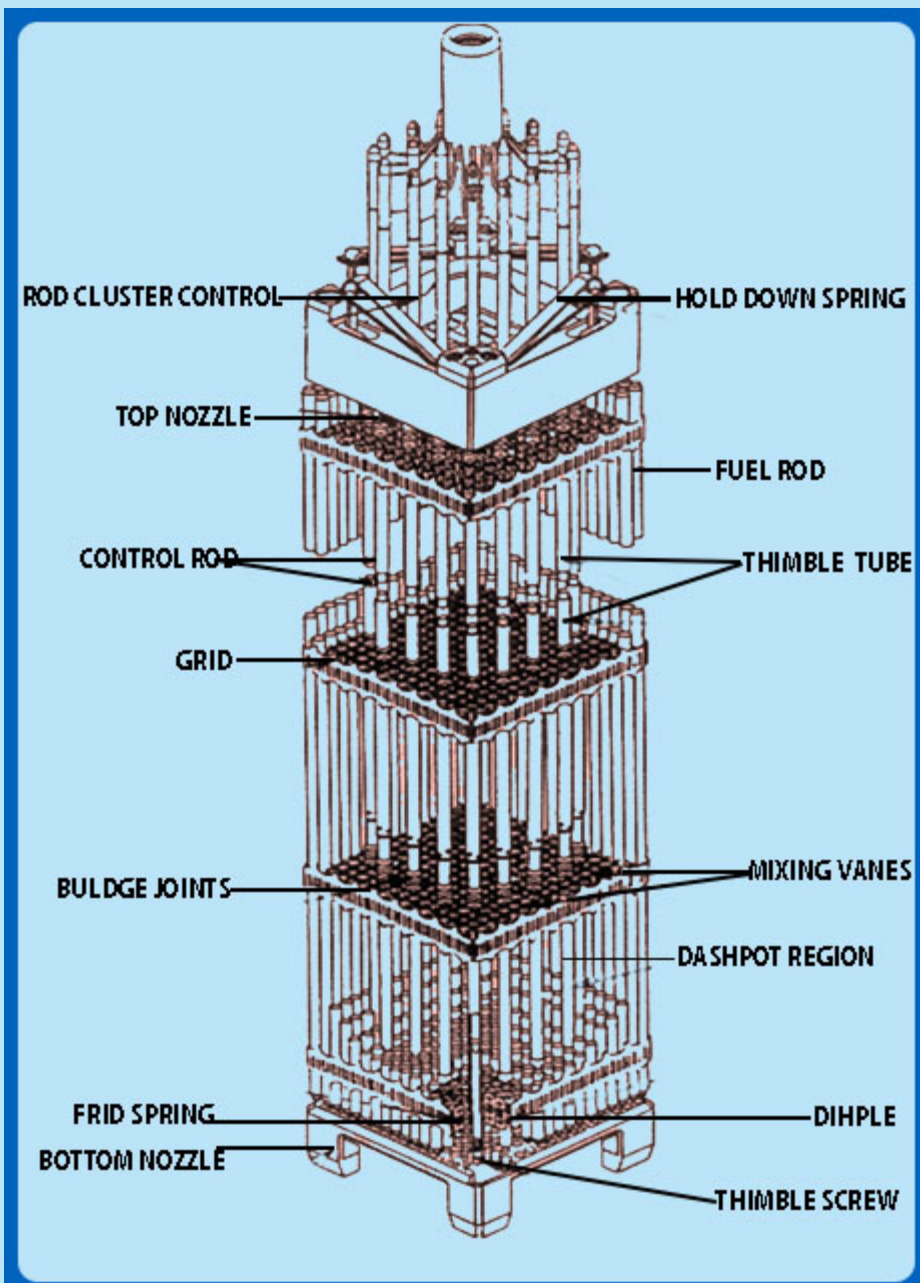


Fig.12 Fuel Assembly

PWR Fuel Assembly

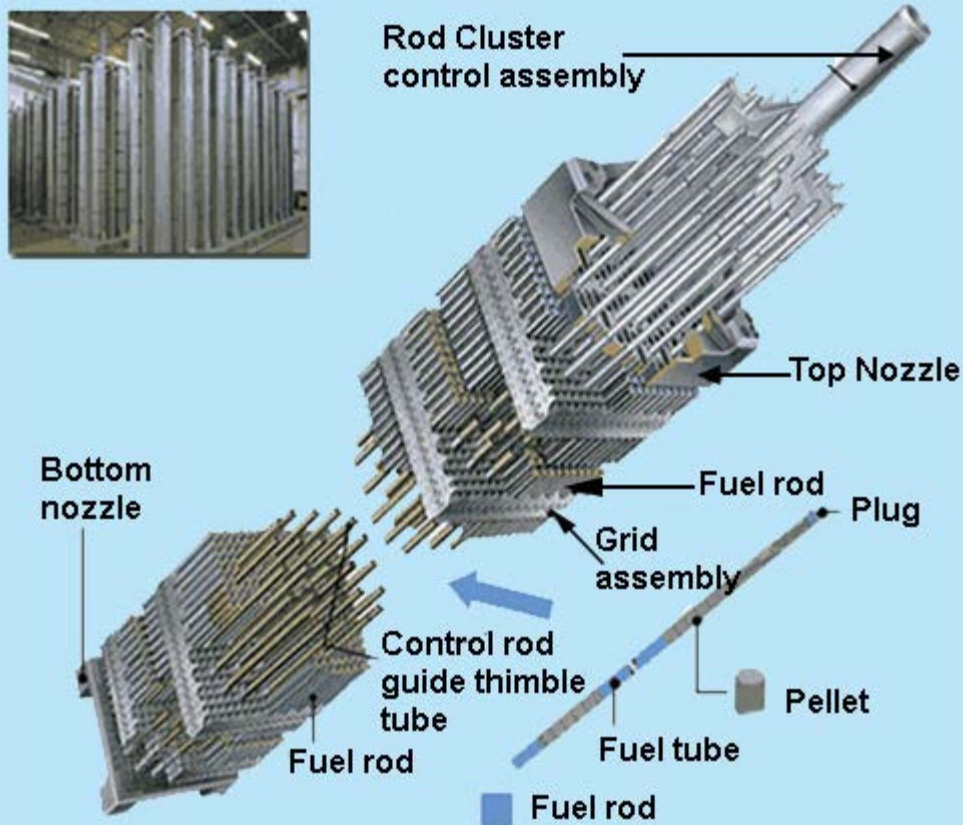


Fig.13 PWR Fuel Assembly

- Figures 12 and 13 shows three dimensional view of the PWR rod bundle.
- The construction of fuel rod is similar to that of a BWR.
- The control rod cluster is also shown in the figure.
- It may be noted that several control rods are moved simultaneously.

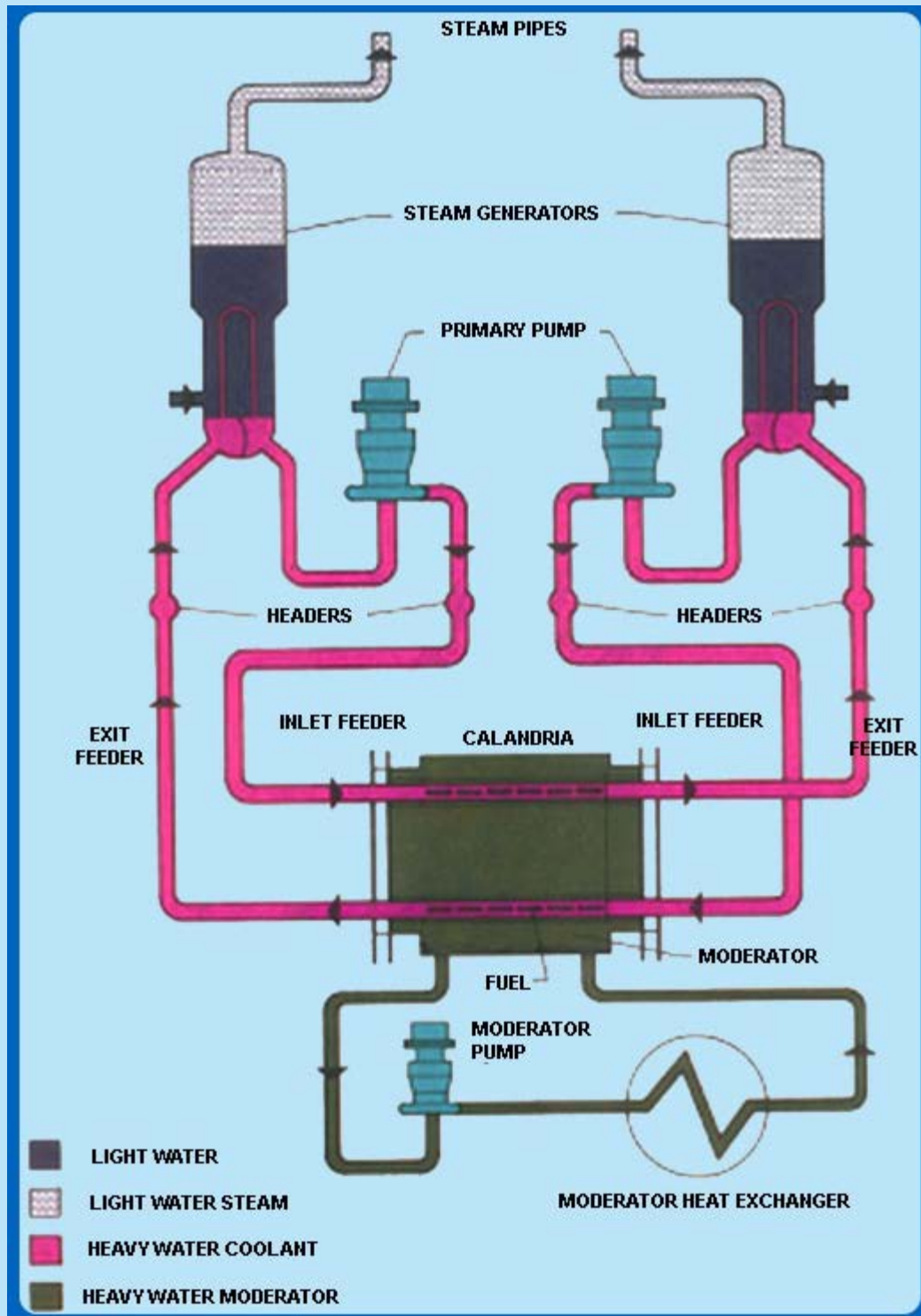


Fig.14 Pressurised Heavy Water Reactor

- A schematic view of a Pressurised Heavy Water Reactor (PHWR) circuit is shown in fig. 14.
- This differs substantially with that of a PWR.
- It employs the pressure tube concept rather than pressure vessel concept employed by PWR and BWR.
- It employs the figure of eight configuration.

- Let us begin tracing the fluid movement from one of the pumps shown in fig.14.
- The fluid is pushed into an inlet header.
- This is also marked as component no. 5 in fig.15.
- It is a long pipe which distributes fluid into many smaller tubes called feeders (component no. 4 in fig. 15).
- Each of this feeder sends the coolant into one of coolant channels in the core. Let us say it goes from right to left in fig. 14.
- A PHWR typically has 300-400 coolant channels.
- Inside the coolant channels, fuel bundles are placed.

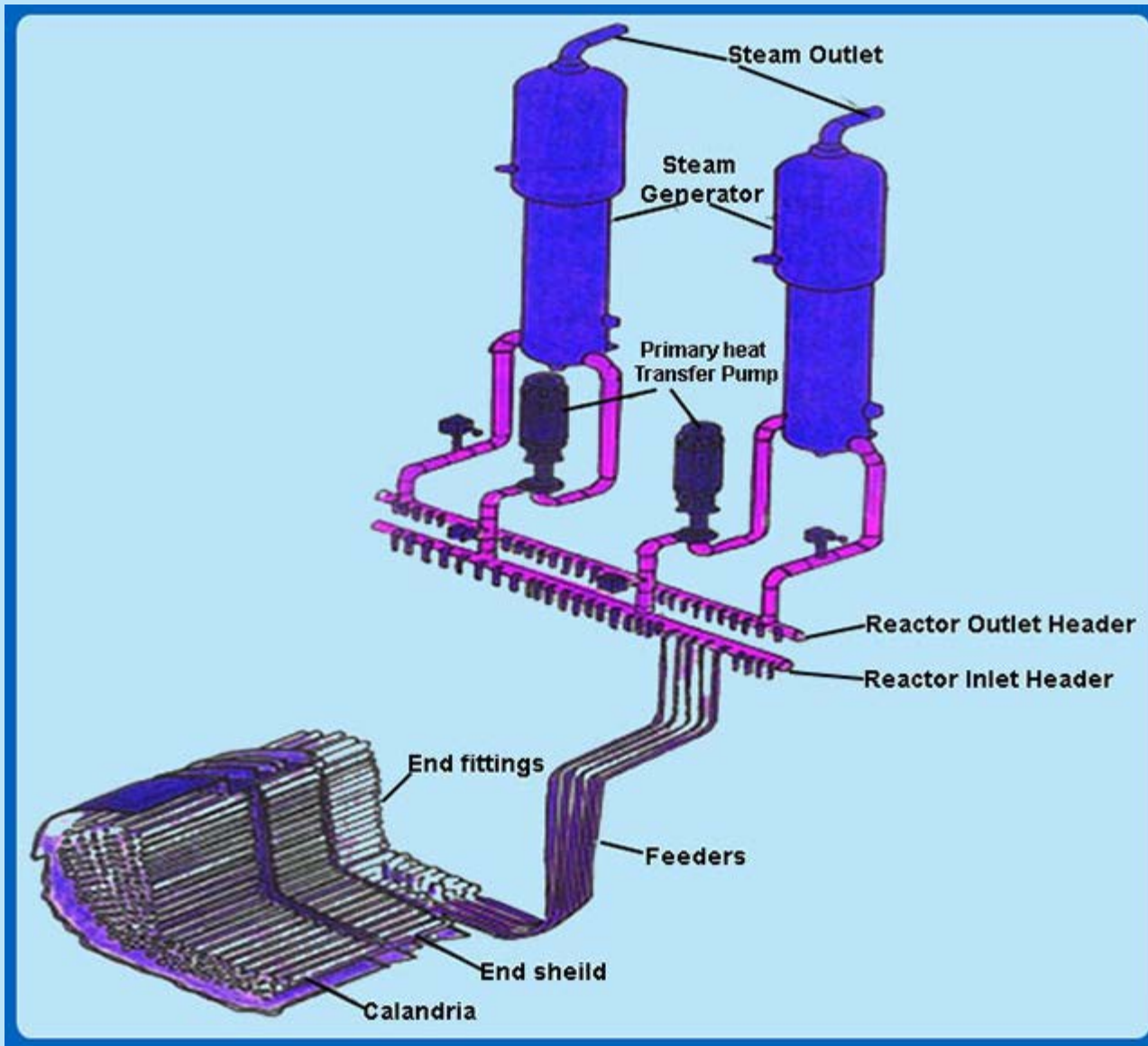


Fig.15 Typical Layout

Fuel Bundle

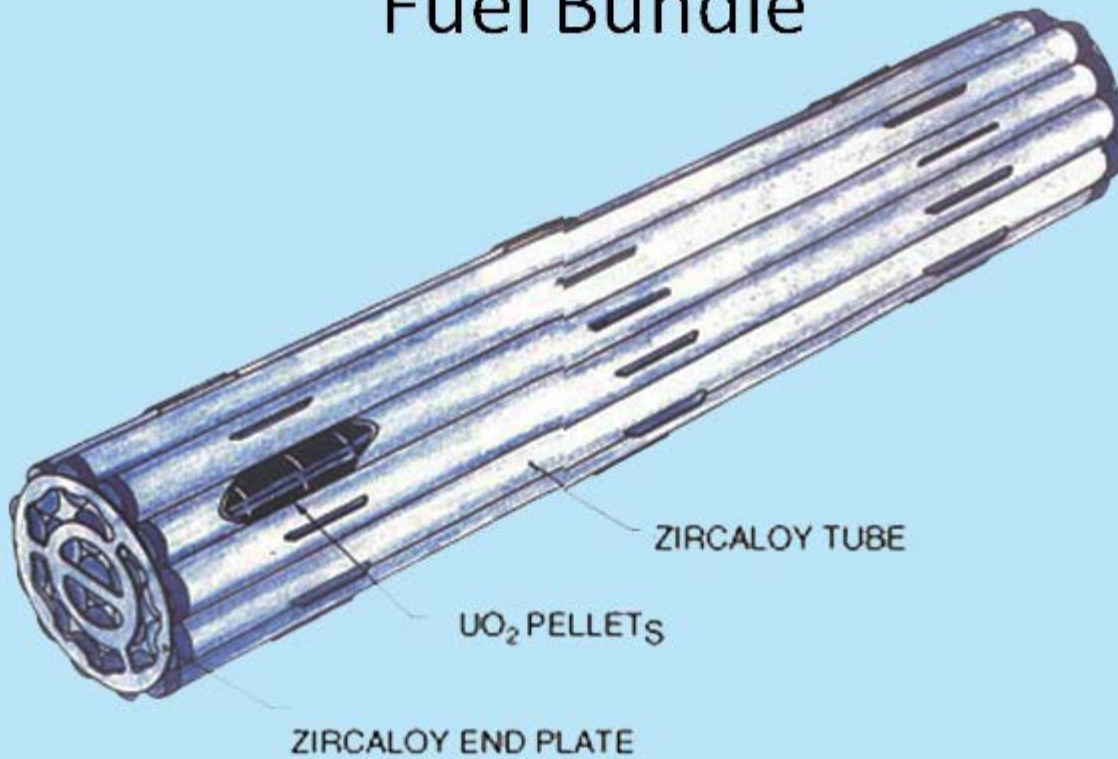


Fig.16 Fuel Bundle

- A schematic view of a fuel bundle is shown in fig. 16.
- It has 19 rods stuck together by end plates.
- Inside each rod, fuel pellet and gas plenum exists as in a PWR.
- Typically there are 10-15 fuel bundles in a coolant channel.
- These are kept short so that they can be removed and inserted easily.
- The heat generated in the fuel bundle increases the temperature of coolant.
- Each coolant channel feeds the hot water into exit feeders which merge into the exit header as shown in fig. 14.
- Note that this feeder is on the other side of the reactor (left in fig. 14).
- This fluid is now sent to the steam generator.
- Here it exchanges heat with secondary coolant (water) and generates steam.
- The coolant after transferring heat in steam generator is directed towards pump.
- The pump increases the pressure of the coolant and sends it to the inlet header.
- From the inlet header, the coolant is sent to coolant channel.
- Now the fluid goes from left to right.
- The coolant is sent to the exit header, then to steam generator and then to the pump, all

located on the right side.

- Thus the circuit is complete. It was noted that the coolant flows first from right to left and then left to right before the circuit is completed.
- This configuration is called figure of eight.
- In such cases, the components are duplicated on either side of the reactor.
- In a PHWR, the coolant channels are surrounded by moderator.
- The moderator is held in a vessel called Calandria.

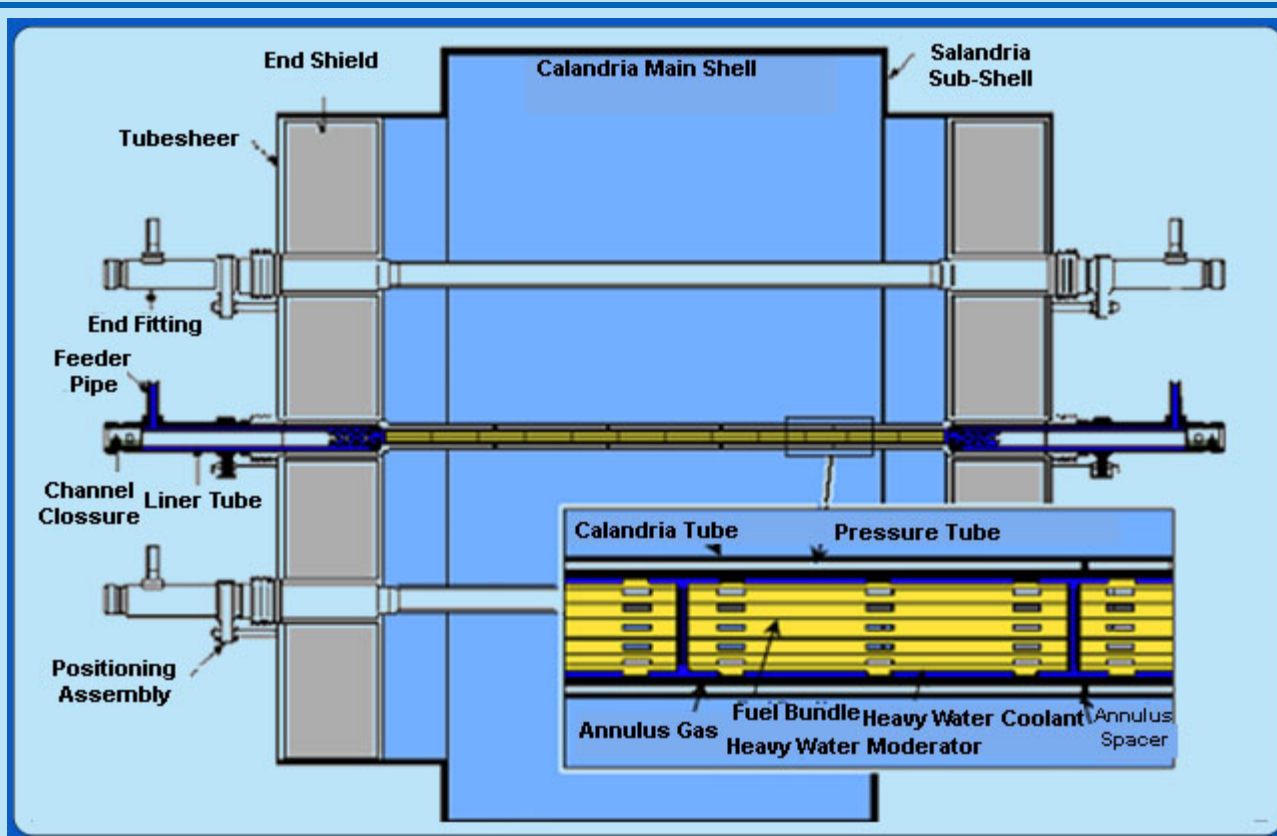


Fig.17 Calandria and Coolant Channel

- Figure 17 shows a schematic of calandria vessel and coolant channel.
- The coolant channel penetrate into the calandria vessel.
- Since the moderator is held around 60°C , while the coolant flows at around 275°C , an insulating air gap is provided.
- This is accomplished by providing a coaxial tube surrounding the coolant channel
- This tube is called the calandria tube.
- The coolant channel tube is often called the pressure tube.
- Thus we have the pressure tube, that is surrounded by calandria tube, with a gap in between the two. This is shown in fig.17.
- CO_2 is circulated in this gap. Outside this calandria tube is the moderator.

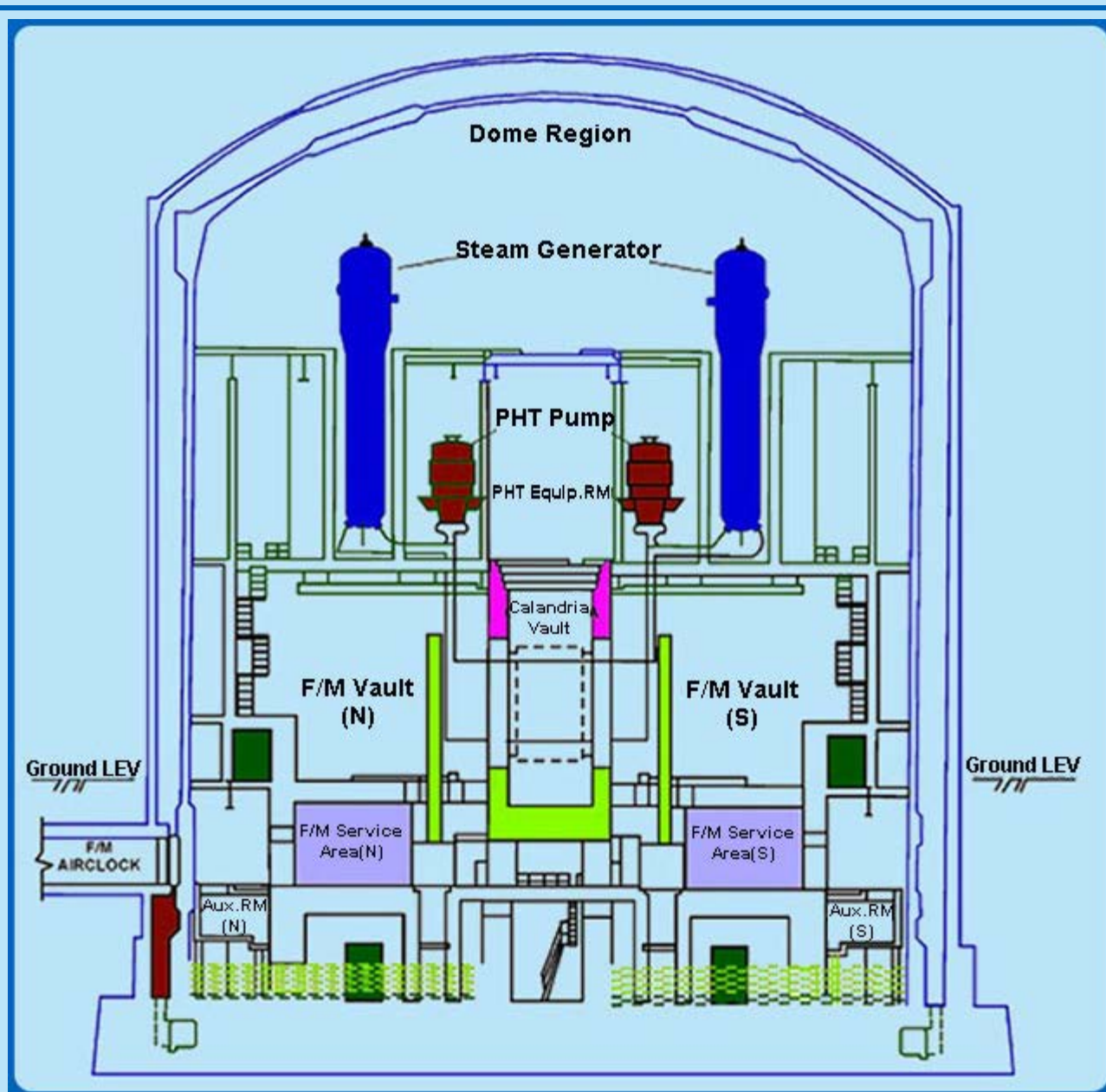


Fig.18 Containment and Equipment Layout

- A typical layout of the components in the containment of a PHWR is shown in fig. 18.

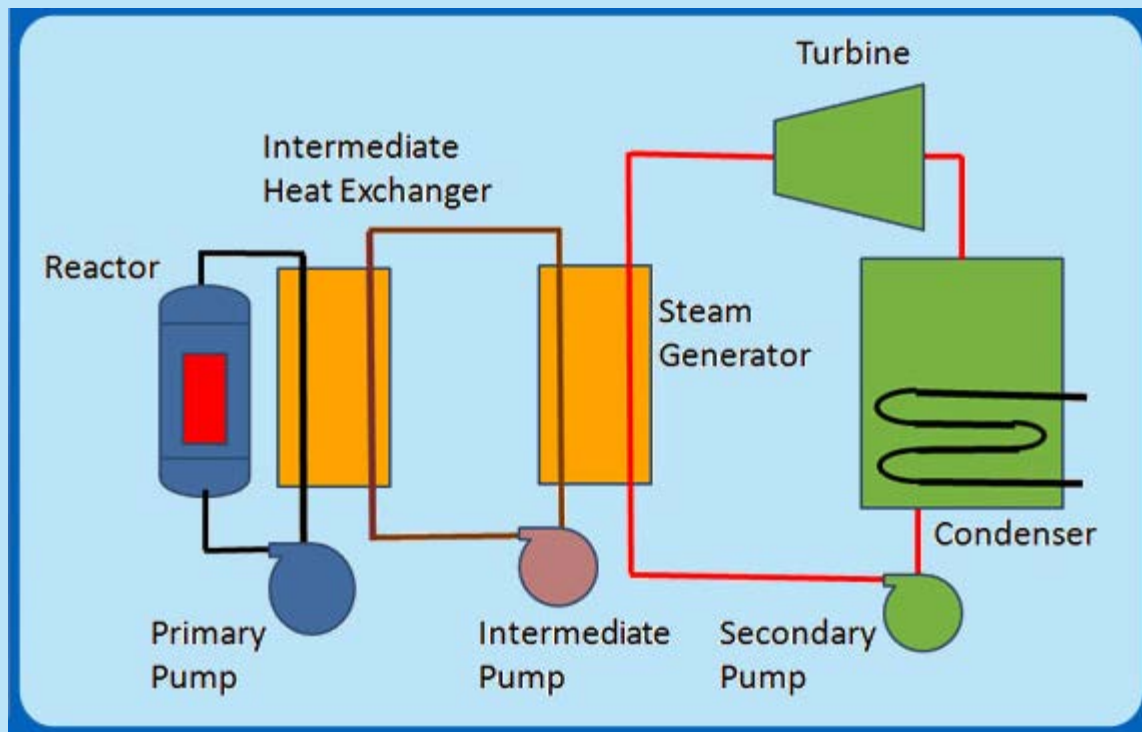


Fig.19 Fast Breeder Reactor

- As discussed earlier, if we do not slow down the neutron fully and allow it to react with fuel, then such a reactor is called fast reactor.
- We shall see later that fast reactors breed fuel and hence they are attractive.
- Since water is not allowed as a coolant, as it will slow down the neutrons, liquid metals are usually used as coolant.
- Sodium is the most commonly used as coolant.
- However as the turbines are run using steam, a sodium water steam generator has to be employed.
- Since sodium violently reacts with water, should there be a leak, fast breeders employ three circuits as shown in fig. 19.
- It consists of a primary loop, an intermediate loop and a secondary loop.
- In primary loop sodium transports the heat from core to the Intermediate Heat Exchanger(IHX).
- The primary pump circulates this fluid.
- The intermediate loop consists of a pump transporting sodium coolant that transports heat from IHX to the steam generator.
- In case of a leak in IHX only primary sodium and secondary sodium mix. Hence there will be no chemical reaction.
- In the secondary circuit, the steam is sent to turbine, expanded, condensed and sent back to the steam generator.
- In case of a leak in steam generator, only non radioactive sodium will react with water.

- This will prevent radioactive sodium leak.
- If IHX were not there, then radioactive sodium can react with water and result in radioactive sodium spill.



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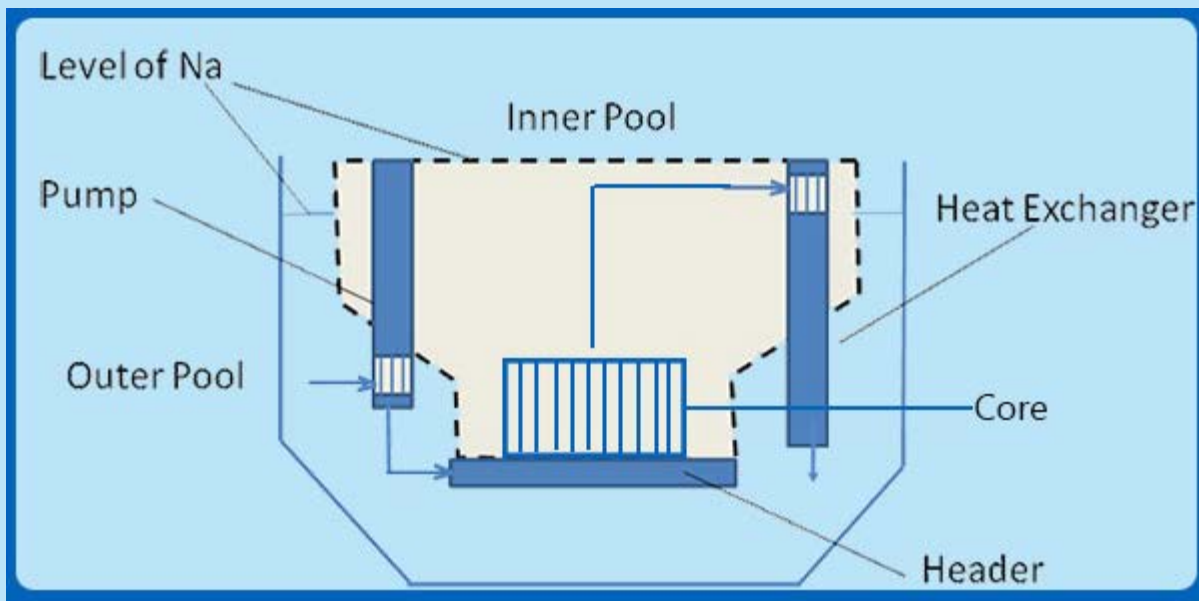


Fig.20 Pool Schematic

- Generally the power reactors use the pool concept. It consists of two pools, an inner pool and an outer pool as shown in fig. 20.
- The primary pump and the IHX are submerged in the pools.
- The pump sucks the fluid from the outer pool into the header and pumps it into the reactor core.
- The heated sodium flows in the inner pool and gets directed towards the IHX.
- It then flows from IHX into the outer pool completing the circuit. This is schematically shown in fig. 20.

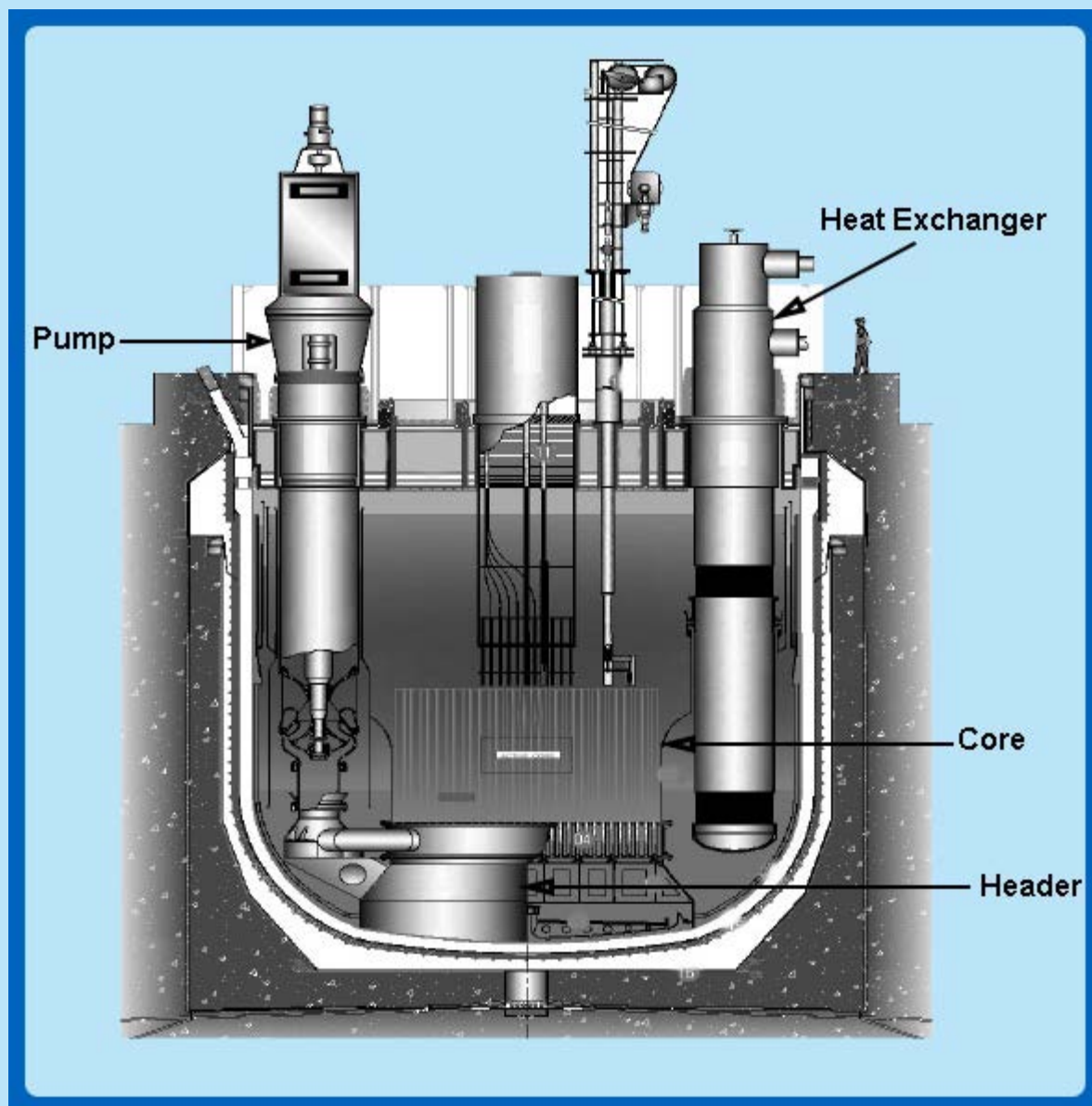


Fig. 21 Pool of Fast Breeder

- A comprehensive schematic view of the fast reactor is shown in fig. 21 with the components approximately to scale.
- Having gotten a feel of the features of the power reactors, we now move on to study the subject in detail.